# Residual Ternary Algebra and a modular foundation: the $s$-series, the Bridge Series $C$, and the 35-component law vector

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**Abstract**

Residual Ternary Algebra (RTA, also called Resonant Algebra) is built from three kinds of written objects: elementary packaging arithmetic (the ternary map $T_3$, the residual cardinality 539, the carrier $\mathcal{O}_{\text{res}}$), the residual series $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s;q^s)_n$, and the Bridge Series $C = P_7^{\text{orth}} - H_7$, an orthant-minus-cone sum on the indefinite lattice $\mathbb{Z}^2$ with Gram matrix $\left(\begin{smallmatrix} 2 & 7 \\ 7 & \end{smallmatrix}\right)$. This paper collects, with complete proofs, what can be established in tying these objects to a modular foundation, and what cannot.

Part I treats the $s$-series $g_s$ (written $g_k$ there): an exact expansion of the coefficients, families of zeros, sign laws for even $k$, and, for every odd $k \leq 33$, a parity-dominance theorem with complete zero sets (computer-assisted in stated steps). Part II treats the Bridge Series $C$: a ray-class decomposition over $\mathbb{Q}(\sqrt{35})$, radial asymptotics, a Baker-type obstruction to every scalar modular or mock residual law, and a vector-valued structure from a Siegel theta lift. Part III is new. It proves that the leading radial profile of $C$ lies outside the span $\mathcal{V}_{35}$ of the 35-component law vector, while the profile of the normalized series $q^{-7/20}C$ lies in $\mathcal{V}_{35}$ and obeys its vector-valued $S$-law; it gives the complete radial expansion of $q^{-7/20}C$ and of 105 companion series through order $t^0$ at every rational point, with closed formulas, an algebraicity statement at order $t^{-1/2}$, and an exact decomposition of that coefficient by quadratic class. The final part states precisely which links between RTA and this framework are proved, which written RTA objects ($T_3$, $\mathcal{O}_{\text{res}}$, 539, the packaging identities) have no written link to it, and where the shadow $\Xi$ would have to enter; the last question is open. No claim is made about the twin prime conjecture or the Riemann hypothesis.

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# Symbol key

| symbol | meaning | where defined |
|---|---|---|
| *General and preliminaries* | | |
| $q = \mathbf{e}(\tau)$, $\mathbf{e}(z)$ | $\mathbf{e}(z) = e^{2\pi i z}$ | §2 |
| $Q$, $B$, $A$ | binary quadratic form $n_1^2 + 7n_1n_2 + \frac{7}{2}n_2^2$, its bilinear form, Gram matrix | §2 |
| $Q_s$ | quadratic form of parameter $s$ | Def. 2.1 |
| $P_s^{\mathrm{orth}}$ | orthant theta series of $Q_s$ | Def. 2.1 |
| $\Pi_s$ | product series $(-q^s;q^s)_\infty g_s$ | Prop. 2.2 |
| $H_s$ | cone series with walls $c_1, c_2$ (never redefined) | §10 |
| $C$ | Bridge Series $P_7^{\mathrm{orth}} - H_7$ | §34 |
| $w(v)$ | cone weight $\frac{1}{2}(\operatorname{sgn} B(c_1,v) - \operatorname{sgn} B(c_2,v))$ | §34 |
| $I$ | $\log(6+\sqrt{35})/(2\sqrt{35})$ | Thm. 34.6 |
| *Part I* | | |
| $g_k$, $g_k(d)$ | residual series $\sum q^{n^2}/(-q^k;q^k)_n$, its coefficients | §3 |
| $P_n$, $\varphi_n(m)$ | $\prod_{j\le n}(1+x^j)^{-1}$ and its coefficients | §3 |
| $A_n(m)$, $Q(m)$ | $\lvert\varphi_n(m)\rvert$; partitions of $m$ into distinct odd parts | §4 |
| $Z_k$ | zero set of $g_k$ on allowed $d$ | §3 |
| $\mathcal{S}_k$, $\mathrm{OccFilt}_k(\infty)$ | square classes mod $k$; classes eventually nonzero | §3 |
| $\Xi_k$ | a finite set of integers (unrelated to the shadow $\Xi$) | §6.6 |
| *Part II* | | |
| $g_s$ | residual series at parameter $s$ | §25.2 |
| $V$ | 42-dim. span of the lift functions | §14 |
| $E$ | ray-class partial theta series | §12 |
| $C_{\mathrm{axis}}$, $C_{\mathrm{third}}$ | pieces of $C$ | §24 |
| $u_L$ | vector of the theta-lift family | §21 |
| $\Delta$ | wall parameter of the cone | §23.1 |
| $H_s^{\mathrm{axes}}$ | axis variant of $H_s$ | §23.3 |
| *Part III* | | |
| $G_{\alpha,\beta}$ | Gauss mean | Def. 33.1 |
| $D$ (set), $a$, $h_A$ | discriminant group; shift; $(\frac{1}{2},0)$ | §33 |
| $K1_\mu,\dots,K4_\mu$ | $G_{a+\mu,a}, G_{a+\mu,0}, G_{\mu,a}, G_{\mu,0}$ | Def. 33.2 |
| $(K2_\mu)$ | the 35-component law vector | Def. 33.2 |
| $K2_0$, $K3_p$ | components at $\mu=0$, $\mu=p$ | Def. 33.2 |
| $\mathcal{V}_{35}$ | 36-dim. span of $K2, K3, K4$ | Def. 33.2, Prop. 33.6 |
| $\mathcal{E}_{36}$, $\mathcal{C}_6$ | 36- and 6-dim. summands of Part II's 42-dim. lift rep. | Prop. 33.7 |
| $S_{\alpha,\beta}$, wt | companion series and weight $1_{\mathrm{orthant}} - w$ | Def. 34.1 |
| $q^{-7/20}C = S_{a,0}$ | normalized Bridge Series | §34 |
| $u_1, u_2$; $K_\pm$ | $(21,-1)$, $(0,1)$; cones split into 21 pieces | Prop. 34.3 |
| $\Lambda_{\alpha,\beta}$ | leading radial profile | Thm. 34.5 |
| Ar, $J_a$, $J_b$, $K(l)$ | Gaussian cone integrals | Lem. 35.3 |
| mean$(f)$, $W_i$, $V_i$, $\mathsf{C}$ | periodic mean, sawtooth, second moment, corner terms | Thm. 35.4 |
| $A_0, A_1, A_2$ | coefficients of $t^{-1}, t^{-1/2}, t^0$ | Thm. 35.6 |
| $D$ (term), $R$, $W[\lambda]$ | drift, ray, sawtooth parts of $A_1$ | Prop. 35.11 |
| $R[\frac{7}{2}]$, $R[\frac{595}{2}]$ | ray terms on $\pm\mathbb{N}(0,1)$, $\pm\mathbb{N}(21,-1)$ | Prop. 35.11 |
| $\lambda \in \{1, \frac{7}{2}, \frac{595}{2}\}$ | $Q$-values $Q(e_1)$, $Q(0,1)$, $Q(21,-1)$ | Prop. 35.11 |
| $Z_0$ | zero set of $A_0$ in $X_{216}$ | Prop. 35.10 |
| $\Xi_\varepsilon$ | sawtooth shadow candidate (Category B) | §36 |
| *Part IV* | | |
| $T_3$ | ternary map $3k\mapsto k$, $3k{+}1\mapsto 4k{+}2$, $3k{+}2\mapsto 2k{+}1$ | Prop. 37.6 |

| symbol | meaning | where defined |
|---|---|---|
| $\mathcal{O}_{\mathrm{res}}$ | carrier $\{21 + 9j : 0 \le j \le 538\}$ | Rem. 37.7 |
| 539 | residual cardinality | Prop. 37.6 |
| $\Xi$ | RTA weight-$\frac{3}{2}$ shadow (not identified) | §37 |

# 1 Introduction

## 1.1 Goal

The aim of this work is to tie Residual Ternary Algebra (RTA) to a foundation: to identify which RTA objects are governed by modular, mock-modular or vector-valued modular structure, and to prove each such statement. RTA, as written, has no algebraic axioms. Its objects are (a) elementary arithmetic of the ternary map $T_3$ and of the residual cardinality 539 (packaging identities, the equitable partition $539 = 3 \cdot 68 + 5 \cdot 67$, the carrier $\mathcal{O}_{\text{res}}$); (b) the residual series $g_s$; (c) the Bridge Series $C$ and its pieces $C_{\text{axis}}$, $C_{\text{third}}$; and (d) a weight-$\frac{3}{2}$ shadow $\Xi$ and a weight-$\frac{1}{2}$ period-function candidate, described in words but not defined as explicit series. A "foundation" here means a theorem placing an object of (a)–(d) inside a structure with known transformation laws under $\mathrm{SL}_2(\mathbb{Z})$.

## 1.2 What is achieved

(1) For the $s$-series (Part I): the coefficient expansion of $g_s$, its zero families, sign laws for even $s$, and complete zero sets and residue-class occupancy for odd $s \leq 33$. In particular the residue law of $g_7$ recorded in RTA follows from the expansion, and its converse holds (Corollary 37.2).

(2) For the Bridge Series (Part II): $C$ and $C_{\text{third}}$ share the orthant remainder $P_7^{\text{orth}}$; up to theta and mock pieces this is a partial theta series of one ray class of $\mathbb{Q}(\sqrt{35})$; a Baker-type argument excludes every scalar modular or mock residual law, including every scalar $W_7$ slash law; the leading radial data carry a vector-valued structure coming from a Siegel theta lift.

(3) For the 35-component law vector (Part III, new): the leading profile of $C$ is $I\,\mathbf{e}(7x/20)K2_0(x)$ and lies outside $\mathcal{V}_{35}$; the normalized series $q^{-7/20}C$ has profile $I\,K2_0 \in \mathcal{V}_{35}$ and satisfies the vector-valued $S$-law of $\mathcal{V}_{35}$; the radial expansion of $q^{-7/20}C$ and of 105 companion series is computed in closed form through order $t^0$ at every rational point.

(4) For the foundation question (§37): every written link between RTA and the framework above is listed with its status.

## 1.3 What is not achieved

(1) No written object links $T_3$, $\mathcal{O}_{\text{res}}$, the number 539 or the packaging identities to $g_s$, $C$ or $\mathcal{V}_{35}$. These parts of RTA are not tied to the modular framework by anything proved here.

(2) The shadow $\Xi$ is not identified. The vector-valued law of $\mathcal{V}_{35}$ holds at leading order only; at order $t^{-1/2}$ no law of the same kind is known, and the numerical evidence (Appendix D) is against closure inside the companion span.

(3) No link to the $g_s$ family is proved beyond the shared definition and the residue law.

(4) Nothing here concerns the twin prime conjecture, the Riemann hypothesis, or any security property of a hash construction.

## 1.4 Status conventions

Every statement in a theorem, proposition, lemma or corollary is **Category A**: proved here; or proved by a finite exact computation (integer, rational, modular or cyclotomic arithmetic, including

ranks modulo a prime together with the argument that transfers them to characteristic 0); or proved by a finite computation in rigorous ball or interval arithmetic (labelled *computer-assisted*); or proved conditional on a cited published theorem, named each time. **Category B** statements (floating-point evidence, fits, singular-value ranks, patterns, interpretations) appear only in remarks labelled "Remark (Category B)", in status boxes marked Category B, or in the appendices. Part II uses status boxes after its results; their wording is part of the record.

### 1.5 Organisation

Section 2 fixes common notation and resolves notational conflicts between the parts. Part I (Sections 3–8) is the $s$-series. Part II (Sections 9–32) is the Bridge Series. Part III (Sections 33–36) is the 35-component law vector and the normalized Bridge Series. Section 37 treats the foundation question and Section 38 lists open problems. Appendices A–C contain the computational records of Parts I–II, and Appendix D the Category B record of Part III.

## 2 Preliminaries and unified notation

### 2.1 Common conventions

$\mathbb{N} = \{0, 1, 2, \dots\}$; $\mathbf{e}(x) = e^{2\pi i x}$; $q = \mathbf{e}(\tau)$ with $\tau \in \mathbb{H}$. "Radial" means $q = \mathbf{e}(x)e^{-t}$ with $x \in \mathbb{Q}$ and $t \to 0^+$, so that $q^\lambda = \mathbf{e}(\lambda x)e^{-\lambda t}$ for real $\lambda$. $(a;q)_n = \prod_{j=0}^{n-1}(1 - aq^j)$. $B_1(y) = y - \frac{1}{2}$, $B_2(y) = y^2 - y + \frac{1}{6}$, $B_3(y) = y^3 - \frac{3}{2}y^2 + \frac{1}{2}y$ are Bernoulli polynomials and $\bar{B}_j(y) = B_j(\{y\})$ their periodic versions.

### 2.2 The lattice

$A = \left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$, $B(x, y) = x^{\mathsf{T}}Ay$, $Q(x) = \frac{1}{2}B(x, x) = x_1^2 + 7x_1x_2 + \frac{7}{2}x_2^2$. The form $Q$ has signature $(1, 1)$ and $\det A = -35$. $D = A^{-1}\mathbb{Z}^2/\mathbb{Z}^2$ has order 35. $a = (\frac{7}{10}, -\frac{1}{5})$, so $Aa = (0, \frac{7}{2})$ and $Q(a) = -\frac{7}{20}$. For general $s$: $A_s = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$ and

$$Q_s(n, k) = n^2 + snk + \tfrac{s}{2}k(k + 1).$$

At $s = 7$, $Q_7(v) = Q(v) + \frac{7}{2}v_2 = Q(v + a) + \frac{7}{20}$.

### 2.3 The residual series and the two series previously both written $P_s$

**Definition 2.1.** For $s \geq 1$:

(i) the *$s$-series* $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s; q^s)_n$ (in Part I written $g_k$, with $k$ in place of $s$);

(ii) the *product-normalised $s$-series* $\Pi_s(q) = (-q^s; q^s)_\infty\, g_s(q)$;

(iii) the *orthant series* $P_s^{\mathrm{orth}}(q) = \sum_{n,k\geq 0} q^{Q_s(n,k)}$.

In the RTA manuscript both $\Pi_s$ and $P_s^{\mathrm{orth}}$ carry the name $P_s$. They are different series:

**Proposition 2.2.** *For every $s \geq 1$, in $\mathbb{Z}[[q]]$,*

$$\Pi_s(q) = \sum_{n,k\geq 0} \frac{q^{Q_s(n,k)}}{(q^s; q^s)_k} = \sum_{k\geq 0} \frac{q^{sk(k+1)/2}}{(q^s; q^s)_k} \sum_{n\geq 0} q^{n^2+skn},$$

*and $\Pi_s - P_s^{\mathrm{orth}}$ has non-negative coefficients and is non-zero; its coefficient of $q^{2s}$ is at least 1.*

*Proof.* Euler's identity $(-z;p)_\infty = \sum_{k\geq 0} p^{k(k-1)/2} z^k/(p;p)_k$ with $p = q^s$ and $z = q^{s(n+1)}$ gives $(-q^s;q^s)_\infty/(-q^s;q^s)_n = (-q^{s(n+1)};q^s)_\infty = \sum_{k\geq 0} q^{sk(k-1)/2+s(n+1)k}/(q^s;q^s)_k$, and $n^2+sk(k-1)/2+s(n+1)k = Q_s(n,k)$. Multiplying by $q^{n^2}$ and summing over $n$ gives the first identity; both double sums converge in $\mathbb{Z}[[q]]$ because $Q_s(n,k) \geq n^2 + \frac{s}{2}k(k+1)$. Since $1/(q^s;q^s)_k - 1$ has non-negative coefficients, so does $\Pi_s - P_s^{\mathrm{orth}} = \sum_{n,k} q^{Q_s(n,k)}(1/(q^s;q^s)_k - 1)$; the term $(n,k) = (0,1)$ contributes $q^s \cdot q^s$ with coefficient 1. □

Throughout, the Bridge Series is built from $P_s^{\mathrm{orth}}$ only. Part II uses $P_s^{\mathrm{orth}}$ for $P_7^{\mathrm{orth}}$ when $s = 7$ is fixed.

### 2.4 The Bridge Series

$c_1 = (-1,1)$, $c_2 = (-4,1)$, $w(v) = \frac{1}{2}(\operatorname{sgn} B(c_1,v) - \operatorname{sgn} B(c_2,v))$ with $\operatorname{sgn} 0 = 0$, and

$$H_s = \sum_{v\in\mathbb{Z}^2} w(v) q^{Q_s(v)}, \qquad C = P_7^{\mathrm{orth}} - H_7.$$

$H_s$ is the cone series of Part II, §10, with the fixed walls $c_1, c_2$ for every $s$; it is not redefined anywhere in this paper. This is the Bridge Series of RTA (Definition 9.1 there), and it is the object $C$ of Part II.

### 2.5 Notational conflicts and their resolution

Each part keeps its own local notation where this causes no ambiguity. The following symbols have different meanings in different parts.

| symbol | meaning | other meaning, and its name here |
|---|---|---|
| $P_s$ | not used | RTA: both $\Pi_s$ and $P_s^{\mathrm{orth}}$ (Proposition 2.2) |
| $P_n(x)$ | Part I: $\prod_{j\leq n}(1+x^j)^{-1}$ | – |
| $V$ | Part II: the 42-dimensional span of the lift functions | Part III: $\mathcal{V}_{35}$, the 36-dimensional span of the 35-component law vector |
| $E$ | Part II: the ray-class partial theta series | Part III: $\mathcal{E}_{36}$, a 36-dimensional summand of a representation |
| $C$ | the Bridge Series | Part III: $\mathcal{C}_6$, a 6-dimensional summand of a representation |
| $\Xi$ | §37: the RTA shadow | Part I: $\Xi_k$, a finite set of integers |
| $Q$ | the quadratic form above | Part I: $Q(m)$, the number of partitions of $m$ into distinct odd parts |
| $N$ | Part III: the period $2m$ | Part II: the label negation map |
| $k$ | Part I: the parameter of $g_k$ | Parts II, III: the second lattice coordinate |
| $G$ | Parts II, III: the Gauss mean of $Q_7$ | Part I, §6.3: an eta-quotient |

# Part I
# The $s$-series

This part treats $g_k(q) = \sum_{n\geq 0} q^{n^2}/(-q^k;q^k)_n$, the $s$-series of RTA with $s = k$. Its notation ($P_n$, $\varphi_n$, $c_m$, $Q(m)$, $E_d$, $O_d$, $\Xi_k$) is local to this part. Results proved with a computer state the computation and what it relies on; the dependence on one cited explicit bound is stated where used.

### 2.5.1 Summary of Part I

| Result | Statement | Nature of proof |
|---|---|---|
| Prop. 3.2 | $g_k(d) = \sum_{n^2\leq d,\, n^2\equiv d\,(k)} \varphi_n\big(\frac{d-n^2}{k}\big)$ | algebraic |
| Lemmas 4.1–4.6 | $\varphi_n(m) = c_m$ for $m \leq n$; $c_m = (-1)^m Q(m)$; $(-1)^m\varphi_n(m) \geq 0$; product form (4.1) | algebraic |
| Prop. 4.7 | odd $k$: $(-1)^d g_k(d) = E_d - O_d$ | algebraic |
| Thm. 5.1 | $k \geq 2$ squarefree $\Rightarrow g_k(k(k+2)) = 0$ | algebraic |
| Thm. 5.3, Cor. 5.4, Thm. 5.5 | for odd prime powers $k$: formula for $g_k(k^2+jk+b^2)$, $b = 1, 2$; zeros at $(k+2)^2-3$ ($k \geq 5$), $k^2+2k+4$ ($k \equiv 3 \bmod 4$, $k \neq 3, 7$), $k^2+3k+4$ ($k \equiv 1 \bmod 4$, $k \neq 5$); exact failures; complete zero classes | algebraic |
| Thm. 6.9 | $\omega(k) = 2$, $k$ odd: $g_k((k+2)^2-3) \neq 0$ except $k = 15, 21$ | algebraic + exact integer certificate for $s \leq 2500$ + cited bound [20] for $s \geq 2501$ |
| Thm. 7.4, Cor. 7.5, Thm. 7.6 | one-sign classes: explicit bound $B_c$ beyond which $g_k \neq 0$ with a definite sign; $k \equiv 2 \pmod 4$: $Z_k$ finite; $g_2$: sign $(-1)^{\lfloor d/2\rfloor}$, $Z_2 = \{2, 8, 10, 26\}$ | algebraic |
| Thm. 7.7 | even $k \geq 8$: $8 \nmid k \Rightarrow g_k((k+2)^2-3) \geq 2$; $k = 2p^a \Rightarrow = 2$; $k = 2^a \Rightarrow < 0$ | algebraic |
| Thm. 8.1 | odd $k \leq 33$: $E_d \leq (R_k + 10^{-6})O_d$ with $R_k < 0.906$ for allowed $d \geq D_0(k)$; $Z_k$ complete; $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$ | analytic reduction + finite computer-assisted verification (interval arithmetic, modular arithmetic) |

**Scope.** Only proved statements are included. For even $k$ no claim is made on classes of $d$ that are not one-sign. For odd $k \geq 35$ no claim is made beyond Sections 4–6.

**Dependence on external results.** Theorem 6.9 for $s \geq 2501$ and the monotonicity of $Q(n)$ for $n \geq 35200$ (Lemma 6.5) rely on the explicit bound of Chern [20] (Theorem 1.1, error term (3.10), Remark 3), cited in Section 6.3. Computer-assisted steps use the Arb ball arithmetic [21] and exact integer arithmetic; they are listed in Section 8.8 and Appendix A.

# 3 Definitions

**Conventions.** $\mathbb{N} = \{0, 1, 2, \dots\}$. All series are formal power series with integer coefficients; $[x^m]F$ is the coefficient of $x^m$ in $F$. For a positive integer $k$ and an integer $n \geq 0$ put $(-q^k; q^k)_n = \prod_{j=1}^{n}(1 + q^{kj})$.

**Definition 3.1.** For $k \geq 1$,

$$g_k(q) = \sum_{n \geq 0} \frac{q^{n^2}}{(-q^k; q^k)_n} = \sum_{d \geq 0} g_k(d)\, q^d.$$

For $n \geq 0$ let

$$P_n(x) = \prod_{j=1}^{n} \frac{1}{1 + x^j} = \sum_{m \geq 0} \varphi_n(m)\, x^m, \qquad P_\infty(x) = \prod_{j \geq 1} \frac{1}{1 + x^j} = \sum_{m \geq 0} \varphi_\infty(m)\, x^m,$$

with $P_0 = 1$.

The series for $g_k$ converges in $\mathbb{Z}[[q]]$ because the $n$-th summand is $q^{n^2} P_n(q^k)$ and has order $n^2$.

**Proposition 3.2 (expansion in $d$).** For all $k \geq 1$ and $d \geq 0$,

$$g_k(d) = \sum_{\substack{n \geq 0,\ n^2 \leq d \\ n^2 \equiv d \pmod{k}}} \varphi_n\!\left(\frac{d - n^2}{k}\right). \tag{3.1}$$

*Proof.* The $n$-th summand of $g_k(q)$ is $q^{n^2} P_n(q^k) = \sum_{m \geq 0} \varphi_n(m) q^{n^2 + km}$. The coefficient of $q^d$ collects the pairs $(n, m)$ with $n^2 + km = d$. ∎

The summand of index $n$ in (3.1) is called the *term* $n$ of $g_k(d)$; it is defined when $n^2 \leq d$ and $k \mid d - n^2$, and $m_n = (d - n^2)/k$ is its *index*.

**Definition 3.3.** Let $\mathcal{S}_k = \{\, n^2 \bmod k : \ n \in \mathbb{Z} \,\} \subseteq \mathbb{Z}/k\mathbb{Z}$. An integer $d \geq 0$ is *allowed* for $k$ if $d \bmod k \in \mathcal{S}_k$.

$$Z_k = \{\, d \geq 0 \text{ allowed for } k : \ g_k(d) = 0 \,\}.$$

$\mathrm{OccFilt}_k(\infty)$ is the set of residue classes $r \in \mathbb{Z}/k\mathbb{Z}$ such that $g_k(d) \neq 0$ for all sufficiently large $d \equiv r \pmod{k}$.

**Proposition 3.4.** (i) If $d$ is not allowed for $k$ then $g_k(d) = 0$. (ii) $\mathrm{OccFilt}_k(\infty) \subseteq \mathcal{S}_k$, with equality if and only if $Z_k$ is finite.

*Proof.* (i) If $d \bmod k \notin \mathcal{S}_k$, no $n$ satisfies $n^2 \equiv d \pmod{k}$, so the sum (3.1) is empty. (ii) A class $r \notin \mathcal{S}_k$ has $g_k(d) = 0$ for all $d \equiv r$, hence $r \notin \mathrm{OccFilt}_k(\infty)$. For $r \in \mathcal{S}_k$, $r \in \mathrm{OccFilt}_k(\infty)$ if and only if $Z_k$ contains only finitely many $d \equiv r$. Since there are finitely many classes, equality holds if and only if $Z_k$ is finite. ∎

By Proposition 3.4, the statement $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$ ("every square class occurs for all large $d$") is the statement that $Z_k$ is finite. Section 8 proves it for every odd $k \leq 33$, with $Z_k$ computed exactly.

# 4 The polynomials $\varphi_n$

## 4.1 Stabilisation, product form, sign rule

**Lemma 4.1 (stabilisation).** For all $n \geq 0$ and $0 \leq m \leq n$, $\varphi_n(m) = \varphi_\infty(m)$.

*Proof.* $P_n(x) = P_\infty(x)\prod_{j>n}(1+x^j)$ and $\prod_{j>n}(1+x^j) \equiv 1 \pmod{x^{n+1}}$. ■

Write $c_m = \varphi_\infty(m)$. By Lemma 4.1, $\varphi_n(m) = c_m$ whenever $m \leq n$.

**Lemma 4.2 (product form).** For $n \geq 0$, as formal power series,

$$\prod_{j=1}^{n} \frac{1}{1+(-x)^j} = \prod_{\substack{j \leq n \\ j \text{ odd}}} (1+x^j) \prod_{\lfloor n/2 \rfloor < l \leq n} \frac{1}{1-x^{2l}}. \tag{4.1}$$

The left side equals $\sum_m (-1)^m \varphi_n(m) x^m$. Consequently, with $a_n(m) = [x^m]$ of the right side,

$$(-1)^m \varphi_n(m) = a_n(m) \geq 0, \qquad a_n(m) = \#\{(\lambda, \mu)\}, \tag{4.2}$$

where $(\lambda, \mu)$ runs over pairs consisting of a set $\lambda$ of distinct odd integers in $[1, n]$ and a multiset $\mu$ of even integers in $(n, 2n]$, with $\sum \lambda + \sum \mu = m$.

*Proof.* Replacing $x$ by $-x$ in $P_n$ gives the left side and $\sum_m \varphi_n(m)(-x)^m$. For odd $j$, $1+(-x)^j = 1-x^j$ and $\frac{1}{1-x^j} = \frac{1+x^j}{1-x^{2j}}$. For $j = 2i$, $\frac{1}{1+x^{2i}} = \frac{1-x^{2i}}{1-x^{4i}}$. Multiplying over $j \leq n$, the numerators contribute $\prod_{j \leq n,\, j \text{ odd}}(1+x^j)\prod_{i \leq \lfloor n/2 \rfloor}(1-x^{2i})$ and the denominators contribute $\prod_{j \leq n,\, j \text{ odd}}(1-x^{2j})\prod_{i \leq \lfloor n/2 \rfloor}(1-x^{4i}) = \prod_{j \leq n}(1-x^{2j})$. The factors $1-x^{2i}$, $i \leq \lfloor n/2 \rfloor$, cancel, which leaves (4.1). The right side of (4.1) is a product of series with nonnegative coefficients; its coefficient count is (4.2), because $l \in (n/2, n]$ corresponds to the even integers $2l \in (n, 2n]$. ■

We write $A_n(m) = |\varphi_n(m)| = a_n(m)$ and $A_n(x) = \sum_m a_n(m) x^m$ (the series (4.1)); the meaning is clear from the argument.

Let $Q(m)$ denote the number of partitions of $m$ into distinct odd parts ($Q(0) = 1$).

**Lemma 4.3.** $P_\infty(x) = \prod_{j \text{ odd}}(1-x^j)$. Hence $c_m = (-1)^m Q(m)$, and $(-1)^m \varphi_n(m) = Q(m)$ for $m \leq n$. The values are

$$(c_0, \ldots, c_{12}) = (1, -1, 0, -1, 1, -1, 1, -1, 2, -2, 2, -2, 3).$$

*Proof.* $\frac{1}{1+x^j} = \frac{1-x^j}{1-x^{2j}}$ and $\prod_{j \geq 1} \frac{1-x^j}{1-x^{2j}} = \prod_{j \text{ odd}}(1-x^j)$, the factors with $j$ even cancelling against the denominators. Expanding, a partition of $m$ into $r$ distinct odd parts contributes $(-1)^r = (-1)^m$ (each odd part is $\equiv 1 \bmod 2$). Lemma 4.1 gives the rest. The values follow from $Q(0..12) = 1, 1, 0, 1, 1, 1, 1, 1, 2, 2, 2, 2, 3$. ■

**Lemma 4.4.** $Q(2) = 0$; $Q(m) \geq 1$ for $m \neq 2$; $Q(m) \geq 2$ for $m \geq 8$.

*Proof.* 2 is not a sum of distinct odd integers. For $m \geq 1$ odd, $\{m\}$ is a partition; for $m \geq 4$ even, $\{1, m-1\}$ is one; $Q(0) = 1$. For $m \geq 9$ odd, $\{m\}$ and $\{1, 3, m-4\}$ are distinct partitions ($m - 4 \geq 5$). For $m \geq 8$ even, $\{1, m-1\}$ and $\{3, m-3\}$ are distinct partitions ($m - 3 \geq 5$, so $m - 3 \neq 3$). ■

**Lemma 4.5 (truncation formula).** For $n \geq 0$ and $0 \leq m \leq 2n+1$,

$$\varphi_n(m) = c_m + \sum_{x=0}^{m-n-1} (-1)^x Q(x), \qquad A_n(m) = Q(m) + (-1)^m \sum_{x=0}^{m-n-1} (-1)^x Q(x), \tag{4.3}$$

where an empty sum is 0. In particular $\varphi_n(m) = c_m$ for $m \leq n$.

*Proof.* $P_n = P_\infty \prod_{j>n}(1+x^j)$. A product of two or more distinct factors $x^j$ with $j > n$ has degree $\geq (n+1)+(n+2) = 2n+3$. Hence modulo $x^{2n+2}$, $\prod_{j>n}(1+x^j) \equiv 1 + \sum_{b=n+1}^{2n+1} x^b$. Therefore

for $m \le 2n+1$, $\varphi_n(m) = c_m + \sum_{b=n+1}^{m} c_{m-b} = c_m + \sum_{x=0}^{m-n-1} (-1)^x Q(x)$. Multiplying by $(-1)^m$ and using (4.2) gives the second formula. ■

**Lemma 4.6 (small $n$).** $\varphi_0(m) = \delta_{m,0}$; $\varphi_1(m) = (-1)^m$; and $\varphi_2(m) = 1, -1, 0, 0$ for $m \equiv 0, 1, 2, 3 \pmod 4$.

*Proof.* $P_1 = 1/(1+x) = \sum (-x)^m$. $(1+x)(1+x^2) = (1-x^4)/(1-x)$, so $P_2 = (1-x)/(1-x^4) = \sum_{i \ge 0} (x^{4i} - x^{4i+1})$. ■

### 4.2 The case of odd $k$

**Proposition 4.7 ($E$–$O$ identity).** Let $k$ be odd and $d \ge 0$. For each term $n$ of $g_k(d)$ with index $m_n$, $(-1)^d \varphi_n(m_n) = (-1)^n a_n(m_n)$. Hence

$$(-1)^d g_k(d) = E_d - O_d, \qquad E_d = \sum_{n \text{ even}} a_n(m_n), \quad O_d = \sum_{n \text{ odd}} a_n(m_n), \tag{4.4}$$

the sums running over the terms $n$ of $g_k(d)$; all summands are $\ge 0$.

*Proof.* $d = n^2 + k m_n$ with $n^2 \equiv n$ and $k m_n \equiv m_n \pmod 2$ ($k$ odd), so $(-1)^d = (-1)^{n+m_n}$. By (4.2), $(-1)^{m_n} \varphi_n(m_n) = a_n(m_n)$. Multiply: $(-1)^d \varphi_n(m_n) = (-1)^n a_n(m_n)$. Summing over $n$ and using (3.1) gives (4.4). ■

In particular, for odd $k$ and allowed $d$: $g_k(d) \ne 0$ if and only if $E_d \ne O_d$, and the term of index $n$ has sign $(-1)^{d+n}$.

## 5 Zero families

### 5.1 Squarefree $k$

**Theorem 5.1.** If $k \ge 2$ is squarefree, then $g_k\big(k(k+2)\big) = 0$.

*Proof.* Let $d = k^2 + 2k$. A term $n$ of $g_k(d)$ has $n^2 \le d < (k+1)^2$, so $n \le k$, and $k \mid n^2$. Since $k$ is squarefree, $k \mid n^2$ implies $k \mid n$, so $n \in \{0, k\}$. For $n = 0$ the index is $m = k + 2 \ge 1$ and $\varphi_0(k+2) = 0$ (Lemma 4.6). For $n = k$ the index is $m = 2 \le k$, so $\varphi_k(2) = c_2 = 0$ (Lemma 4.1, Lemma 4.3). Hence $g_k(d) = 0$. ■

*Remark 5.2.* The hypothesis $k \ge 2$ is needed: $g_1(3) = \varphi_1(2) = 1$. Exact evaluation of $g_k(k(k+2))$ for every $k = 2, \dots, 1001$ gives 0 for all 608 squarefree $k$ and a nonzero value for each of the 392 non-squarefree $k$ (for $k \le 30$ non-squarefree: $g_4 = -1$, $g_8 = 1$, $g_9 = 2$, $g_{12} = -2$, $g_{16} = -2$, $g_{18} = 6$, $g_{20} = -5$, $g_{24} = 6$, $g_{25} = 6$, $g_{27} = 8$, $g_{28} = -8$ at $d = k(k+2)$). This is a computation (Appendix A.1).

### 5.2 Values at $d = k^2 + jk + b^2$

**Theorem 5.3.** Let $b \in \{1, 2\}$, $j \in \mathbb{Z}$, and let $k = p^a$ be an odd prime power with

$$k \ge K := \max(j + 3b,\ 2b + 1,\ -j).$$

Then for $d = k^2 + jk + b^2$,

$$g_k(d) = \varphi_b(k + j) + [\, j \ge -2b \,]\, c_{j+2b} + [\, j \ge 2b \,]\, c_{j-2b}, \tag{5.1}$$

where $[\cdot]$ is the Iverson bracket and $c_m$ is as in Lemma 4.3 (with $c_m$ for $m \ge 0$ only occurring when its bracket is 1).

*Proof.* Let $n$ be a term of $g_k(d)$: $n^2 \equiv d \equiv b^2 \pmod{p^a}$, so $p^a \mid (n-b)(n+b)$. A common prime factor of $n - b$ and $n + b$ divides $2b \in \{2, 4\}$, hence is not $p$. Thus $p^a$ divides one of the two factors:

$n \equiv \pm b \pmod{k}$. The candidates are $n \in \{b,\ k-b,\ k+b,\ 2k-b,\ 2k+b, \dots\}$ (as $k > 2b$, these are distinct and increasing in this order, and $b \geq 1$).

*Large $n$ do not occur.* For $n \geq 2k-b$ we would need $(2k-b)^2 \leq d$, i.e. $3k \leq j+4b$. But $k \geq j+3b$ gives $j+4b \leq k+b$, and $3k \leq k+b$ contradicts $k \geq 2b+1$.

*The three remaining candidates.*

- $n = b$: $b^2 \leq d$ iff $k+j \geq 0$, true since $k \geq -j$. Index $m = (d-b^2)/k = k+j$; the term is $\varphi_b(k+j)$.
- $n = k-b$ $(\geq 1)$: $(k-b)^2 \leq d$ iff $j \geq -2b$. Index $m = (d-(k-b)^2)/k = j+2b$. Since $k \geq j+3b$, $m = j+2b \leq k-b = n$, so the term is $c_{j+2b}$ (Lemma 4.1).
- $n = k+b$: $(k+b)^2 \leq d$ iff $j \geq 2b$. Index $m = j-2b$, and $m \leq n$ since $k \geq j+3b$; the term is $c_{j-2b}$.

Summing gives (5.1). ■

**Corollary 5.4 (three families).** Let $k$ be an odd prime power.

1. $g_k\big((k+2)^2-3\big) = 0$ for every $k \geq 5$. For $k=3$: $g_3(22) = -1$.
2. If $k \equiv 3 \pmod 4$: $g_k(k^2+2k+4) = 0$ for every $k \neq 3, 7$. Values: $g_3(19) = -1$, $g_7(67) = +1$.
3. If $k \equiv 1 \pmod 4$: $g_k(k^2+3k+4) = 0$ for every $k \neq 5$. Value: $g_5(44) = -1$.

*Proof.* (1) $(k+2)^2-3 = k^2+4k+1$: $b=1$, $j=4$, $K=7$. By (5.1), for odd $k \geq 7$: $g = \varphi_1(k+4) + c_6 + c_2 = (-1)^{k+4}+1+0 = 0$. For $k=5$ the terms are $(n, m, \varphi_n(m)) = (1,9,-1), (4,6,+1), (6,2,0)$ (the term $n=4$ has $m=6>4$; by (4.3), $\varphi_4(6) = c_6 + \sum_{x=0}^{1}(-1)^x Q(x) = 1+(1-1) = 1$), so $g_5(46) = 0$. For $k=3$: $d=22$, terms $(1,7,-1), (2,6,0), (4,2,0)$, so $g_3(22) = -1$.

(2) $k^2+2k+4$: $b=2$, $j=2$, $K=8$. For $k \geq 9$, (5.1) gives $g = \varphi_2(k+2)+c_6 = \varphi_2(k+2)+1$, which is 0 iff $\varphi_2(k+2) = -1$ iff $k+2 \equiv 1 \pmod 4$ (Lemma 4.6) iff $k \equiv 3 \pmod 4$. The odd prime powers $k \equiv 3 \pmod 4$ below 9 are $3, 7$. For $k=7$: $d=67$, terms $(2,9,-1), (5,6,+2)$ [$\varphi_5(6) = c_6 + Q(0) = 2$], so $g_7(67) = +1$. For $k=3$: $d=19$, terms $(1,6,+1), (2,5,-1), (4,1,-1)$, so $g_3(19) = -1$.

(3) $k^2+3k+4$: $b=2$, $j=3$, $K=9$. For $k \geq 9$, (5.1) gives $g = \varphi_2(k+3)+c_7 = \varphi_2(k+3)-1$, which is 0 iff $\varphi_2(k+3) = 1$ iff $k+3 \equiv 0 \pmod 4$ iff $k \equiv 1 \pmod 4$. The only odd prime power $k \equiv 1 \pmod 4$ below 9 is 5: $d=44$, terms $(2,8,+1), (3,7,-2)$, so $g_5(44) = -1$. ■

(The term lists were computed by exact integer arithmetic. For all odd prime powers $k \leq 1001$ the exact values of $g_k$ at these $d$ were also computed directly: the only nonzero values are those listed. Appendix A.1.)

**Theorem 5.5 (zero classes for $b \in \{1,2\}$).** Let $b \in \{1,2\}$, $j \in \mathbb{Z}$, and $k \geq K$ an odd prime power as in Theorem 5.3. Then $g_k(k^2+jk+b^2) = 0$ in exactly the following cases.

- $b=1$: $j \in \{-2,-1,1,4\}$ (all $k$).
- $b=2$, $-4 \leq j \leq 3$: $k \equiv 1,3,1,1,1,3,3,1 \pmod 4$ for $j = -4,-3,-2,-1,0,1,2,3$ respectively.
- $b=2$, $j \leq -5$: $k \equiv 1 \pmod 4$ if $j \equiv 1,2 \pmod 4$ and $k \equiv 3 \pmod 4$ if $j \equiv 0,3 \pmod 4$.
- $b=2$, $j \geq 4$: never.

*Proof.* Let $k$ be odd. We use (5.1) and the values $c_0, \dots, c_7 = 1,-1,0,-1,1,-1,1,-1$ and $c_m = (-1)^m Q(m)$.

$b = 1$. $\varphi_1(k+j) = (-1)^{k+j} = -(-1)^j$. For $j \le -3$ only this term is present, so $g = \pm 1 \neq 0$. For $-2 \le j \le 1$: $j = -2$: $-1 + c_0 = 0$; $j = -1$: $+1 + c_1 = 0$; $j = 0$: $-1 + c_2 = -1$; $j = 1$: $+1 + c_3 = 0$. For $j \ge 2$: $g = -(-1)^j + (-1)^j\big(Q(j+2) + Q(j-2)\big) = (-1)^j\big(Q(j+2) + Q(j-2) - 1\big)$ (here $c_{j+2}$ and $c_{j-2}$ have the sign $(-1)^j$). This vanishes iff $Q(j+2) + Q(j-2) = 1$. Since $Q(j+2) \ge 1$ and $Q(j-2) \ge 1$ unless $j - 2 = 2$ (Lemma 4.4), this holds iff $j = 4$ ($Q(6) + Q(2) = 1 + 0$). For $j = 2, 3$: $Q(4) + Q(0) = 2$, $Q(5) + Q(1) = 2$, nonzero.

$b = 2$, $j \ge 4$. $\varphi_2(k+j) \in \{1, -1, 0, 0\}$ and $c_{j+4} + c_{j-4} = (-1)^j\big(Q(j+4) + Q(j-4)\big)$ with $Q(j+4) \ge 2$ ($j + 4 \ge 8$) and $Q(j-4) \ge 0$, so $|c_{j+4} + c_{j-4}| \ge 2 > |\varphi_2(k+j)|$: $g \neq 0$.

$b = 2$, $-4 \le j \le 3$. Here $g = \varphi_2(k+j) + c_{j+4}$, which vanishes iff $\varphi_2(k+j) = -c_{j+4}$. For $j + 4 = 0, \dots, 7$ we have $-c_{j+4} = -1, 1, 0, 1, -1, 1, -1, 1$. By Lemma 4.6, $\varphi_2(m) = -1$ iff $m \equiv 1$, $= 1$ iff $m \equiv 0$, $= 0$ iff $m \equiv 2, 3 \pmod 4$. So: $j = -4$: $k - 4 \equiv 1$, $k \equiv 1$. $j = -3$: $k - 3 \equiv 0$, $k \equiv 3$. $j = -2$: $k - 2 \in \{2, 3\} \bmod 4$, $k$ odd gives $k \equiv 1$. $j = -1$: $k - 1 \equiv 0$, $k \equiv 1$. $j = 0$: $k \equiv 1$. $j = 1$: $k + 1 \equiv 0$, $k \equiv 3$. $j = 2$: $k + 2 \equiv 1$, $k \equiv 3$. $j = 3$: $k + 3 \equiv 0$, $k \equiv 1$.

$b = 2$, $j \le -5$. Then $j < -2b$ and $j < 2b$, so $g = \varphi_2(k+j)$, zero iff $k + j \equiv 2, 3 \pmod 4$. With $k$ odd: $j$ even requires $k + j \equiv 3$, i.e. $k \equiv 3 - j$; $j$ odd requires $k + j \equiv 2$, i.e. $k \equiv 2 - j$. These give $k \equiv 1$ for $j \equiv 2, 1$ and $k \equiv 3$ for $j \equiv 0, 3 \pmod 4$. ■

(For $b = 2$ and $j \le -5$, the condition $k \ge K$ reads $k \ge -j$.) The cases $(b, j) = (1, 4)$, $(2, 2)$, $(2, 3)$ are the three families of Corollary 5.4. Appendix A.1 lists the exact values of (5.1) for all $(b, j)$ with $-8 \le j \le 16$ and all odd prime powers $k \le 500$ with $k \ge K$ against the exact coefficients of $g_k$: 5121 cases, no discrepancy.

# 6 The value of $g_k$ at $(k+2)^2 - 3$ for odd $k$ with two prime factors

Throughout this section $d_k = (k+2)^2 - 3 = k^2 + 4k + 1$.

## 6.1 Reduction to one pair of terms

For a positive integer $s$ and an integer $k > 2s$ with $k \mid s^2 - 1$ put

$$t = \frac{s^2 - 1}{k}, \quad m_1 = k + 4 - t, \quad m_2 = 4 + 2s - t, \quad n_2 = k - s, \quad u = k - 2s,$$

$$A_1 = A_s(m_1), \qquad A_2 = A_{n_2}(m_2)$$

(so $m_1 - m_2 = u$).

**Lemma 6.1 (pair reduction).** Let $k \ge 7$ and $\Xi_k = \{s : 1 < s < k/2,\ s^2 \equiv 1 \pmod k\}$. Then

$$g_k(d_k) = (-1)^k + 1 + \sum_{s \in \Xi_k} \big[\varphi_s(m_1) + \varphi_{k-s}(m_2)\big], \tag{6.1}$$

and $\varphi_s(m_1) = (-1)^{m_1} A_1$, $\varphi_{k-s}(m_2) = (-1)^{m_2} A_2$. For odd $k$ the bracket equals $(-1)^{t+1}(A_1 - A_2)$.

*Proof.* A term $n$ of $g_k(d_k)$ has $n^2 \le k^2 + 4k + 1 < (k+2)^2$, so $n \le k + 1$, and $n^2 \equiv d_k \equiv 1 \pmod k$. The residues $r \in \{0, \dots, k-1\}$ with $r^2 \equiv 1$ include 1 and $k - 1$ ($r = 0$ is excluded since $k \ge 2$), and the others $1 < r < k - 1$ come in pairs $\{s, k - s\}$ with $s \in \Xi_k$ ($s = k/2$ is not a root for $k \ge 4$: if $k = 2h$ with $h$ odd then $h^2 \equiv 1 \bmod 2h$ forces $h = 1$, and if $4 \mid k$ then $(k/2)^2 \equiv 0 \bmod k$). For $n \le k + 1$ the possible $n$ are therefore $n = 1$, $n = k - 1$, $n = k + 1$, and $n \in \{s, k - s\}$, $s \in \Xi_k$ (note $r + k > k + 1$ for $r \ge 2$).

- $n = 1$: $m = (d_k - 1)/k = k + 4$; term $\varphi_1(k+4) = (-1)^k$.

- $n = k-1$: $m = \big(k^2+4k+1-(k-1)^2\big)/k = 6 \le k-1$, so the term is $c_6 = 1$ (Lemma 4.1, 4.3).
- $n = k+1$: $m = \big(k^2+4k+1-(k+1)^2\big)/k = 2$; the term is $c_2 = 0$.
- $n = s$: $m = (k^2+4k+1-s^2)/k = k+4-t = m_1$; term $\varphi_s(m_1)$.
- $n = k-s$: $m = \big(k^2+4k+1-(k-s)^2\big)/k = 4+2s-t = m_2$; term $\varphi_{k-s}(m_2)$.

This is (6.1). By (4.2), $\varphi_n(m) = (-1)^m A_n(m)$. For odd $k$: $m_1 \equiv 1+t$ and $m_2 \equiv t \pmod 2$, so the bracket is $(-1)^{t+1}A_1 + (-1)^t A_2$. ■

**Lemma 6.2 (root geometry).** Let $k$ be odd, $s \ge 3$, $k > 2s$, $k \mid s^2-1$. Then: (i) $1 \le t < s/2$; (ii) $u = k-2s$ is odd and $u \ge 3$, $u^2 \ge k+4$, $m_1 \le u^2-1$ and hence $u \ge \sqrt{m_1+1}$; (iii) $\frac{3}{2}s+4 < m_2 \le 2s+3$ and $m_2 < m_1$; (iv) $n_2 \ge s+3$, and with $y_2 = m_2-n_2-1$: $y_2 \le s-1$ whenever $m_2 > n_2$.

*Proof.* (i) $t \ge 1$ since $s^2-1 \ge k > 0$ is a positive multiple of $k$; and $k > 2s$ gives $t < (s^2-1)/(2s) < s/2$. (ii) $u$ is odd since $k$ is odd. $u \equiv -2s \pmod k$ gives $u^2 \equiv 4s^2 \equiv 4 \pmod k$. If $u = 1$ then $k \mid -3$, i.e. $k = 3$, contradicting $k > 2s \ge 6$. So $u \ge 3$, $u^2-4 > 0$ is a positive multiple of $k$, whence $u^2 \ge k+4$. Then $m_1 = k+4-t \le k+3 \le u^2-1$. (iii) $m_2 = 2s+4-t \le 2s+3$ and $m_2 > 2s+4-s/2$. $m_1 - m_2 = u > 0$. (iv) $k = 2s+u \ge 2s+3$. If $m_2 > n_2$: $y_2 = (2s+4-t)-(k-s)-1 = s+3-t-u \le s+3-1-3$. ■

## 6.2 Estimates for $A_1$ and $A_2$

**Lemma 6.3.** Let $s \ge 3$, $k$ odd, $k > 2s$, $k \mid s^2-1$, and assume that $Q$ is nondecreasing on $[2,\infty)$ (Lemma 6.5 below).

1. For every $m \ge s+4$: $A_s(m) \;\ge\; Q(m) - \dfrac{m-s+1}{2}\,Q(m-s-1)$.
2. Let $e_0 = s+1$ if $s$ is odd and $e_0 = s+2$ if $s$ is even. If $D \ge 0$ is a sum of integers from the set $\{e \text{ even}: \; s < e \le 2s\}$, then $A_s(m) \ge A_s(m-D)$. A number $D$ is such a sum if $D = 0$ or if $D$ is even, $D \ge e_0$ and: ($s$ odd) always; ($s$ even, $s \ge 4$) $D \ne 2s+2$.
3. $A_2 \le Q(m_2) + Q(s)$, and $A_2 = Q(m_2)$ if $m_2 \le n_2$.
4. If $m_1 \ge 5s+8$, then $n_2 > m_2$ and $A_2 = Q(m_2) \le Q(2s+3)$.

*Proof.* (1) By (4.2), $A_s(m)$ is at least the number of partitions of $m$ into distinct odd parts all $\le s$ (take $\mu = \emptyset$). The number of partitions of $m$ into distinct odd parts having at least one part $j > s$ is at most $\sum_{j \text{ odd},\ s<j\le m} Q(m-j)$ (remove the part $j$). The arguments $x = m-j$ have the parity of $m-1$ and satisfy $0 \le x \le y := m-s-1$. For $y \ge 3$, $Q(x) \le Q(y)$ for $2 \le x \le y$ (monotonicity) and for $x \in \{0,1\}$ (as $Q(x) = 1 \le Q(y)$, Lemma 4.4). The number of such $x$ is at most $\lfloor y/2 \rfloor + 1 \le (m-s+1)/2$. Hence $A_s(m) \ge Q(m) - \frac{m-s+1}{2}Q(m-s-1)$.

(2) By (4.2), adjoining a fixed multiset of even parts in $(s, 2s]$ to $\mu$ is an injective map from the pairs counted by $A_s(m-D)$ to those counted by $A_s(m)$. For $j$ parts, the attainable sums are exactly all even numbers in $[je_0, 2js]$ (each part ranges over all even integers in $[e_0, 2s]$). For $s$ odd, $e_0 = s+1$, and consecutive intervals overlap or touch: $(j+1)(s+1) \le 2js+2$ iff $s-1 \le j(s-1)$. So every even $D \ge s+1$ is attainable. For $s \ge 4$ even, $e_0 = s+2$: for $j = 1, 2$ the intervals are $[s+2, 2s]$ and $[2s+4, 4s]$, missing only $2s+2$ between them; for $j \ge 2$, $(j+1)(s+2) \le 2js+2$ iff $s(j-1) \ge 2j$, true for $s \ge 4$.

(3) By Lemma 6.2 (iv), $n_2 \ge s+3$ so $m_2 \le 2s+3 \le 2n_2+1$ and (4.3) applies: $A_2 = Q(m_2) + (-1)^{m_2}\sum_{x=0}^{y_2}(-1)^x Q(x)$ for $y_2 = m_2-n_2-1 \ge 0$ (empty sum if $m_2 \le n_2$). Since $Q(0) = Q(1)$, the sum equals $\sum_{x=2}^{y_2}(-1)^x Q(x)$ for $y_2 \ge 1$, an alternating sum of a nondecreasing nonnegative

sequence, which has absolute value at most $Q(y_2)$; for $y_2 = 0$ it equals $1 \le Q(s)$. As $y_2 \le s-1$, $A_2 \le Q(m_2) + Q(s)$.

(4) $k = m_1 + t - 4 \ge 5s+5$, so $n_2 = k - s \ge 4s+5 > 2s+3 \ge m_2$ and $A_2 = Q(m_2) \le Q(2s+3)$ by monotonicity. ■

### 6.3 Chern's bound for $Q(n)$

For $n \ge 1$ let $v(n) = 24n-1$, $x(n) = \pi\sqrt{v(n)}/12$, and

$$\psi(n) = \frac{\sqrt{6}\, e^{x(n)}}{v(n)^{3/4}}, \qquad M(n) = \frac{\pi}{\sqrt{v(n)}}\, I_1\big(x(n)\big),$$

with $I_1$ the modified Bessel function.

**Cited result.** Let $G(q) = \prod_{r=1}^{R} (q^{m_r}; q^{m_r})_\infty^{\delta_r} = \sum g(n) q^n$ with distinct positive integers $m_r$ and nonzero integers $\delta_r$, $\sum \delta_r \ge 0$. Put $\Delta_1 = -\frac{1}{2}\sum \delta_r$, $\Delta_2 = \sum m_r \delta_r$, $\Delta_3(\ell) = -\sum \delta_r \gcd^2(m_r,\ell)/m_r$, $\Delta_4(\ell) = \prod (m_r/\gcd(m_r,\ell))^{-\delta_r/2}$, $L = \operatorname{lcm}(m_r)$, $L_{>0} = \{1 \le \ell \le L : \Delta_3(\ell) > 0\}$. S. Chern [20, Theorem 1.1] states: if $\Delta_1 \le 0$ and $\min_r \gcd^2(m_r,\ell)/m_r \ge \Delta_3(\ell)/24$ for all $1 \le \ell \le L$, then for all integers $n > -\Delta_2/24$,

$$g(n) = E(n) + \sum_{\ell \in L_{>0}} 2\pi \Delta_4(\ell) \Big(\frac{24n+\Delta_2}{\Delta_3(\ell)}\Big)^{-\frac{\Delta_1+1}{2}} \sum_{\substack{1 \le k \le N \\ k \equiv \ell \ (L)}} \frac{I_{-\Delta_1-1}\big(\frac{\pi}{6k}\sqrt{\Delta_3(\ell)(24n+\Delta_2)}\big)}{k} \sum_{\substack{0 \le h < k \\ (h,k)=1}} \omega_{h,k}\, e\Big(-\frac{nh}{k}\Big), \quad (6.2)$$

with $N = \lfloor \sqrt{2\pi(n+\Delta_2/24)} \rfloor$ and $|\omega_{h,k}| = 1$. The explicit bound for $E(n)$ is [20, (3.10)] (announced in [20, Remark 3 after Theorem 1.1]); with $\nu = n + \Delta_2/24$, $\varrho = 1/N^2$, and $\Delta_1 = 0$ (so $\Xi_{\Delta_1}(N) = 1$):

$$|E(n)| \le \frac{N^2}{\pi\nu} e^{2\pi\varrho\nu} \sum_{\ell \in L_{>0}} e^{\pi\Delta_3(\ell)/3} + 2e^{2\pi\varrho\nu}\Big\{ \sum_{\ell=1}^{L} \Delta_4(\ell) \exp\Big(\frac{\pi\Delta_3(\ell)}{24} + \sum_r \frac{|\delta_r| e^{-\pi \gcd^2(m_r,\ell)/m_r}}{(1 - e^{-\pi \gcd^2(m_r,\ell)/m_r})^2}\Big) - \sum_{\ell \in L_{>0}} \Delta_4(\ell) e^{\pi\Delta_3(\ell)/24}\Big\}. \quad (6.3)$$

(Numbering as in arXiv:1805.08759v1.)

**Application.** $\sum_n Q(n) q^n = (-q;q^2)_\infty = \dfrac{(q^2;q^2)_\infty^2}{(q;q)_\infty (q^4;q^4)_\infty}$ (since $(-q;q^2)_\infty = (q^2;q^4)_\infty/(q;q^2)_\infty$ and $1/(q;q^2)_\infty = (q^2;q^2)_\infty/(q;q)_\infty$). Here $m = (1,2,4)$, $\delta = (-1,2,-1)$, $\sum \delta_r = 0$, $\Delta_1 = 0$, $\Delta_2 = -1$, $L = 4$, and

$$(\Delta_3(\ell))_{\ell=1..4} = (\tfrac{1}{4}, -2, \tfrac{1}{4}, 1), \quad (\Delta_4(\ell)^2)_{\ell=1..4} = (1,2,1,1), \quad L_{>0} = \{1,3,4\}.$$

The inequality $\min_r \gcd^2(m_r,\ell)/m_r \ge \Delta_3(\ell)/24$ holds for $\ell = 1,2,3,4$ (the minima are $\frac{1}{4}, 1, \frac{1}{4}, 1$; exact rational comparison in `chern_margins.py`). So (6.2) holds for all $n \ge 1$. The term $k = 1$ of (6.2) is $2\pi \cdot 1 \cdot (4v)^{-1/2} I_1\big(\frac{\pi}{6}\sqrt{v/4}\big) = M(n)$ ($\omega_{1,1} = 1$), where $v = v(n)$ and $I_{-1} = I_1$.

**Lemma 6.4 (Bessel bounds).** For $y > 0$:

$$\Big(1 - \frac{3}{4y}\Big)\frac{e^y}{\sqrt{2\pi y}} \;\le\; I_1(y) \;\le\; \min\Big(\frac{ye^y}{2},\; \frac{e^y}{\sqrt{2\pi y}}\Big) \le 0.4302\, e^y.$$

Moreover $I_1$ is increasing and $I_1'(y) = I_0(y) - I_1(y)/y$, $I_0 \ge I_1$.

*Proof.* $I_1(y) = \frac{y}{\pi}\int_{-1}^{1}\sqrt{1-t^2}\,e^{yt}\,dt$. Substituting $t = 1 - w$ gives $I_1(y) = \frac{y}{\pi}e^y\int_0^2\sqrt{w(2-w)}\,e^{-yw}dw$. *Upper:* $\sqrt{w(2-w)} \le \sqrt{2w}$ and $\int_0^\infty\sqrt{2w}e^{-yw}dw = \sqrt{2}\,\Gamma(\frac{3}{2})y^{-3/2}$ give $I_1 \le e^y/\sqrt{2\pi y}$; also $e^{yt} \le e^y$ and $\int_{-1}^{1}\sqrt{1-t^2}dt = \pi/2$ give $I_1 \le ye^y/2$. Since $y/2$ is increasing and $(2\pi y)^{-1/2}$ is decreasing, $\min(y/2, (2\pi y)^{-1/2})$ is at most the common value at the crossing point $y = (2/\pi)^{1/3}$, namely $\frac{1}{2}(2/\pi)^{1/3} = 0.43012\ldots < 0.4302$. *Lower:* for $w \in [0,2]$, $\sqrt{w(2-w)} = \sqrt{2w}\sqrt{1-w/2} \ge \sqrt{2w}(1-w/2)$, and $\sqrt{2w}(1-w/2) < 0$ for $w > 2$, so $\int_0^2\sqrt{2w}(1-\frac{w}{2})e^{-yw}dw \ge \int_0^\infty\sqrt{2w}(1-\frac{w}{2})e^{-yw}dw = \sqrt{2}\big[\Gamma(\frac{3}{2})y^{-3/2} - \frac{1}{2}\Gamma(\frac{5}{2})y^{-5/2}\big]$. With $\Gamma(\frac{3}{2}) = \frac{\sqrt{\pi}}{2}$, $\Gamma(\frac{5}{2}) = \frac{3\sqrt{\pi}}{4}$ this gives the lower bound. The series $I_1(y) = \sum_{i\ge0}(y/2)^{2i+1}/(i!(i+1)!)$ has positive coefficients, so $I_1$ is increasing; $I_1' = I_0 - I_1/y$ is a standard identity; $I_0(y) - I_1(y) = \frac{1}{\pi}\int_0^\pi e^{y\cos\theta}(1-\cos\theta)d\theta \ge 0$. ■

**Lemma 6.5.** Let $C_E = 138.1955$ and $K = 1.3828$ (the numbers are upper roundings, see below), and $\bar B(n) = C_E + Ke^{x(n)/2}$, $\bar\beta(n) = \bar B(n)/\psi(n)$.

1. For every integer $n \ge 300$: $|Q(n) - M(n)| \le \bar B(n)$. Consequently, for $n \ge 300$,

$$\Big(1 - \frac{3}{4x(n)} - \bar\beta(n)\Big)\psi(n) \;\le\; Q(n) \;\le\; \big(1 + \bar\beta(n)\big)\psi(n). \tag{6.4}$$

2. $\bar\beta$ is nonincreasing on $[300,\infty)$, and $\bar\beta(300) \le 6.7\cdot10^{-3}$, $\bar\beta(1000) \le 1.7\cdot10^{-6}$, $\bar\beta(1804) \le 2.6\cdot10^{-9}$, $\bar\beta(6000) \le 1.2\cdot10^{-18}$.
3. $Q(n+1) \ge Q(n)$ for all $n \ge 2$.

*Proof.* (1) Let $n \ge 300$, $\nu = n - \frac{1}{24} \ge 299.9$, $N = \lfloor\sqrt{2\pi\nu}\rfloor \ge 43$. By the definition of $N$, $N^2 \le 2\pi\nu < (N+1)^2$, hence $N^2/(\pi\nu) \le 2$ and $e^{2\pi\varrho\nu} = e^{2\pi\nu/N^2} < e^{(1+1/N)^2} \le e^{(44/43)^2}$. In (6.3) all other quantities depend only on the data of $Q$. Evaluating them in interval arithmetic ($\sum_{\ell\in L_{>0}} e^{\pi\Delta_3(\ell)/3} = 5.448\ldots$ and the curly bracket $= 18.803\ldots$) gives $|E(n)| \le C_E$ with $C_E = 2e^{(44/43)^2}(5.4482 + 18.8031) \le 138.1955$ (value of the interval upper end: $138.19547\ldots$).

The terms $k \ge 3$ in (6.2) with $k \equiv 1, 3$ (4) have $\Delta_3 = \frac{1}{4}$, $\Delta_4 = 1$ and size at most $\frac{\pi}{\sqrt{v}}I_1(x/k)\cdot\frac{\#\{h\}}{k} \le \frac{\pi}{\sqrt{v}}I_1(x/k)$ (using $|\omega_{h,k}| = 1$, $\#\{h\} \le k$, $I_1$ increasing); those with $4 \mid k$ have $\Delta_3 = 1$ and size at most $\frac{2\pi}{\sqrt{v}}I_1(2x/k)$, $2x/k \le x/2$. In both cases the Bessel argument is at most $x/2$, so by Lemma 6.4 each such term is at most $\frac{2\pi}{\sqrt{v}}\cdot 0.4302\,e^{x/2}$. There are at most $N \le \sqrt{2\pi\nu} \le \sqrt{\pi v/12}$ terms (since $2\pi\nu = \pi v/12$). So their sum is at most $2\pi\sqrt{\pi/12}\cdot 0.4302\,e^{x/2} \le Ke^{x/2}$ with $K = 1.3828$ (evaluation: $1.38280\ldots$). Together with (6.2), $|Q(n) - M(n)| \le C_E + Ke^{x/2} = \bar B(n)$. Lemma 6.4 gives $(1 - \frac{3}{4x})\frac{e^x}{\sqrt{2\pi x}} \le I_1(x) \le \frac{e^x}{\sqrt{2\pi x}}$, and $\frac{\pi}{\sqrt{v}}\frac{e^x}{\sqrt{2\pi x}} = \psi(n)$ (because $2\pi x = \pi^2\sqrt{v}/6$). Hence $(1 - \frac{3}{4x})\psi \le M \le \psi$, which with the bound on $|Q - M|$ gives (6.4).

(2) $\frac{d}{dn}\ln\psi = \frac{\pi}{\sqrt{v}} - \frac{18}{v} > 0$ for $\sqrt{v} > 18/\pi$, so $C_E/\psi$ is decreasing; $Ke^{x/2}/\psi = \frac{K}{\sqrt{6}}v^{3/4}e^{-x/2}$ and $\frac{d}{dn}\ln(v^{3/4}e^{-x/2}) = \frac{18}{v} - \frac{\pi}{2\sqrt{v}} < 0$ for $\sqrt{v} > 36/\pi$. The numerical values are interval evaluations (`chern_margins.py`).

(3) For $2 \le n \le 35199$ this is checked on the exact table of $Q$ (Appendix A.2). Let $n \ge 35200$, $x = x(n) \ge 240$, $v = v(n)$. Differentiating, $M'(\xi) = \pi^2 v(\xi)^{-1}\big[I_0(x(\xi)) - 2I_1(x(\xi))/x(\xi)\big]$ (use $\frac{dx}{d\xi} = \pi/\sqrt{v}$, $12/\sqrt{v} = \pi/x$ and $I_1' = I_0 - I_1/x$), which is $\ge \pi^2 v(\xi)^{-1}(1 - 2/x(\xi))I_1(x(\xi))$ by $I_0 \ge I_1$. On $[n, n+1]$: $v(\xi) \le v(n+1)$, and $(1 - 2/x)$, $I_1$ are increasing, so with Lemma 6.4,

$$M(n+1) - M(n) \;\ge\; \Lambda(n) := \pi\Big(1 - \tfrac{2}{x}\Big)\Big(1 - \tfrac{3}{4x}\Big)\frac{\psi(n)\sqrt{v(n)}}{v(n+1)} \;\ge\; 3\,\frac{\psi(n)}{\sqrt{v}} = 3\sqrt{6}\,e^x v^{-5/4}$$

(using $x \ge 240$, $v(n+1) \le 1.0001\,v$). On the other hand $x(n+1) - x(n) \le \pi/\sqrt{v} < 0.0035$ so $\bar{B}(n) + \bar{B}(n+1) \le 2C_E + 2.004\,K e^{x/2}$. By (1), $Q(n+1) - Q(n) \ge \Lambda(n) - \bar{B}(n) - \bar{B}(n+1)$, which is positive as soon as

$$2C_E\, v^{5/4} e^{-x} + 2.004\,K\, v^{5/4} e^{-x/2} < 3\sqrt{6}.$$

The left side is nonincreasing in $n$ (for $\sqrt{v} > 60/\pi$, $\frac{d}{dn}\ln(v^{5/4}e^{-x/2}) = \frac{30}{v} - \frac{\pi}{2\sqrt{v}} < 0$, and similarly for $e^{-x}$), and at $n = 35200$ it is $< 10^{-40}$ (interval evaluation). ■

### 6.4 The inequality $A_1 > A_2$

**Theorem 6.6.** Let $k$ be odd, $s \ge 3$, $k > 2s$, $k \mid s^2 - 1$. Then $A_1 > A_2$, except for $(s,k) = (4,15)$ and $(8,21)$, where $A_1 = A_2 = 2$ and $A_1 = A_2 = 6$ respectively.

The proof has two parts: a finite exact computation for $3 \le s \le 2500$ (Proposition 6.7) and an analytic argument for $s \ge 2501$ (Proposition 6.8).

**Proposition 6.7 (finite range; computer-assisted).** For every pair $(s,k)$ with $3 \le s \le 2500$, $k$ odd, $k > 2s$, $k \mid s^2 - 1$ (there are 15 616 pairs), $A_1 > A_2$ except for $(4,15)$ and $(8,21)$.

*Proof.* All arithmetic is exact integer arithmetic. The table of $Q(0), \dots, Q(35200)$ is computed exactly. For each pair, $A_2 = A_{n_2}(m_2)$ is computed by (4.3) (valid since $m_2 \le 2n_2 + 1$, Lemma 6.2). A lower bound $\mathrm{LB} \le A_1$ is computed as follows: choose $m'' \le m_1$ with $m'' \equiv m_1 \pmod 2$ such that $D = m_1 - m''$ is a sum of even integers in $(s, 2s]$ (this is decided exactly by the criterion of Lemma 6.3(2): $D = 0$, or $D$ even and $\lceil D/2s \rceil e_0 \le D$), and set $\mathrm{LB} = A_s(m'')$ computed by (4.3) if $m'' \le 2s + 1$, and otherwise $\mathrm{LB} = Q(m'') - \sum_{j \text{ odd},\ s < j \le m''} Q(m'' - j)$. By Lemma 6.3(2) and the proof of Lemma 6.3(1), $\mathrm{LB} \le A_s(m'') \le A_1$ (the inequality $Q(x) \le Q(y)$ is not used here: the exact sum is subtracted). The candidates are the integers $m''$ with $m_2 < m'' \le \min(m_1, 35200)$ satisfying the two conditions above (they are tried in the order $m_1$, $4s+3, 4s+2, 4s+1, 4s$, then downward from $\min(m_1, 8s, 35200)$; the order only affects speed), and the first with $\mathrm{LB} > A_2$ is accepted. The result: 15 587 pairs are certified ($\mathrm{LB} > A_2$, so $A_1 > A_2$). The remaining 29 pairs all have $s \le 32$; they are listed in the output file and are settled by the exact computation of $A_1 = [x^{m_1}] \prod_{j \text{ odd} \le s}(1 + x^j) \prod_{s/2 < i \le s}(1 - x^{2i})^{-1}$ (Lemma 4.2), which was carried out for **all** pairs with $s \le 300$ (995 pairs): $A_1 > A_2$ for every pair except $(4,15)$ and $(8,21)$; the smallest ratio $A_1/A_2$ among the others is 1.7811 at $(s,k) = (34,77)$. For the two exceptions the terms of $g_k(d_k)$ are listed in Section 6.6; they give $A_1 = A_2 = 2$ for $(4,15)$ and $A_1 = A_2 = 6$ for $(8,21)$. ■

**Proposition 6.8 (analytic range).** Under Lemma 6.5, $A_1 > A_2$ for all $(s,k)$ as in Theorem 6.6 with $s \ge 2501$.

*Proof.* Let $s \ge 2501$. All arguments of $Q$ below are $\ge 300$, so (6.4) applies, and $\bar{\beta}$ is nonincreasing. Define

$$\lambda(n) = 1 - \frac{3}{4x(n)} - \bar{\beta}(n),$$

which is increasing in $n$.

**Case 1:** $m_1 \le 5s+7$**.** Put $y_1 = m_1 - s - 1$, $n_a = \frac{3}{2}s + 4$. By Lemma 6.2, $m_1 > m_2 > n_a$ and $y_1 > s/2 + 3$. By Lemma 6.3 (1),(3),

$$A_1 - A_2 \;\ge\; Q(m_1) - \tfrac{m_1-s+1}{2} Q(y_1) - Q(m_2) - Q(s).$$

Divide by $\psi(m_1)$ and use (6.4):

$$\frac{A_1 - A_2}{\psi(m_1)} \;\ge\; \lambda(m_1) - \tfrac{m_1-s+1}{2}\big(1+\bar\beta(y_1)\big)\frac{\psi(y_1)}{\psi(m_1)} - \big(1+\bar\beta(m_2)\big)\frac{\psi(m_2)}{\psi(m_1)} - \big(1+\bar\beta(s)\big)\frac{\psi(s)}{\psi(m_1)}.$$

We bound the four terms.

*First term.* $\lambda(m_1) \ge \lambda(n_a) =: L$.

*Third term* ($m_2 = m_1 - u$). Write $\psi(m_2)/\psi(m_1) = e^{f(u)}$ with $v_1 = v(m_1)$ and

$$f(u) = -\big(x(m_1) - x(m_1-u)\big) + \tfrac{3}{4}\ln\frac{v_1}{v_1 - 24u}.$$

Then $f'(u) = \dfrac{18 - \pi\sqrt{v_1 - 24u}}{v_1 - 24u} < 0$ (as $\sqrt{v(m_2)} > 18/\pi$), so $f$ is decreasing in $u$ and, with $u_0 = \sqrt{m_1+1} \le u$ (Lemma 6.2), $f(u) \le f(u_0)$. By concavity of $x(\cdot)$, $x(m_1) - x(m_1-u_0) \ge u_0 x'(m_1) = \pi u_0/\sqrt{v_1} > \pi/\sqrt{24}$ (as $(m_1+1)/(24m_1-1) > 1/24$). The logarithmic term is at most $\frac{3}{4}\ln\big(1 - 24u_0/v_1\big)^{-1}$, which is decreasing in $m_1$, hence at most its value at $m_1 = n_a$. Therefore

$$\frac{\psi(m_2)}{\psi(m_1)} \le r_0 := e^{-\pi/\sqrt{24}}\Big(1 - \frac{24\sqrt{n_a+1}}{24n_a - 1}\Big)^{-3/4}, \qquad \text{third term} \le (1+\bar\beta(n_a))\, r_0 =: T.$$

*Second term.* $m_1 - s + 1 \le 4s+8$. By concavity, $x(m_1) - x(y_1) \ge (s+1)x'(m_1) = \pi(s+1)/\sqrt{v_1} \ge \pi(s+1)/\sqrt{120s+167}$ (since $m_1 \le 5s+7$). Also $v(m_1)/v(y_1) = \frac{24m_1-1}{24m_1-24s-25}$ is decreasing in $m_1$, so $\le \frac{36s+95}{12s+71} < 3$ at $m_1 = n_a$. Hence the second term is at most

$$\theta(s) = (2s+4)\big(1+\bar\beta(\tfrac{s}{2}+3)\big)3^{3/4}e^{-\pi(s+1)/\sqrt{120s+167}}.$$

*Fourth term.* $x(m_1) - x(s) \ge (m_1 - s)x'(m_1) = \pi\frac{m_1-s}{\sqrt{24m_1-1}} \ge \pi\frac{s/2+4}{\sqrt{120s+167}}$ (numerator minimal at $m_1 = n_a$, denominator maximal at $m_1 = 5s+7$), and $v(m_1)/v(s) \le (120s+167)/(24s-1)$. So the fourth term is at most

$$U(s) = \big(1+\bar\beta(s)\big)e^{-\pi(s/2+4)/\sqrt{120s+167}}\Big(\frac{120s+167}{24s-1}\Big)^{3/4}.$$

*Monotonicity in $s$.* $L$ is increasing and $T$ is decreasing in $s$ (both $1+\bar\beta(n_a)$ and $r_0$ decrease as $n_a$ grows). For $\theta$: $\bar\beta$ factor nonincreasing, and $\frac{d}{ds}\ln\big[(s+2)e^{-\pi(s+1)/\sqrt{120s+167}}\big] = \frac{1}{s+2} - \frac{\pi(60s+107)}{(120s+167)^{3/2}} < 0$ for $s \ge 100$ (the second fraction is $\ge 0.139\, s^{-1/2}$, the first is $\le 0.1\, s^{-1/2}$). For $U$: $\frac{d}{ds}\big[-\pi(\frac{s}{2}+4)(120s+167)^{-1/2}\big] = -\pi\frac{30s-156.5}{(120s+167)^{3/2}} < 0$ and $\frac{d}{ds}\frac{3}{4}\ln\frac{120s+167}{24s-1} < 0$; the $\bar\beta$ factor is nonincreasing. Hence the lower bound $\mathrm{M}_1(s) = L - T - \theta - U$ is nondecreasing in $s$, and at $s = 2501$ interval evaluation gives $L \ge 0.99046$, $T \le 0.53316$, $\theta \le 6.73\cdot 10^{-3}$, $U \le 2.52\cdot 10^{-3}$, so $\mathrm{M}_1(2501) \ge 0.448 > 0$ (`chern_margins.py`). Hence $A_1 > A_2$ in Case 1 for all $s \ge 2501$.

**Case 2:** $m_1 \ge 5s+8$**.** By Lemma 6.3 (4), $A_2 \le Q(2s+3)$. Choose $m'' \in \{4s, 4s+1, 4s+2, 4s+3\}$ with $m'' \equiv m_1 \pmod 2$ and $D = m_1 - m'' \ne 2s+2$ (of the two admissible values, the corresponding $D$ differ by 2, so one of them is not $2s+2$). Then $D \ge m_1 - (4s+3) \ge s+5$ is even, so by Lemma

6.3 (2), $A_1 \geq A_s(m'')$. By Lemma 6.3 (1) and monotonicity, $A_s(m'') \geq Q(4s) - (\frac{3}{2}s+2)Q(3s+2)$ (as $m'' \geq 4s$, $\frac{m''-s+1}{2} \leq \frac{3}{2}s+2$, $m''-s-1 \leq 3s+2$). Hence

$$\frac{A_1 - A_2}{\psi(4s)} \;\geq\; \lambda(4s) - c_1(s) - c_2(s),$$

$$c_1 = (\tfrac{3}{2}s+2)\big(1+\bar\beta(3s+2)\big)\frac{\psi(3s+2)}{\psi(4s)}, \qquad c_2 = \big(1+\bar\beta(2s+3)\big)\frac{\psi(2s+3)}{\psi(4s)}.$$

By concavity $x(4s) - x(3s+2) \geq (s-2)\pi/\sqrt{96s-1}$ and $x(4s) - x(2s+3) \geq (2s-3)\pi/\sqrt{96s-1}$, and $v(4s)/v(3s+2) = \frac{96s-1}{72s+47}$, $v(4s)/v(2s+3) = \frac{96s-1}{48s+71}$. Thus $c_1 \leq (\frac{3}{2}s+2)(1+\bar\beta(3s+2))e^{-\pi(s-2)/\sqrt{96s-1}}\big(\frac{96s-1}{72s+47}\big)^{3/4}$ and $c_2 \leq (1+\bar\beta(2s+3))e^{-\pi(2s-3)/\sqrt{96s-1}}\big(\frac{96s-1}{48s+71}\big)^{3/4}$. These bounds are decreasing in $s$ for $s \geq 2501$ (the logarithmic derivatives of the elementary factors are $\frac{1}{s+4/3} - \frac{\pi(48s+95)}{(96s-1)^{3/2}} + O(s^{-2}) < 0$ and $-\frac{\pi(96s+142)}{(96s-1)^{3/2}} + O(s^{-2}) < 0$, and the $\bar\beta$ factors are nonincreasing), while $\lambda(4s)$ is increasing. At $s = 2501$: $\lambda(4s) \geq 0.99415$, $c_1 \leq 5.13\cdot 10^{-4}$, $c_2 \leq 2.1\cdot 10^{-14}$, so the margin is $\geq 0.9936 > 0$. ■

*Proof of Theorem 6.6.* Propositions 6.7 and 6.8 cover all $s \geq 3$. ■

*Dependence.* Proposition 6.8 depends on [20] only through (6.2)–(6.3) (Lemma 6.5 (1)). Proposition 6.7 and the exact tables do not depend on [20]. The margins in Proposition 6.8 are large: if every occurrence of $\bar\beta$ is replaced by a constant 0.25, the lower bounds at $s = 2501$ are still positive (0.06 in Case 1), whereas $\bar\beta(1253) \leq 2\cdot 10^{-7}$.

### 6.5 The theorem

**Theorem 6.9.** Let $k$ be an odd integer with exactly two distinct prime factors. Then $g_k\big((k+2)^2-3\big) \neq 0$, except for $k = 15$ and $k = 21$, for which $g_k\big((k+2)^2-3\big) = 0$.

For odd prime powers $k \geq 5$, $g_k\big((k+2)^2-3\big) = 0$ (Corollary 5.4).

*Proof.* Let $k = p^a q^b$ with $p < q$ odd primes; $k \geq 15$. The solutions of $x^2 \equiv 1 \pmod k$ are, by the Chinese remainder theorem, exactly four residues $\pm 1, \pm s_0$ with $s_0 \not\equiv \pm 1$. Hence $\Xi_k$ consists of exactly one element $s$ (one of $s_0, k-s_0$ lies in $(1, k/2)$), and $s \geq 3$ since $s^2 \equiv 1$, $s = 2$ would give $k \mid 3$. By Lemma 6.1 and $(-1)^k + 1 = 0$,

$$g_k\big((k+2)^2-3\big) = (-1)^{t+1}(A_1 - A_2).$$

By Theorem 6.6, $A_1 \neq A_2$ unless $(s,k) \in \{(4,15),(8,21)\}$, and for these two pairs $A_1 = A_2$, so $g_k(d_k) = 0$. ■

### 6.6 The cases $k = 15$ and $k = 21$

For $k = 15$, $d = 286$: the terms $(n, m, \varphi_n(m))$ are $(1,19,-1)$, $(4,18,+2)$, $(11,11,-2)$, $(14,6,+1)$, $(16,2,0)$ ($s = 4$, $t = 1$, $A_1 = A_4(18) = 2$, $A_2 = A_{11}(11) = 2$). For $k = 21$, $d = 526$: $(1,25,-1)$, $(8,22,+6)$, $(13,17,-6)$, $(20,6,+1)$, $(22,2,0)$ ($s = 8$, $t = 3$, $A_1 = A_8(22) = 6$, $A_2 = A_{13}(17) = 6$). In both cases the sum is 0.

*Computer-assisted items in this section:* the table of $Q$ up to 35200 and its monotonicity; the interval evaluation of $C_E$, $K$, $\bar\beta$ and of the margins at $s = 2501$; the finite certificate of Proposition 6.7 (Appendix A.2).

# 7 Even $k$

## 7.1 Signs of the terms and finiteness of the zero set in "one-sign" classes

**Lemma 7.1.** Let $k \geq 1$ and $d \geq 0$. The term $n$ of $g_k(d)$ equals $(-1)^{m_n} A_n(m_n)$. Two terms $n, n'$ of $g_k(d)$ have the same sign factor $(-1)^{m_n} = (-1)^{m_{n'}}$ if and only if $n^2 \equiv n'^2 \pmod{2k}$.

*Proof.* The first statement is (4.2). Also $m_n - m_{n'} = (n'^2 - n^2)/k$, which is even if and only if $2k \mid n'^2 - n^2$. ■

*Definition 7.2.* A residue class $c \in \mathcal{S}_k$ is a *one-sign class* if $n^2 \equiv n'^2 \pmod{2k}$ whenever $n^2 \equiv n'^2 \equiv c \pmod{k}$. For such a class and every $d \equiv c \pmod{k}$, all terms of $g_k(d)$ have the same sign factor $\varepsilon_d = (-1)^{m_n}$ and

$$g_k(d) = \varepsilon_d \sum_n A_n(m_n), \qquad \text{so } g_k(d) \neq 0 \iff A_n(m_n) \geq 1 \text{ for some term } n. \tag{7.1}$$

**Lemma 7.3.** If $k \equiv 2 \pmod 4$, every class $c \in \mathcal{S}_k$ is a one-sign class.

*Proof.* Let $k = 2h$, $h$ odd, $n^2 \equiv n'^2 \pmod k$. Then $n \equiv n' \pmod 2$, so $n^2 - n'^2 = (n-n')(n+n')$ is a product of two even numbers and $4 \mid n^2 - n'^2$; also $h \mid n^2 - n'^2$. As $\gcd(4, h) = 1$, $4h = 2k$ divides $n^2 - n'^2$. ■

**Theorem 7.4.** Let $k \geq 2$ and let $c$ be a one-sign class. Let $n_0$ be the least integer $n \geq 3$ with $n^2 \equiv c \pmod k$, $L = \lfloor n_0/2 \rfloor + 1$, and

$$B_c = n_0^2 + k\big(2L(L-1) + 1\big).$$

Then $g_k(d) \neq 0$ for every $d \equiv c \pmod k$ with $d \geq B_c$, and $g_k(d)$ has the sign $\varepsilon_d$.

*Proof.* $n_0$ exists since if $n^2 \equiv c$ then also $(n+k)^2 \equiv c$ and one of $n, n+k, n+2k$ is $\geq 3$. Let $d \geq B_c$. Then $n_0$ is a term of $g_k(d)$, with index $m = (d - n_0^2)/k \geq 2L(L-1) + 1$. By (7.1) it suffices to show $A_{n_0}(m) \geq 1$.

The even integers $2L$ and $2L+2$ lie in $(n_0, 2n_0]$: $2L = 2\lfloor n_0/2 \rfloor + 2 > n_0$ and $2L + 2 \leq 2n_0$ iff $\lfloor n_0/2 \rfloor + 2 \leq n_0$, true for $n_0 \geq 3$. Every integer $N \geq L(L-1)$ is of the form $aL + b(L+1)$ with $a, b \geq 0$: write $N = qL + r$, $0 \leq r < L$; then $q \geq L - 1 \geq r$ and $N = (q - r)L + r(L+1)$. Hence every even integer $\geq 2L(L-1)$ is a sum of parts $2L$ and $2L+2$. If $m$ is even, take $\lambda = \emptyset$ and $\mu$ such a multiset with sum $m$ ($m \geq 2L(L-1) + 2$). If $m$ is odd, take $\lambda = \{1\}$ ($1 \leq n_0$) and $\mu$ with sum $m - 1 \geq 2L(L-1)$. By (4.2), $A_{n_0}(m) \geq 1$. ■

**Corollary 7.5.** If $k \equiv 2 \pmod 4$ then $Z_k$ is finite, $Z_k \subset [0, \max_c B_c)$, and $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$.

The sets $Z_k$ for $k \equiv 2 \pmod 4$, $k \leq 34$, are computed exactly in Appendix B.2 ($d < \max_c B_c$ is a finite range; the program evaluates $g_k(d)$ exactly for all $d \leq \max_c B_c + 3000$). For example:

$$Z_2 = \{2, 8, 10, 26\}, \qquad Z_6 = \{3, 6, 12, 18, 21, 24, 30, 46, 48, 78\},$$

$$Z_{10} = \{5, 6, 10, 15, 20, 24, 29, 30, 34, 40, 45, 50, 60, 70, 80, 90, 120\},$$

$$Z_{14} = \{2, 7, 8, 11, 14, 21, 22, 28, 32, 35, 37, 42, 44, 46, 53, 56, 70,$$
$$77, 84, 86, 88, 98, 102, 112, 126, 140, 154, 168, 182, 224\},$$

with $\max_c B_c = 42, 186, 710, 1778$ and $|Z_k| = 4, 10, 17, 30$ respectively.

### 7.2 The case $k = 2$

**Theorem 7.6.** For every $d \geq 0$, $g_2(d) = 0$ or $\operatorname{sign} g_2(d) = (-1)^{\lfloor d/2 \rfloor}$. Moreover $g_2(d) \neq 0$ for all even $d \geq 42$ and all odd $d \geq 19$, and the zero set is

$$Z_2 = \{2, 8, 10, 26\}.$$

*Proof. Sign.* Let $n$ be a term of $g_2(d)$, $m = (d - n^2)/2$. If $d$ is even, $n = 2j$ and $m = d/2 - 2j^2 \equiv d/2 \pmod 2$. If $d$ is odd, $n$ is odd and $n^2 \equiv 1 \pmod 8$, so $m = \frac{d-1}{2} - \frac{n^2-1}{2} \equiv \frac{d-1}{2} \pmod 2$ (as $4 \mid \frac{n^2-1}{2}$). In both cases $(-1)^m = (-1)^{\lfloor d/2 \rfloor}$, independent of $n$, and by (4.2) every term $\varphi_n(m) = (-1)^m A_n(m)$ has that sign (or is 0).

*Nonvanishing.* The two classes $c = 0$ ($n$ even) and $c = 1$ ($n$ odd) are one-sign classes (Lemma 7.3). For $c = 0$: $n_0 = 4$, $L = 3$, $B_0 = 16 + 2(2 \cdot 3 \cdot 2 + 1) = 42$. For $c = 1$: $n_0 = 3$, $L = 2$, $B_1 = 9 + 2(2 \cdot 2 \cdot 1 + 1) = 19$. By Theorem 7.4, $g_2(d) \neq 0$ for even $d \geq 42$ and odd $d \geq 19$.

*Finite range.* The exact values $g_2(d)$ for $0 \leq d \leq 41$ are:

| $d$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $g_2(d)$ | 1 | 1 | 0 | -1 | 1 | 1 | -1 | -1 | 0 | 2 | 0 | -2 | 1 | 1 |

| $d$ | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $g_2(d)$ | -1 | -2 | 1 | 3 | -1 | -2 | 1 | 2 | -2 | -3 | 1 | 4 | 0 | -4 |

| $d$ | 28 | 29 | 30 | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 | 41 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $g_2(d)$ | 2 | 3 | -2 | -4 | 1 | 5 | -2 | -5 | 3 | 5 | -3 | -5 | 2 | 7 |

The zeros in $d < 42$ are $2, 8, 10, 26$; all nonzero values have the sign $(-1)^{\lfloor d/2 \rfloor}$. ■

### 7.3 The value at $(k+2)^2 - 3$ for even $k$

**Theorem 7.7.** Let $k \geq 8$ be even and $d_k = (k+2)^2 - 3$.

1. $g_k(d_k) = 2 + \sum_{s \in \Xi_k} (-1)^t (A_1 + A_2)$ with $t = (s^2 - 1)/k$, and every $s \in \Xi_k$ is odd.
2. If $8 \nmid k$, every $t$ is even and $g_k(d_k) \geq 2$.
3. If $k = 2p^a$ with $p$ an odd prime ($k \geq 10$), then $\Xi_k = \emptyset$ and $g_k(d_k) = 2$.
4. If $k = 2^a$ with $a \geq 3$, then $\Xi_k = \{k/2 - 1\}$ and $g_k(d_k) = 2 - A_1 - A_2 < 0$. Values: $g_8(97) = -3$, $g_{16}(321) = -8$, $g_{32}(1153) = -31$, $g_{64}(4353) = -236$, $g_{128}(16897) = -5174$.

*Proof.* (1) Lemma 6.1 with $(-1)^k + 1 = 2$; for even $k$, $m_1 \equiv m_2 \equiv t \pmod 2$, so the bracket is $(-1)^t(A_1 + A_2)$. A root $s$ of $s^2 \equiv 1 \pmod k$ with $k$ even is odd.

(2) $s$ odd gives $8 \mid s^2 - 1 = tk$, and $v_2(k) \leq 2$ forces $v_2(t) \geq 1$. All pair terms are $\geq 0$.

(3) Modulo $p^a$ the roots are $\pm 1$ and modulo 2 the root is 1, so $s^2 \equiv 1 \pmod{2p^a}$ implies $s \equiv \pm 1 \pmod{2p^a}$: only $s = 1, k - 1$, which are excluded from $\Xi_k$.

(4) The solutions of $x^2 \equiv 1 \pmod{2^a}$, $a \geq 3$, are $\pm 1$ and $2^{a-1} \pm 1$. So $\Xi_k = \{s\}$ with $s = k/2 - 1$ ($> 1$), and $t = \frac{(s-1)(s+1)}{k} = \frac{(k/2-2)(k/2)}{k} = \frac{k}{4} - 1$, which is odd since $k/4$ is even. Hence $g_k(d_k) = 2 - (A_1 + A_2)$. Here $n_2 = k - s = \frac{k}{2} + 1$ and $m_2 = 4 + 2s - t = \frac{3k}{4} + 3$, which is odd (as $\frac{3k}{4}$ is even). For $k \geq 16$ and $o \in \{1, 3, 5\}$, the pair $\lambda = \{o\}$, $\mu = \{m_2 - o\}$ is admissible in (4.2) for $A_{n_2}(m_2)$: $o \leq n_2$; $m_2 - o$ is even; $m_2 - o \leq \frac{3k}{4} + 2 \leq k + 2 = 2n_2$; and $m_2 - o > \frac{k}{2} + 1 = n_2$ iff $o < \frac{k}{4} + 2$, true for $o \leq 5$ when $k \geq 16$. These are three different pairs, so $A_2 \geq 3$, $A_1 \geq 0$ and $g_k(d_k) \leq -1$. For $k = 8$ the terms of $g_8(97)$ are $(1, 12, +1)$, $(3, 11, -2)$, $(5, 9, -3)$, $(7, 6, +1)$, $(9, 2, 0)$, with sum $-3$. The values for $k = 16, \ldots, 128$ are exact evaluations (Appendix A.1). ■

*Scope.* Lemma 7.1 shows that for $4 \mid k$ there are classes $c$ in which two roots $n, n'$ have $n^2 \not\equiv n'^2 \pmod{2k}$; in such a class the terms of $g_k(d)$ have different signs and Theorem 7.4 does not apply. For $k = 4, 8, 12$ these classes are: $c \equiv 0$ ($k = 4$), $c \equiv 1$ ($k = 8$), $c \equiv 0, 4$ ($k = 12$) (computed by enumeration of $n$ modulo $2k$). Nothing is claimed for such classes.

# 8 Parity dominance for odd $k \leq 33$

Throughout this section $k$ is odd, $3 \leq k \leq 33$, and $d$ is allowed for $k$ (Definition 3.3). The notation $a_n(m)$, $A_n(x)$, $E_d$, $O_d$ is that of Section 4: $a_n(m) = |\varphi_n(m)|$, $A_n(x) = \sum_m a_n(m) x^m$ is the product (4.1), and by (4.4)

$$(-1)^d g_k(d) = E_d - O_d, \qquad E_d = \sum_{n \text{ even}} a_n(m_n), \quad O_d = \sum_{n \text{ odd}} a_n(m_n).$$

## 8.1 Statement

For $k$ in the table of Section 8.7, $R_k < 1$ and $D_0(k)$ are the numbers computed there (Table 8.2). Put $\eta_t = 10^{-6}$.

**Theorem 8.1 (parity dominance).** Let $k \in \{3, 5, \ldots, 33\}$.

1. For every allowed $d \geq D_0(k)$: $O_d > 0$ and $E_d \leq (R_k + \eta_t)\, O_d$. Consequently $(-1)^d g_k(d) \leq -(1 - R_k - \eta_t) O_d < 0$; in particular $g_k(d) \neq 0$ and $\operatorname{sign} g_k(d) = (-1)^{d+1}$.
2. $Z_k$ is a finite set, $\max Z_k < D_0(k)$, and $Z_k$ is the explicit set of Appendix B.1 (cardinalities and maxima in Table 8.5).
3. $\mathrm{OccFilt}_k(\infty) = \mathcal{S}_k$: the residue classes mod $k$ that occur for all large $d$ are exactly the squares mod $k$.

Part 3 follows from Part 2 and Proposition 3.4. Part 2 follows from Part 1 and the finite certificate of Section 8.7. Part 1 is proved in Sections 8.2 to 8.5: it combines (a) a local limit estimate for a family of integer-valued random variables attached to $A_n$ (Sections 8.2, 8.3), (b) a comparison of each even term with the two odd terms $n \pm k$ (Section 8.4) and (c) an upper bound for the even terms far from the saddle, and a lower bound for one odd term (Section 8.5).

Computer-assisted steps: the finite computations of Algorithms C (cells), R (block bound), T (tail) and the modular certificate. Each is specified below and reproduced by the code listed in Appendix A.3; they use interval (ball) arithmetic with the Arb library through python-flint [21], at 128-bit working precision, except for the modular certificate, which uses exact integer arithmetic modulo two primes.

## 8.2 The tilted model

Let $n \geq 1$ and $0 < r < 1$. Put $t = -\ln r$ and

$$p_j = \frac{r^j}{1+r^j} = \frac{1}{1+e^{jt}} \ (j \text{ odd},\ j \leq n), \qquad w_l = r^{2l} = e^{-2lt} \ \big(\tfrac{n}{2} < l \leq n\big).$$

Let $B_j$ ($j$ odd $\leq n$) be independent Bernoulli variables with $P(B_j = 1) = p_j$, and $N_l$ ($n/2 < l \leq n$, $l$ integer) independent geometric variables with $P(N_l = i) = (1-w_l)w_l^i$, $i \geq 0$, all independent, and

$$S = S_n(r) = \sum_{j \text{ odd} \leq n} jB_j + \sum_{n/2 < l \leq n} 2l\, N_l \in \mathbb{N}.$$

**Lemma 8.2.** $E\, z^S = A_n(rz)/A_n(r)$ for $|z| \leq 1$. Hence $P(S = m) = a_n(m)\, r^m / A_n(r)$ for all integers $m$ (both sides vanish for $m < 0$).

*Proof.* $E\, z^{jB_j} = 1 - p_j + p_j z^j = \frac{1+(rz)^j}{1+r^j}$ and $E\, z^{2lN_l} = \frac{1-w_l}{1-w_l z^{2l}} = \frac{1-r^{2l}}{1-(rz)^{2l}}$. Multiply and use (4.1). ■

Put $\mu = ES$, $\sigma^2 = \operatorname{Var} S$, $\kappa_3$ the third cumulant, and $\varphi(\theta) = Ee^{i\theta S}$. With $v_j = p_j(1-p_j) = \big(4\cosh^2(jt/2)\big)^{-1}$ and $\lambda_l = \frac{4w_l}{(1-w_l)^2} = \sinh^{-2}(lt)$:

$$\varphi(\theta) = \prod_{j \text{ odd} \leq n} \big(1 - p_j + p_j e^{ij\theta}\big) \prod_{n/2 < l \leq n} \frac{1-w_l}{1-w_l e^{2il\theta}}, \tag{8.1}$$

$$\mu = \sum_j jp_j + \sum_l \frac{2lw_l}{1-w_l}, \quad \sigma^2 = \sum_j j^2 v_j + \sum_l l^2 \lambda_l, \quad \kappa_3 = \sum_j j^3 v_j(1-2p_j) + \sum_l (2l)^3 \frac{w_l(1+w_l)}{(1-w_l)^3}. \tag{8.2}$$

(The cumulants of $jB_j$ are $jp_j$, $j^2p_j(1-p_j)$, $j^3p_j(1-p_j)(1-2p_j)$; those of $2lN_l$ are $\frac{2lw}{1-w}$, $\frac{4l^2w}{(1-w)^2}$, $\frac{8l^3w(1+w)}{(1-w)^3}$.)

In all applications $u > 0$ is given and $r = e^{-u/n}$, i.e. $t = u/n$. Then $\mu/n^2$ and $\sigma^2/n^3$ are bounded above and below by the enclosures of Table 8.1 (Lemma 8.5).

*Monotonicity of the mean.* For fixed $n$, $\mu_n(r)$ is continuous and strictly increasing in $r \in (0,1)$ ($\frac{d\mu}{dr} = \sigma^2/r > 0$ from Lemma 8.2: $\mu = r\, A_n'(r)/A_n(r)$), with $\mu \to 0$ as $r \to 0$ and $\mu \to \infty$ as $r \to 1$ (the geometric means $2lw_l/(1-w_l) \to \infty$). Equivalently $u \mapsto \mu_n(e^{-u/n})$ is strictly decreasing and onto $(0,\infty)$. So for every real $m > 0$ there is exactly one $u > 0$ with $\mu_n(e^{-u/n}) = m$.

## 8.3 The local limit estimate

### 8.3.1 Fourier inversion

**Lemma 8.3.** Let $0 < \theta_0 \leq \pi$ and put $\tilde\varphi(\theta) = \varphi(\theta)e^{-i\mu\theta}$, $g(\theta) = e^{-\sigma^2\theta^2/2}$. For every integer $m$,

$$\Big|\sqrt{2\pi}\,\sigma P(S = m) - e^{-(m-\mu)^2/(2\sigma^2)}\Big| \leq M + T + Q,$$

$$M = \frac{\sigma}{\sqrt{2\pi}} \int_{|\theta| \leq \theta_0} |\tilde\varphi - g|\, d\theta, \quad T = \operatorname{erfc}\Big(\frac{\sigma\theta_0}{\sqrt{2}}\Big), \quad Q = \frac{2\sigma}{\sqrt{2\pi}} \int_{\theta_0}^{\pi} |\varphi(\theta)|\, d\theta.$$

*Proof.* $S$ is integer valued, so $P(S = m) = \frac{1}{2\pi}\int_{-\pi}^{\pi} \varphi(\theta) e^{-im\theta} d\theta$. With $x = m - \mu$, $e^{-im\theta}\varphi = \tilde\varphi e^{-ix\theta}$, and $e^{-x^2/(2\sigma^2)} = \frac{\sigma}{\sqrt{2\pi}} \int_{\mathbb{R}} g\, e^{-ix\theta} d\theta$. Hence the difference is $\frac{\sigma}{\sqrt{2\pi}}\big[\int_{|\theta|\leq\pi} (\tilde\varphi - g) e^{-ix\theta} d\theta -$

$\int_{|\theta|>\pi} g\, e^{-ix\theta} d\theta\big]$. Split $|\theta| \le \theta_0$ and $\theta_0 < |\theta| \le \pi$; use $|\tilde{\varphi}| = |\varphi|$, $|\varphi(-\theta)| = |\varphi(\theta)|$ and $\int_{\theta_0<|\theta|\le\pi} g + \int_{|\theta|>\pi} g = \int_{|\theta|>\theta_0} g = \frac{\sqrt{2\pi}}{\sigma}\operatorname{erfc}(\sigma\theta_0/\sqrt{2})$. ■

We use these elementary facts: (E1) $1 - x^2/2 \le \cos x \le 1 - x^2/2 + x^4/24$ for real $x$ (integrate $\sin x \le x$ repeatedly); (E2) Taylor's formula with integral remainder: for $F \in C^r[0,\theta]$, $|F(\theta) - \sum_{s<r} F^{(s)}(0)\theta^s/s!| \le \sup|F^{(r)}|\,|\theta|^r/r!$; (E3) $|e^z - 1| \le |z| e^{\max(0,\operatorname{Re} z)}$; (E4) $\operatorname{erfc}(z) \le e^{-z^2}/(z\sqrt{\pi})$ for $z > 0$ (since $\operatorname{erfc}(z) = \frac{2}{\sqrt{\pi}}\int_z^\infty e^{-s^2} ds \le \frac{2}{\sqrt{\pi}}\int_z^\infty \frac{s}{z} e^{-s^2} ds$).

### 8.3.2 The logarithm of $\varphi$ and its third derivative

For $|\theta| \le \pi$ every factor in (8.1) has positive real part after normalisation: $1 - p_j + p_j e^{iy} = (1-p_j)(1 + q_j e^{iy})$ with $q_j = p_j/(1-p_j) = e^{-jt} < 1$, so $\operatorname{Re}(1 + q_j e^{iy}) \ge 1 - q_j > 0$; and $\operatorname{Re}(1 - w_l e^{iy}) \ge 1 - w_l > 0$. Let $f(\theta) = \operatorname{Log}\varphi(\theta)$, the sum of the principal logarithms of the factors; $f$ is real-analytic on $\mathbb{R}$, $e^f = \varphi$, $f(0) = 0$, and $f^{(s)}(0) = i^s \kappa_s$.

**Lemma 8.4 (third derivative).** (a) Let $0 < p \le \frac{1}{2}$, $v = p(1-p)$, $|y| \le t' < \pi$ and $h = \dfrac{pe^{iy}}{1 - p + pe^{iy}}$. Then

$$|h(1-h)(1-2h)| \le \frac{v\,[(1-2p) + p|y|]}{\cos(y/2)^3} \le \frac{v(1-2p) + t'vp}{\cos(t'/2)^3}.$$

(b) For $0 < w < 1$, $y \in \mathbb{R}$ and $g = \dfrac{we^{iy}}{1 - we^{iy}}$: $|g(1+g)(1+2g)| \le \dfrac{w(1+w)}{(1-w)^3}$. (c) Let $n \ge 1$, $u > 0$, $0 < \tau_0 < \pi$, $\theta_0 = \tau_0/n$. Then $\sup_{|\theta|\le\theta_0} |f'''(\theta)| \le n^4 \kappa(\tau_0)$ with

$$\kappa(\tau_0) = \frac{K_3 + \tau_0 K_4}{\cos(\tau_0/2)^3} + 8K_G,$$

$$K_3 \ge n^{-4} \sum_{j \text{ odd} \le n} j^3 v_j (1 - 2p_j),\ K_4 \ge n^{-5} \sum_j j^4 v_j p_j,\ K_G \ge n^{-4} \sum_l l^3 \frac{w_l(1+w_l)}{(1-w_l)^3}.$$

*Proof.* (a) With $e = e^{iy}$: $1 - h = \frac{1-p}{1-p+pe}$, $1 - 2h = \frac{1-p-pe}{1-p+pe}$, so $h(1-h)(1-2h) = \dfrac{v\,e\,(1-p-pe)}{(1-p+pe)^3}$. Numerator: $|1 - p - pe| = |(1-2p) + p(1-e)| \le (1-2p) + p|y|$ since $|1 - e^{iy}| = 2|\sin(y/2)| \le |y|$. Denominator: $|1 - p + pe|^2 = (1-p)^2 + p^2 + 2p(1-p)\cos y = 1 - 4v\sin^2(y/2) \ge 1 - \sin^2(y/2) = \cos^2(y/2)$ since $4v \le 1$. Finally $\cos(y/2) \ge \cos(t'/2) > 0$ and $|y| \le t'$. (b) $1 + g = \frac{1}{1-we}$, $1 + 2g = \frac{1+we}{1-we}$, so $|g(1+g)(1+2g)| = \frac{w|1+we|}{|1-we|^3} \le \frac{w(1+w)}{(1-w)^3}$. (c) $f(\theta) = \sum_j [\ln(1-p_j) + B_j(ij\theta)] + \sum_l [\ln(1 - w_l) + G_l(2il\theta)]$ with $B_j(z) = \ln(1 + q_j e^z)$, $G_l(z) = -\ln(1 - w_l e^z)$. $B_j'''(z) = h(1-h)(1-2h)$ with $h = q_j e^z/(1+q_j e^z)$, and $G_l'''(z) = g(1+g)(1+2g)$ with $g = w_l e^z/(1-w_l e^z)$ (differentiate $B' = h$, $h' = h(1-h)$, and $G' = g$, $g' = g(1+g)$). By the chain rule $f''' = \sum_j (ij)^3 B_j'''(ij\theta) + \sum_l (2il)^3 G_l'''(2il\theta)$. For $|\theta| \le \theta_0$ and odd $j \le n$, $|j\theta| \le t_j := j\tau_0/n \le \tau_0 < \pi$, so by (a), $|B_j'''| \le \dfrac{v_j(1-2p_j) + t_j v_j p_j}{\cos(\tau_0/2)^3}$; summing $j^3|B_j'''|$ gives at most $\cos(\tau_0/2)^{-3}\big[\sum j^3 v_j(1-2p_j) + \frac{\tau_0}{n}\sum j^4 v_j p_j\big] \le n^4 \cos(\tau_0/2)^{-3}(K_3 + \tau_0 K_4)$. By (b), $\sum_l (2l)^3 |G_l'''| \le 8\sum_l l^3 \frac{w_l(1+w_l)}{(1-w_l)^3} \le 8n^4 K_G$. ■

### 8.3.3 Lattice sums

**Lemma 8.5 (lattice sums).** Let $n_0 \ge 1$, let $f \in C^1[0, 1 + 1/n_0]$ and $n \ge n_0$. Write $TV$ for the total variation and sup for the supremum norm of $f$ on the indicated interval.

- (odd) $\left|\frac{1}{n}\sum_{j\ \mathrm{odd}\le n} f(j/n) - \frac{1}{2}\int_0^1 f\right| \le \frac{TV_{[0,1+1/n_0]}f + \frac{1}{2}\sup|f|}{n}$.
- (even-index geometric) $\left|\frac{1}{n}\sum_{n/2<l\le n} f(l/n) - \int_{1/2}^1 f\right| \le \frac{TV_{[1/2,1+1/n_0]}f + \sup|f|}{n}$.

*Proof.* (odd) The intervals $[(j-1)/n, (j+1)/n]$, $j$ odd $\le n$, are disjoint with union $[0,b]$, $b = 1$ ($n$ even) or $1+1/n$ ($n$ odd). On each, $f(j/n)$ and the mean value $\frac{n}{2}\int f$ both lie between the minimum and maximum of $f$, so they differ by at most the variation of $f$ on the interval. Summing and dividing by $n$: $|\frac{1}{n}\sum f(j/n) - \frac{1}{2}\int_0^b f| \le TV/n$; and $|\frac{1}{2}\int_1^b f| \le \frac{\sup|f|}{2n}$. (geometric) The intervals $[(l-\frac{1}{2})/n, (l+\frac{1}{2})/n]$ are disjoint with union $[a, 1+\frac{1}{2n}]$, $a = \frac{1}{2}+\frac{1}{2n}$ ($n$ even) or $\frac{1}{2}$ ($n$ odd); the same argument gives $TV/n$, and the endpoint mismatch with $[\frac{1}{2}, 1]$ costs at most $\sup|f|/n$. $TV$ and sup over $[0, 1+1/n]$ are bounded by those over $[0, 1+1/n_0]$. ■

The sums needed are of the form $n^{-(q+1)}\sum_j j^q F(cuj/n) = \frac{1}{n}\sum f(j/n)$ with $f(y) = y^q F(cuy)$. The nine sums are listed in Table 8.1 (with $z = cuy$ and $w = e^{-z}$, $p(z) = \frac{1}{1+e^z}$, $v(z) = p(1-p)$, $\lambda(z) = \frac{4w}{(1-w)^2}$):

| name | sum | $q$ | $c$ | $F(z)$ | summation |
|---|---|---|---|---|---|
| $SB$ | $n^{-3}\sum j^2 v_j$ | 2 | 1 | $v(z)$ | odd $j \le n$ |
| $SG$ | $n^{-3}\sum l^2 \lambda_l$ | 2 | 2 | $\lambda(z)$ | $\frac{n}{2} < l \le n$ |
| $V4$ | $n^{-5}\sum j^4 v_j$ | 4 | 1 | $v(z)$ | odd |
| $QG$ | $n^{-5}\sum l^4 \lambda_l$ | 4 | 2 | $\lambda(z)$ | geometric |
| $K3a$ | $n^{-4}\sum j^3 v_j(1-2p_j)$ | 3 | 1 | $v(1-2p)$ | odd |
| $K4a$ | $n^{-5}\sum j^4 v_j p_j$ | 4 | 1 | $vp$ | odd |
| $KGr$ | $n^{-4}\sum l^3 \frac{w_l(1+w_l)}{(1-w_l)^3}$ | 3 | 2 | $\frac{w(1+w)}{(1-w)^3}$ | geometric |
| $MB$ | $n^{-2}\sum j p_j$ | 1 | 1 | $p(z)$ | odd |
| $MG$ | $n^{-2}\sum \frac{l w_l}{1-w_l}$ | 1 | 2 | $\frac{w}{1-w}$ | geometric |

By (8.2) and the table: $\sigma^2/n^3 = SB + SG$, $\mu/n^2 = MB + 2MG$, and $\kappa_3/n^4 = K3a + 8KGr$.

**Algorithm E (enclosure of one lattice sum).** Input: $F$ (with its derivative formula), $q$, $c$, a ball $U = [u_a, u_b]$ for $u$, $n_0$, an integer $Q_e$ (we use $Q_e = 600$), and the flag odd/geometric. Put $[y_a, y_b] = [0,1]$ (odd) or $[\frac{1}{2}, 1]$ (geometric) and $y_e = 1 + 1/n_0$. (1) $J$: split $[y_a, y_b]$ into $Q_e$ equal sub-intervals $Y_i$ and take the ball sum $\sum_i \mathrm{width}\cdot\big(Y_i^q\, F(cUY_i)\big)$ (ball arithmetic: the product encloses $f(y) = y^q F(cuy)$ for all $y \in Y_i$, $u \in U$). (2) Split $[y_a, y_e]$ into $Q_e$ sub-intervals $Y_i'$; $TV$ is the sum of $\mathrm{width}\cdot\sup_{Y_i'\times U}|f'|$ with $f'(y) = qy^{q-1}F(cuy) + y^q\, cu\, F'(cuy)$, and sup is the maximum over $i$ of the upper bound of $|f|$ on $Y_i' \times U$. (3) Return the interval $[\,J/2 - e,\ J/2 + e\,]$ with $e = (TV + \sup/2)/n_0$ (odd) or $[\,J - e,\ J + e\,]$ with $e = (TV + \sup)/n_0$ (geometric). The derivative formulas are $v' = -(1-2p)v$, $p' = -v$, $(v(1-2p))' = -v((1-2p)^2 - 2v)$, $(vp)' = -v(p(1-2p) + v)$, $\lambda' = -\frac{4w(1+w)}{(1-w)^3}$, $\big(\frac{w(1+w)}{(1-w)^3}\big)' = -\frac{w(1+4w+w^2)}{(1-w)^4}$, $\big(\frac{w}{1-w}\big)' = -\frac{w}{(1-w)^2}$ (in $z$, $w = e^{-z}$; verified symbolically).

By Lemma 8.5, the interval returned in Algorithm E contains $n^{-(q+1)}\sum_j j^q F(cuj/n)$ for **every** $n \ge n_0$ and every $u \in U$. We write $X_{lo}, X_{hi}$ for its endpoints, $X \in \{SB, \dots, MG\}$.

### 8.3.4 The main arc

**Proposition 8.6.** Let $n \ge n_0$, $u \in U$, $0 < \tau_0 < \pi$, $\theta_0 = \tau_0/n$, $S_{lo} = SB_{lo} + SG_{lo}$ (so $\sigma^2 \ge S_{lo} n^3$), $\kappa = \kappa(\tau_0)$ computed from $K_3 = K3a_{hi}$, $K_4 = K4a_{hi}$, $K_G = KGr_{hi}$, and suppose $c := \tau_0\kappa/S_{lo} < 3$.

Then

$$M \le D_1 := \frac{\kappa}{6\sqrt{2\pi}\, S_{lo}^{3/2}\sqrt{n_0}\,(\frac12-\frac{c}{6})^2}, \qquad T \le D_2 := \frac{e^{-z^2}}{z\sqrt{\pi}},\ z = \tau_0\sqrt{S_{lo}n_0/2}.$$

*Proof.* By (E2) with $r=3$ and Lemma 8.4(c), $f(\theta) = i\mu\theta - \frac{\sigma^2\theta^2}{2} + R(\theta)$ with $|R| \le K_3'|\theta|^3/6$, $K_3' = n^4\kappa$. Then $\tilde\varphi = ge^R$, and by (E3), $|\tilde\varphi - g| \le g|R|e^{|R|}$. For $|\theta| \le \theta_0$: $|R| \le \frac{K_3'\theta_0}{6}\theta^2$ and $K_3'\theta_0 = n^3\kappa\tau_0 \le c\,\sigma^2$ (as $\sigma^2 \ge S_{lo}n^3$). Hence $ge^{|R|} \le e^{-a\theta^2}$ with $a = \sigma^2(\frac12 - \frac{c}{6}) > 0$, and

$$\int_{|\theta|\le\theta_0} |\tilde\varphi - g| \le \frac{K_3'}{6}\cdot 2\int_0^\infty \theta^3 e^{-a\theta^2}d\theta = \frac{K_3'}{6a^2}.$$

Multiplying by $\sigma/\sqrt{2\pi}$: $M \le \frac{K_3'}{6\sqrt{2\pi}\sigma^3(1/2-c/6)^2} \le \frac{n^4\kappa}{6\sqrt{2\pi}S_{lo}^{3/2}n^{9/2}(\dots)^2} = \frac{\kappa}{6\sqrt{2\pi}S_{lo}^{3/2}\sqrt{n}(1/2-c/6)^2} \le D_1$. For $T$: $\sigma\theta_0/\sqrt{2} = \sigma\tau_0/(n\sqrt{2}) \ge \tau_0\sqrt{S_{lo}n/2} \ge z$ and erfc is decreasing, $e^{-z^2}/(z\sqrt{\pi})$ is decreasing in $z$; (E4). ■

### 8.3.5 Minor arcs

**Lemma 8.7 (modulus).** For $\theta \in \mathbb{R}$,

$$|\varphi(\theta)| \le \exp\Big(-\sum_{j\ \text{odd}\le n} v_j(1-\cos j\theta)\Big)\prod_{n/2<l\le n}\big(1+\lambda_l\sin^2(l\theta)\big)^{-1/2}.$$

*Proof.* $|1-p+pe^{iy}|^2 = 1-2v(1-\cos y) \le e^{-2v(1-\cos y)}$. Also $\big|\frac{1-w}{1-we^{iy}}\big|^2 = \frac{(1-w)^2}{(1-w)^2+4w\sin^2(y/2)} = \big(1+\lambda\sin^2(y/2)\big)^{-1}$; put $y = 2l\theta$. ■

Fix $n \ge n_0$, a cell $U = [u_a, u_b] \ni u$, the enclosures of Algorithm E, and $\tau_0$. Put $\tau = n\theta$ and write $\mathrm{pref} = \sqrt{2\pi S_{hi}}/\pi$, where $S_{hi} = SB_{hi} + SG_{hi}$, so that $\sigma \le \sqrt{S_{hi}}\,n^{3/2}$ and $\frac{2\sigma}{\sqrt{2\pi}} = \sigma\sqrt{2/\pi} \le \mathrm{pref}\cdot n^{3/2}$. Split $[\theta_0,\pi]$ into $A = [\tau_0/n,\ 1.5/n]$, $B = [1.5/n,\ \pi - 1.5/n]$, $C = [\pi - 1.5/n,\ \pi]$ (these are non-empty since $n \ge n_0 > 3$).

**Lemma 8.8 (region A).** Let $[\tau_l,\tau_h] \subset [\tau_0, 1.5]$, $\tau_l \ge 0$, $Z = \lambda(u_a)\min(1,\tau_h^2)$ and $\ell = \ln(1+Z)/Z$ (here $\lambda(u) = 4e^{-u}/(1-e^{-u})^2$). For $\tau = n\theta \in [\tau_l,\tau_h]$, $|\varphi(\theta)| \le e^{-nb}$ with $b = b_o^+ + b_g^+$ ($x^+ = \max(x,0)$),

$$b_o = \tfrac12\tau_l^2 SB_{lo} - \tfrac{1}{24}\tau_h^4 V4_{hi}, \qquad b_g = \tfrac{\ell}{2}\big(\tau_l^2 SG_{lo} - \tfrac13\tau_h^4 QG_{hi}\big).$$

*Proof.* By Lemma 8.7 it suffices to bound the two sums. By (E1), $\sum_j v_j(1-\cos j\theta) \ge \sum_j v_j\big(\frac{(j\theta)^2}{2} - \frac{(j\theta)^4}{24}\big) \ge n\big(\frac{\tau^2}{2}SB - \frac{\tau^4}{24}V4\big)$ (as $\sum j^2 v_j = n^3 SB$, $\sum j^4 v_j = n^5 V4$, $\theta = \tau/n$), and this is $\ge nb_o$. Since $z_l = \lambda_l\sin^2(l\theta) \le \lambda_l\min(1,(l\theta)^2) \le Z$ (here $l\theta \le \tau \le \tau_h$, and $\lambda_l = \lambda(2lu/n) \le \lambda(u) \le \lambda(u_a)$ because $2l/n > 1$ and $\lambda$ is decreasing), concavity of $\ln(1+z)$ gives $\ln(1+z_l) \ge \ell z_l$. And $\sin^2 x \ge x^2 - x^4/3$ ($\sin^2 x = \frac{1-\cos 2x}{2}$, (E1)). Hence $\frac12\sum_l \ln(1+z_l) \ge \frac{\ell}{2}\sum_l \lambda_l\big((l\theta)^2 - \frac{(l\theta)^4}{3}\big) \ge nb_g$, using $\sum l^2\lambda_l = n^3 SG$, $\sum l^4\lambda_l = n^5 QG$. Both exponents are nonnegative, so each may be replaced by its positive part. ■

**Lemma 8.9 (Abel step and region B).** (a) Let $\sin\theta \ne 0$, $X = 1/|\sin\theta|$, and let $v_1 \ge v_3 \ge \cdots \ge v_J \ge 0$ ($J$ odd). Then $\sum_{j\ \text{odd}\le J} v_j(1-\cos j\theta) \ge \sum_{j\ \text{odd},\,X+1\le j\le J} v_j$. (b) Let $Y_0 = \frac{3}{n_0} + \big[1.5\,(1-\frac{1.5^2}{6n_0^2})\big]^{-1} < 1$ and $b_B = \frac12\cdot\frac{1-Y_0}{M}\sum_{i=1}^M v\big(u_b(Y_0 + i\frac{1-Y_0}{M})\big)$ with $v(z) = (2\cosh(z/2))^{-2}$, $M = 300$. For $\theta \in [1.5/n, \pi - 1.5/n]$, $n \ge n_0$, $u \in U$: $|\varphi(\theta)| \le e^{-nb_B}$.

*Proof.* (a) Let $N(j) = \frac{j+1}{2}$ (the number of odd integers in $[1,j]$), $c_i = 1-\cos i\theta$ and $C_j = \sum_{i\ \text{odd}\le j} c_i = N(j) - s_j$ with $s_j = \sum_{i\ \text{odd}\le j}\cos i\theta = \frac{\sin((j+1)\theta)}{2\sin\theta}$ (telescoping $2\sin\theta\cos i\theta = \sin((i+$

$1)\theta)-\sin((i-1)\theta))$. Since $|s_j|\le\min(N(j),X/2)$, $C_j\ge h(j):=(N(j)-X/2)^+$. Put $v_{J+2}=0$. By Abel summation, $\sum_j v_jc_j=\sum_j(v_j-v_{j+2})C_j\ge\sum_j(v_j-v_{j+2})h(j)=\sum_j v_j\big(h(j)-h(j-2)\big)$ with $h(-1)=0$. Here $h$ is non-decreasing, so every difference is $\ge 0$, and $h(j)-h(j-2)=1$ whenever $N(j-2)\ge X/2$, i.e. $j\ge X+1$. This gives (a). (b) For $\theta\in[1.5/n,\pi-1.5/n]$, $|\sin\theta|\ge\sin(1.5/n)\ge\frac{1.5}{n}\big(1-\frac{1.5^2}{6n^2}\big)$, so $(X+3)/n\le Y_0$ for $n\ge n_0$. Drop the second product in Lemma 8.7 (each factor is $\le 1$). By (a) with $J$ the largest odd integer $\le n$, $\sum_j v_j(1-\cos j\theta)\ge\sum_{j\ \mathrm{odd}\in[X+1,J]}v_j$. As $v(\cdot)$ is even and decreasing on $[0,\infty)$, $v_j\ge\frac12\int_j^{j+2}v(yu/n)dy$; the smallest odd $j\ge X+1$ is $<X+3$, and $J+2>n$, so the sum is $\ge\frac12\int_{X+3}^{n}v(yu/n)dy=\frac{n}{2}\int_{(X+3)/n}^{1}v(ux)dx\ge\frac{n}{2}\int_{Y_0}^{1}v(u_bx)dx\ge nb_B$, because a right Riemann sum underestimates the integral of a decreasing function. ■

**Lemma 8.10 (region C).** Let $b_C=\frac12\cdot\frac{1/2-1/n_0}{M}\sum_{i=1}^{M}v\big(u_b(\frac{1}{n_0}+i\frac{1/2-1/n_0}{M})\big)$. For $\theta=\pi-\phi$, $0\le\phi\le 1.5/n$, $n\ge n_0$, $u\in U$: $|\varphi(\theta)|\le e^{-nb_C}$.

*Proof.* For odd $j\le n/2$: $j\phi\le 0.75<\pi/2$, so $\cos j\theta=-\cos j\phi<0$ and $1-\cos j\theta\ge 1$. By Lemma 8.7, $|\varphi|\le\exp(-\sum_{j\ \mathrm{odd}\le n/2}v_j)$. With $J$ the largest odd integer $\le n/2$ we have $J+2>n/2$, and $v_j\ge\frac12\int_j^{j+2}v(yu/n)dy$, so $\sum\ge\frac12\int_1^{n/2}v(yu/n)dy=\frac{n}{2}\int_{1/n}^{1/2}v(ux)dx\ge\frac{n}{2}\int_{1/n_0}^{1/2}v(u_bx)dx\ge nb_C$. ■

**Proposition 8.11 (local limit theorem on a cell).** Let $U=[u_a,u_b]$, $n_0\ge 4$ and $\tau_0\in(0,1.5]$ be such that: $c=\tau_0\kappa(\tau_0)/S_{lo}<2.9$; $n_0b\ge\frac12$ for each sub-cell of Lemma 8.8 when $[\tau_0,1.5]$ is cut into 240 equal parts; $n_0b_B\ge\frac32$ and $n_0b_C\ge\frac32$. Put

$$D_3=\operatorname{pref}\Big[\sqrt{n_0}\sum_i(\tau_{i+1}-\tau_i)e^{-n_0b_i}+\pi n_0^{3/2}e^{-n_0b_B}+1.5\sqrt{n_0}\,e^{-n_0b_C}\Big],\quad \delta=D_1+D_2+D_3.$$

Then for all $n\ge n_0$, $u\in U$, integers $m$:

$$\Big|\sqrt{2\pi}\,\sigma P(S_n(e^{-u/n})=m)-e^{-(m-\mu)^2/(2\sigma^2)}\Big|\le\delta.$$

Moreover $S_{lo}n^3\le\sigma^2\le S_{hi}n^3$ and $M_{lo}n^2\le\mu\le M_{hi}n^2$, where $M=MB+2MG$.

*Proof.* Lemma 8.3 with $\theta_0=\tau_0/n$ gives the bound $M+T+Q$. $M\le D_1$ and $T\le D_2$ by Proposition 8.6 (all quantities there were enclosed uniformly in $n\ge n_0$, $u\in U$). For $Q=\frac{2\sigma}{\sqrt{2\pi}}\int_{\theta_0}^{\pi}|\varphi|$ we use $\frac{2\sigma}{\sqrt{2\pi}}\le\operatorname{pref}\,n^{3/2}$, $\operatorname{pref}=\sqrt{2\pi S_{hi}}/\pi=\sqrt{2S_{hi}/\pi}$. Region $A$ ($\theta=\tau/n$, $d\theta=d\tau/n$): $\int_A|\varphi|\le\frac1n\sum_i(\tau_{i+1}-\tau_i)e^{-nb_i}$, so its contribution to $Q$ is $\le\operatorname{pref}\sqrt{n}\sum_i(\tau_{i+1}-\tau_i)e^{-nb_i}$. Region $B$: length $\le\pi$ and Lemma 8.9: contribution $\le\operatorname{pref}\,\pi n^{3/2}e^{-nb_B}$. Region $C$: length $1.5/n$ and Lemma 8.10: $\le\operatorname{pref}\,1.5\sqrt{n}\,e^{-nb_C}$. Each of the three terms has the form $n^ae^{-n\beta}$ with $a\in\{\frac12,\frac32\}$, which is non-increasing for $n\ge a/\beta$; the hypotheses give $n_0\beta\ge a$. So the three contributions at $n\ge n_0$ are bounded by their values at $n=n_0$. Finally $D_1\propto n^{-1/2}$ and $D_2$ is decreasing in $n$ (only $S_{lo}n_0$ enters, as a lower bound for $n$). The bounds on $\sigma^2,\mu$ are (8.2) and Table 8.1. ■

**Algorithm C (cells).** Fix $n_0=1467$. For $i=19,\dots,331$ let $u_i$ be the double-precision number nearest $i/100$. The 312 cells are $U_i=[u_i,u_{i+1}]$, $i=19,\dots,330$; they are contiguous and cover $[u_{19},u_{331}]$. For each cell: compute the nine enclosures by Algorithm E ($Q_e=600$); for each $\tau_0\in\{0.05i:\ i=2,\dots,24\}$ test the hypotheses of Proposition 8.11 and compute $\delta(\tau_0)$; keep the admissible $\tau_0$ with least $\delta$. All operations are in ball arithmetic at 128 bits; $\delta$ is rounded up to double precision. All later uses of the cells take $u$-intervals whose endpoints are double-precision numbers inside $[u_{19},u_{331}]$.

**Theorem 8.12 (result of Algorithm C).** All 312 cells admit an admissible $\tau_0$. The following hold: $\max\delta=0.08727$ (on $[3.30,3.31]$), $\min\delta=0.02847$; $\tau_0\in[0.10,0.55]$; $\max D_1=0.06539$, $\max D_2=2.21\cdot10^{-3}$, $\max D_3=1.99\cdot10^{-2}$; $\max c=1.1870$. Consequently the conclusion of Proposition 8.11 holds with $n_0=1467$ on every cell. The cell-wise values are stored in

`data/pd/cells56_sharp.json` (Appendix A.3), and Table 8.4 of Section 8.7 lists the first and last cells.

The proof is the computation. (The code is `code/pd/clt56.py`, `make_cells56.py`; a full regeneration produced byte-identical data, see Appendix A.3.)

## 8.4 The block inequality

Fix odd $k \in [3, 33]$, $N_1 = 1500$ and the window parameters $U_A < U_B$ of Table 8.3 (the same numbers for all $k$ are listed there). Call an even $n \geq N_1$ **admissible for** $d$ if $m_n = (d - n^2)/k$ is a positive integer. For admissible $n$ put $e_n = a_n(m_n)$; $n_\pm = n \pm k$ are odd, $n_\pm^2 \equiv n^2 \pmod k$, and $m_\pm = m_n - \frac{n_\pm^2 - n^2}{k}$, i.e. $m_+ = m_n - 2n - k$, $m_- = m_n + 2n - k$; $o_\pm = a_{n_\pm}(m_\pm)$ whenever $m_\pm \geq 0$ (otherwise there is no such term). By Section 8.2 let $u = u_n$ be the unique number with $\mu_n(e^{-u/n}) = m_n$ and put $r = e^{-u/n}$.

The finite differences. For even $n$ and $j = n + t$, $(-1)^j = (-1)^t$. By (4.1), $A_N(r) = \prod_{j \leq N}(1 + (-r)^j)^{-1}$. Put $s = 1/n$, $\xi_t = 1 + ts$, $z_t = (-1)^t r^{\,n+t} = (-1)^t e^{-u\xi_t}$, and for $\varepsilon \in \{+1, -1\}$ let $T_+ = \{1, \dots, k\}$, $T_- = \{-k + 1, \dots, 0\}$. Define

$$\Lambda_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \big(-\log(1 + z_t)\big), \quad \Delta\mu_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \Big(-\xi_t \frac{z_t}{1 + z_t}\Big), \quad \Delta\sigma_\varepsilon = \varepsilon \sum_{t \in T_\varepsilon} \Big(-\xi_t^2 \frac{z_t}{(1 + z_t)^2}\Big). \tag{8.3}$$

Then

$$\log \frac{A_{n+\varepsilon k}}{A_n}(r) = \Lambda_\varepsilon, \qquad \mu_{n+\varepsilon k}(r) - \mu_n(r) = n\,\Delta\mu_\varepsilon, \qquad \sigma^2_{n+\varepsilon k}(r) - \sigma^2_n(r) = n^2 \Delta\sigma_\varepsilon. \tag{8.4}$$

Indeed $\log A_N = -\sum_{j \leq N} \log(1 + (-r)^j)$; with $z = (-r)^j$ one has $r\,\frac{dz}{dr} = jz$, so $r\frac{d}{dr}[-\log(1 + z)] = -\frac{jz}{1+z}$ and $r\frac{d}{dr}\big[-\frac{jz}{1+z}\big] = -\frac{j^2 z}{(1+z)^2}$; summing these over $j \leq N$ gives $\log A_N$, $\mu_N$, $\sigma_N^2$ (Lemma 8.2: $\mu_N = r\frac{d}{dr}\log A_N$, $\sigma_N^2 = r\frac{d}{dr}\mu_N$), and passing from $n$ to $n + k$ adds the terms $j = n + 1, \dots, n + k$, while passing to $n - k$ removes $j = n - k + 1, \dots, n$, with $j = n\xi_t$.

**Lemma 8.13 (block inequality).** Let $n \geq N_1$ be even and admissible, $U = [u_a, u_b] \ni u_n$ a box with $[u_a(1 - k/N_1),\, u_b(1 + k/N_1)] \subset [u_{19}, u_{331}]$, $\delta_U$ the maximum of $\delta$ over the cells meeting that interval, $S_b$ a lower bound of $S_{lo}$ over the cells meeting $U$, and $S_\varepsilon$ a lower bound of $S_{lo}$ over the cells meeting the $u$-range of the partner $n + \varepsilon k$ (namely $[u_a(1 - k/N_1), u_b]$ for $\varepsilon = -1$ and $[u_a, u_b(1 + k/N_1)]$ for $\varepsilon = +1$). Let $s_b = 1/N_1$, $\gamma \geq (1 - k s_b)^{-3}$. Let $\Lambda_\varepsilon, \Delta\mu_\varepsilon, \Delta\sigma_\varepsilon$ be enclosed for $u \in U$, $s \in [0, s_b]$, and put

$$\rho_\varepsilon = \exp\big(\Lambda_\varepsilon - u(2\varepsilon + ks)\big)\big(1 + q_\varepsilon\big)^{-1/2}, \quad q_\varepsilon = \frac{s_b \sup|\Delta\sigma_\varepsilon|}{S_b}, \quad X_\varepsilon = -(2\varepsilon + ks) - \Delta\mu_\varepsilon,$$

$$\beta_\varepsilon = 1 - \frac{\sup X_\varepsilon^2 \; s_b\,\gamma}{2S_\varepsilon} - \delta_U, \qquad R_U = \frac{1 + \delta_U}{\frac{1}{2}\sum_{\varepsilon = \pm 1} \inf \rho_\varepsilon \cdot \beta_\varepsilon}$$

(the suprema and infima are over $u \in U$, $s \in [0, s_b]$). If $\beta_\pm > 0$ then $m_\pm \geq 0$ and $e_n \leq R_U \cdot \frac{1}{2}(o_{n-k} + o_{n+k})$.

(Here $(n_\varepsilon^2 - n^2)/(kn) = 2\varepsilon + ks$ for $n_\varepsilon = n + \varepsilon k$.)

*Proof.* Write $a_N(m) = A_N(r) r^{-m} P_N(S_N(r) = m)$ (Lemma 8.2) for $N \in \{n, n_\pm\}$ with the same $r$; $S_{n_\varepsilon}$ is the model of Section 8.2 at level $u_\varepsilon = u n_\varepsilon / n \in \{u(1 + \varepsilon k s)\}$, which lies in the range of the partner. Then

$$\frac{o_\varepsilon}{e_n} = \frac{A_{n_\varepsilon}}{A_n}(r)\, r^{m_n - m_\varepsilon}\, \frac{P_{n_\varepsilon}(m_\varepsilon)}{P_n(m_n)}, \qquad r^{m_n - m_\varepsilon} = r^{(n_\varepsilon^2 - n^2)/k} = e^{-u(2\varepsilon + ks)},$$

because $(n_\varepsilon^2-n^2)/k=(2\varepsilon kn+k^2)/k=n(2\varepsilon+ks)$ and $r=e^{-us}$. Apply Proposition 8.11 at $(n,u)$ with $m=m_n=\mu_n$ and at $(n_\varepsilon,u_\varepsilon)$ with $m=m_\varepsilon$ ($n_\varepsilon\ge N_1-k\ge 1467=n_0$; the $u$-values lie in cells by the hypothesis):

$$P_n(m_n)\le\frac{1+\delta_U}{\sqrt{2\pi}\sigma_n},\qquad P_{n_\varepsilon}(m_\varepsilon)\ge\frac{e^{-x_\varepsilon^2/(2\sigma_\varepsilon^2)}-\delta_U}{\sqrt{2\pi}\sigma_\varepsilon}\ge\frac{1-x_\varepsilon^2/(2\sigma_\varepsilon^2)-\delta_U}{\sqrt{2\pi}\sigma_\varepsilon},$$

where $\sigma_\varepsilon=\sigma_{n_\varepsilon}(r)$ and $x_\varepsilon=m_\varepsilon-\mu_{n_\varepsilon}(r)=-\frac{n_\varepsilon^2-n^2}{k}-(\mu_{n_\varepsilon}-\mu_n)=nX_\varepsilon$ by (8.4) (as $\mu_n=m_n$). Since $\sigma_\varepsilon^2=\sigma_n^2+n^2\Delta\sigma_\varepsilon\le\sigma_n^2(1+s\sup|\Delta\sigma_\varepsilon|/S_b)\le\sigma_n^2(1+q_\varepsilon)$ (using $\sigma_n^2\ge S_bn^3$), $\sigma_n/\sigma_\varepsilon\ge(1+q_\varepsilon)^{-1/2}$. Also $x_\varepsilon^2/(2\sigma_\varepsilon^2)\le\frac{X_\varepsilon^2n^2}{2S_\varepsilon n_\varepsilon^3}=\frac{X_\varepsilon^2 s}{2S_\varepsilon}\big(\frac{n}{n_\varepsilon}\big)^3\le\frac{\sup X_\varepsilon^2 s_b\gamma}{2S_\varepsilon}$ since $n_\varepsilon\ge n(1-ks)$. Combining: if $\beta_\varepsilon>0$ then $P_{n_\varepsilon}(m_\varepsilon)>0$, so $m_\varepsilon$ is in the support of $S_{n_\varepsilon}$, hence $m_\varepsilon\ge 0$ and $o_\varepsilon$ is a term; and $o_\varepsilon\ge e_n\,\rho_\varepsilon\beta_\varepsilon/(1+\delta_U)$. Adding the two inequalities: $\frac12(o_++o_-)\ge e_n\frac{\frac12\sum\rho_\varepsilon\beta_\varepsilon}{1+\delta_U}$, i.e. $e_n\le R_U\frac12(o_++o_-)$. ■

**Algorithm R.** Input $k$, $U_A,U_B$ (Table 8.3), $H=0.0025$, the cells of Theorem 8.12. Let $s_b$ be the upper end of the ball $1/N_1$, $\gamma$ the upper end of $(1-ks_b)^{-3}$, $n_{box}=\mathrm{round}((U_B-U_A)/H)$, and for $i<n_{box}$ the box $U_i=[U_A+iH,\ U_A+(i+1)H]$ (the last one ends at $U_B$). For each box compute $\delta_U$, $S_b$, $S_\varepsilon$ from the cells (an exact comparison of double-precision endpoints; the program asserts that the cells meeting the range are contiguous and cover it), enclose $\Lambda_\varepsilon,\Delta\mu_\varepsilon,\Delta\sigma_\varepsilon$ by interval arithmetic in $u\in U_i$, $s\in[0,s_b]$ (the $2k$ terms $t\in T_\pm$ are summed with interval arithmetic in the variables $(u,s)$), check $\beta_\varepsilon>0$, and compute $R_{U_i}$ (rounded up). Return $R_k=\max_i R_{U_i}$ (rounded up).

**Proposition 8.14 (block bound).** For each odd $k\in[3,33]$ Algorithm R succeeds (all $\beta_\varepsilon>0$, all required cells exist) and returns the value $R_k$ of Table 8.2; in all cases $R_k<1-10^{-6}$. Consequently, for $d$ allowed and every even $n\ge N_1$ admissible for $d$ with $u_n\in[U_A,U_B]$: $e_n\le R_k\cdot\frac12(o_{n-k}+o_{n+k})$, and

$$\sum_{n\text{ even admissible},\ u_n\in[U_A,U_B]} e_n\le R_k\,O_d.$$

*Proof.* The computation is the content of the first assertion (code `code/pd/blocks56.py`). For the second, each $u_n\in[U_A,U_B]$ lies in a box $U_i$, and Lemma 8.13 applies with $R_{U_i}\le R_k$ ($n\ge N_1$ gives $s\le s_b$). Every odd $N$ occurs at most twice among $\{n\pm k\}$ ($n=N-k$ as $n_+$ and $n=N+k$ as $n_-$) with weight $\frac12$ each, so $\sum_n\frac12(o_{n-k}+o_{n+k})\le\sum_{N\text{ odd}}a_N(m_N)=O_d$. ■

## 8.5 The tails

Let $\Phi_1(u)=\frac12\int_0^1\log(1+e^{-uy})dy+\frac12\int_1^2\big[-\log(1-e^{-uy})\big]dy$, $g(u)=-\log(1-e^{-u})$, and for $\alpha=n/\sqrt{d}$

$$F_k(\alpha,u)=\alpha\Phi_1(u)+\frac{(1-\alpha^2)u}{k\alpha},\qquad \rho(\alpha)=\frac{1/\alpha^2-1}{k}=\frac{m_n}{n^2}.$$

**Lemma 8.15.** (a) $\Phi_1(u)=\big[\mathrm{Li}_2(-e^{-u})+\frac{\pi^2}{12}+\mathrm{Li}_2(e^{-u})-\mathrm{Li}_2(e^{-2u})\big]/(2u)$ for $u>0$. (b) For even $n\ge2$ and $u>0$: $\log A_n(e^{-u/n})\le n\Phi_1(u)+\log 2$. (c) For $n\ge1$, $u>0$: $\log A_n(e^{-u/n})\ge n\Phi_1(u)-\frac12\log 2-g(u)$. (d) For $n$ even with $n^2\le d$, $m=(d-n^2)/k\in\mathbb{Z}$ and any $u_c>0$: $a_n(m)\le A_n(e^{-u_c/n})e^{u_cm/n}\le 2\exp\big(\sqrt{d}\,F_k(n/\sqrt{d},u_c)\big)$. (e) $\partial_\alpha^2F_k=2u/(k\alpha^3)>0$; hence for fixed $u$, $F_k(\cdot,u)$ on an interval $[x,y]$ is at most $\max(F_k(x,u),F_k(y,u))$, and for $\alpha$ in an interval with midpoint $c$ and half-width $h$, $F_k(\alpha,u)\ge F_k(c,u)-h\,|\partial_\alpha F_k(c,u)|$.

*Proof.* (a) $\frac{d}{dy}\mathrm{Li}_2(-e^{-uy})=u\log(1+e^{-uy})$ and $\frac{d}{dy}\mathrm{Li}_2(e^{-uy})=u\log(1-e^{-uy})$ (from $\mathrm{Li}_2'(z)=-\log(1-z)/z$), and $\mathrm{Li}_2(-1)=-\pi^2/12$. Integrating, $\int_0^1\log(1+e^{-uy})dy=[\mathrm{Li}_2(-e^{-u})+\pi^2/12]/u$ and $\int_1^2[-\log(1-e^{-uy})]dy=[\mathrm{Li}_2(e^{-u})-\mathrm{Li}_2(e^{-2u})]/u$. By (4.1), with $r=e^{-u/n}$, $\log A_n=\sum_{j\text{ odd}\le n}f_B(j/n)+\sum_{n/2<l\le n}f_G(l/n)$, $f_B(x)=\log(1+e^{-ux})$, $f_G(x)=-\log(1-e^{-2ux})$, both

decreasing, and $\Phi_1 = \frac{1}{2}\int_0^1 f_B + \int_{1/2}^1 f_G$. (b) For odd $j \ge 3$, $f_B(j/n) \le \frac{n}{2}\int_{(j-2)/n}^{j/n} f_B$; these intervals are disjoint in $[1/n, 1]$, and $f_B(1/n) \le \log 2$. So $\sum_{\text{odd}} f_B \le \log 2 + \frac{n}{2}\int_0^1 f_B$. For $n$ even, the integers $l \in (n/2, n]$ give the intervals $[(l-1)/n, l/n]$ with union $[1/2, 1]$, and $f_G(l/n) \le n\int_{(l-1)/n}^{l/n} f_G$. Add. (c) $f_B(j/n) \ge \frac{n}{2}\int_{j/n}^{(j+2)/n} f_B$; the intervals for odd $j \le n$ cover $[1/n, 1]$, so $\sum f_B \ge \frac{n}{2}\int_{1/n}^1 f_B \ge \frac{n}{2}\int_0^1 f_B - \frac{1}{2}\log 2$ ($f_B \le \log 2$). $f_G(l/n) \ge n\int_{l/n}^{(l+1)/n} f_G$ for $l = \lfloor n/2 \rfloor + 1, \dots, n$; the union contains $[\frac{1}{2} + \frac{1}{n}, 1]$ and $f_G \le f_G(\frac{1}{2}) = g(u)$ on $[\frac{1}{2}, 1]$, so $\sum f_G \ge n\int_{1/2}^1 f_G - g(u)$. (d) All $a_n(m') \ge 0$ (Lemma 4.2), so $a_n(m) r^m \le A_n(r)$; with $r = e^{-u_c/n}$ and (b): $\log a_n(m) \le n\Phi_1(u_c) + \log 2 + u_c m/n$, and $u_c m/n = \sqrt{d}\, u_c(1-\alpha^2)/(k\alpha)$, $n = \alpha\sqrt{d}$. (e) $F_k = \alpha\Phi_1 + \frac{u}{k}(\frac{1}{\alpha} - \alpha)$ and the first term is linear in $\alpha$; $\partial_\alpha^2 = \frac{u}{k}\cdot\frac{2}{\alpha^3}$. Convex functions lie below the chord's maximum and above their tangents. ■

**Lemma 8.16 (small $\alpha$).** Let $3 \le k \le 33$, $n \ge 2$ even, $\alpha = n/\sqrt{d} \le 0.005$ and $u_c = k\alpha^2/2$. Then

$$F_k(\alpha, u_c) \le F_{\text{tiny}}(\alpha) := \frac{\alpha}{2}\Big(2 - \log k - 2\log\alpha + \frac{3k\alpha^2}{8}\Big), \quad g(u_c) \le \log\frac{2d}{k},$$

$F_{\text{tiny}}$ is increasing on $(0, 1/\sqrt{k})$. Hence $\log a_n(m) \le \sqrt{d}\, F_{\text{tiny}}(0.005) + \log 2 + \log(2d/k)$.

*Proof.* $1 - e^{-x} = 2e^{-x/2}\sinh(x/2) \ge xe^{-x/2}$, so $g(u) \le -\log u + u/2$. Bound $\int_0^1 \log(1 + e^{-uy})dy \le \log 2$ and $-\log(1 - e^{-uy}) \le -\log(uy) + uy/2$ on $[1, 2]$, using $\int_1^2 \log y\, dy = 2\log 2 - 1$: $\Phi_1(u) \le \frac{1}{2}(1 - \log 2 - \log u + \frac{3u}{4})$. With $u = u_c = k\alpha^2/2$: $-\log 2 - \log u_c = -\log k - 2\log\alpha$, $\frac{3u_c}{4} = \frac{3k\alpha^2}{8}$, and $\frac{(1-\alpha^2)u_c}{k\alpha} \le \frac{\alpha}{2}$; add. Monotonicity: $F'_{\text{tiny}}(\alpha) = -\log(\sqrt{k}\alpha) + \frac{9k\alpha^2}{16} > 0$ for $\sqrt{k}\alpha < 1$. Finally $g(u_c) \le -\log u_c + \frac{u_c}{2} = \log\frac{2d}{kn^2} + \frac{k\alpha^2}{4} \le \log\frac{2d}{k} - \log 4 + 0.001$. ■

**Lemma 8.17 (central odd term).** Let $A_0 \in (0, 1)$, $d$ allowed, $d \ge D$ where $A_0\sqrt{D} \ge N_1$, $h = k/\sqrt{d}$. There is an odd integer $n^* \in [A_0\sqrt{d},\, A_0\sqrt{d} + 2k]$ with $k \mid d - n^{*2}$; $m^* = (d - n^{*2})/k \ge 0$ if $A_0 + 2h < 1$. Suppose $\underline{M}(U_{LO}) \ge (1/A_0^2 - 1)/k$ and $\overline{M}(U_{HI}) \le (1/(A_0 + 2h)^2 - 1)/k$, where $\underline{M}(u) \le \mu_n(e^{-u/n})/n^2 \le \overline{M}(u)$ are the enclosures of Table 8.1 valid for all $n \ge N_1$. Then the saddle $u^*$ of $n^*$ satisfies $U_{LO} \le u^* \le U_{HI}$. Let $\delta_c = \max\delta$ over the cells meeting $[U_{LO}, U_{HI}]$ and $S_{hi} \ge \sigma^2/n^3$ for $n \ge N_1$, $u \in [U_{LO}, U_{HI}]$. Then $o^* := a_{n^*}(m^*)$ satisfies

$$\log o^* \ge \sqrt{d}\,\Psi_{\text{low}} - C^*, \quad C^* = \tfrac{1}{2}\log 2 + g(U_{LO}) + \tfrac{1}{2}\log 2\pi + \tfrac{3}{2}\log(A_0\sqrt{d} + 2k) + \tfrac{1}{2}\log S_{hi} - \log(1 - \delta_c),$$

where $\Psi_{\text{low}} \le \min\{F_k(\alpha, u) : \alpha \in [A_0, A_0 + 2h],\, u \in [U_{LO}, U_{HI}]\}$.

*Proof.* The interval contains $2k$ consecutive integers, so one in every class mod $2k$; as $k$ is odd, CRT gives an odd $n^*$ in the class $n_0 \pmod k$ where $n_0^2 \equiv d \pmod k$ ($d$ allowed). $\alpha^* = n^*/\sqrt{d} \in [A_0, A_0 + 2h]$ and $n^* \ge A_0\sqrt{D} \ge N_1$. Let $\rho^* = m^*/n^{*2} = \rho(\alpha^*) \in [(1/(A_0 + 2h)^2 - 1)/k, (1/A_0^2 - 1)/k]$. Then $\mu_{n^*}(e^{-U_{LO}/n^*})/n^{*2} \ge \rho^* \ge \mu_{n^*}(e^{-U_{HI}/n^*})/n^{*2}$ and since $u \mapsto \mu_{n^*}(e^{-u/n^*})$ is strictly decreasing, $u^* \in [U_{LO}, U_{HI}]$. At $m = \mu$, Proposition 8.11 gives $P(S = m^*) \ge (1 - \delta_c)/(\sqrt{2\pi}\sigma)$, with $\sigma \le \sqrt{S_{hi}}\, n^{*3/2}$. Lemma 8.15(c) with $g(u^*) \le g(U_{LO})$: $a_{n^*}(m^*) = A(r^*)r^{*-m^*}P \ge \exp(n^*\Phi_1(u^*) + u^*m^*/n^* - \frac{1}{2}\log 2 - g(U_{LO}))\cdot\frac{1-\delta_c}{\sqrt{2\pi}\sqrt{S_{hi}}\, n^{*3/2}}$, and $n^*\Phi_1(u^*) + u^*m^*/n^* = \sqrt{d}\, F_k(\alpha^*, u^*)$, $n^* \le A_0\sqrt{d} + 2k$. ■

**Lemma 8.18 (window).** Let $\alpha_1 \ge (1 + k\underline{M}(U_A))^{-1/2}$, $\alpha_2 \le (1 + k\overline{M}(U_B))^{-1/2}$, $D_0 \ge (N_1/\alpha_1)^2$. For $d \ge D_0$ and every even admissible $n$ with $\alpha = n/\sqrt{d} \in [\alpha_1, \alpha_2]$: $n \ge N_1$ and $u_n \in [U_A, U_B]$.

*Proof.* $n = \alpha\sqrt{d} \ge \alpha_1 N_1/\alpha_1 = N_1$. $\rho(\alpha)$ is decreasing, $\rho(\alpha_1) \le \underline{M}(U_A)$ and $\rho(\alpha_2) \ge \overline{M}(U_B)$. The saddle satisfies $\mu_n/n^2 = \rho(\alpha)$; so $\mu_n(e^{-u_n/n})/n^2 \le \underline{M}(U_A) \le \mu_n(e^{-U_A/n})/n^2$ gives $u_n \ge U_A$ (monotonicity), and similarly $u_n \le U_B$. ■

**Algorithm T.** Input $k$ and the numbers of Table 8.3 ($U_A, U_B, U_{LO}, U_{HI}, A_0$), the constants $\underline{M}(U_A)$, $\overline{M}(U_B)$, $\underline{M}(U_{LO})$, $\overline{M}(U_{HI})$, $S_{hi}$ (from Algorithm E on $[U_{LO}, U_{HI}]$ cut into 35 equal pieces, $n_0 = 1500$) and $\delta_c$. Set $\eta_t = 10^{-6}$, $\alpha_1, \alpha_2, D_0$ as in Lemma 8.18 (rounded in the safe direction),

$h = k/\sqrt{D_0}$, $\alpha_m = A_0 + h$. 1. (Window condition) verify $A_0\sqrt{D_0} \ge N_1$ and the two hypotheses of Lemma 8.17 for $d = D_0$. 2. $\Psi_{\text{low}}$: cut $[U_{LO}, U_{HI}]$ into boxes $V_i$ of width $10^{-4}$ and take the minimum over $i$ of the lower end of $F_k(\alpha_m, V_i) - h|\partial_\alpha F_k(\alpha_m, V_i)|$ (Lemma 8.15(e)). 3. $\alpha$-boxes: $[0.005 + 0.0025i,\ 0.005 + 0.0025(i+1)]$ for $i \ge 0$, the last one truncated at $\alpha_1$ (left tail), and $[\alpha_2 + 0.0025i, \alpha_2 + 0.0025(i+1)]$ truncated at 1 (right tail). For each box $[x, y]$ choose $u_c$ (a double near the minimiser of $F_k(\frac{x+y}{2}, \cdot)$; any positive value is valid) and put $\text{Fup} = \max(F_k(x, u_c), F_k(y, u_c))^{\uparrow}$, $\text{gap} = \text{Fup} - \Psi_{\text{low}}$, $w_{box} = \sqrt{D_0}\,\text{gap} + g(u_c)$. 4. Tiny part: $\text{tiny_gap} = F_{\text{tiny}}(0.005) - \Psi_{\text{low}}$, $w_{\text{tiny}} = \sqrt{D_0}\,\text{tiny_gap} + \log(2D_0/k) + 1$. 5. $C^*(D_0)$ as in Lemma 8.17 at $d = D_0$, and $T(D_0) = \log\sqrt{D_0} + \log 2 + \max(\max_{box} w_{box}, w_{\text{tiny}}) + C^*(D_0) - \log\eta_t$. 6. Check $\max \text{gap} \le -2.5/\sqrt{D_0}$ and $\text{tiny_gap} \le -4.5/\sqrt{D_0}$ and $T(D_0) \le 0$.

**Proposition 8.19 (tail bound).** Suppose Algorithm T succeeds. Then for every allowed $d \ge D_0$,

$$\sum_{\substack{n \text{ even}, 2 \le n \le \sqrt{d} \\ n/\sqrt{d} \notin [\alpha_1, \alpha_2],\ k \mid d - n^2}} a_n\Big(\frac{d - n^2}{k}\Big) \le \eta_t\, o^* \le \eta_t\, O_d.$$

*Proof.* Fix $d \ge D_0$, put $x = \sqrt{d}$. By Lemma 8.17 $\log o^* \ge x\Psi_{\text{low}} - C^*(d)$: the window $[A_0, A_0 + 2k/x]$ is contained in the one at $D_0$, the $u^*$-hypotheses at $d$ follow from those at $D_0$ (the right side of the second inequality increases with $d$), and $\Psi_{\text{low}}$ is a lower bound on the larger window. For an even $n$ with $\alpha = n/x \in [0.005, \alpha_1] \cup [\alpha_2, 1]$, $\alpha$ lies in a box $[x_b, y_b]$ and Lemma 8.15(d),(e) give $\log a_n \le x \max(F_k(x_b, u_c), F_k(y_b, u_c)) + \log 2 \le x\text{Fup} + \log 2$ (and $+g(u_c)$ may be added, harmlessly). For $\alpha < 0.005$, Lemma 8.16 gives $\log a_n \le xF_{\text{tiny}}(0.005) + \log 2 + \log(2d/k)$. There are at most $x/2 \le x$ even $n$. Hence the tail sum divided by $o^*$ is at most $\exp(\tau(x))$ with

$$\tau(x) = \log x + \log 2 + \max\Big(\max_{box}[x(\text{Fup} - \Psi_{\text{low}}) + g(u_c)],\ x(F_{\text{tiny}}(0.005) - \Psi_{\text{low}}) + \log(2x^2/k)\Big) + C^*(x).$$

It suffices to show $\tau(x) \le \log\eta_t$ for $x \ge \sqrt{D_0}$; by hypothesis $\tau(\sqrt{D_0}) = T(D_0) + \log\eta_t \le \log\eta_t$. The function of $x$ in the box part is $\log x + x\,\text{gap} + \frac{3}{2}\log(A_0 x + 2k) + \text{const}$ with derivative $\le \text{gap} + \frac{1}{x} + \frac{3A_0}{2(A_0 x + 2k)} \le \text{gap} + \frac{2.5}{x} \le 0$ for $x \ge \sqrt{D_0}$ (as $\text{gap} \le -2.5/\sqrt{D_0} \le -2.5/x$). The tiny part is $x\,\text{tiny_gap} + \log x + \log(2x^2/k) + \frac{3}{2}\log(A_0 x + 2k) + \text{const}$, with derivative $\le \text{tiny_gap} + \frac{1+2+1.5}{x} \le 0$. A maximum of non-increasing functions is non-increasing. Finally $o^* \le O_d$. ■

**Proposition 8.20 (results of Algorithms R and T).** For every odd $k \in \{3, \dots, 33\}$ both algorithms succeed with the parameters of Table 8.3 and give the values of Table 8.2 (in particular $R_k < 1 - 2\eta_t$, $T(D_0) < 0$, and $\max \text{gap} \cdot \sqrt{D_0}$ as listed).

This is a computation (Appendix A.3). Its inputs are the cells (Theorem 8.12) and the constants of Table 8.3.

## 8.6 Proof of Theorem 8.1

**Part 1.** Let $k$ be odd, $3 \le k \le 33$, $d \ge D_0(k)$ allowed. The even $n$ with $n^2 \le d$, $k \mid d - n^2$ are split by $\alpha = n/\sqrt{d}$. The terms with $\alpha \in [\alpha_1, \alpha_2]$ satisfy $n \ge N_1$ and $u_n \in [U_A, U_B]$ (Lemma 8.18), so by Proposition 8.14 their sum is $\le R_k O_d$. The remaining terms are bounded by Proposition 8.19 by $\eta_t O_d$; the term $n = 0$ vanishes for $d > 0$. Therefore $E_d \le (R_k + \eta_t)O_d$. By Lemma 8.17, $O_d \ge o^* > 0$. With (4.4), $(-1)^d g_k(d) = E_d - O_d \le -(1 - R_k - \eta_t)O_d < 0$ (Table 8.2: $R_k < 0.906$).

**Part 2 (finite range).** Section 8.7 describes the modular certificate for $0 \le d \le 6 \cdot 10^7$ and the exact evaluation of the candidates, giving Table 8.5 and Appendix B.1: the set of allowed $d \le 6 \cdot 10^7$ with $g_k(d) = 0$. Since $6 \cdot 10^7 > D_0(k)$ for each $k$, Part 1 and the finite computation together give $Z_k$ in full; and $\max Z_k < D_0(k)$ (Table 8.5).

**Part 3.** By Proposition 3.4.

## 8.7 The finite certificate, the tables, and the zero sets

### 8.7.1 The modular certificate

**Lemma 8.21.** Let $k$ be odd and $D \ge 0$. Put $a_0[m] = \delta_{m,0}$ and, for $n \ge 1$ and $0 \le m \le (D-n^2)/k$, $a_n[m] = a_{n-1}[m] - (-1)^n a_n[m-n]$ (with $a_n[m'] = 0$ for $m' < 0$). Then $a_n[m] = a_n(m)$ and for $0 \le d \le D$

$$G[d] := \sum_{n \ge 0,\ n^2 \le d,\ k \mid d-n^2} (-1)^n a_n\Big(\frac{d-n^2}{k}\Big) = (-1)^d g_k(d).$$

If $G[d] \not\equiv 0 \pmod p$ for some integer modulus $p \ge 2$, then $g_k(d) \ne 0$.

*Proof.* $A_n(x)(1+(-x)^n) = A_{n-1}(x)$ by (4.1); compare coefficients of $x^m$. The identity for $G$ is (4.4). The recursion for $a_n[m]$ needs only $m \le (D-n^2)/k$, a range that decreases with $n$, so the entries computed at step $n-1$ suffice. Reduction mod $p$ commutes with the integer recursion. ■

**Algorithm M.** Input $k$, $D = 6 \cdot 10^7$ and a modulus $p < 2^{63}$; compute $G[d] \bmod p$ for all $0 \le d \le D$ by Lemma 8.21 in 64-bit unsigned arithmetic (sums of two residues $< 2p < 2^{64}$ do not overflow), and print the allowed $d$ with $G[d] \equiv 0$. (`cert/modcert_copy.c`; $O(D^{3/2}/k)$ operations.) Run with the two primes $p_1 = 2^{61} - 1 = 2305843009213693951$ and $p_2 = 4611686018427387847$ for each odd $k \in [3, 33]$ (the moduli need not be prime for the argument; both are).

**Theorem 8.22 (zero sets).** For each odd $k \in \{3, \dots, 33\}$ let $C_k$ be the set of allowed $d \le 6 \cdot 10^7$ with $G[d] \equiv 0$ modulo both $p_1$ and $p_2$. Then every $d \in C_k$ has $g_k(d) = 0$ (exact evaluation in integer arithmetic), so $Z_k \cap [0, 6 \cdot 10^7] = C_k$. The cardinalities and maxima are in Table 8.5, the elements in Appendix B.1. In every case $\max Z_k < D_0(k) \le 6 \cdot 10^7$.

*Proof.* If $d$ is allowed, $d \le D$ and $d \notin C_k$ then $G[d] \not\equiv 0$ for $p_1$ or $p_2$, so $g_k(d) \ne 0$ by Lemma 8.21. For $d \in C_k$, $g_k(d)$ is computed exactly by `code/zero_sets.py` and vanishes (the program also evaluates $g_k(d)$ exactly for all allowed $d \le \max C_k$ and asserts that the zeros are precisely $C_k$). ■

### 8.7.2 Tables

**Table 8.2.** Results of Algorithms R and T ($N_1 = 1500$, $\eta_t = 10^{-6}$, $H = 0.0025$). $\eta_k = 1 - R_k - 10^{-6}$ truncated to five decimals. Entries are printed to the digits shown; the stored values (`data/pd/pd56_results.json`) are directed-rounded doubles, and $D_0$ is exact.

| $k$ | $R_k$ | $\max\delta$ on block cells | $\eta_k$ | $\alpha_1$ | $\alpha_2$ | $D_0$ | $T(D_0)$ | $\max\text{gap}\cdot\sqrt{D_0}$ |
|---|---|---|---|---|---|---|---|---|
| 3 | 0.5077783 | 0.05933 | 0.49222 | 0.36739 | 0.74689 | 16,669,607 | -69.25 | -108.45 |
| 5 | 0.5601190 | 0.04374 | 0.43988 | 0.33225 | 0.72095 | 20,382,618 | -54.18 | -92.85 |
| 7 | 0.5989585 | 0.03885 | 0.40104 | 0.30672 | 0.70327 | 23,916,479 | -49.60 | -88.00 |
| 9 | 0.6311271 | 0.04069 | 0.36887 | 0.28726 | 0.68958 | 27,265,747 | -47.21 | -85.45 |
| 11 | 0.6601314 | 0.04215 | 0.33986 | 0.27013 | 0.67591 | 30,835,058 | -48.67 | -86.81 |
| 13 | 0.6867128 | 0.04462 | 0.31328 | 0.26126 | 0.66711 | 32,963,377 | -40.85 | -78.86 |
| 15 | 0.7098506 | 0.04621 | 0.29014 | 0.25133 | 0.65684 | 35,620,105 | -38.58 | -76.51 |
| 17 | 0.7316365 | 0.04789 | 0.26836 | 0.24350 | 0.64926 | 37,947,145 | -35.41 | -73.27 |
| 19 | 0.7529213 | 0.04973 | 0.24707 | 0.23414 | 0.64067 | 41,043,238 | -36.87 | -74.70 |
| 21 | 0.7740846 | 0.05103 | 0.22591 | 0.22915 | 0.63381 | 42,849,645 | -32.55 | -70.32 |
| 23 | 0.7949990 | 0.05225 | 0.20499 | 0.22225 | 0.62687 | 45,550,651 | -32.75 | -70.50 |
| 25 | 0.8160826 | 0.05402 | 0.18391 | 0.21627 | 0.62111 | 48,102,974 | -32.56 | -70.28 |

| $k$ | $R_k$ | $\max \delta$ on block cells | $\eta_k$ | $\alpha_1$ | $\alpha_2$ | $D_0$ | $T(D_0)$ | $\max \text{gap} \cdot \sqrt{D_0}$ |
|---|---|---|---|---|---|---|---|---|
| 27 | 0.8379773 | 0.05526 | 0.16202 | 0.21361 | 0.61493 | 49,312,289 | -27.62 | -65.28 |
| 29 | 0.8596991 | 0.05641 | 0.14029 | 0.20890 | 0.60963 | 51,557,554 | -26.95 | -64.58 |
| 31 | 0.8819648 | 0.05761 | 0.11803 | 0.20472 | 0.60507 | 53,684,094 | -26.10 | -63.72 |
| 33 | 0.9052174 | 0.05905 | 0.09478 | 0.20099 | 0.59981 | 55,698,896 | -25.05 | -62.64 |

**Table 8.3.** Window parameters (inputs; chosen from the critical point of $F_k$, any values passing the checks would do). In the continuation, $\underline{M}, \overline{M}$ are enclosures of $\mu_n/n^2$ at the indicated points and $S_{hi}$ is the maximum of $SB_{hi} + SG_{hi}$ over 35 equal pieces of $[U_{LO}, U_{HI}]$.

| $k$ | UA | UB | ULO | UHI | A0 |
|---|---|---|---|---|---|
| 3 | 0.21 | 1.1 | 0.346 | 0.519 | 0.508 |
| 5 | 0.27 | 1.38 | 0.435 | 0.653 | 0.466 |
| 7 | 0.31 | 1.59 | 0.501 | 0.751 | 0.438 |
| 9 | 0.34 | 1.76 | 0.554 | 0.83 | 0.417 |
| 11 | 0.36 | 1.89 | 0.598 | 0.896 | 0.401 |
| 13 | 0.39 | 2.02 | 0.636 | 0.953 | 0.387 |
| 15 | 0.41 | 2.12 | 0.669 | 1.004 | 0.375 |
| 17 | 0.43 | 2.22 | 0.699 | 1.049 | 0.365 |
| 19 | 0.44 | 2.3 | 0.726 | 1.089 | 0.356 |
| 21 | 0.46 | 2.38 | 0.751 | 1.127 | 0.348 |
| 23 | 0.47 | 2.45 | 0.774 | 1.161 | 0.341 |
| 25 | 0.48 | 2.52 | 0.795 | 1.193 | 0.334 |
| 27 | 0.5 | 2.58 | 0.815 | 1.223 | 0.329 |
| 29 | 0.51 | 2.64 | 0.834 | 1.251 | 0.323 |
| 31 | 0.52 | 2.7 | 0.851 | 1.277 | 0.318 |
| 33 | 0.53 | 2.75 | 0.868 | 1.302 | 0.313 |

**Table 8.3 (continued).** Lattice-sum constants at $n \geq 1500$.

| $k$ | $\underline{M}(U_A)$ | $\overline{M}(U_B)$ | $\underline{M}(U_{LO})$ | $\overline{M}(U_{HI})$ | $S_{hi}$ |
|---|---|---|---|---|---|
| 3 | 2.136238 | 0.264198 | 1.210238 | 0.744639 | 4.13818 |
| 5 | 1.611788 | 0.184786 | 0.920200 | 0.553556 | 2.59855 |
| 7 | 1.375649 | 0.145980 | 0.772751 | 0.458497 | 1.94612 |
| 9 | 1.235346 | 0.122548 | 0.680379 | 0.399027 | 1.58216 |
| 11 | 1.154952 | 0.108081 | 0.616503 | 0.357888 | 1.35068 |
| 13 | 1.050030 | 0.095924 | 0.568702 | 0.327282 | 1.18826 |
| 15 | 0.988744 | 0.087856 | 0.531774 | 0.303090 | 1.06913 |
| 17 | 0.933259 | 0.080721 | 0.501364 | 0.283879 | 0.97518 |
| 19 | 0.907444 | 0.075594 | 0.476250 | 0.268270 | 0.90042 |
| 21 | 0.859252 | 0.070921 | 0.454692 | 0.254579 | 0.83830 |
| 23 | 0.836728 | 0.067164 | 0.436159 | 0.243176 | 0.78640 |
| 25 | 0.815164 | 0.063686 | 0.420230 | 0.233110 | 0.74291 |

| $k$ | $\underline{M}(U_A)$ | $\overline{M}(U_B)$ | $\underline{M}(U_{LO})$ | $\overline{M}(U_{HI})$ | $S_{hi}$ |
|---|---|---|---|---|---|
| 27 | 0.774688 | 0.060908 | 0.405871 | 0.224213 | 0.70460 |
| 29 | 0.755671 | 0.058301 | 0.392911 | 0.216345 | 0.67074 |
| 31 | 0.737406 | 0.055853 | 0.381839 | 0.209391 | 0.64236 |
| 33 | 0.719850 | 0.053924 | 0.371233 | 0.203005 | 0.61565 |

**Table 8.4.** Sample of the 312 cells of Theorem 8.12 (all cells: `data/pd/cells56_sharp.json`). Over all cells: $\max\delta = 0.08727$ (cell $[3.30, 3.31]$, $\tau_0 = 0.55$), $\min\delta = 0.02847$, $\tau_0 \in [0.10, 0.55]$, $\max D_1 = 0.06539$, $\max D_2 = 2.21\cdot10^{-3}$, $\max D_3 = 1.99\cdot10^{-2}$, $\max c = 1.1870$; every monotonicity flag of Proposition 8.11 holds.

| $u$-cell | $\tau_0$ | $c$ | $D_1$ | $D_2$ | $D_3$ | $\delta$ | $S_{lo}$ | $S_{hi}$ | $M_{lo}$ | $M_{hi}$ |
|---|---|---|---|---|---|---|---|---|---|---|
| [0.19,0.20] | 0.10 | 1.187 | 0.0641 | 2.00e-41 | 3.32e-21 | 0.06412 | 12.3906 | 13.8664 | 2.2538 | 2.3995 |
| [0.49,0.50] | 0.10 | 0.441 | 0.0302 | 1.01e-07 | 7.19e-07 | 0.03021 | 1.9378 | 2.0371 | 0.7746 | 0.8003 |
| [0.79,0.80] | 0.15 | 0.419 | 0.0308 | 1.02e-06 | 3.80e-06 | 0.03080 | 0.7263 | 0.7530 | 0.4165 | 0.4276 |
| [1.19,1.20] | 0.20 | 0.393 | 0.0331 | 3.02e-05 | 6.32e-05 | 0.03319 | 0.2983 | 0.3073 | 0.2286 | 0.2341 |
| [1.69,1.70] | 0.25 | 0.375 | 0.0376 | 5.66e-04 | 8.21e-04 | 0.03903 | 0.1310 | 0.1347 | 0.1286 | 0.1315 |
| [2.19,2.20] | 0.35 | 0.438 | 0.0457 | 5.11e-04 | 7.28e-04 | 0.04697 | 0.0679 | 0.0698 | 0.0810 | 0.0828 |
| [2.69,2.70] | 0.45 | 0.490 | 0.0547 | 7.12e-04 | 9.96e-04 | 0.05641 | 0.0390 | 0.0401 | 0.0550 | 0.0563 |
| [3.19,3.20] | 0.55 | 0.536 | 0.0646 | 1.15e-03 | 7.47e-03 | 0.07320 | 0.0242 | 0.0249 | 0.0395 | 0.0405 |
| [3.30,3.31] | 0.55 | 0.523 | 0.0654 | 1.98e-03 | 1.99e-02 | 0.08727 | 0.0219 | 0.0226 | 0.0370 | 0.0379 |

**Table 8.5.** Zero sets of Theorem 8.22.

| $k$ | 3 | 5 | 7 | 9 | 11 | 13 | 15 | 17 | 19 | 21 | 23 | 25 | 27 | 29 | 31 | 33 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $\#Z_k$ | 9 | 17 | 24 | 20 | 43 | 54 | 55 | 80 | 89 | 88 | 105 | 76 | 68 | 121 | 127 | 88 |
| $\max Z_k$ | 43 | 176 | 247 | 118 | 273 | 545 | 565 | 1137 | 766 | 2055 | 837 | 936 | 1375 | 1934 | 1712 | 1587 |

## 8.8 Computer-assisted steps and what each relies on

The following steps are carried out by programs; everything else in Section 8 is proved by hand above.

1. **Lattice-sum enclosures** (Algorithm E) for the 312 cells and for the window points: Arb ball arithmetic, 128 bits, through python-flint [21].
2. **Cells** (Algorithm C, Theorem 8.12): the same arithmetic. Output: $\delta$ for 312 cells.

3. **Block bound** (Algorithm R, Proposition 8.14), for 16 values of $k$ and about $(U_B - U_A)/0.0025$ boxes each: interval arithmetic over Arb balls.
4. **Tail bound** (Algorithm T, Propositions 8.19, 8.20): Arb balls, with the dilogarithm of Arb.
5. **Modular certificate** (Algorithm M) and the exact evaluation of $C_k$ (Theorem 8.22): integer arithmetic only (a C program with 64-bit residues; exact Python integers).

The only places where floating-point numbers enter as inputs are: the double-precision endpoints of cells and boxes (each is an exactly representable binary number, converted without rounding to a ball), the choice of $\tau_0$, $u_c$, $U_A, \dots, A_0$ (any value is valid; correctness never depends on how they were chosen), and output roundings, which are all directed (upward for upper bounds, downward for lower bounds). The programs were re-run from their inputs for this paper: every output file (window parameters, cells, constants, results) is identical to the stored one (Appendix A.3). This checks reproducibility, not independence of the method.

### 8.9 Remarks

*An independent derivation of the block step* (separate code, a different decomposition of the circle into five regimes, cells of width 0.005 on $[0.40, 2.90]$, a fourth-derivative remainder) reports, for $k = 27, 29, 31, 33$, block ratios 0.79055, 0.80677, 0.82294, 0.83906, below the values $R_k$ of Table 8.2. These numbers play no role in the proofs above and were not recomputed for this paper.

*What is proved and what is not.* Theorem 8.1 is proved for the sixteen values $k = 3, \dots, 33$ through the finite data above. It is a statement about each fixed $k$; no bound uniform in $k$ is claimed, and no extension to $k \geq 35$ or to even $k$ is asserted.

# Part II
# The Bridge Series $C$

## 9 Introduction to Part II

### 9.1 Setting and question

The series studied here arise as the "Bridge Series $C$" of the author's $S^2$-11D$M^2$ET-X programme. The question posed there was whether $C$ satisfies a *residual law*: a modular part plus a controlled remainder, or an exact weight-$\frac{1}{2}$ slash relation. This part is self-contained. All objects are defined from scratch in §10, and nothing in it depends on, or is claimed to bear on, other parts of that programme.

### 9.2 Summary

In one sentence: $C$ and $C_{\text{third}}$ share the remainder $P_s^{\text{orth}}$; up to theta and mock pieces, $P_s^{\text{orth}}$ is one ray class partial theta series of $\mathbb{Q}(\sqrt{35})$; its regulator-type singularity rules out every scalar modular or mock residual law (Baker); and the obstruction is resolved not by a scalar law but by a vector-valued weight-1 quantum modular structure, coming from a Siegel theta lift to weight-0 Maass forms, whose cocycle is computed in closed form. The precise statements are collected in §11.

### 9.3 What is not claimed

No claim is made about the twin prime conjecture or the Riemann hypothesis, and nothing here concerns any termination claim of the wider programme. We do not claim that $C$, $C_{\text{third}}$ or $P_s^{\text{orth}}$ is modular, mock modular or quantum modular as a scalar $q$-series. The quantum-modular statements concern the fundamental-domain cone-sum vector of §15. The K3 statements of §26 concern a shared representation, not a shared modular form.

### 9.4 Organisation

Section 10 fixes conventions and notation, and §11 states the main theorems. Sections 12–18 prove them: the ray class decomposition, the radial asymptotics and the Baker obstruction, the profile law and the theta lift, the Lewis–Zagier period formula, the Eisenstein/cusp split, the Herglotz closed form, and the $L(0)$ data for the cusp rows. Sections 19–21 treat the lift family for general $s$ (tensor structure, dimensions, the cusp content of $u_L$). Sections 22–25 treat $H_s$, the cone, $C_{\text{third}}$ and the dependence on $s$. Section 26 gives the K3 realisation, §27 the uniqueness of the vector completion, and §28 the obstructions for coefficient-emitting maps. Sections 29–32 collect consequences, numerical checks, dependencies and open problems. Appendix C lists the scripts that carry out the exact and numerical verifications.

## 10 Conventions and notation

### 10.1 Conventions on status

Every result carries one of two labels.

- **Category A**: proved in this paper; or *exact*, meaning a finite verification in rational, integer, modular or cyclotomic arithmetic over a stated finite set (including exact ranks, and ranks modulo a prime $\ell$ together with a stated argument that transfers them to characteristic 0); or proved *conditional on* a named published theorem, stated each time. Where a written proof is long and only its structure is reproduced here, it is marked "Proof (outline)".
- **Category B**: statements whose only support is floating-point or multiprecision numerics (including singular-value ranks, FFT character counts and fits), PSLQ recognition, identification by citation only, interpretation, analogy, patterns observed on finitely many cases, and conjectures.

Numerical checks reported inside a Category A status box are supporting evidence for a statement that is proved or exact. The scripts behind every exact and numerical claim are listed in Appendix C. Section 31 lists the cited results on which proofs depend. *Computer-assisted proofs.* A finite verification carried out in rigorous ball (interval) arithmetic (Arb), in which every quantity entering the final inequality is a proven enclosure of the exact value, counts as Category A and is labelled "Category A (computer-assisted)". *Naming.* Results are numbered by section, continuously with the rest of the paper.

### 10.2 The lattice and the quadratic polynomial

Write $\mathbf{e}(x) = e^{2\pi i x}$ and $q = \mathbf{e}(\tau)$, $\tau \in \mathbb{H}$. Let

$$A = \begin{pmatrix} 2 & 7 \\ 7 & 7 \end{pmatrix}, \qquad B(x,y) = x^{\mathsf{T}} A y, \qquad Q(x) = \tfrac{1}{2} B(x,x) = x_1^2 + 7x_1x_2 + \tfrac{7}{2}x_2^2.$$

Then $\det A = -35$, $Q$ has signature $(1,1)$, $L = \mathbb{Z}^2$ is an *odd* lattice ($Q(L) \not\subset \mathbb{Z}$), $L' = A^{-1}\mathbb{Z}^2$ with $A^{-1} = \begin{pmatrix} -1/5 & 1/5 \\ 1/5 & -2/35 \end{pmatrix}$, $|L'/L| = 35$, and the level is 70 (the least $N$ with $NA^{-1}$ even integral). Put

$$Q_7(n,k) = Q(n,k) + \tfrac{7}{2}k = n^2 + 7nk + \tfrac{7}{2}k(k+1), \qquad Q_-(n,k) = n^2 + 7nk + \tfrac{7}{2}k(k-1) = Q_7(-n,-k).$$

With $a = (\frac{7}{10}, -\frac{1}{5})$ one has $Aa = (0, \frac{7}{2})$, $Q(a) = -\frac{7}{20}$, hence

$$Q_7(v) = Q(v+a) + \tfrac{7}{20} \qquad (v \in \mathbb{Z}^2). \tag{10.1}$$

The characteristic vector of the odd lattice is $w_c = A^{-1}(0, \frac{1}{2}) = (\frac{1}{10}, -\frac{1}{35})$; it satisfies $\mathbf{e}(Q(n)) = \mathbf{e}(B(n, w_c))$ for $n \in \mathbb{Z}^2$ and $w_c \equiv a \pmod{L'}$. Let $\varepsilon = 6 + \sqrt{35}$, the fundamental unit of $K = \mathbb{Q}(\sqrt{35})$ ($\mathrm{N}\varepsilon = +1$), and

$$I = \frac{\log \varepsilon}{2\sqrt{35}} = 0.209419820314714631473\ldots$$

### 10.3 The series

Let $c_1 = (-1, 1)$, $c_2 = (-4, 1)$; then $Q(c_1) = -\frac{5}{2}$, $Q(c_2) = -\frac{17}{2}$ (negative norm), and for $v = (n,k)$, $B(c_1, v) = 5n$, $B(c_2, v) = -n - 21k$. With $\operatorname{sgn}(0) = 0$ put $w(v) = \frac{1}{2}\big(\operatorname{sgn} B(c_1, v) - \operatorname{sgn} B(c_2, v)\big)$ (so $w = \frac{1}{2}$ on the walls). Define

$$P_s^{\mathrm{orth}} = \sum_{n,k \ge 0} q^{Q_7(n,k)}, \qquad H_s = \sum_{v \in \mathbb{Z}^2} w(v)\, q^{Q_7(v)}, \qquad C := P_s^{\mathrm{orth}} - H_s,$$

$$C_{\mathrm{axis}} = \tfrac{3}{2} + \sum_{n \ge 1} q^{n^2} + \sum_{k \ge 1} q^{7k(k+1)/2}, \quad C_{\mathrm{third}} = \sum_{n,k \ge 1} q^{Q_-(n,k)}, \qquad C_{\mathrm{mixed}} := C - C_{\mathrm{axis}} - C_{\mathrm{third}}.$$

Further, $\theta_3 = \sum_{n \in \mathbb{Z}} q^{n^2}$, $R := \sum_{n \ge 0,\, k \ge 1} q^{Q_7(n,k)}$,

$$\Theta_{\mathrm{hol}} := \sum_{n \ge 1,\, k \ge 0} q^{Q_-(n,k)} - \sum_{n \le 0,\, k \le -1} q^{Q_-(n,k)},$$

the holomorphic part of a Zwegers indefinite theta function [18], and $f_{a,b,c}$ is the Hecke-type double sum of Hickerson–Mortenson [8, Def. 0.2], $f_{a,b,c}(x,y,q) = \sum_{\operatorname{sgn}(r) = \operatorname{sgn}(s)} \operatorname{sgn}(r)(-1)^{r+s} x^r y^s q^{a\binom{r}{2} + brs + c\binom{s}{2}}$, with $\operatorname{sgn}(r) = 1$ for $r \ge 0$ and $-1$ for $r < 0$ in this formula.

Finally, for a rational $x = h/m$ ($\gcd(h,m) = 1$) define the Gauss mean

$$G(x) = (2m)^{-2} \sum_{(n,k) \bmod 2m} \mathbf{e}\big(x Q_7(n,k)\big).$$

Throughout, "radial" refers to $q = \mathbf{e}(x)e^{-t}$ with $x$ rational and $t \to 0^+$, and $t^r$ coefficients are those of the resulting asymptotic expansion.

### 10.4 Notation for the lift family

For an integer $s \ge 5$ let $A_s = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$, $Q_s(n,k) = n^2 + snk + \frac{s}{2}k(k+1)$, $a = A_s^{-1}(0, \frac{s}{2})$, $c_0 = s/(4(s-2))$ and $\gamma = \left(\begin{smallmatrix} -1 & -s \\ 2 & 2s-1 \end{smallmatrix}\right)$ (so that $Q_s(v) = Q(v+a) + c_0$; at $s = 7$ these are the objects above).

- $n_s := s(s-2)$ (the radicand: $35, 48, 99$ at $s = 7, 8, 11$); $D_s$ is the discriminant of the order of the form $(140, 48, 396)$; $\varepsilon_s = (s-1) + \sqrt{n_s}$.
- $N_s$ is the number of labels of the lift family; $N_s = 4n_s$ for odd $s$ (so $N_7 = 140$).
- $S_{\mathrm{lab}}$, $T$ are the label matrices of the $S$- and $T$-laws (Theorem 11.6); $P = S_{\mathrm{lab}}^2$ is the label involution and $W_P := \mathrm{Fix}(P)$ ($\dim W_P = 72$ at $s = 7$). The representation of $\mathrm{SL}_2(\mathbb{Z})$ in the standard presentation is $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$ (Remark 19.1); on $W_P$, $S_{\mathrm{lab}} = S_{\mathrm{lab}}^{-1}$.
- $V$ is the span of the values of the lift vector (the function span); $V_{\mathrm{Eis}}$ is the $\mathrm{SL}_2(\mathbb{Z})$-cyclic span of the constant-term vector, $d_{\mathrm{Eis}} := \dim V_{\mathrm{Eis}}$, $d_{\mathrm{cusp}} := \dim V - d_{\mathrm{Eis}}$; $V^{\perp}$ is the complement of $V$ in $W_P$.
- $N$ denotes the negation map $u_{\alpha,\beta} \mapsto u_{-\alpha,-\beta}$ on labels (not to be confused with $N_s$); $Z_p$ is the local reflection $x \mapsto -x$ on a $p$-primary factor, and $Z_{(m)} = \bigotimes_{p|m} Z_p$.
- $W_p^{\pm}$ are the even and odd parts of a local Weil representation; $A_3$, $B_1$ are the 3- and 1-dimensional summands of the 2-adic factor $\rho_2$ (Theorem 19.3); $X^{\pm} = W_5^{\pm} \otimes W_7^{\pm}$ at $s = 7$.
- $H_s$ always denotes the cone series of §10 with the fixed walls $c_1 = (-1, 1)$, $c_2 = (-4, 1)$ (for every $s$). The axes-cone series $H_s^{\mathrm{axes}}$ of §23 is a separate object. We write $g := \gcd(24, s-8)$ and $\Delta := (s-2)(7s-32)$.

# 11 Main results

Each theorem is followed by its status; proofs are in the sections indicated.

**Theorem 11.1** (Shared remainder)**.** *As formal $q$-series,*

*(i)* $C_{\mathrm{third}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + P_s^{\mathrm{orth}}$;

*(ii)* $C - C_{\mathrm{third}} = -H_s - q^8 f_{2,7,7}(-q^{10}, -q^{14}, q)$;

*(iii)* $\Theta_{\mathrm{hol}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + \theta_3 = C_{\mathrm{third}} + \sum_{n \geq 1} q^{n^2} - R$, *and* $P_s^{\mathrm{orth}} = R + \frac{1}{2}(1 + \theta_3)$;

*(iv)* $C_{\mathrm{mixed}} = -H_s - q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) - C_{\mathrm{axis}}$ *vanishes through* $q^{293}$ *but not identically:* $C_{\mathrm{mixed}} = -\frac{1}{2}q^{294} + \frac{1}{2}q^{301} - q^{330} + q^{337} - q^{368} + q^{375} - \cdots$.

**Status (Category A).** (i) is proved by reindexing, and (ii)–(iv) follow from (i) and the definitions (§12.1). All identities are also checked with exact rational coefficients through $q^{3000}$; the first nonzero terms in (iv) are an exact computation.

**Theorem 11.2** (Real-quadratic decomposition and ray class)**.** *Let $\mathcal{K}$ be the narrow ray class modulo $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ of $K = \mathbb{Q}(\sqrt{35})$ containing $\mathfrak{p}_7(\alpha_1)$ for any $\alpha_1 \equiv 1 \bmod {}^{\times}\mathfrak{p}_2\mathfrak{p}_5$ with signs $(-,+)$. Then*

$$2P_s^{\mathrm{orth}} = E + 1 + \theta_3 - \Theta_{\mathrm{hol}}, \qquad E(q) = \sum_{\mathfrak{a} \in \mathcal{K}} q^{(\mathrm{N}\mathfrak{a}+7)/20},$$

*and $E$ is the sum of $q^{Q_7(v)}$ over $v + a$ in one fundamental cone for the action of the automorph $\gamma = \begin{pmatrix} -1 & -7 \\ 2 & 13 \end{pmatrix}$ (eigenvalues $6 \pm \sqrt{35}$) on the positive sector.*

**Status (Category A).** Proved in §§12.2–12.3; the proof uses that $\mathbb{Z}[\sqrt{35}]$ is the maximal order and that its unit group is $\pm\varepsilon^{\mathbb{Z}}$. The cone identity is also checked exactly to $q^{3000}$ for four fundamental cones.

**Theorem 11.3** (Radial asymptotics and partial zeta values)**.** *As $t \to 0^+$, with $q = e^{-t}$, $P_s^{\mathrm{orth}} = I/t + a_P\, t^{-1/2} + \frac{11-7I}{20} + O(t^{1/2})$ with $a_P = \sqrt{\pi}(49-2\sqrt{14})/140 = 0.5256172043\ldots$; $E = 2I/t + \frac{1-7I}{10} + O(t)$ with no half-integral powers; $\zeta(0,\mathcal{K}) = \frac{1}{10}$, $\zeta(-1,\mathcal{K}) = \frac{29}{60}$, $\operatorname{Res}_{s=1}\zeta(s,\mathcal{K}) = \log\varepsilon/(20\sqrt{35})$; and $C = I/t + a_C\, t^{-1/2} + \frac{11-7I}{20} + O(t^{1/2})$ with*

$$a_C = \sqrt{\pi}\left(833 + 204\sqrt{14} - 14\sqrt{1190}\right)/2380 = 0.8291426171\ldots$$

*The full table of coefficients is Table 13.*

**Status (Category A).** The rows for $P_s^{\mathrm{orth}}$, $R$, $E$ and $a_P$ are exact, by symbolic two-dimensional Euler–Maclaurin (Proposition 13.1). $\zeta(0,\mathcal{K})$ and $\zeta(-1,\mathcal{K})$ are exact rational arithmetic, confirmed by five independent Shintani cone decompositions. $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7)$ are the $s = 7$ values of the closed forms of Theorem 22.1; they agree symbolically with the radical forms above. The constant term 0 of $H_s$ is part of Theorem 22.1.

**Theorem 11.4** (Gauss-mean pattern and zero set)**.** *For $\gcd(h,m) = 1$: $G(h/m) = 0$ if $2 \mid m$ or $5 \mid m$; $|G(h/m)| = 1/m$ if $\gcd(m,70) = 1$; $|G(h/m)| = \sqrt{7}/m$ if $7 \mid m$ and $\gcd(m,10) = 1$. Moreover $\lim_{t\to 0^+} t\, S(\mathbf{e}(x)e^{-t}) = I \cdot G(x)$ for $S \in \{P_s^{\mathrm{orth}}, C, C_{\mathrm{third}}\}$ and every rational $x$.*

**Status (Category A).** The pattern is proved in §13.5 and checked on all 3374 coprime pairs with $m \le 105$. The limit follows from leading-order Euler–Maclaurin (Proposition 13.1); it is checked numerically down to $t = 6.25 \cdot 10^{-4}$.

**Theorem 11.5** (Baker obstruction: scalar no-go)**.** *(i) (Additive.) There is no identity $S = F + \mathcal{E}$ with $S \in \{C, C_{\mathrm{third}}\}$, $F$ in the $\overline{\mathbb{Q}}$-span $\mathcal{M}$ of §13 (holomorphic modular forms with algebraic coefficients, signature-$(1,1)$ Zwegers holomorphic parts with rational characteristics, unary/partial/false theta series), and $\mathcal{E}(e^{-t}) = o(1/t)$.*

*(ii) (Multiplicative.) For holomorphic modular forms $g \not\equiv 0$ and $F$ of any weights, levels and multipliers, with complex coefficients, and any $q^r$ factor: $C_{\mathrm{third}} \cdot g \neq F$ and $C \cdot g \neq F$.*

*(iii) (Scalar $W_7$.) For no $k$ and no constant $\lambda$ is $C(\tau) - \lambda(7\tau)^{-k}C(-1/(7\tau))$ radially bounded at all rationals.*

**Status (Category A).** (i) is proved *conditional on* Baker's theorem [1] and on the $q$-expansion principle (algebraicity of expansions at all cusps), both cited. (ii) is proved; for $C$ the nonvanishing input is the closed form $a_C = \sqrt{\pi}(833 + 204\sqrt{14} - 14\sqrt{1190})/2380 \neq 0$ (Theorem 11.3). (iii) is an exact finite computation: 178 of 270 test rationals (denominator $< 30$) have exactly one of $G(x)$, $G(-1/(7x))$ zero, including $x = 5/7$; the zero pattern is that of Theorem 11.4.

**Theorem 11.6** (Theta lift and vector-valued $S/T$ law)**.** *Let $u_{\alpha,\beta}$ ($\alpha \in (\{0,a\}+L')/\mathbb{Z}^2$, $\beta \in \{0,-a\}$; 140 labels) be the Siegel-theta lifts of §14. Each $u_{\alpha,\beta}$ is a weight-0 Laplace eigenfunction with $\lambda = \frac{1}{4}$, and*

$$u_{\alpha,\beta}(-1/w) = \frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}} \sum_{p \in L'/\mathbb{Z}^2} u_{p-\beta,\alpha}(w), \qquad u_{\alpha,\beta}(w+1) = \mathbf{e}\big(-Q(\alpha) - B(\alpha,w_c)\big)\, u_{\alpha,\alpha+\beta+w_c}(w).$$

*The resulting $S$, $T$ matrices are unitary, $S^2 = P$ (the label involution $(\alpha,\beta) \mapsto (-\alpha,-\beta)$), $(ST)^3 = I$, and on the 72-dimensional $P$-invariant subspace $S^2 = (ST)^3 = I$: a 72-dimensional*

*representation of* $\mathrm{PSL}_2(\mathbb{Z})$ *on label space; we write* $W_P := \mathrm{Fix}(P)$. *The functions* $u_{\alpha,\beta}$ *span a* 42*-dimensional subspace* $V \subset W_P$, *invariant under* $S$ *and* $T$ *(Theorem 11.8, §16.3). Here* $S$ *is the label matrix* $S_{\mathrm{lab}}$; *the representation of* $\mathrm{SL}_2(\mathbb{Z})$ *in the standard presentation is* $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$ *(Remark 19.1).*

**Status (Category A).** The $S$-law is proved by Poisson summation (§14.3), independently of [3]; the $T$-law is elementary. The relations $S^2 = P$, $P^2 = I$, $[P,T] = 0$ and $(ST)^3 = I$ on all of $\mathbb{C}^{140}$ are exact (verified at all 48 Galois conjugates over $\mathbb{Q}(\zeta_{140})$). $\dim V = 42$ is exact (Proposition 19.4). The label representation is exactly $\rho_{2x^2/5} \otimes \rho_{x^2/7} \otimes \rho_2$ (Theorem 19.3) and is isomorphic to the Weil representation of the even lattice $N_2$ (Theorem 26.2). The $S/T$ laws are also verified numerically on all 140 rows ($\le 1.2 \cdot 10^{-15}$; supporting).

**Theorem 11.7** (Lewis–Zagier period formula). *For* $z \in \mathbb{H}$ *let* $\Psi(z) = E_L(-1/z) - (z/\sqrt{35})E_R(z)$ *and* $\Psi^-(z) = E_L(-1/z) + (z/\sqrt{35})E_R(z)$ *(§15.1), and for* $z \in \mathbb{C} \setminus i\mathbb{R}$ *put*

$$\Omega(z) = \frac{z}{\pi\sqrt{35}}\Big(\int_0^{i\infty} [u_R, R_z^{1/2}] - \frac{\log \varepsilon}{2}\Big),$$

*the integral taken along* $i\mathbb{R}_{>0}$ *with the principal square root.*

*(i)* $\Omega$ *is holomorphic on* $\mathbb{C} \setminus i\mathbb{R}$; *on* $\mathbb{H} \cap \{\mathrm{Re}\, z > 0\}$ *it equals* $\Psi$, *and on* $\mathbb{H} \cap \{\mathrm{Re}\, z < 0\}$ *it equals* $-\Psi^-$. *In particular* $\Psi$ *continues holomorphically to* $U = \mathbb{H} \cup \{\mathrm{Re}\, z > 0\}$.

*(ii) Let* $\psi$ *be the order-*$t^0$ *radial* $S$*-cocycle of §15.1. For rational* $x > 0$, $\psi(x) = \Omega(x)$; *for rational* $x < 0$, $\psi(x) = -\Omega(x) = \frac{|x|}{\pi\sqrt{35}}\big(\int_0^{i\infty}[u_R, R_x^{1/2}] - \frac{\log\varepsilon}{2}\big)$.

*(iii) The analogous closed form holds on every row of the* 140*-label vector cocycle* $\Phi$, *which is real-analytic on* $\mathbb{R} \setminus \{0\}$ *and satisfies the three-term relation* (15.2).

**Status (Category A).** Proved in §15.4, given Lemma 15.1, the branch lemma (Lemma 15.2), the $S$-law of Theorem 11.6, the sector-uniform Euler–Maclaurin expansion (Proposition 13.1) and the Fourier interchange R2 (§15.6). Numerically: the period formula agrees with exact Bernoulli sums at 84 rationals (denominators $\le 60$, both signs) to $1.1 \cdot 10^{-13}$; $|\Omega - \Psi| \approx 10^{-16}$ at three points with $\mathrm{Re}\, z > 0$ and $|\Omega + \Psi^-| \approx 10^{-16}$ at three points with $\mathrm{Re}\, z < 0$ (where $|\Omega - \Psi| = 0.076, 0.069, 0.539$); all 140 rows to $2.3 \cdot 10^{-14}$; the three-term relation to $7 \cdot 10^{-15}$. The phrase "quantum modular form in Zagier's sense" for the real factor $|cx + d|$ is terminology (Category B).

**Theorem 11.8** (Eisenstein/cusp split). $u_L = u_{a,0}$ *and* $u_R$ *are pure Eisenstein (combinations of the* 8 *genus series attached to* $\psi \circ \mathrm{N}$, $\psi \bmod 20$*). The* 42*-dimensional function span* $V$ *splits* $\mathrm{SL}_2(\mathbb{Z})$*-equivariantly as* $V_{\mathrm{Eis}} \oplus V_{\mathrm{cusp}}$ *with* $\dim V_{\mathrm{Eis}} = 36$ *(from the* 116 *Eisenstein components) and* $\dim V_{\mathrm{cusp}} = 6$ *(from the* 24 *cusp components, §16). The cusp components are exact combinations of* 12 *dihedral Maass forms* ($\lambda = \frac{1}{4}$) *from the ray class characters of conductor* $(2\sqrt{35})$ *of pure infinity type; these* 12 *forms occur only in* 6 *fixed linear combinations. Consequently* $\psi_{\mathrm{cusp}} \equiv 0$ *for the scalar cocycle.*

**Status (Category A).** $u_L$ is Eisenstein by a hand proof independent of PARI (§16.1), with an exact coefficient match for all $n \le 60\,000$. The dimensions 36 and 6 and the $S,T$-invariance of the split are exact: $V_{\mathrm{Eis}} = A_3 \otimes X^+$ and $V_{\mathrm{cusp}} = B_1 \otimes X^-$ (Proposition 19.4). $\psi_{\mathrm{cusp}} \equiv 0$ follows because $\psi$ involves only $u_L$ and $u_R$.

**Category B.** The expression of the cusp components through the 12 named dihedral forms rests on floating-point fits (§16.2); the rank 6 of the 12 forms inside the cusp components is floating, and is explained exactly by $\dim(B_1 \otimes X^-) = 6$. Cuspidality $\Leftrightarrow \chi \ne \chi \circ \sigma$, the level 19600, the nebentypus and newform status are standard theory [11], cited.

**Theorem 11.9** (Herglotz closed form)**.** *On* $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$,

$$\Psi(z) = \frac{1}{8} \sum_{j\in\{0,2,4,6\}} \sum_{l\in\{0,1\}} \mathbf{e}(-j/8)\, R_{j,l}(z), \qquad R_\chi(z) = \sum_{m\geq 1} a_\chi(m) \sum_{s=1}^{140} \rho_s^\chi\, \psi_\Gamma\Big(\frac{s}{140} + \frac{m}{20z}\Big),$$

*with* $\psi_\Gamma = \Gamma'/\Gamma$ *and the data of §17. In particular (given R1)* $\psi(x)$ *for rational* $x > 0$ *equals this expression.*

**Theorem 11.10** (Closed form of $E_R$)**.**

$$E_R(z) = \frac{\sqrt{35}}{140} \sum_\chi w_\chi \sum_{n,k\geq 1} \widehat{b}_\chi(n)\, \check{a}_\chi(k)\, \mathbf{e}(nkz/140), \qquad w_\chi = \mathbf{e}(-j/8)/16.$$

**Status (Category A).** Theorems 11.9 and 11.10 are proved in §17 from the Eisenstein expansion of $E_L$ (§16.1), the $S$-law (Theorem 11.6), Lemma 15.1 and Theorem 11.7(i), which is used on $\operatorname{Re} z > 0$ only. The proof is given in outline; the full estimates are in the companion report listed in Appendix C. Numerically, Theorem 11.9 holds to $5.5\cdot 10^{-35}$ on $\mathbb{H}$, to $\leq 9.2\cdot 10^{-16}$ off $\mathbb{H}$ and to $5.7\cdot 10^{-35}$ at rationals; Theorem 11.10 holds for all $N \leq 14\,000$ (627 nonzero coefficients identical as elements of $\mathbb{Z}[\phi]$). No arithmetic (Gauss-sum or ideal-theoretic) proof of Theorem 11.10 is claimed.

**Remark (Category B) 11.11** (Cusp rows and $L(0)$)**.** For every cusp row $c$ and each of the seven rationals $x \in \{1, \frac{1}{3}, 2, \frac{3}{5}, \frac{7}{4}, -1, -\frac{2}{3}\}$, the radial constants $\kappa_c(h/k)$ entering $\Phi_c(x)$ are explicit finite combinations of ray-class Hecke $L$-values $L_{\mathfrak{m}_k}(0, \chi')$ of $K$; only totally odd characters contribute, and the $L(0)$ values lie in $\mathbb{Z}[\zeta_{12}]$. Replacing $L(0,\cdot)$ by $L(\frac{1}{2},\cdot)$ fails by up to 0.32.

**Category B.** Verified numerically at the seven points for all 24 cusp rows, to $4\cdot 10^{-14}$ with PARI `lfun` values and to $1.2\cdot 10^{-14}$ with PARI-free inputs (§18). Not proved.

**Status (Category A).** The 12 values $L(0,\chi) \in \mathbb{Z}[\zeta_{12}]$ are exact (PARI-free Shintani sums), and the structure $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2} \cong \mathbb{Z}/2 \times \mathbb{Z}/2 \times \mathbb{Z}/24$ with its $\sigma$-action, types and conductors is certified by hand (§18).

# 12 The real-quadratic decomposition and the ray class

## 12.1 Elementary identities (Theorem 11.1)

*Proof of Theorem 11.1.* (i). In $q^8 f_{2,7,7}(-q^{10}, -q^{14}, q)$ put $n = r+1$, $k = s+1$. The signs $(-1)^{r+s}$ from $x = -q^{10}$, $y = -q^{14}$ cancel those in $f$, and on $r, s \geq 0$ the exponent is $8 + 10r + 14s + r(r-1) + 7rs + \frac{7}{2}s(s-1) = Q_-(n,k)$; this orthant gives $C_{\text{third}}$. On $r, s < 0$, after $(r,s) = (-1-n, -1-k)$ the exponent is $Q_7(n,k)$ with sign $-1$; this gives $-P_s^{\text{orth}}$.

(ii) is (i) combined with $C = P_s^{\text{orth}} - H_s$. For (iii): in the second sum defining $\Theta_{\text{hol}}$ substitute $(n,k) = (-n', -k')$ with $n' \geq 0$, $k' \geq 1$; since $Q_-(-v) = Q_7(v)$ it equals $R$. The first sum is $C_{\text{third}} + \sum_{n\geq 1} q^{n^2}$ (split $k \geq 1$ and $k = 0$). Hence $\Theta_{\text{hol}} = C_{\text{third}} + \sum_{n\geq 1} q^{n^2} - R$. The $k = 0$ row of $P_s^{\text{orth}}$ gives $P_s^{\text{orth}} = R + \sum_{n\geq 0} q^{n^2} = R + \frac{1}{2}(1 + \theta_3)$, and then (i) gives $\Theta_{\text{hol}} = q^8 f + \theta_3$. (iv) is the definition combined with (ii). □

**Status (Category A).** All identities in Theorem 11.1 were in addition checked with exact rational coefficients through $q^{3000}$; the $H_s$ summation box was enlarged by 30 with identical output.

**Remark 12.1** (Origin of $C_{\text{mixed}}$)**.** At $q^{294}$ the point $(21,-1)$ lies on the $c_2$-wall $n+21k=0$, has $w=\frac{1}{2}$ and lies outside the orthant, contributing $-\frac{1}{2}$. Points such as $(22,-1)$ ($q^{330}$) and $(-22,1)$ ($q^{337}$) lie in the $H_s$ cones but outside the orthant. Thus "$C = C_{\text{axis}} + C_{\text{third}}$" holds only through $q^{293}$. The coefficient set $\{0,1,\frac{3}{2}\}$ holds through $q^{101}$ only (first coefficient 2 at $q^{102}$, first 3 at $q^{144}$, first $\frac{1}{2}$ at $q^{294}$); through $q^{3000}$ the set is $\{0,\frac{1}{2},1,\frac{3}{2},2,3\}$. The support $\{0,1,2,4\}$ mod 7 holds for all terms since $Q_7 \equiv n^2 \pmod 7$. (Category A, exact.)

Consequently, every part of $C$ and of $C_{\text{third}}$ other than $P_s^{\text{orth}}$ is a signature-$(1,1)$ mock/indefinite theta object or a unary theta series, and the residual-law question is a question about $P_s^{\text{orth}}$ alone. Note also that $R$ is a cone sum of the *indefinite* form $Q_7$ (positive only on the closed orthant), so it cannot be split by inclusion–exclusion into full lattice theta series: $\sum_{\mathbb{Z}^2} q^{Q_7}$ diverges.

## 12.2 Automorphs and the fundamental cone

In the coordinates $\nu = v + a$, let $\rho_1 = \begin{pmatrix} 1 & 7 \\ 0 & -1 \end{pmatrix}$, $\rho_2 = \begin{pmatrix} -1 & 0 \\ 2 & 1 \end{pmatrix}$. Both are integral reflections preserving $Q$; $\rho_1$ fixes the ray $\nu_2 = 0$, $\rho_2$ fixes $\nu_1 = 0$, the two walls of $C_0 = \{\nu_1 > 0, \nu_2 > 0\}$. Their product $\gamma = \rho_2\rho_1$ has eigenvalues $6 \pm \sqrt{35}$. The reflections swap $a + \mathbb{Z}^2$ and $-a + \mathbb{Z}^2$; $\gamma$ preserves $a + \mathbb{Z}^2$. Coset points never lie on the walls ($\nu_1 \equiv 0.7$, $\nu_2 \equiv 0.8 \bmod 1$). Hence $C_0 \cup \rho_2 C_0$ is a fundamental domain $F$ for $\langle\gamma\rangle$ on the positive sector $S^+$, and

$$E := R + C_{\text{third}} + \sum_{n\geq 1} q^{n^2} = \sum_{v:\ v+a\in F} q^{Q_7(v)} \tag{12.1}$$

for any $\gamma$-fundamental cone $F = [r, \gamma r) \subset S^+$. With Theorem 11.1(iii) this gives $R - (C_{\text{third}} + \sum_{n\geq 1} q^{n^2}) = -\Theta_{\text{hol}}$ and, adding, $2P_s^{\text{orth}} = E + 1 + \theta_3 - \Theta_{\text{hol}}$.

**Status (Category A).** Proved; reflections and coset behaviour checked symbolically; (12.1) checked exactly to $q^{3000}$ for $r = (1,0), (1,1), (7,-1), (1,2)$. Also $E_{-a} = E_a$, so the sign-twisted partner $E_a - E_{-a}$ (a weight-1 Hecke form in Zwegers' $c_2 = \gamma c_1$ set-up) vanishes for this coset; this holds for every $s \geq 5$ (Theorem 24.1(ii)).

## 12.3 Ray-class identification (Theorem 11.2)

Put $(X,Y) = 10\nu \in (7,-2) + 10\mathbb{Z}^2$ and $\beta = (2X+7Y) + Y\sqrt{35} \in \mathbb{Z}[\sqrt{35}]$. Then $Q(\nu) = \mathrm{N}(\beta)/400$, $\beta(\gamma\nu) = \varepsilon\beta(\nu)$, $\beta$ runs over $\beta_0 + 10\mathfrak{p}_2$ with $\beta_0 = -2\sqrt{35}$ and $\mathfrak{p}_2 = (2, 1+\sqrt{35})$, and $\nu \in S^+ \Leftrightarrow \beta \gg 0$. With $\mathfrak{d}' = \gcd((\beta_0), 10\mathfrak{p}_2) = \mathfrak{p}_2^2\mathfrak{p}_5$ and $\mathfrak{f} = \mathfrak{p}_2\mathfrak{p}_5$, the ideals $\mathfrak{a} = (\beta)\mathfrak{d}'^{-1}$ are exactly $\alpha\mathfrak{p}_7$ with $\alpha \equiv 1 \bmod {}^\times\mathfrak{f}$ and $\operatorname{sgn}\alpha = (-,+)$: one narrow ray class $\mathcal{K}$ modulo $\mathfrak{f}\infty_1\infty_2$. Totally positive units $\equiv 1 \bmod {}^\times\mathfrak{f}$ are $\varepsilon^{\mathbb{Z}}$ ($\varepsilon \equiv 1 \bmod \mathfrak{p}_2$ and $\bmod \mathfrak{p}_5$), so $\gamma$-orbits correspond bijectively to ideals in $\mathcal{K}$, and $\mathrm{N}\mathfrak{a} = 20Q_7(v) - 7$. This yields Theorem 11.2. Sanity check: the first norms are $13, 73, 133 = 7\cdot 19, 153 = 9\cdot 17, 173, \ldots$, all $\equiv 13 \bmod 20$.

**Status (Category A).** Proved: hand algebra with symbolic checks of $Q = \mathrm{N}(\beta)/400$ and $\beta(\gamma\nu) = \varepsilon\beta$. The region-to-class bijection was also checked exactly against PARI's independent ideal enumeration up to norm 60 000.

**Remark 12.2** (Group data)**.** $\mathrm{Cl}(K) = \mathbb{Z}/2$, $\mathrm{Cl}^+(K) = (\mathbb{Z}/2)^2$; $|\mathrm{Cl}^+_{\mathfrak{f}\infty_1\infty_2}| = 16$, with structure $\mathbb{Z}/8 \times \mathbb{Z}/2$. Category A: $\mathrm{Cl}(K)$ and the order 16 are hand-certified (Minkowski bound $\sqrt{140}/2 \approx 5.92$; $h\varphi(\mathfrak{f})2^2/[U : U_{\mathfrak{f},1}] = 2\cdot 4\cdot 4/2$); the group structure $\mathbb{Z}/8 \times \mathbb{Z}/2$ and the narrow class group are from PARI (`bnfcertify`$= 1$).

### 12.4 Analytic consequences

By Hecke, $\zeta(s,\mathcal{K}) = \sum_{\mathfrak{a}\in\mathcal{K}} \mathrm{N}\mathfrak{a}^{-s}$ continues meromorphically with a single simple pole at $s=1$, so

$$E(e^{-t}) = e^{-7t/20}\Big[\frac{20\,\mathrm{Res}_{s=1}\,\zeta(s,\mathcal{K})}{t} + \sum_{j\ge 0}\zeta(-j,\mathcal{K})\frac{(-t/20)^j}{j!}\Big]$$

has an expansion in integral powers of $t$; by Siegel–Klingen the $\zeta(-j,\mathcal{K})$ are rational. The exact computation gives $\mathrm{Res}_{s=1}\,\zeta(s,\mathcal{K}) = \log\varepsilon/(20\sqrt{35})$, $\zeta(0,\mathcal{K}) = \frac{1}{10}$ $\zeta(-1,\mathcal{K}) = 20\,\zeta_E(-1) = \frac{29}{60}$, where $\zeta_E(s) = \sum_{\nu\in F\cap(a+\mathbb{Z}^2)} Q(\nu)^{-s}$, $\zeta_E(0) = \frac{1}{10}$, $\zeta_E(-1) = \frac{29}{1200}$.

**Status (Category A).** The exact values are Euler–Maclaurin computations in rational arithmetic; the half-integral coefficients vanish identically (symbolic); independently confirmed by five Shintani cones $[r,\gamma r)$, $r\in\{(1,0),(1,1),(1,2),(2,1),(3,2)\}$ ($|\det| = 2,23,58,43,130$), each giving $\frac{1}{10}$ and $\frac{29}{1200}$. The appeal to Hecke and Siegel–Klingen [7, 15] is to standard theorems cited by name; rationality is used only as a sanity check.

**Category B.** The residue matches the narrow ray-class residue formula $R_{\mathfrak{f}}/(w\sqrt{d_K}\,\mathrm{N}\mathfrak{f})$ with $d_K = 140$, $\mathrm{N}\mathfrak{f} = 10$, $w = 1$; this was not checked against a published statement. The Mellin transform of $P_s^{\mathrm{orth}}$ is a Shintani-type cone zeta whose residue is "class-number-like".

## 13 Asymptotics and the Baker obstruction

### 13.1 Two-dimensional Euler–Maclaurin

All radial expansions in this paper rest on the following expansion for cone sums.

**Proposition 13.1** (Cone Euler–Maclaurin)**.** *For every rational $x$, the shifted-lattice cone sums of $e^{-tQ}$ that make up $H_s(\mathbf{e}(x)e^{-t})$ and $E_L(x+h)$ have a full asymptotic expansion in powers of $t^{1/2}$. The coefficients are area terms, edge Bernoulli sums with the half-weight rule $\tilde{B}_r$ at the edges, and corner terms. The remainders are explicit integrals of periodic Bernoulli functions and are bounded uniformly in every sector $|\arg t| \le \pi/2 - \delta$.*

**Status (Category A).** Proved, by a one-dimensional lemma (Zagier's asymptotic lemma with the Euler–Maclaurin remainder $\frac{(-1)^{p+1}}{p!}\int_0^\infty \tilde{B}_p(y-u)f^{(p)}(y)\,dy$), a quadrant lemma, and a decomposition of the closed cone into translated quadrants. The sector-uniform bound is order-3 two-variable Euler–Maclaurin with every remainder bounded; only $|\tilde{B}_p|$ enters the bounds. The written proof is in the companion report (Appendix C); its structure is reproduced here.

In particular we use the following consequence, which we call **R1**: for rational $x$, any cone sum $f$ of this type and $|\arg t| \le \pi/2-\delta$, $|t|\le 1$,

$$f(\mathbf{e}(x)e^{-t}) = A/t + h\,t^{-1/2} + \kappa + O(|t|^{1/2}),$$

with the coefficients of the real-$t$ expansion. Thus radial expansions hold non-tangentially.

### 13.2 Radial expansions

Table 13 lists the coefficients of $t^{-1}, t^{-1/2}, t^0$ at $q = e^{-t}$. The method is two-dimensional Euler–Maclaurin on shifted quadrants; for $H_s$ the closed cone $\langle(0,1),(21,-1)\rangle$ is decomposed into 21 residue classes $(s,0)+\mathbb{N}v_1+\mathbb{N}v_2$ minus half-walls (decomposition exact to $q^{2000}$). Consistency: $2P_s^{\mathrm{orth}} = E+1+\theta_3-\Theta_{\mathrm{hol}}$ forces the $t^{-1/2}$ coefficient of $\Theta_{\mathrm{hol}}$ to be $\sqrt{\pi}-2a_P = \sqrt{\pi}(21+2\sqrt{14})/70$. For $Z(s) = \sum_{(n,k)\ge 0,\ne 0} Q_7^{-s}$: simple poles at $s=1$ (residue $I$) and $s=\frac{1}{2}$ (residue $(49-2\sqrt{14})/140$), and $Z(0) = -(9+7I)/20$; the $\log\varepsilon$ in $Z(0)$ comes entirely from the shift $\frac{7}{20}$.

| series | $t^{-1}$ | $t^{-1/2}$ | $t^0$ |
|---|---|---|---|
| $P_s^{\mathrm{orth}}$ | $I$ | $\sqrt{\pi}(49-2\sqrt{14})/140 = 0.5256172043$ | $(11-7I)/20 = 0.4767030629$ |
| $C_{\mathrm{third}}$ | $I$ | $-\sqrt{\pi}(49-2\sqrt{14})/140$ | $(11-7I)/20$ |
| $R$ | $I$ | $-0.3606097211$ | $(1-7I)/20$ |
| $E$ | $2I$ | $0$ | $(1-7I)/10$ |
| $H_s$ | $0$ | $a_H = \sqrt{\pi}(\sqrt{1190}-17\sqrt{14})/170 = -0.3035254127$ | $0$ |
| $C$ | $I$ | $a_C = \sqrt{\pi}(833+204\sqrt{14}-14\sqrt{1190})/2380 = 0.8291426171$ | $(11-7I)/20$ |

Table 13: Radial coefficients at $q = e^{-t}$. Status as in Theorem 11.3; the $H_s$ and $C$ entries are the closed forms of Theorem 22.1.

**Lemma 13.2** (Zwegers parts carry no $1/t$ term)**.** *The holomorphic part of any signature-*$(1,1)$ *indefinite theta* $\vartheta_{a,b}^{c_1;c_2}$ *with* $c_1, c_2$ *of negative norm and rational* $a, b$ *has* $1/t$ *coefficient* $0$ *at* $q = e^{-t}$*. This covers* $H_s$*,* $\Theta_{\mathrm{hol}}$ *and* $q^8 f_{2,7,7}$*.*

*Sketch.* The weight is $+1$ on a cone $\mathcal{C}$ and $-1$ on $-\mathcal{C}$ (walls contribute $O(t^{-1/2})$). The mean of $\mathbf{e}(B(\nu, b))$ over the coset is the same on $\mathcal{C}$ and $-\mathcal{C}$, and $Q$ is even, so the leading integrals cancel. □

**Status (Category A).** Proved (symmetry argument; null cones excluded). Numerically $t\,H_s \to 0$ (fit $-0.0002$).

### 13.3 The additive obstruction (Theorem 11.5(i))

Let $\mathcal{M}$ be the $\overline{\mathbb{Q}}$-span of: (i) $q^r f$, $r \in \mathbb{Q}$, $f$ a holomorphic weight-1 form on some $\Gamma(N)$ with $\overline{\mathbb{Q}}$ coefficients; (ii) $q^r f$ with $f$ of half-integral weight, and $f$ of integral weight $k \geq 2$ without $q^r$ shift; (iii) $q^r$ times Zwegers holomorphic parts as in Lemma 13.2; (iv) unary, partial and false theta series $\sum_{n \geq n_0} \psi(n) n^\epsilon q^{an^2+bn}$ with $\psi$ periodic and $\overline{\mathbb{Q}}$-valued, $\epsilon \in \{0, 1\}$, $a > 0$.

*Proof of Theorem 11.5(i).* Multiply by $t$ and let $t \to 0^+$. Class (i) contributes $2\pi\beta$ with $\beta \in \overline{\mathbb{Q}}$ (algebraic expansion of $f|S$); (ii) contributes 0; (iii) contributes 0 by Lemma 13.2; (iv) contributes 0 or $\mathrm{mean}(\psi)/(2a) \in \overline{\mathbb{Q}}$. So $I = 2\pi\beta + \gamma'$ with $\beta, \gamma' \in \overline{\mathbb{Q}}$, i.e. $\log(6+\sqrt{35}) + 4\sqrt{35}\, i\beta \log(-1) - 2\sqrt{35}\gamma' = 0$, a nontrivial $\overline{\mathbb{Q}}$-linear relation among $1$, $\log(6+\sqrt{35})$, $\log(-1) = i\pi$. Since $\log(6+\sqrt{35})$ (real, nonzero) and $i\pi$ are $\mathbb{Q}$-linearly independent, Baker's theorem [1] says $1, \log\alpha_1, \log\alpha_2$ are $\overline{\mathbb{Q}}$-linearly independent: contradiction. □

**Status (Category A).** Proved, conditional on Baker's theorem [1] and on the algebraicity of cusp expansions (Shimura's $\mathbb{Q}(\zeta_N)$-rationality, i.e. the $q$-expansion principle), both cited. The radial expansion used is R1 (§13.1).

**Category B.** Reading: the remainder of $C$ is "weight-1 sized" at $q \to 1$ with a regulator as coefficient; weight-$\frac{1}{2}$ false thetas are $O(t^{-1/2})$ and cannot carry it. An Appell–Lerch formula for $E$ or $P_s^{\mathrm{orth}}$ with algebraic data would likewise be excluded, but this needs the $q \to 1$ asymptotics of Appell–Lerch sums, which were not verified; that obstruction is *conditional*.

### 13.4 The multiplicative obstruction (Theorem 11.5(ii))

*Proof.* Along $q = e^{-t}$, $F/g = c\,t^{-j} e^{-\alpha/t}(1 + O(e^{-\beta/t}))$ times a power series in $t$, so only integer-step powers of $t$ occur; but $C_{\mathrm{third}}$ has the nonzero term $-a_P t^{-1/2}$ and $C$ the nonzero term $a_C t^{-1/2}$ (Table 13). □

**Status (Category A).** Proved. For $C$ the nonvanishing of $a_C$ rests on its closed form (Theorem 11.3). This covers every space with $k \in \{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ and $N \mid 140$ or 280; no Sturm-bound search is needed.

### 13.5 Gauss means and the scalar $W_7$ obstruction

*Proof of Theorem 11.4 (pattern).* If $2 \mid m$, shift $k \mapsto k+m$: $(Q_7(n,k+m) - Q_7(n,k))/m = 7n + 7k + \frac{7}{2}(m+1)$ and the phase $\mathbf{e}(7h(m+1)/2) = -1$, so $G = 0$. If $m$ is odd, $Q_7$ is $m$-periodic and 2 is invertible mod $m$; with $n' = n + 7 \cdot 2^{-1}k$, $hQ_7 \equiv h[n'^2 - \frac{35}{4}k^2 + \frac{7}{2}k]$ and $G = m^{-2}g_n g_k$ with $|g_n| = \sqrt{m}$. If $5 \mid m$, the shift $(n,k) \mapsto (n - m/5, k + m/5)$ changes the constant part by $j/5$, $j \equiv 1 \bmod 5$, so $G = 0$. If $\gcd(m, 35) = 1$, completing the square gives $|g_k| = \sqrt{m}$. If $7^e \| m$, $5 \nmid m$, by CRT the 7-part of $g_k$ is $7\sum_{k \bmod 7^{e-1}} \mathbf{e}\big(h(-\frac{5}{4}k^2 + \frac{k}{2})/7^{e-1}\big)$, of modulus $7^{(e+1)/2}$, so $|G| = \sqrt{7}/m$. □

**Status (Category A).** Proved; verified on all 3374 coprime pairs with $m \le 105$, zero mismatches. For $x = h/7$, $G = \big(\frac{h}{7}\big) i/\sqrt{7}$.

For the scalar $W_7$ question: a singularity $\sim G(x)I/(2\pi\varepsilon')$ at $x$ (with $\tau = x + i\varepsilon'$) maps under $W_7$ to $\sim |7x|^2 G(-1/(7x))I/(2\pi\varepsilon')$, so boundedness of $C - \lambda(7\tau)^{-k}C(-1/(7\tau))$ would force $G(x) = \lambda(7x)^{-k}|7x|^2 G(-1/(7x))$ for all $x$. This fails at the 178 zero-mismatch points; on the 60 pairs of common support the moduli obey an exact *weight-1* relation with $|\lambda| = 7^{-1/2}$ but the phase varies ($\pi/2$ at $1/7$ and 1, $-\pi/6$ at $1/3$ and $3/7$); for $W_{35}, W_{70}, W_{140}$ there are no pairs with both values nonzero. The map $\tau \mapsto -1/(7\tau)$ does not normalise $\Gamma_0(70)$.

**Status (Category A).** Zero/nonzero pattern and moduli: exact, by Theorem 11.4 (the floating-point exponential sums, nonzero values $\ge 0.034$ and zeros $< 10^{-9}$, agree).

**Category B.** The phase values quoted above are floating-point (with a wide margin) and are not separately proved.

**Category B.** The scalar failure is *vector mixing*: the leading profiles obey a vector-valued law (§14.1), under which $W_7$-type maps mix components; this is not a contradiction with modularity of the vector.

## 14 The theta lift and the vector-valued $S/T$ law

### 14.1 Leading order: radial profiles

For $\alpha, \beta \in \mathbb{Q}^2$ let $G_{\alpha,\beta}(x)$ be the mean over $\nu \in \alpha + \mathbb{Z}^2$ of $\mathbf{e}(xQ(\nu) + B(\nu,\beta))$; this is the $1/t$ radial profile (divided by the cone measure) of any cone sum over that coset and twist (leading-order Euler–Maclaurin, as in Theorem 11.4 and R1). For rational $\alpha, \beta$ and rational $x$ the summand is periodic on $\alpha + \mathbb{Z}^2$, and the mean is the average over one period. Write $A^* = (A_{11}, A_{22}) = (2, 7)$, so $\frac{1}{2}A^{-1}A^* = (\frac{1}{2}, 0)$.

**Theorem 14.1** (Profile $S/T$ law)**.** *For all $\alpha, \beta \in \mathbb{Q}^2$ and all $x \in \mathbb{Q} \setminus \{0\}$,*

$$G_{\alpha,\beta}(-1/x) = \frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}\,|x|} \sum_{p \in L'/\mathbb{Z}^2} G_{\beta+p,-\alpha}(x),$$

$$G_{\alpha,\beta}(x+1) = \mathbf{e}\big(-Q(\alpha) - B(\alpha, w_c)\big)\, G_{\alpha,\,\alpha+\beta+w_c}(x).$$

*The $T$-law may equivalently be written with $\frac{1}{2}A^{-1}A^* = (\frac{1}{2}, 0)$ in place of $w_c$ (both in the phase and in the label), since $(\frac{1}{2}, 0) - w_c \in L'$ and $G_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))G_{\alpha,\beta}$ for $\mu \in L'$.*

*Proof. T-law.* For $\nu=\alpha+n$, $Q(\nu)=Q(\alpha)+B(\alpha,n)+Q(n)$ and $\mathbf{e}(Q(n))=\mathbf{e}(B(n,w_c))$ (indeed $B(n,w_c)=n_2/2$ and $Q(n)\equiv\frac{7}{2}n_2^2\equiv n_2/2 \bmod 1$). Hence $\mathbf{e}(Q(\nu))=\mathbf{e}(-Q(\alpha)-B(\alpha,w_c))\,\mathbf{e}(B(\nu,\alpha+w_c))$; average over the coset.

*S-law.* (1) *Gaussian-weighted sums.* Let $M\subset\mathbb{R}^2$ be a lattice, $c\in\mathbb{R}^2$, $\Pi$ a function on $c+M$ periodic modulo a sublattice $\Lambda\subset M$ of finite index, and $N$ complex symmetric with $\operatorname{Re}N>0$. Splitting $c+M$ into $\Lambda$-cosets and applying Poisson summation on each,

$$\sum_{\eta\in c+M}\Pi(\eta)e^{-\pi\delta\,\eta^{\mathsf T}N\eta}=\frac{\operatorname{mean}_{c+M}(\Pi)}{\operatorname{vol}(M)\,\delta\,(\det N)^{1/2}}+O\big(\delta^{-1}e^{-\kappa/\delta}\big)\qquad(\delta\to0^+),$$

since $\operatorname{Re}N^{-1}>0$; here $(\det N)^{1/2}$ is the branch on the (convex) Siegel domain that is positive on real $N$, and $\kappa>0$ and the $O$-constant are uniform for $N$ in a compact neighbourhood of $I$. (2) *Poisson.* Fix $x\neq0$, $\epsilon>0$, put $Z_\epsilon=\epsilon I+\frac{i}{x}A$ ($\operatorname{Re}Z_\epsilon=\epsilon I>0$) and $g(\nu)=\mathbf{e}(B(\nu,\beta))e^{-\pi\nu^{\mathsf T}Z_\epsilon\nu}=\mathbf{e}(-Q(\nu)/x+B(\nu,\beta))e^{-\pi\epsilon|\nu|^2}$, a Schwartz function. By (1) with $M=\mathbb{Z}^2$, $N=I$, $\delta=\epsilon$: $\sum_{\nu\in\alpha+\mathbb{Z}^2}g(\nu)=\epsilon^{-1}G_{\alpha,\beta}(-1/x)+O(\epsilon^{-1}e^{-\kappa/\epsilon})$. On the other hand, $\sum_n g(\alpha+n)=\sum_{k\in\mathbb{Z}^2}\widehat g(k)\mathbf{e}(k\cdot\alpha)$ with $\widehat g(k)=(\det Z_\epsilon)^{-1/2}\exp(-\pi(k-A\beta)^{\mathsf T}Z_\epsilon^{-1}(k-A\beta))$. Write $k=A\mu$, $\mu\in L'$, $\eta=\mu-\beta\in-\beta+L'$, so $\mathbf{e}(k\cdot\alpha)=\mathbf{e}(B(\alpha,\beta))\mathbf{e}(B(\eta,\alpha))$. With $N_\epsilon=(I-i\epsilon xA^{-1})^{-1}$ one has $Z_\epsilon=\frac{i}{x}A\,N_\epsilon^{-1}$ and $AZ_\epsilon^{-1}A=-ixA+\epsilon x^2N_\epsilon$, so

$$\sum_{\nu\in\alpha+\mathbb{Z}^2}g(\nu)=\frac{\mathbf{e}(B(\alpha,\beta))}{(\det Z_\epsilon)^{1/2}}\sum_{\eta\in-\beta+L'}\mathbf{e}\big(xQ(\eta)+B(\eta,\alpha)\big)\,e^{-\pi\epsilon x^2\eta^{\mathsf T}N_\epsilon\eta}.$$

By (1) with $M=L'$ ($\operatorname{vol}=1/35$), $N=N_\epsilon\to I$, $\delta=\epsilon x^2$, and $\operatorname{mean}_{-\beta+L'}=\frac{1}{35}\sum_{p\in L'/\mathbb{Z}^2}G_{p-\beta,\alpha}(x)$, the right side is $\mathbf{e}(B(\alpha,\beta))\,[(\det Z_\epsilon)(\det N_\epsilon)]^{-1/2}(\epsilon x^2)^{-1}\sum_p G_{p-\beta,\alpha}(x)$ up to $O(\epsilon^{-1}e^{-\kappa'/\epsilon})$. (3) *The factor $|x|$.* $(\det Z_\epsilon)(\det N_\epsilon)=\det(\frac{i}{x}A)=35/x^2$ for all $\epsilon$. Along $Z_s=\epsilon I+s\frac{i}{x}A$, $s\in[0,1]$, the eigenvalues $\epsilon+is\lambda_j/x$ ($\lambda_1>0>\lambda_2$ the eigenvalues of $A$, $\lambda_1\lambda_2=-35$) stay in the right half-plane, so $(\det Z_\epsilon)^{1/2}=\prod_j(\epsilon+i\lambda_j/x)^{1/2}$ (principal roots) $\to(i\lambda_1/x)^{1/2}(i\lambda_2/x)^{1/2}=\sqrt{35}/|x|$, the arguments $\pm\pi/4$ cancelling for either sign of $x$ (signature $(1,1)$); and $(\det N_\epsilon)^{1/2}\to1$. The continuous product, whose square is constant, therefore equals $\sqrt{35}/|x|$. Comparing the coefficients of $\epsilon^{-1}$ gives $G_{\alpha,\beta}(-1/x)=\mathbf{e}(B(\alpha,\beta))(\sqrt{35}|x|)^{-1}\sum_p G_{p-\beta,\alpha}(x)$. Finally $G_{-\gamma,-\delta}=G_{\gamma,\delta}$ ($\nu\mapsto-\nu$, $Q$ even) and $p\mapsto-p$ permutes $L'/\mathbb{Z}^2$, so $\sum_p G_{p-\beta,\alpha}=\sum_p G_{\beta+p,-\alpha}$. □

**Status (Category A).** Proved above; the analytic inputs are Poisson summation for Schwartz functions and the complex Gaussian Fourier transform. Numerically verified independently of the proof: the $S$-law on all 140 labels at $x\in\{1,-1,\frac13,-\frac23\}$ (560 tests, 280 with nonzero value), on 13 labels at 12 values of $x$ (both signs), and on $40+60$ random rational pairs $(\alpha,\beta)$ outside the family ($3+60$ with nonzero value), with maximal deviation $3.3\cdot10^{-15}$; the $T$-law on all 140 labels to $5.6\cdot10^{-16}$. With the signed factor $x$ in place of $|x|$ the ratio is exactly $-1$ at $x<0$.

**Corollary 14.2** (Consequences of Theorem 14.1)**.** *(i) The* 140*-vector* $\mathbf{G}(x)=(G_{\alpha,\beta}(x))$ *satisfies* $P\mathbf{G}=\mathbf{G}$, $\mathbf{G}(-1/x)=|x|^{-1}S\mathbf{G}(x)$ *and* $\mathbf{G}(x+1)=T\mathbf{G}(x)$ *with the matrices of Theorem 11.6; hence* $\mathbf{G}(\gamma x)=|cx+d|^{-1}\rho(\gamma)\mathbf{G}(x)$ *for all* $\gamma\in\mathrm{SL}_2(\mathbb{Z})$ *and rational* $x\neq-d/c$. *(ii)* $G_{a,0}\big(x/(140x+1)\big)\,|140x+1|=G_{a,0}(x)$ *for all rational* $x\neq0,-\frac{1}{140}$. *(iii) For* $\gcd(m,70)=1$, $m\,\mathbf{e}(-7/(20m))\,G(1/m)=m\,G_{a,0}(1/m)$ *depends only on* $m \bmod 70$. *(iv) For the fundamental-domain vector (all components are cone sums over the same cone measure), the* $1/t$ *coefficients* $A(x)$ *satisfy* $|x|\,A(-1/x)=S\,A(x)$ *at every rational* $x\neq0$.

*Proof.* (i) The reductions $G_{\alpha+n,\beta} = G_{\alpha,\beta}$, $G_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))G_{\alpha,\beta}$ ($\mu \in L'$), $G_{-\alpha,-\beta} = G_{\alpha,\beta}$ are those of the $u_{\alpha,\beta}$, so the laws of Theorem 14.1 have exactly the matrices $S, T$; $|cx+d|$ is a cocycle and $\rho$ is well defined on $\mathrm{Fix}(P)$. (ii) Apply the $S$-law at $x' = -1/x-140$, then $G_{p,-a}(y-140) = G_{p,-a}(y)$ (since $70Q(\nu) \in \mathbb{Z}$ for $\nu \in L'$), then the $S$-law at $-1/x$. (iii) The $S$-law gives $m\,G_{a,0}(1/m) = 35^{-1/2}\sum_p G_{p,-a}(-m)$, and $mQ(\nu) \bmod 1$ depends only on $m \bmod 70$ for $\nu \in L'$. (iv) Each $1/t$ coefficient is (cone measure)$\times G$, by leading-order Euler–Maclaurin (R1); apply (i). □

**Status (Category A).** Proved given Theorem 14.1 ((iv) also uses R1). Numerically: (i) for random words in $S, T$ to $6.6 \cdot 10^{-16}$, (ii) at 6 rationals to $3 \cdot 10^{-15}$, (iii) for all $m < 1200$ (spread $1.2 \cdot 10^{-13}$). Part (ii) is the Weil-kernel invariance used in Theorem 24.6(ii), and (iii) is the Gauss-sum input of Theorem 24.4.

At the cusp pair $(0,\infty)$ no exact vector-valued *modular* law with algebraic data can hold: at $x = 0$ each cone sum has singular part $IG/t \neq 0$, while at $\infty$ there is no $q^0$ term ($Q$ has no nonzero rational zeros since 35 is not a square); an exact law would force $I \in \pi\overline{\mathbb{Q}} + \overline{\mathbb{Q}}$, excluded by Baker (Category A, conditional on Baker). At order $t^{-1/2}$ the *quadrant*-cone vector fails the weight-1 $S$-law ($|\mathrm{LHS}|/|\mathrm{RHS}| \in [0.49, 2.97]$ over 16 tests, either wall convention), while the *fundamental-domain* vector has vanishing $t^{-1/2}$ coefficients ($\leq 5 \cdot 10^{-15}$) (Category B: floating-point ratios and residuals). For the scalar rows the vanishing that is actually needed is proved in the course of the proof of Theorem 11.7 (§15.4).

## 14.2 Siegel theta and Poisson summation

Fix a majorant (a point of the $\gamma$-geodesic) with $Q = Q_+ + Q_-$ and put $G_\tau(\nu) = \mathbf{e}(\tau Q_+(\nu) + \bar\tau Q_-(\nu))$, $\Theta_{\alpha,\beta}(\tau) = \sum_{\nu\in\alpha+\mathbb{Z}^2} \mathbf{e}(B(\nu,\beta))G_\tau(\nu)$. Poisson summation, with $\widehat{G}_{-1/\tau}(A\eta) = (|\tau|/\sqrt{35})G_\tau(\eta)$ (the two square roots of $\det(M/i) = 35|\tau|^2$ are complex conjugate), gives

$$\Theta_{\alpha,\beta}(-1/\tau) = |\tau|\,\frac{\mathbf{e}(B(\alpha,\beta))}{\sqrt{35}} \sum_{p\in L'/\mathbb{Z}^2} \Theta_{p-\beta,\alpha}(\tau). \tag{14.1}$$

## 14.3 The lift

Integrating over the closed geodesic of $\langle\gamma^k\rangle$ ($k = 2$ is the common stabiliser order on all labels), unfolding with $\int_{\mathbb{R}} e^{-a\cosh 2w}\,dw = K_0(a)$ and geodesic length $\log\varepsilon$, and multiplying by $\sqrt{y}$ ($w = x + iy$) gives

$$u_{\alpha,\beta}(w) = \sqrt{y}\Big[\log\varepsilon\cdot\delta_{\alpha\in\mathbb{Z}^2} + \tfrac{1}{2}\sum_{\nu \bmod \gamma^2} \mathbf{e}(B(\nu,\beta))\,K_0(2\pi|Q(\nu)|y)\,\mathbf{e}(Q(\nu)x)\Big],$$

the sum over $\nu \in \alpha + \mathbb{Z}^2$, $\nu \neq 0$, in all four sign regions $\pm S^+$, $\pm T^+$ (negative-norm vectors give negative frequencies; the fundamental cones are $[e_1, \gamma e_1)$, $[f, \gamma f)$, $f = (1,-1)$, and their negatives). Each term $\sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$, and $\sqrt{y}$, is a $\Delta$-eigenfunction with $\lambda = \frac{1}{4}$. The scalar rows are $u_L = u_{a,0}$ and $u_R = \sum_p u_{p,-a}$; for these the formula reduces to $u_L = \sqrt{y}\sum_{\nu\in(a+\mathbb{Z}^2)/\langle\gamma\rangle} K_0(\cdot)\mathbf{e}(\cdot)$ and $u_R = \sqrt{y}[\log\varepsilon + \sum_{0\neq\nu\in L'/\langle\gamma\rangle} \mathbf{e}(B(\nu,-a))K_0(\cdot)\mathbf{e}(\cdot)]$. Reductions: $u_{\alpha+n,\beta} = u_{\alpha,\beta}$ ($n \in \mathbb{Z}^2$), $u_{\alpha,\beta+\mu} = \mathbf{e}(B(\alpha,\mu))u_{\alpha,\beta}$ ($\mu \in L'$), $u_{-\alpha,-\beta} = u_{\alpha,\beta}$.

*Proof of Theorem 11.6.* Integrate (14.1) over the geodesic and use $\sqrt{\mathrm{Im}(-1/\tau)} = \sqrt{y}/|\tau|$; the factor $|\tau|$ cancels and gives the weight-0 $S$-law; the constant term $\sqrt{y}\log\varepsilon$ comes from $\nu = 0$ and appears exactly when $\alpha \in \mathbb{Z}^2$. For $(a,0)$ the right side is $\frac{1}{\sqrt{35}}\sum_p u_{p,a} = u_R/\sqrt{35}$ using $u_{p,a} = u_{-p,-a}$, i.e. $u_L(-1/w) = u_R(w)/\sqrt{35}$. The $T$-law follows from $\mathbf{e}(Q(n)) = \mathbf{e}(B(n,w_c))$; the naive even-lattice

$T$-law fails by $O(1)$. With labels $\alpha \in (\{0,a\}+L')/\mathbb{Z}^2$ (70 classes) and $\beta \in \{0,-a\}$ the family is closed; each row of $S$ has exactly 35 entries of modulus $1/\sqrt{35}$, $T$ is monomial. □

**Status (Category A).** $S$-law proved by Poisson summation; $T$-law derived. Numerically on all 140 rows at three points with $y \approx 1$: $|u(-1/w) - Su(w)| \le 1.2 \cdot 10^{-15}$, $|u(w+1) - Tu(w)| \le 7 \cdot 10^{-16}$, $|Pu - u| \le 6 \cdot 10^{-16}$; on the 72-dimensional $P$-invariant subspace $S^2 = I$, $(ST)^3 = I$ to $1.3 \cdot 10^{-15}$. Borcherds [3, Thm. 4.1] gives the same automorphy factor $(c\tau+d)^{b^+/2}(c\bar\tau+d)^{b^-/2}$; it is not needed. The $S$-law $u_L(-1/w) = u_R(w)/\sqrt{35}$ was also checked to $6 \cdot 10^{-11}$, and $u_L(w+1) = \mathbf{e}(13/20)u_L(w)$ exactly.

**Remark 14.3.** The cyclic $S,T$-span of the row $e_{(a,0)}$ has dimension 60 in label space $\mathbb{C}^{140}$; after projection to $\mathrm{Fix}(P)$ (rows multiplied by $\frac{1}{2}(I+P)$, which is what acts on the $P$-invariant vector $u$) its rank is 36. The positive Fourier coefficients satisfy $f_{u_L} = 2E_L$, $f_{u_R} = 2E_R$ (the automorph $-\rho_1$ maps $S^+$ to $S^-$ preserving the coset and phase), checked for all norms $\le 25$ (Category A for the automorph argument; the numerical check is supporting).

**Remark 14.4** (Normalisation of the lift)**.** The lift can be written as a sum over $\langle\gamma\rangle$-orbits or as $\frac{1}{2}\sum_{\bmod \gamma^2}$. For the scalar rows these agree exactly: $\gamma$ maps $a+\mathbb{Z}^2$ to itself ($\gamma a - a = (0,-1)$) and preserves $Q$; it fixes no nonzero vector (hyperbolic), so every $\langle\gamma\rangle$-orbit is the disjoint union of exactly two $\langle\gamma^2\rangle$-orbits with equal $Q$; and the phase $\mathbf{e}(B(\nu,\beta))$ is $\gamma$-invariant for $\beta = 0$, and for $\beta = -a$ on $L'$ because $\gamma^{-1}a - a \in \mathbb{Z}^2$ and $B(L',\mathbb{Z}^2) \subset \mathbb{Z}$. Hence $\frac{1}{2}\sum_{\nu \bmod \gamma^2} = \sum_{\nu \bmod \gamma}$ termwise for $u_L$ and $u_R$ (Category A, elementary). For labels whose coset is not $\gamma$-stable the $\frac{1}{2}\sum_{\bmod \gamma^2}$ form is the definition.

# 15 The Lewis–Zagier cocycle and quantum modularity

## 15.1 Definitions

Let $F = [e_1, \gamma e_1)$ (half-open, vertex excluded). For labels $(\alpha,\beta)$ set $E_{\alpha,\beta}(\tau) = \sum_{\nu \in F \cap (\alpha+\mathbb{Z}^2)} \mathbf{e}(B(\nu,\beta))\mathbf{e}(\tau Q(\nu))$, $E_L = E_{a,0}$ and $E_R = \sum_p E_{p,-a}$. Thus $E_L(\tau) = \sum_{\mathfrak{a}\in\mathcal{K}} \mathbf{e}(\tau \mathrm{N}\mathfrak{a}/20)$ and $E = q^{7/20}E_L$. At rational $x$, $E(x+iy) = 2I\,G(x)/t + \kappa(x) + O(t)$ with no $t^{-1/2}$ term. Write $\kappa_L(x)$ and $\kappa_{p,-a}(x)$ for the radial constant terms of $E_L$ and $E_{p,-a}$ at $x$. The *scalar cocycle* is

$$\psi(x) = \kappa_L(-1/x) - \frac{|x|}{\sqrt{35}} \sum_p \kappa_{p,-a}(x) \qquad (x \in \mathbb{Q}\setminus\{0\}),$$

a finite Shintani-type sum of $B_2$, $B_1B_1$ and roots of unity at each rational; hence $\psi(h/m) \in \mathbb{Q}(\zeta_N, \sqrt{35})$. The real factor $|x|$ is the one dictated by the profile law (Theorem 14.1). On $\mathbb{H}$ we use

$$\Psi(z) = E_L(-1/z) - \frac{z}{\sqrt{35}}E_R(z), \qquad \Psi^-(z) = E_L(-1/z) + \frac{z}{\sqrt{35}}E_R(z),$$

whose radial constant terms at rational $x > 0$, respectively $x < 0$, are $\psi(x)$ (Theorem 11.7(ii)). For the vector, let $f_{\alpha,\beta}$ be the $\gamma$-averaged positive-frequency parts of $u_{\alpha,\beta}$, $\kappa$ the vector of their radial constant terms, and $\Phi(x) = \kappa(-1/x) - |x|\,S\kappa(x)$. Normalisation: $\Phi_{(a,0)} = 2\psi$ (since $f_{u_L} = 2E_L$).

## 15.2 The Green-form lemma

Let $R_z(w) = \mathrm{Im}\, w/((w-z)(\bar w - z))$, $[u,v] = u_w v\,dw + uv_{\bar w}\,d\bar w$ and $T_n(w) = \sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$.

**Lemma 15.1** (Green-form lemma)**.** *For $z \in \mathbb{H}$, $F_n(z) := \int_z^{z+i\infty}[T_n, R_z^{1/2}]$ (vertical ray) equals $-\frac{\pi}{2}\mathbf{e}(nz)$ for $n > 0$ and $0$ for $n < 0$, and $\int_z^{z+i\infty}[\sqrt{y}, R_z^{1/2}] = \frac{1}{2}$.*

*Proof.* $\Delta_w R_z^{1/2} = \frac{1}{4} R_z^{1/2}$ and the form is closed (a direct computation). $R_z$ is holomorphic in $z$ and the endpoint term vanishes ($R^{-1/2} \to 0$ like $|w-z|^{1/2}$), so $F_n$ is holomorphic (in detail: the integrand is $O(s^{-1/2})$ at the lower end, so one works with translation covariance gives $F_n = C_n \mathbf{e}(nz)$. For $n < 0$ boundedness forces $C_n = 0$. For $n > 0$, Laplace's method as $y \to \infty$ with $K_\nu(X) = \sqrt{\pi/(2X)} e^{-X}(1 + O(1/X))$ (DLMF 10.40.2) gives the limit $(\sqrt{\pi a}/2) \int_0^\infty s^{-1/2} e^{-as} ds = \pi/2$ with $a = 2\pi n$, and an overall minus sign. For $n = 0$ the integral is $\frac{1}{4} B(\frac{1}{2}, 1) = \frac{1}{2}$. □

$F_{n,\eta}(z) = \int_{z+i\eta}^{z+i\infty}$; by closedness and the $z$-holomorphy of $R_z$, $\partial_{\bar z} F_{n,\eta}(z) = -T_n \, \partial_{\bar w} R_z^{1/2}$ at $w = z + i\eta$, which tends to 0 locally uniformly, while $F_{n,\eta} \to F_n$ locally uniformly);

**Status (Category A).** Proved; the symbolic ingredients are checked by computer algebra. Independent 30-digit quadrature at $z \in \{0.3 + 0.8i, -0.17 + 0.45i, 1.1 + 1.3i\}$, $n \in \{\pm 1, \pm 2, 3\}$ and $n = 0$ agrees to $\le 7 \cdot 10^{-22}$. This is the $s = \frac{1}{2}$ case of the Lewis–Zagier construction [10, Ch. II §2]; the proof does not depend on the citation. By translation covariance the lemma holds for every real $n \neq 0$.

## 15.3 The branch lemma

For $z \in \mathbb{C} \setminus i\mathbb{R}$ write $\varsigma(z) = \operatorname{sgn} \operatorname{Re} z$.

**Lemma 15.2** (Branch lemma)**.** *Let $z \in \mathbb{H} \setminus i\mathbb{R}$ and let $C_z$ be the image of the vertical ray $-1/z + i\mathbb{R}_{\ge 0}$ under $w' \mapsto -1/w'$, an arc from $z$ to 0. On the ray take the positive root of $R_{-1/z}$; on $C_z$ take the root of $R_z$ continued from the principal root on $i\mathbb{R}_{>0}$ near 0. Then, for $w \in C_z$,*

$$R_{-1/z}(-1/w)^{1/2} = \varsigma(z)\, z\, R_z(w)^{1/2}.$$

*Proof.* Since $-1/w + 1/z = (w - z)/(wz)$ and $\operatorname{Im}(-1/w) = \operatorname{Im} w/|w|^2$, one has $R_{-1/z}(-1/w) = z^2 R_z(w)$. Both roots are continuous and nonvanishing along $C_z$, so the two sides agree up to a sign that is constant on $C_z$. Compare at $w \to 0$ along $C_z$, where $C_z$ is tangent to $i\mathbb{R}$. There $R_z(w) = \operatorname{Im} w \, z^{-2}(1 + o(1))$, and the principal root of $z^{-2}$ is $\varsigma(z)/z$ (the root with positive real part), so $\varsigma(z)\, z\, R_z(w)^{1/2} = (\operatorname{Im} w)^{1/2}(1 + o(1)) > 0$. On the other side $-1/w \to i\infty$ on the ray, where $R_{-1/z}(-1/w) > 0$ and its positive root is taken. Hence the sign is $+$. □

**Status (Category A).** Elementary. Checked numerically at points of both half-planes (the sign is $+1$ for $\operatorname{Re} z > 0$ and $-1$ for $\operatorname{Re} z < 0$).

## 15.4 Proof of the period formula

*Proof of Theorem 11.7. (i) Holomorphy.* On the path $w = it$, $R_z(it) = t/(t^2 + z^2)$. For $z \notin i\mathbb{R}$ and $t > 0$ we have $t^2 + z^2 \notin (-\infty, 0]$, so the principal roots of $R_z(it)$ and of its $\bar w$-derivative are holomorphic in $z$. The integrand decays exponentially as $t \to 0$ (since $u_R(it) = \sqrt{35}\, u_L(i/t)$ and $u_L$ has no constant term) and is $O(t^{-2})$ as $t \to \infty$ (the $1/t$ terms of the constant-term integrand cancel). The convergence is locally uniform in $z$, so $\Omega$ is holomorphic on $\mathbb{C} \setminus i\mathbb{R}$.

*Identification on $\mathbb{H}$.* Fix $z \in \mathbb{H} \setminus i\mathbb{R}$ and put $\varsigma = \varsigma(z)$. Lemma 15.1 at the point $-1/z$, applied to $u_L$ (no constant term, $f_{u_L} = 2E_L$), gives

$$\int_{-1/z}^{-1/z + i\infty} [u_L, R_{-1/z}^{1/2}] = -\pi E_L(-1/z).$$

Pull back by the involution $g : w \mapsto -1/w$, which maps the vertical ray from $-1/z$ onto the arc $C_z$ from $z$ to 0. The Green form is natural: $g^*[u, v] = [u \circ g, v \circ g]$. By Theorem 11.6, $u_L \circ g = u_R/\sqrt{35}$, and by Lemma 15.2, $R^{1/2}_{-1/z} \circ g = \varsigma\, z\, R^{1/2}_z$ on $C_z$. By bilinearity,

$$-\pi E_L(-1/z) = \frac{\varsigma z}{\sqrt{35}} \int_{C_z} [u_R, R^{1/2}_z].$$

Lemma 15.1 at $z$, applied to $u_R$ (constant term $\sqrt{y}\log\varepsilon$, $f_{u_R} = 2E_R$), gives $\int_z^{z+i\infty}[u_R, R^{1/2}_z] = -\pi E_R(z) + \frac{1}{2}\log\varepsilon$. The form $[u_R, R^{1/2}_z]$ is closed on $\mathbb{H}\setminus\{z\}$ (both entries are $\Delta$-eigenfunctions with eigenvalue $\frac{1}{4}$), $R_z$ has no zeros on $\mathbb{H}$, and the continuation of $R^{1/2}_z$ around the loop formed by the vertical ray from $z$, the imaginary axis and $C_z$ is single-valued (it is the positive root on the vertical ray and near $i\infty$, and the principal root on $i\mathbb{R}_{>0}$). At $w = z$ the integrand is $O(|w - z|^{-1/2})$, so small arcs around $z$ contribute $o(1)$; the contribution at height $Y \to \infty$ is $O(1/Y)$; at 0 the arc $C_z$ is tangent to $i\mathbb{R}$ and $u_R$ decays exponentially. Hence

$$\int_{C_z} [u_R, R^{1/2}_z] = \int_z^{z+i\infty} [u_R, R^{1/2}_z] - \int_0^{i\infty} [u_R, R^{1/2}_z] = -\pi E_R(z) + \tfrac{1}{2}\log\varepsilon - \int_0^{i\infty} [u_R, R^{1/2}_z].$$

Combining the last two displays and multiplying by $\varsigma/\pi$ (note $\varsigma^2 = 1$),

$$\Omega(z) = \varsigma\, E_L(-1/z) - \frac{z}{\sqrt{35}} E_R(z),$$

which is $\Psi(z)$ for $\operatorname{Re} z > 0$ and $-\Psi^-(z)$ for $\operatorname{Re} z < 0$. In particular $\Psi$ and $\Omega$ agree on $\mathbb{H}\cap\{\operatorname{Re} z > 0\}$, so $\Psi$ continues holomorphically to $U$.

*(ii) Boundary values.* Let $x > 0$ be rational and let $z = x + iy \to x$ non-tangentially in $\mathbb{H}$. With $t = 2\pi y$ and $t'' = -2\pi i(-1/z + 1/x) = t/(xz)$ (so that $\mathbf{e}(-1/z) = \mathbf{e}(-1/x)e^{-t''}$, and $t'' \to 0$ non-tangentially), R1 (§13.1) gives

$$E_L(-1/z) = \frac{A_L(-1/x)}{t''} + \kappa_L(-1/x) + O(|t|^{1/2}), \qquad E_{p,-a}(z) = \frac{A_p(x)}{t} + \frac{h_p(x)}{t^{1/2}} + \kappa_{p,-a}(x) + O(|t|^{1/2}),$$

where $E_L$ has no $t''^{-1/2}$ term (Lemma 24.3). Since $1/t'' = xz/t$, the $1/t$ part of $\Psi(z)$ is

$$\frac{z}{t}\Big(x\, A_L(-1/x) - \frac{1}{\sqrt{35}}\sum_p A_p(x)\Big) = 0$$

*exactly*, by Corollary 14.2(iv) on the row $(a, 0)$, whose $S$-row is $\frac{1}{\sqrt{35}}\sum_p$ over the labels $(p, -a)$. Therefore

$$\Psi(z) = -\frac{z}{\sqrt{35}}\Big(\sum_p h_p(x)\Big)t^{-1/2} + \kappa_L(-1/x) - \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) + o(1).$$

By (i), $\Psi = \Omega$ near $x$ and $\Omega$ is holomorphic at $x$, so $\Psi$ is bounded as $z \to x$. Hence $\sum_p h_p(x) = 0$, and the limit is $\Omega(x) = \kappa_L(-1/x) - \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) = \psi(x)$.

For rational $x < 0$ the same computation applies to $\Psi^-$: its $1/t$ part is $\frac{z}{t}\big(xA_L(-1/x) + \frac{1}{\sqrt{35}}\sum_p A_p(x)\big) = \frac{z}{t}\big(-|x|A_L(-1/x) + \frac{1}{\sqrt{35}}\sum_p A_p(x)\big) = 0$ by Corollary 14.2(iv); $\Psi^- = -\Omega$ is bounded near $x$, so the $t^{-1/2}$ coefficient vanishes, and $-\Omega(x) = \kappa_L(-1/x) + \frac{x}{\sqrt{35}}\sum_p \kappa_{p,-a}(x) = \psi(x)$.

*(iii) Rows.* For a row $u = u_{\alpha,\beta}$ with $S$-row combination $U$, the same argument (Lemma 15.1 with the constant terms $a_0(u), a_0(U)$, the $S$-law of Theorem 11.6, Lemma 15.2, Corollary 14.2(iv) and R2 for the termwise integration) gives the split-contour formula (15.1) below. The functions $J_U(x)$ and $J_u(-1/x)$ there are holomorphic in $x$ off $i\mathbb{R}$, so $\Phi$ is real-analytic on $\mathbb{R}\setminus\{0\}$. The three-term relation follows from $f(z + 1) = Tf(z)$ and $(ST)^3 = I$. □

The order of the argument in (ii) matters: the profile law removes the $1/t$ singularity exactly; boundedness, which comes from the holomorphic continuation (i), then removes the $t^{-1/2}$ term and identifies the limit.

The split-contour form used numerically is $\psi(x) = -\frac{1}{\pi}\big[\frac{x}{\sqrt{35}}(\frac{1}{2}\log\varepsilon - J_R(x)) + J_L(-1/x)\big]$ for $x > 0$, $J(z) = \int_i^{i\infty}[u, R_z^{1/2}]$. For every row $u = u_{\alpha,\beta}$, with $U$ the corresponding $S$-row combination and $a_0$ the constant term,

$$\Phi_u(x) = -\frac{2}{\pi}\Big[\,|x|\frac{a_0(U)}{2} - \frac{a_0(u)}{2} - |x|J_U(x) + J_u(-1/x)\Big]. \tag{15.1}$$

From $f(z+1) = Tf(z)$ and $(ST)^3 = I$,

$$\Phi(z) = T\,\Phi\big(z/(z+1)\big) + \frac{z}{z+1}\,TST\,\Phi(z+1). \tag{15.2}$$

**Category B.** Supporting numerics for (ii)–(iii): a holomorphic fit from $\operatorname{Im} z \geq 0.01$ reproduces the real-axis values to $2\cdot 10^{-12}$ at 169 rationals; negative side with the factor $|x|$ to $1.6\cdot 10^{-14}$. All 140 rows at $x \in \{1, \frac{1}{3}, 2, \frac{3}{5}, \frac{7}{4}, -1, -\frac{2}{3}\}$: singular cancellation $9.0\cdot 10^{-16}$, no $t^{-1/2}$ terms $5.7\cdot 10^{-16}$, $P\kappa = \kappa$ to $5.3\cdot 10^{-15}$, closed form to $2.3\cdot 10^{-14}$; (15.2) at $z \in \{1, \frac{1}{2}, \frac{2}{3}, \frac{3}{2}\}$ to $7\cdot 10^{-15}$.

**The automorphy factor.** With the holomorphic factor $x$ in place of $|x|$ the cocycle at $x < 0$ is unbounded (it grows like $1/y$); the bounded object on the negative side is $\Psi^-$, i.e. the real factor $|x|$. Numerically $\psi(-x) = \overline{\psi(x)}$ to $8\cdot 10^{-14}$ (Category B; this symmetry is not proved here and is not used). Near 0, $\psi(x) \approx -0.0549i\,x^2 + 0.0206\,x^3$ (from $x = 0.01, 0.05$); $\psi$ is not analytic at 0. Since $T$ acts by phases and $S, T$ generate, every $h_\gamma$ is real-analytic off $\gamma^{-1}\infty$. Example: $\psi(1) = 0.00688399376041986919032158 7 - 0.03825971221697307224258855\,i$.

**Category B.** Interpretation: the 72-dimensional fundamental-domain vector is a vector-valued quantum modular form of weight 1 in Zagier's sense for the real automorphy factor $|cx+d|$ ("weight $(\frac{1}{2}, \frac{1}{2})$"), i.e. the $f$-function of a $\lambda = \frac{1}{4}$ Maass vector, with exceptional set $\{0\}$ for $S$; $E$ itself is then a "renormalised" weight-1 quantum modular object with the singular parts $I \cdot G(x)$ removed. The Lewis–Zagier period-function reading [10] and the broader context of [4] are cited for context only.

## 15.5 The cocycle of the $E$-part

**Theorem 15.3** (Cocycle of the $E$-part)**.** *Let $s = 7$, and let $u_L$ and $E_L$ be as in §15.*

*(i) (Constant) For $u = \sqrt{y}\sum A_n K_0(2\pi|n|y)\mathbf{e}(nx) + b\sqrt{y}$, $\int_z^{i\infty}[u, R_z^{1/2}] = -\frac{\pi}{2}f_u(z) + \frac{b}{2}$, so $C = -\pi/2$ and $E_L(z) = -\frac{1}{\pi}\int_z^{i\infty}[u_L, R_z^{1/2}]$.*

*(ii) (Boundary step, both signs of $cx+d$) If $u_L \circ \gamma' = u_L$, then for rational $x \neq \gamma'^{-1}\infty$,*

$$\kappa_L(x) - |cx+d|^{-1}\kappa_L(\gamma' x) = -\frac{1}{\pi}\int_{\gamma'^{-1}\infty}^{i\infty}[u_L, R_x^{1/2}],$$

*with branch sign $\operatorname{sgn}(cx+d)$, and the singular profile obeys $A(x) = |cx+d|\,A(\gamma' x)$.*

*(iii) (Vector law) For all 140 labels $\lambda = (\alpha, \beta)$, $u \circ \gamma' = \rho(\gamma')u$ for every $\gamma' \in \mathrm{SL}_2(\mathbb{Z})$, with $\rho$ generated by the $S$- and $T$-matrices of Theorem 11.6. The constant terms $b_\lambda = \log\varepsilon \cdot \delta_{\alpha\in\mathbb{Z}^2}$ (here $\varepsilon = 6 + \sqrt{35}$) are carried inside the theta identity. The vector cocycle with $b$ holds for general $\gamma'$, and on the $L$-row at $\gamma' = S$ it reduces exactly to the $S$-formula for $\psi$ (§15.1).*

**Status (Category A).** (i): the integral is holomorphic and translation-covariant, so it equals $C\,\mathbf{e}(nz)$; the asymptotics as $y \to \infty$ give $C = -\pi/2$ (as in Lemma 15.1; the constant term contributes $b/2$). (ii): closedness of the Green form, the non-tangential expansion R1 and the positive-root branch comparison (Lemma 15.2 is the case $\gamma' = S$). (iii): Poisson summation at a fixed majorant, the $T$-law, the reductions, exact $\gamma^2$-periodicity and cycle integration, label by label.

**Category B.** The vector identity at $\gamma' = STS$, $ST^2S$, $ST^3S$ agrees to $\le 3.3 \cdot 10^{-12}$; the scalar cocycle at 45 rationals to $7 \cdot 10^{-13}$; the quadrature check of $C$ to $10^{-18}$; non-tangential $\kappa$ fits in 4 directions.

### 15.6 Interchange of sum and integral

**R2.** The Fourier expansion of $u$ may be integrated termwise against $R_z^{1/2}$ along the paths used above: the frequencies satisfy $|Q| \ge n_0 > 0$ ($n_0 = 4/140$ for $u_R$, 0.35 for $u_L$), there are $O(X)$ $\gamma$-orbits with $|Q| \le X$, and dominated convergence applies.

**Status (Category A).** Proved (elementary estimates).

## 16 The Eisenstein/cusp split

### 16.1 $u_L$ is Eisenstein

Modulo $\mathfrak{f}\infty_1\infty_2$ ($\mathfrak{f} = \mathfrak{p}_2\mathfrak{p}_5$; $\mathrm{Cl}_{\mathfrak{f}}^{+} = \mathbb{Z}/8 \times \mathbb{Z}/2$) each sign region of $u_L$ is one ray class: $S^+ \mapsto (5,0)$, $S^- \mapsto (1,0)$, $T^+ \mapsto (1,1)$, $T^- \mapsto (5,1)$, with lattice counts equal to ideal counts for every $n \le 60\,000$. With $d = (4,0)$ the class of a totally negative element $\equiv 1 \bmod \mathfrak{f}$, $S^- = S^+ + d$ and $T^- = T^+ + d$, so each character coefficient carries $(1 + \bar\chi(d))$ and the characters $\chi_{j,l}$ with $j$ odd (exactly the non-$\sigma$-invariant ones) drop out:

$$u_L(w) = \tfrac{1}{4} \sum_{j \in \{0,2,4,6\}} \mathbf{e}(-j/8)\big[U^{\cos}_{\chi_{j,0}}(w) + i\,U^{\sin}_{\chi_{j,1}}(w)\big],$$

$$U_\chi^{\cos\,/\,\sin} = \sqrt{y} \sum_{(\mathfrak{a},10)=1} \chi(\mathfrak{a}) K_0\Big(\frac{2\pi \mathrm{N}\mathfrak{a}\, y}{20}\Big)\{2\cos, 2\sin\}\Big(\frac{2\pi \mathrm{N}\mathfrak{a}\, x}{20}\Big).$$

Each of the 8 characters is a genus character $\psi \circ \mathrm{N}$, $\psi$ of conductor dividing 20, normalised by $\psi(7) = \chi(\mathfrak{p}_7)$ (Table 14), with coefficients $\lambda_\chi(n) = 1_{(n,10)=1} \sum_{d|n} \psi(d)(\psi\chi_{140})(n/d)$.

| $\chi$ | $\psi$ | $\psi\chi_{140}$ | type |
|---|---|---|---|
| $\chi_{0,0}$ | $1$ | $\chi_{140}$ | $(0,0)$ |
| $\chi_{0,1}$ | $\chi_{-4}$ | $\chi_{-35}$ | $(1,1)$ |
| $\chi_{4,0}$ | $\chi_5^2 = (\frac{5}{\cdot})$ | $\chi_{28}$ | $(0,0)$ |
| $\chi_{4,1}$ | $\chi_{-20}$ | $\chi_{-7}$ | $(1,1)$ |
| $\chi_{2,0}$ | $\chi_5^3\chi_{-4}$ | $\chi_5\chi_{-7}$ | $(0,0)$ |
| $\chi_{2,1}$ | $\chi_5^3$ | $\chi_5\chi_{-4}\chi_{-7}$ | $(1,1)$ |
| $\chi_{6,0}$ | $\chi_5\chi_{-4}$ | $\chi_5^3\chi_{-7}$ | $(0,0)$ |
| $\chi_{6,1}$ | $\chi_5$ | $\chi_5^3\chi_{-4}\chi_{-7}$ | $(1,1)$ |

Table 14: The genus characters $\chi_{j,l}(c) = \mathbf{e}(jc_1/8 + lc_2/2)$; $\chi_5$ is the quartic character mod 5 with $\chi_5(2) = i$.

**Status (Category A).** Hand proof, independent of PARI: $\mathrm{Cl}(K) = \mathbb{Z}/2$; $|\mathrm{Cl}_{\mathfrak{f}}^{+}| = 16$; $S^- = S^+ + d$ by $\beta(-\nu) = -\beta(\nu)$, $d \neq 0$ since $-\varepsilon^k \equiv -1 \bmod \mathfrak{p}_5$; the 8 characters $\psi \circ \mathrm{N}$ ($\psi \bmod 20$) are distinct, satisfy $\psi \circ \mathrm{N}(d) = 1$, and hence are exactly the characters with $\chi(d) = 1$. Exact coefficient match $A(n) = \#(S^+ \cup S^-)$, $B(n) = \#(T^+ \cup T^-)$ for every $n \leq 60\,000$ (error $10^{-15}$). Calling the pieces Maass–Eisenstein series $E_{\psi,\psi\chi_{140}}(\cdot, \frac{1}{2})$ is standard (Category B).

## 16.2 The Eisenstein/cusp split $(36 + 6)$ and the dihedral forms

Every one of the 140 components is exactly a ray class theta series modulo $\mathfrak{m} = \mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ (fit over $n$ coprime to 70 up to 54 880; max residual $4.4 \cdot 10^{-14}$ over 280 fits). Expanding in characters: mixed-type content $\leq 10^{-29}$; 116 components have cusp content $\leq 3 \cdot 10^{-30}$ (pure Eisenstein); 24 components have Eisenstein content $\leq 8 \cdot 10^{-31}$ (pure cusp). *Function-space dimensions.* Since distinct frequencies give independent mode functions, $\dim \mathrm{span}\{u_{\alpha,\beta}\}$ is the rank of the label×frequency coefficient matrix. This rank is 42; the Eisenstein components span 36 and the cusp components 6 dimensions, and both subspaces are $S$- and $T$-invariant. These statements are exact (Proposition 19.4); the floating-point rank computations agree (invariance residuals $\leq 8 \cdot 10^{-16}$; with independent code at $|Q| \leq 12$ and $|Q| \leq 25$, gap 6.4 versus $6 \cdot 10^{-15}$). The 24 cusp components are the $\beta = -a$ labels with $70\alpha_1 \in \{7, 21, 49, 63\}$ and $7 \mid$ denominator of $\alpha_2$, excluding the four labels $70\alpha = (49, 56), (63, 42), (7, 28), (21, 14)$, which give identically zero functions. In all, 11 labels (all with $\beta = -a$) give identically vanishing $u_{\alpha,\beta}$. This vanishing is exact: for the first five ($\alpha \in \{(\frac{7}{10}, \frac{4}{5}), (\frac{9}{10}, \frac{3}{5}), (\frac{1}{10}, \frac{2}{5}), (\frac{3}{10}, \frac{1}{5}), (\frac{1}{2}, 0)\}$) the automorph $\gamma$, for the remaining six ($\alpha = (\frac{1}{2}, \frac{j}{7})$, $1 \leq j \leq 6$; also $j = 0$) the automorph $-\gamma$, maps $\alpha + \mathbb{Z}^2$ onto itself, commutes with $\gamma^2$, leaves the kernel of the lift unchanged (it depends on $\nu$ evenly and $\gamma$-covariantly, and the closed geodesic is $\gamma$-stable), and multiplies the phase $\mathbf{e}(B(\nu, \beta))$ by $-1$ (because, with $g = \pm\gamma$ the relevant automorph, $g^{-1}\beta - \beta = \mu \in L'$ with $B(\alpha, \mu) \equiv \frac{1}{2}$); so the sum over $\nu$ modulo $\gamma^2$ equals its own negative. Checked exactly (rational arithmetic) over all 140 labels: these are exactly 11 labels. (Category A.)

Modulo $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ the non-$\sigma$-invariant characters all have *mixed* type (weight-1 holomorphic, cancelling in $u$); pure-type (Maass) cusp characters first appear at modulus $(2\sqrt{35})\infty_1\infty_2 = \mathfrak{p}_2^2\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$. There, of 96 characters, 24 are non-$\sigma$-invariant of pure type, all of conductor $(2\sqrt{35})$ (12 of type $(0, 0)$, 12 of type $(1, 1)$), 48 are mixed and 24 are genus. With $\chi = \chi_{j,l,1}$, $j$ odd mod 24, $\sigma : j \mapsto 13j$, one obtains one form per $\sigma$-orbit and each $l$:

$$u_\chi(z) = \sqrt{y} \sum_{n \geq 1} \lambda_\chi(n) K_0(2\pi n y) \cdot \{2\cos(2\pi n x)\ (l = 0);\ 2i \sin(2\pi n x)\ (l = 1)\}, \qquad \lambda_\chi(n) = \sum_{\mathrm{N}\mathfrak{a} = n} \chi(\mathfrak{a}),$$

6 even and 6 odd, of level $140 \cdot \mathrm{N}(\mathfrak{f}_\chi) = 19600$, nebentypus $\chi_{140}\omega_0$ with $\omega_0(n) = \chi((n))$ even mod 35 (consistent on 27 primes). Each cusp component is $u_c(w) = \sum_\chi 2\widehat{F}_c(\chi) u_\chi(w/140)$ with $\widehat{F}_c(\chi) = \frac{1}{48}\mathbf{e}(k_{c,\chi}/60)$; e.g. for $c = 73$, $k = 7, 57, 47, 37, 27, 17$ for $j = 1, 3, 5, 7, 9, 11$ ($l = 0$), shifted by 30 for $l = 1$.

**Category B.** The component fits and the Eisenstein/cusp classification of components are floating-point (PARI ray-class data for $\mathfrak{m}$; base field certified with `bnfcertify`$= 1$); phases recognised to $6 \cdot 10^{-9}$ from $10^{-14}$-accurate fits.

**Status (Category A).** The structure of $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2}$ is hand-certified (§18); the structure at the fitting modulus $\mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ is PARI-only. The dimensions 36 and 6, the $S, T$-invariance of the split, and the fact that the 24 cusp components are exactly the nonzero labels of odd theta characteristic are exact (Proposition 19.4).

**Category B.** Maass's construction [11] of dihedral $\lambda = \frac{1}{4}$ forms from pure-type Hecke characters; cuspidality $\Leftrightarrow \chi \neq \chi \circ \sigma$; the level formula $d_K \mathrm{N}(\mathfrak{f}_\chi)$; newform status; Hecke's theory [7] cited generically.

### 16.3 Function span, $S$ and $T$ spectra

Let $V = \mathrm{span}\{u_{\alpha,\beta}(w)\} \subset \mathbb{C}^{140}$ be the span of the values of the vector $u$ (equivalently the column span of the coefficient matrix). Since $u(-1/w) = Su(w)$ and $u(w+1) = Tu(w)$, $V$ is $S,T$-invariant, and $V \subset \mathrm{Fix}(P)$.

**Proposition 16.1.** *(i)* $\dim V = 42 = 36 + 6$, $V = V_{\mathrm{Eis}} \oplus V_{\mathrm{cusp}}$, *both summands* $S,T$*-invariant. (ii) On* $\mathrm{Fix}(P)$ *(*$\dim 72$*),* $S$ *has eigenvalues* $\pm 1$ *with multiplicities* $36/36$ *(*$\mathrm{tr}\, S|_{\mathrm{Fix}(P)} = 0$*); on* $V$ *they are* $21/21$*, namely* $18/18$ *on* $V_{\mathrm{Eis}}$ *and* $3/3$ *on* $V_{\mathrm{cusp}}$*. (iii)* $T$ *has order* $140$*. On* $\mathrm{Fix}(P)$ *its eigenvalues are* $\mathbf{e}(r)$ *with* $36$ *distinct* $r \in \frac{1}{140}\mathbb{Z}/\mathbb{Z}$*, of multiplicities* $1$ *(*$12$ *values),* $2$ *(*$18$ *values) and* $4$ *(*$6$ *values:* $r = \frac{19}{140}, \frac{31}{140}, \frac{59}{140}, \frac{111}{140}, \frac{131}{140}, \frac{139}{140}$*). On* $V_{\mathrm{Eis}}$ *each of the* $36$ *values occurs once; on* $V_{\mathrm{cusp}}$ *the eigenvalues are the* $6$ *values of multiplicity* $4$*. Hence* $\mathrm{Fix}(P) = V \oplus V^\perp$ *with a* $30$*-dimensional* $T$*-stable complement* $V^\perp$ *(multiplicities* $0, 1, 2$*), which the lift functions do not realise. (iv) The* $24$ *cusp characters form* $12$ $\sigma$*-orbits of size* $2$ *and give* $12$ *distinct dihedral forms, all with nonzero content; the* $24 \times 48$ *content matrix has rank* $6$*, so the* $12$ *forms occur only in* $6$ *fixed linear combinations.*

**Status (Category A).** (i) by Proposition 19.4; (ii) by the exact Gauss-sum traces of the local factors (Theorem 19.3), which give $n_\pm = d/2$ on each of the four sectors; (iii) order 140 exact (monomial $T$, rational phases), multiplicities from the exact phases and the sector decomposition. Floating-point computations agree (singular-value gap $17.1 \to 1.8 \cdot 10^{-13}$, invariance residuals $\leq 8 \cdot 10^{-16}$).

**Category B.** (iv): the rank 6 of the content matrix is floating-point. Its value is explained exactly by $V_{\mathrm{cusp}} = B_1 \otimes X^-$ (dim = 6). The complement $V^\perp = B_1 \otimes X^+ \oplus A_3 \otimes X^-$ is killed by the lift exactly (Proposition 19.4); the positive-frequency partners $f_{\alpha,\beta}$ span the same $V$. Weight-1 vectors lie in $\mathrm{Fix}(-P)$ (Remark 25.3(e)), so they cannot realise $V^\perp$ either.

**Consequence for $\psi$.** Both representations of the scalar $\psi$ involve only $u_L$ and $u_R$, which are Eisenstein; hence $\psi_{\mathrm{cusp}} \equiv 0$, $\psi = \psi_{\mathrm{Eis}}$ (Category A given the above). The 24 cusp rows of $\Phi$ are nonzero (e.g. $\Phi_{73}(1) = 0.120446470003 + 0.176437422094\, i$); their identification as Lewis–Zagier period functions of the cusp vector is Category B.

## 17 The Herglotz closed form (Theorems 11.9 and 11.10)

The proof below is given in outline; the full estimates are in the companion report listed in Appendix C.

### 17.1 Data

For each genus character $\chi = \chi_{j,l}$ (Table 14) with its $\psi \bmod 20$: $a(m) = \psi(m)1_{(m,10)=1}$ (period 20, $a(0) = 0$); $b(k) = \psi(k)\chi_{140}(k)$ (period 140, $b(0) = b(140) = 0$); parities $\eta_a = \psi(-1)$, $\eta_b = \psi(-1)\chi_{140}(-1) = \eta_a$, so $\eta_a\eta_b = 1$ ($\chi_{140}$ is even, $K$ being real). Gauss sums $\widehat{b}(n) = \sum_{r \bmod 140} b(r)\mathbf{e}(rn/140)$, $\check{a}(k) = \sum_{r \bmod 20} a(r)\mathbf{e}(kr/20)$; $\rho_n = -\widehat{b}(n)/(2\pi i \cdot 140)$, which is 140-periodic with $\rho_0 = 0$, $\rho_{-n} = \eta_b\rho_n$, $\sum_{s=1}^{140} \rho_s = 0$. Put $G_\chi(\tau) = \sum_{m,k\geq 1} a(m)b(k)\mathbf{e}(mk\tau)$, $\varphi(X) = \sum_s \rho_s \psi_\Gamma(s/140 + X)$, $R_\chi(z) = \sum_{m\geq 1} a(m)\varphi(m/(20z))$, $w_\chi = \mathbf{e}(-j/8)/16$. Equivalently $R_\chi(z) = -140z \sum_{m,n\geq 1} a(m)\rho_n/(7m + nz)$. When $\psi' = \psi\chi_{140}$ is primitive mod 140, $\rho_s = -\tau(\psi')\overline{\psi'(s)}/(2\pi i \cdot 140)$.

**Inputs.** (I1) $E_L = \sum_\chi w_\chi G_\chi(\cdot/20)$ (§16; $w_\chi = \bar\chi(S^+)/16$). (I2) The Maass $S$-law (Theorem 11.6). (I3) Lemma 15.1 and Theorem 11.7(i) *on* $\operatorname{Re} z > 0$ *only*: there $\Omega = \Psi$, so $\Psi$ continues holomorphically to $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$. Input (I1) was also checked exactly for $20Q \le 2000$.

**Remark 17.1** (Use of Theorem 11.7 in this section)**.** Every step below takes place in $U$. The functions $R_\chi$ are holomorphic on $U$; the difference $D$ of the Liouville step is used on $\{\operatorname{Re} z > 0\}$; the growth bound is on $S = \{1 \le \operatorname{Re} z \le 141,\ \operatorname{Im} z \le -1\} \subset \{\operatorname{Re} z > 0\}$; and the 140-translates of $U$ cover $\mathbb{C}$, which is how $P = D/z$ becomes entire. No step uses Theorem 11.7 on $\operatorname{Re} z < 0$, where $\Omega = -\Psi^- \ne \Psi$.

## 17.2 Proof outline

**Lemma 17.2** (Mittag-Leffler)**.** $B(u) := \sum_{k\ge1} b(k)\mathbf{e}(ku)$ *continues to* $P(\mathbf{e}(u))/(1-\mathbf{e}(140u))$, $\deg P \le 139$, *and for* $u \notin \frac{1}{140}\mathbb{Z}$, $B(u) = \sum_n \rho_n/(u - n/140)$ *(symmetric summation)* $= \sum_{s=1}^{140} \rho_s \pi \cot \pi(u - s/140)$*; the one-sided series converge.*

(Same poles and residues, both vanish at $\pm i\infty$, Liouville in $\mathbf{e}(u)$; one-sided convergence by Abel summation.)

**Lemma 17.3.** *For* $X \notin (-\infty, -\frac{1}{140}]$*:* $\varphi(X) = -140\sum_{n\ge1} \rho_n/(n + 140X)$. *With* $\beta_1 = \sum_s \rho_s B_1(s/140)$ *(*$= 0$ *if* $\eta_b = +1$*), uniformly in* $|\arg X| \le \pi - \delta$, $\varphi(X) = \beta_1/X + O_\delta(|X|^{-2})$ *for* $|X| \ge 1$*. Consequently* $R$ *is holomorphic on* $U = \mathbb{H} \cup \{\operatorname{Re} z > 0\}$ *(since* $\arg X_m = -\arg z \in (-\pi, \pi/2)$*), and* $|R(z)| \le C(1 + |z|)$ *on* $S = \{1 \le \operatorname{Re} z \le 141,\ \operatorname{Im} z \le -1\}$ *(odd tail by Abel summation).*

**Proposition 17.4.** *For* $z \in \mathbb{H}$*:* $G_\chi(-1/(20z)) = 2R_\chi(z) + T_\chi(z)$ *with* $T_\chi(z) = -\lim_M \sum_{0<|m|\le M} a(m)\varphi(m/(20z))$*. Hence Theorem 11.9 is equivalent to*

$$\sum_\chi w_\chi T_\chi(z) = (z/\sqrt{35})\, E_R(z) \qquad (z \in \mathbb{H}). \tag{17.1}$$

**Proposition 17.5.** *For every genus* $\chi$ *and* $z \in \mathbb{H}$, $T_\chi(z) = \frac{z}{140}\sum_{n,k\ge1} \widehat{b}(n)\check{a}(k)\mathbf{e}(nkz/140)$*, absolutely convergent.*

For nontrivial $\psi$ this is two-variable summation by parts (kernel $F(m,n) = 1/(nz + 7m)$ with uniform $O(|w|^{-3})$ bounds on $\Delta_m\Delta_n F$, $D_n^2 F$) followed by the cotangent evaluation $\lim_M \sum_{|m|\le M} a(m)/(7m + nz) = -\frac{2\pi i}{140}\sum_k \check{a}(k)\mathbf{e}(nkz/140)$. For trivial $\psi$ the $n$-outer sum diverges and a second-order argument is needed; the constants $\pm\frac{2}{5}\sum_n \rho_n/(nz)$ cancel. They are genuinely nonzero: $\sum \rho_n/n = i\log\varepsilon/(70\pi)$ to 15 digits (Category B context via the class number formula; not used).

**Liouville step.** Let $D = \Psi - 2\sum_\chi w_\chi R_\chi$, holomorphic on $\{\operatorname{Re} z > 0\}$ by (I3) and the lemma. On $\mathbb{H}$, $D = zP$ with $P(z) = \sum_{N\ge1} p_N \mathbf{e}(Nz/140)$ ($E_R$ has no constant term since $\nu = 0 \notin F$; its frequencies lie in $\frac{1}{70}\mathbb{Z}$). $P$ extends to an entire 140-periodic function, holomorphic in $q' = \mathbf{e}(z/140)$ on $\mathbb{C}^*$ with Laurent coefficients $p_N$, $N \ge 1$. A growth bound on $S$ (proved in the companion report: $|\Psi| \le C(1 + |z|)^2$) gives $|P| \le C'(1 + \log|q'|)$ for $|q'| \ge e^{2\pi/140}$; Cauchy's estimates force $p_N = 0$. Hence $P \equiv 0$, which is (17.1) (Theorem 11.10) and $D \equiv 0$ (Theorem 11.9). Periodicity and absence of a constant term alone would not suffice ($\mathbf{e}(z/140)$); the $o(|q'|)$ growth is essential.

**Arithmetic form of Theorem 11.10.** With $\nu \in L'$, $(u,v) = A\nu$, $\theta = (7u-2v) + u\sqrt{35}$: $\nu \mapsto \theta$ is a bijection $L' \to \mathfrak{p}_2$, $Q(\nu) = -\mathrm{N}(\theta)/140$, $B(\nu,a) = \mathrm{Re}\,\theta/10$, and the coefficient of $\mathbf{e}(Nz/140)$ in $E_R$ is $c_R(N/140) = \frac{\sqrt{35}}{140}\sum_\chi w_\chi \sum_{nk=N} \widehat{b}(n)\check{a}(k)$. For rational $x > 0$ it follows, by Theorem 11.7(ii), that $\psi(x) = \kappa_L(-1/x) - (x/\sqrt{35})\sum_p \kappa_{p,-a}(x)$ is the boundary value of the right-hand side of Theorem 11.9.

**Category B.** Reading of $R_\chi$ as the Lewis–Zagier Eisenstein period function at $s = \frac{1}{2}$ for $(\psi, \psi')$ [10, Ch. IV]; "twisted Herglotz/Kronecker-limit" terminology. Contrast with Radchenko–Zagier [14], whose untwisted $F(x) = \sum(\psi_\Gamma(nx) - \log nx)/n$ has dilogarithmic rational values; here no log-regularisation is needed and the combination takes algebraic values at rationals. Radchenko–Zagier also record Herglotz's evaluation at the same unit $6+\sqrt{35}$ (suggestive only). Zagier's Kronecker limit formula [17] is context only. Individual $R_\chi(x)$ were not tested for algebraicity.

## 18 $L(0)$ identities for the cusp rows

For a cusp row $c$, $\kappa_c(x)$ is the $t^0$ coefficient of $f_c(\mathbf{e}(x)e^{-t})$; by Mellin, $\kappa_c(x) = Z_c(0;x)$ with $Z_c(w;x) = \sum A_Q \mathbf{e}(Qx)Q^{-w}$ entire. For $x = h/k$ with $k$ composed of $2, 5, 7$, $\mathbf{e}(h\mathrm{N}\mathfrak{a}/(140k))$ is a ray class function mod $\mathfrak{m}_k$ (checked on all classes to norm 60 000); for $3 \mid k$ the ideals divisible by the inert prime 3 give one extra ray-class sum. This yields

$$\kappa_c(h/k) = \sum_{\chi'} \widehat{G}_{c,h/k}(\chi')\, L_{\mathfrak{m}_k}(0,\chi'),$$

and, for $3 \nmid k$,

$$\kappa_c(h/k) = \frac{1}{\varphi(140k)} \sum_{\psi \bmod 140k} \tau(\bar{\psi})\psi(h) \sum_\chi \widehat{F}_c(\chi)\, L(0, u_\chi \otimes \psi).$$

Only totally odd characters contribute; for those $L(0,\chi) = W(\chi)\cdot 140 \cdot L(1,\bar{\chi})/\pi^2$ (checked to $2\cdot 10^{-15}$). Type-$(0,0)$ and mixed-type characters have $L(0) = 0$ (Category A from the exact PARI-free Shintani sums of Appendix 18.1; the PARI values agree).

**Category B.** Numerically verified, conditional on PARI `lfun` (38 digits; primitive characters via `bnrmap`, imprimitive Euler factors restored): all 24 cusp rows at all 7 points to $4\cdot 10^{-14}$ (Table 21). Not proved. Central-value hypothesis rejected: replacing $L(0)$ by $L(\frac{1}{2})$ gives $\Phi_{73}(1) = 0.0349 + 0.0748i$, max deviation 0.32 (0.31 with only the 12 totally odd central values). At $x = 1$ the 24 values lie in a 6-dimensional space.

**Hand certification of $\mathrm{Cl}^+_{(2\sqrt{35})\infty_1\infty_2}$.** From $1 \to U/U_{\mathfrak{m},1} \to (\mathcal{O}/\mathfrak{m})^\times \times \{\pm\}^2 \to \mathrm{Cl}^+_{\mathfrak{m}} \to \mathrm{Cl} \to 1$ with $(\mathcal{O}/(2))^\times = \langle\sqrt{35}\rangle \cong \mathbb{Z}/2$, $\mathbb{F}_5^\times = \langle 2\rangle$, $\mathbb{F}_7^\times = \langle 3\rangle$, $U = \langle -1, \varepsilon\rangle$, $\mathrm{Cl} = \mathbb{Z}/2$ generated by $\mathfrak{p}_{13}$ ($\mathfrak{p}_{13}^2 = (22+3\sqrt{35})$), the Smith normal form gives invariants $[2,2,24]$, order $96 = 2\cdot 48\cdot 4/4$. The $\sigma$-action, infinity types and conductors are read off and match PARI. (Category A, PARI-free.)

### 18.1 PARI-free certification

**Status (Category A).** An exact, finite computation, or a standard theorem applied to exact data. Hand inputs: $\varepsilon = 6+\sqrt{35}$ is the fundamental unit and $\mathrm{Cl} = \mathbb{Z}/2$ (generated by $\mathfrak{p}_{13}$, $\mathfrak{p}_{13}^2 = (22+3\sqrt{35})$).

- The ray class group mod $\mathfrak{m} = \mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$, constructed by hand from $U \to (\mathcal{O}/\mathfrak{m})^\times \times \{\pm\}^2 \to \mathrm{Cl}^+_{\mathfrak{m}} \to \mathrm{Cl} \to 1$, has order 96 and structure $\mathbb{Z}/24 \times \mathbb{Z}/2 \times \mathbb{Z}/2$; the same holds for $\mathfrak{f} = (2\sqrt{35})\infty_1\infty_2$.

| $\chi$ ($\sigma$-pair) | type | $L(\frac{1}{2}, u_\chi)$ | $L(0)$ |
|---|---|---|---|
| $(1,\cdot,1) \sim (13,\cdot,1)$ | $(0,0)$ | $1.109240202417 - 0.205788403698i$ | $0$ |
| | $(1,1)$ | $0.131513747745 + 0.708885114750i$ | $-(6+4\sqrt{3}) + (6\sqrt{3}-4)i$ |
| $(3,\cdot,1) \sim (15,\cdot,1)$ | $(0,0)$ | $1.057777176883 - 0.300492329005i$ | $0$ |
| | $(1,1)$ | $-0.123002454086 - 0.432986722370i$ | $-12 + 4i$ |
| $(5,\cdot,1) \sim (17,\cdot,1)$ | $(0,0)$ | $0.514021161380 + 0.466537408032i$ | $0$ |
| | $(1,1)$ | $1.122786592523 - 1.237062791397i$ | $(4\sqrt{3}-6) - (4+6\sqrt{3})i$ |

Table 15: Central values and $L(0)$ for six of the twelve dihedral forms; the other six ($j = 7, 9, 11 \sim 19, 21, 23$) are complex conjugates. All 12 central values are nonzero. PARI `lfun`, 38 digits; $L(0)$ values recognised to $10^{-10}$.

- The independent class assignment of all 140 components reproduces the PARI fit exactly: the 22 empty (label, sign) slots, every content $g$, the Eisenstein/cusp/mixed norms (differences $\leq 6.7 \cdot 10^{-16}$), the same 24 cusp components with coefficients $\frac{1}{48}\mathbf{e}(k/60)$, and the label-row ranks 60 and 12 (in function space 36 and 6, §16.3). The theta identity holds for all $n$ with $|Q| \leq 400$ (residual $\leq 8.9 \cdot 10^{-16}$); the statement for all $n$ is Category B.
- Exact Shintani cone sums give the 12 values $L(0, \chi)$ of the totally odd, non-$\sigma$-invariant characters of conductor $(2\sqrt{35})$ in $\mathbb{Z}[\zeta_{12}]$: $-(6+4\sqrt{3}) \pm (6\sqrt{3}-4)i$, $-12 \pm 4i$, $(4\sqrt{3}-6) \mp (4+6\sqrt{3})i$, each twice (one per $\sigma$-pair). Validations: $\sum_c \zeta(0, c) = 0$; $L(0, \chi) = 0$ for all 72 non-totally-odd characters; $L(\chi) = L(\chi \circ \sigma)$; a genus test exact for all 8 characters mod 20; two independent moduli agree exactly.
- All $7 \times 24$ cusp-row identities of Theorem 11.11 hold to $1.2 \cdot 10^{-14}$ with PARI-free inputs. No discrepancy was found.

*Not* redone without PARI: $L(\frac{1}{2}, \chi)$, $L(1, \chi)$, root numbers. Characters are determined only up to $\chi \leftrightarrow \chi \circ \sigma$. Note that the group mod $\mathfrak{m}$ (order 96) is not the group mod $\mathfrak{p}_2\mathfrak{p}_5\infty_1\infty_2$ (order 16, $\mathbb{Z}/8 \times \mathbb{Z}/2$) used in §12.

## 19 The theta-lift family: tensor structure

For $s \geq 5$ let $A_s$, $Q_s$, $a$, $c_0$ and $\gamma$ be as in §10.4, and let $u_{\alpha,\beta}$ be the theta lifts of §14 built from $A_s$ (for $s = 7$ these are the 140 functions of Theorem 11.6).

**Remark 19.1** (The $S$ versus $S^{-1}$ label convention)**.** Let $S_{\mathrm{lab}}$ be the label matrix of Theorem 11.6, i.e. $u(-1/w) = S_{\mathrm{lab}} u(w)$. For $s = 7, 8, 11$ one checks exactly: $S_{\mathrm{lab}}^2 = P$, $(S_{\mathrm{lab}}T)^3 = I$ on all labels, $(S_{\mathrm{lab}}T)^3 \neq S_{\mathrm{lab}}^2$, and $(S_{\mathrm{lab}}^{-1}T)^3 = S_{\mathrm{lab}}^{-2}$. So the representation of $\mathrm{SL}_2(\mathbb{Z})$ in the standard presentation $S^4 = 1$, $(ST)^3 = S^2$ is $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho(T) := T$. On $W_P$ one has $S_{\mathrm{lab}} = S_{\mathrm{lab}}^{-1}$. (Category A, exact.)

**Remark 19.2** (Label operators)**.** Label maps (negation, $\gamma$, the local reflections $Z_p$) are realised as monomial matrices $Y$ with $Y[i, j(i)] = \mathbf{e}(\mathrm{ph}_i)$, the convention of the permutation matrix of the family. With this convention neg $= \rho(-I) = S^2$, and neg, $\gamma$, $Z_p$ ($p \mid s$) and $Z_q$ ($q \mid s-2$) commute exactly with $S$ and $T$ (exact exponent arithmetic modulo $2D^2$ at $s = 5, 7, 9, 11, 13, 15, 25$). (Category A.)

### 19.1 Exact tensor factorisation at $s = 7$

Let $g = (14, 6)/70$, a generator of $L'/\mathbb{Z}^2 \cong \mathbb{Z}/35$, and $a_2 = (\frac{1}{2}, 0)$, so that $a = a_2 + a_{\mathrm{odd}}$ with $a_{\mathrm{odd}} = (\frac{1}{5}, -\frac{1}{5}) \in L'$. In the adapted basis $u'_{x,\epsilon,\delta} := u_{xg+\epsilon a_2,\, -\delta a_2}$ ($x \in \mathbb{Z}/35 \cong \mathbb{Z}/5 \times \mathbb{Z}/7$ by CRT, $\epsilon, \delta \in \{0, 1\}$) put $F_p[x, y] = \mathbf{e}(4xy/p)$, $T_p = \operatorname{diag} \mathbf{e}(2x^2/p)$, and let $S_2$, $T_2$ be the $4 \times 4$ monomial matrices on the theta characteristics $(\epsilon, \delta)$: $S_2 : 00 \mapsto 00$, $01 \leftrightarrow 10$, $11 \mapsto -11$; $T_2 : 00 \leftrightarrow 01$, $10 \mapsto i \cdot 10$, $11 \mapsto i \cdot 11$.

**Theorem 19.3** (Tensor factorisation, $s = 7$)**.** *(i) In the adapted basis $u'_{x,\epsilon,\delta}$ built on the generator $g = (14, 6)/70 = (7, 3)/35$, exactly $S_{\mathrm{lab}} = (F_5/\sqrt{5}) \otimes (F_7/\sqrt{7}) \otimes S_2$ and $T = T_5 \otimes T_7 \otimes T_2$. (ii) $\rho = \rho_5 \otimes \rho_7 \otimes \rho_2$, where $\rho_p$ is the Weil representation of $(\mathbb{Z}/p, q_p)$ with $q_5 = 2x^2/5$ and $q_7 = x^2/7$, and $\rho_2 = (\mathbf{e}(3/4)S_2^{-1}, T_2)$ acting on the 4-dimensional space of theta characteristics, is a faithful representation of $\mathrm{SL}_2(\mathbb{Z}/4)$ (image of order 48). Hence $\rho$ factors through $\mathrm{SL}_2(\mathbb{Z}/140) = \mathrm{SL}_2(\mathbb{Z}/4) \times \mathrm{SL}_2(\mathbb{F}_5) \times \mathrm{SL}_2(\mathbb{F}_7)$. (iii) $\rho_2 = A_3 \oplus B_1$, where $A_3$ is the span of the three even characteristics $00, 01, 10$ (irreducible, $T$-spectrum $\{1, -1, i\}$) and $B_1$ is the odd characteristic 11 ($S, T \mapsto \mathbf{e}(1/4)$). (iv) $P = \rho(-I) = Z_5 \otimes Z_7 \otimes I_4$, where $Z_p$ is the reflection $x \mapsto -x$. Hence $W_P = (W_5^+ \otimes W_7^+ \oplus W_5^- \otimes W_7^-) \otimes \mathbb{C}^4$; the mixed local parities are absent. The restrictions are $16W_5^+ + 12W_5^-$, $12W_7^+ + 8W_7^-$ and $18(A_3 + B_1)$. (v) $\operatorname{tr} \rho_5(S)|_{W_5^\pm} = (-1, 0)$, $\operatorname{tr} \rho_7(S)|_{W_7^\pm} = (0, -1)$, $\operatorname{tr} \rho_2(S)|_{A_3} = -i$, $\operatorname{tr} \rho_2(S)|_{B_1} = i$, so $\operatorname{tr} S = 0$ on each of the four sectors below and $n_\pm = 36/36$ on $W_P$.*

*Proof outline.* Every entry of $S_{\mathrm{lab}}$ is one term $\mathbf{e}(r)/\sqrt{35}$ with $r \in \mathbb{Q}$, and $T$ is monomial, so (i) is a finite identity over $\mathbb{Q}(\zeta_{140})$, checked exactly. For (ii), $(\overline{F_p} T_p)^3 = c\, Z_p$ exactly in $\mathbb{Z}[\zeta_p]$ with $c^2 = 125$ ($p = 5$) and $c^2 = -343$ ($p = 7$); with Gauss's sign of the quadratic Gauss sum this gives the scalars $\gamma_5 = -1$, $\gamma_7 = \mathbf{e}(3/4)$, so $\rho_p$ is exactly the Weil representation of $q_p$; by the congruence theorem for Weil representations of finite quadratic modules of odd level [16], $\rho_p$ factors through $\mathrm{SL}_2(\mathbb{Z}/p)$. $\rho_2$ satisfies $S^4 = I$, $(ST)^3 = S^2 = -I$, $T^4 = I$ exactly, and the pair group $(\rho_2(g), g \bmod 4)$ has exactly 48 elements. The scalars cancel in the product, which equals $(S_{\mathrm{lab}}^{-1}, T)$. (iv) and (v) follow from the local Gauss sums. □

**Status (Category A).** Exact (rational and cyclotomic arithmetic), conditional on the cited congruence theorem for the odd local factors [16] (hypotheses: exact match with the Weil formulas, odd level). The local forms are determined as follows: a 140th-root exponent $a$ decomposes as $a = 35x_4 + 28x_5 + 20x_7$, i.e. $x_4 \equiv 3a \bmod 4$, $x_5 \equiv 2a \bmod 5$, $x_7 \equiv 6a \bmod 7$, which gives $q_5 = 2x^2/5$ and $q_7 = x^2/7$. (Since 2 is a non-square mod 5 and 3 is a non-square mod 7, these are not isometric to $x^2/5$ and $3x^2/7$.)

**Proposition 19.4** ($V$ is the matched-sector space; local reflections)**.** *Let $X^\pm = W_5^\pm \otimes W_7^\pm$. At $s = 7$: (i) negation acts on $u'$ as $Z_5 \otimes Z_7 \otimes I_4$, and the unit $\gamma$ ($\varepsilon = 6 + \sqrt{35}$) acts as $I_5 \otimes Z_7 \otimes \operatorname{diag}(1, 1, 1, -1)$; (ii) $V = \mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma) = A_3 \otimes X^+ \oplus B_1 \otimes X^-$, of dimension $36 + 6 = 42$, and the lift kills $V^\perp = B_1 \otimes X^+\,(12) \oplus A_3 \otimes X^-\,(18)$; (iii) $V_{\mathrm{Eis}} = A_3 \otimes X^+$ and the cuspidal part of $V$ is $V_{\mathrm{cusp}} = B_1 \otimes X^-$. The 24 cusp components of §16 are exactly the nonzero labels with $(\epsilon, \delta) = (1, 1)$, and the 11 identically vanishing labels are exactly the 6 inconsistent orbits ($x_5 x_7 = 0$, $\epsilon = \delta = 1$); (iv) each of the 42 classes is a free $\langle N, \gamma \rangle$-orbit (sizes $1, 2, 4$: $3, 15, 24$ classes), with proportionality constants $\pm 1$ in the $u'$ basis; (v) the local reflections $Z_5$, $Z_7$ and $Z_2 := \operatorname{diag}(1, 1, 1, -1)$ commute with $\rho$. Their signs $(Z_5, Z_7, Z_2)$ are $(+, +, +)$ on $V_{\mathrm{Eis}}$, $(-, -, -)$ on $V_{\mathrm{cusp}}$, $(+, +, -)$ on $B_1 \otimes X^+$ and $(-, -, +)$ on $A_3 \otimes X^-$. Thus $Z_5 Z_7 = +1$ on $W_P$ (this is $P$) and $Z_7 Z_2 = +1$ exactly on $V$ (this is $\gamma$). $S$ commutes with all three, and $n_\pm = d/2$ on every joint eigenspace.*

*Proof outline.* (i) The identities $u_{-\alpha,-\beta} = u_{\alpha,\beta}$ and $u_{\gamma\alpha,\gamma\beta} = u_{\alpha,\beta}$ hold as identities of functions, and their actions on $u'$ are computed exactly. (ii) Hence $V \subseteq \mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma)$, which has dimension 42. The exact Fourier matrix (sums of 4900th roots of unity, $|Q| \le 25$) has rank 42 modulo a prime

above $\ell = 29401 \equiv 1 \bmod 4900$; rank cannot increase under reduction, so $\dim V \geq 42$. (iii) The constant-term vector is $\delta_{x=0} \otimes (e_{00} + e_{01})$, a pure tensor with 2-adic part in $A_3$; its cyclic span is $A_3 \otimes X^+$ (the argument is that of Theorem 20.1(iii)). The cuspidal functions of the span are those orthogonal to $V_{\mathrm{Eis}}$ (the $\lambda = \frac{1}{4}$ constant-term criterion); by unitarity this is $B_1 \otimes X^-$. □

**Status (Category A).** Exact: (i), (ii), (iv), (v) and the label statements of (iii). The mechanism that kills $V^\perp$ is the $\gamma$ phase flip: the unit acts by $-1$ at 7 and by $-1$ on the odd characteristic, the analogue of the exact vanishing of the 11 labels (§16.2). Part (iii) uses the cited congruence theorem and the standard $\lambda = \frac{1}{4}$ constant-term criterion.

**Category B.** Calling $Z_5, Z_7, Z_2$ the Atkin–Lehner operators $W_5, W_7, W_4$ on the lifted forms (not checked on the forms). The Hecke-character reading: a character contributes only if $\psi(\varepsilon) = \psi(-1) = 1$; genus characters are even at the ramified primes and land in $A_3 \otimes X^+$; characters odd at 5 and 7 are not genus characters, hence not $\sigma$-invariant, and give Maass cusp forms in $B_1 \otimes X^-$. The identification of $V_{\mathrm{cusp}}$ with specific dihedral Hecke eigenforms rests on the floating fits of §16.

## 19.2 General odd $s$

**Theorem 19.5** (Tensor factorisation for odd $s$)**.** *Let $s \geq 5$ be odd and $n = n_s$. Then $L'/\mathbb{Z}^2 \cong \mathbb{Z}/n$ is generated by $g = A_s^{-1} e_2$, $a = a_2 + a_{\mathrm{odd}}$ with $a_2 = (\frac{1}{2}, 0)$ and $a_{\mathrm{odd}} = (1, -1)/(s-2) \in L'$, and in the basis $u'_{x,\epsilon,\delta} := u_{xg + \epsilon a_2, -\delta a_2}$ ($N_s = 4n$ labels):*

$$T = \mathrm{diag}_x\, \mathbf{e}(-x^2/n) \otimes T_2, \qquad S_{\mathrm{lab}} = \left[\mathbf{e}(-2xy/n)/\sqrt{n}\right]_{x,y} \otimes S_2,$$

*with $S_2 : (\epsilon, \delta) \mapsto (\delta, \epsilon)$ with phase $(-1)^{\epsilon\delta}$, and $T_2$ as above; $S_2, T_2$ do not depend on $s$. (i) With $\rho(S) := S_{\mathrm{lab}}^{-1}$, $\rho = \bigotimes_{p^e \| n} \rho_{q_p} \otimes \rho_2$, where $\rho_{q_p}$ is the Weil representation of $(\mathbb{Z}/p^e, q_p)$, $q_p(x) = -v_p x^2/p^e$, $v_p = (n/p^e)^{-1} \bmod p^e$, and $\rho_2 = (\mathbf{e}(3/4) S_2^{-1}, T_2)$ is the 4-dimensional $s = 7$ factor. The Weil scalar is $\mathbf{e}(1/4)$ for every odd $s$, since $n \equiv 3 \bmod 4$. $\rho$ factors through $\mathrm{SL}_2(\mathbb{Z}/4n) = \mathrm{SL}_2(\mathbb{Z}/4) \times \prod \mathrm{SL}_2(\mathbb{Z}/p^e)$, so $N_s = 4s(s-2)$ is the number of labels and the order of $T$. (ii) Negation acts as $\bigotimes_p Z_p \otimes I_4$; $\gamma$ acts as $x \mapsto (s-1)x$ with phase $(-1)^{\epsilon\delta}$, i.e. as $\bigotimes_{p|s} Z_p \otimes \bigotimes_{p|s-2} I \otimes Z_2$. In fact $\gamma \equiv (s-1) \cdot I$ on all of $L'/\mathbb{Z}^2$, because $(\gamma - (s-1)I) A_s^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$; and $s - 1 \equiv -1 \bmod p^e$ for $p \mid s$, $\equiv +1$ for $p \mid s-2$ (the reduction of $\varepsilon_s = (s-1) + \sqrt{n_s}$ at the ramified primes). (iii) $P = S_{\mathrm{lab}}^2 = Z_n \otimes I_4$, so $\dim W_P = 4 \cdot \frac{n+1}{2} = 2(s-1)^2$ (this is 72 at $s = 7$). (iv) $\mathrm{Fix}(N) \cap \mathrm{Fix}(\gamma) = \{Z_{(s)} Z_{(s-2)} = +1,\ Z_{(s)} Z_2 = +1\}$ has dimension $3E(s)E(s-2) + O(s)O(s-2) = s(s-1)$, with $E(m) = \frac{m+1}{2}$, $O(m) = \frac{m-1}{2}$ and $Z_{(m)} = \bigotimes_{p|m} Z_p$. Hence $\dim V \leq s(s-1)$. There are exactly $2s-3$ forced-zero labels, and the odd-characteristic sector $B_1 \otimes (\mathbb{Z}/s)^- \otimes (\mathbb{Z}/(s-2))^-$ has dimension $(s-1)(s-3)/4$.*

**Status (Category A).** Symbolic proof for all odd $s$ (the $T$- and $S$-entries from $B(g,g) = -2/n$, $B(g, a_2) = 0$, $w_c = g/2$, and the CRT split of $-x^2/n$); exact verification entry by entry at $s = 5, 7, 9, 11, 13, 15, 21, 25$ (including the prime powers 9 and 25). The congruence statement in (i) is conditional on the cited theorem [16]; its convention was checked against the exact $s = 7$ relations. Note $\dim W_P = 2(s-1)^2$, while $3\prod(p^e + 1)/2$ is the bound for $d_{\mathrm{Eis}}$.

**Remark 19.6** (The two generators at $s = 7$)**.** Theorem 19.3 uses $g = (7, 3)/35$, Theorem 19.5 uses $A_7^{-1} e_2 = (7, -2)/35 = 11 \cdot (7, 3)/35$ in $L'/\mathbb{Z}^2$. By CRT the local factors of Theorem 19.3(i) combine to the $\mathbb{Z}/35$-phases $\mathbf{e}(13xy/35)/\sqrt{35}$ for $S_{\mathrm{lab}}$ ($\frac{4}{5} + \frac{4}{7} = \frac{48}{35}$) and $\mathbf{e}(24x^2/35)$ for $T$ ($\frac{2}{5} + \frac{2}{7} = \frac{24}{35}$). Under $x \mapsto 11x$ these become $\mathbf{e}(-2xy/35)$ and $\mathbf{e}(-x^2/35)$, since $121 \cdot 13 \equiv -2$ and $121 \cdot 24 \equiv -1 \pmod{35}$; these are the phases of Theorem 19.5 at $n = 35$. The two descriptions are the same representation in two bases.

### 19.3 Sectors: parity and the orbit lemma

**Proposition 19.7** (Odd-parity sectors are cusp)**.** *If $\rho(T)$ has no eigenvalue 1 on a $\rho$-invariant subspace $\sigma$, the projection of the lift onto $\sigma$ is cuspidal. Consequently $d_{\mathrm{cusp}} \geq (s-1)(s-3)/4$ wherever the $B_1$ sector lies in $V$, and $d_{\mathrm{Eis}} \leq 3\prod(p^e+1)/2$ whenever no odd parity is carried only by primes with $e \geq 2$ (every tested odd $s \leq 99$; the first undecided cases are $s = 225, 1225$).*

**Status (Category A).** Proved: no constant term at any cusp. *Category B:* equality $d_{\mathrm{Eis}} = 3\prod(p^e+1)/2$, i.e. that the all-even sector carries no cusp content.

**Proposition 19.8** (Parity fact)**.** *For every odd $s$, every $\Delta$ label has even second coordinate $x_2$. Hence $Z_{\mathrm{all}}$ acts as negation on $\Delta$, fixes every $\langle\gamma,-1\rangle$-orbit, and every sector with $\varepsilon_{\mathrm{all}} = -1$ vanishes.*

**Status (Category A).** Symbolic: $D\cdot A^{-1} = -2\,\mathrm{adj}A$ gives even second coordinates on $L'$; $a = (s^2, -2s)$; $D$ is even.

**Proposition 19.9** (Orbit lemma)**.** *For every odd $s$, $c_\varepsilon(O_0, q_{\min}) = (1+\varepsilon_{\mathrm{all}})N_{O_0}(q_{\min})$, where $O_0 = \{\pm\nu_0, \pm\gamma\nu_0\}$ and $\nu_0 = (s,-2)$. This is nonzero on every sector with $\varepsilon_{\mathrm{all}} = +1$. Extra reduced classes do occur, at $s = 27, 245, 363, 847, 2025, \ldots$ (16 cases with $s \leq 30001$), but they never land in $Z_S O_0$.*

**Status (Category A).** Symbolic (Nagell bound, $\gamma^2\nu_0 \equiv \nu_0$, stabiliser $\{1, Z_{\mathrm{all}}\}$, exclusion of extra classes). Also checked exactly for odd $s \leq 401$ and at all 16 exceptions. At $s = 27$ the sector run certifies $\dim V = 702$ (exact), in agreement with Theorem 20.1. Combined with Proposition 19.8, the matched sectors with $\varepsilon_{\mathrm{all}} = +1$ have a nonzero lift. Injectivity therefore follows wherever the sector is irreducible (squarefree $s(s-2)$). *Open:* $\dim V = s(s-1)$ when $e \geq 2$.

## 20 The theta-lift family: dimensions

### 20.1 Dimensions of $V$, $V_{\mathrm{Eis}}$ and the cusp part

**Theorem 20.1** (Dimensions)**.** *(i)* Orbit bound. $\dim V \leq \#\{$*consistent $\langle\gamma,-1\rangle$-orbits*$\}$*, and the number of orbits is $s(s-1)$ for odd $s$ and $(s/2)(s/2+1)$ for even $s$ (Burnside, with $|\mathrm{Fix}\,\gamma| = 2(s-2)$, $|\mathrm{Fix}(-\gamma)| = 2s$ and $|\Delta[2]| = 4$, using (ii) of Theorem 19.5). (ii)* Exact ranks. *For every $s = 5, \ldots, 51$ and for $s = 72$, the exact rational rank of the integer orbit-count matrix equals the upper bound (for $s = 8, 18, 32, 50, 72$ the half-cone bound of (iv)). Hence $\dim V$ is known exactly there; for odd $s \leq 51$, $\dim V = s(s-1)$. (iii)* Eisenstein dimension. *For odd $s$, $d_{\mathrm{Eis}} = 3\prod_{p^e \| n_s}(p^e+1)/2$, at every odd $s$ whose local factors with $e \geq 2$ are certified ($p^e \in \{9, 25, 27, 49\}$ suffices for odd $s \leq 51$). For even $s = 2t$, with $2^k \| 2t(t-1)$ and $v$ the odd part of $2t(t-1)$, $d_{\mathrm{Eis}} = d_2(k,v)\prod_{p^e\|v}(p^e+1)/2$, where $d_2(k,v) = 2^{k-1}+4$ ($k$ even), $2^{k-1}+6$ ($k$ odd, $v \equiv 1 \bmod 4$), $2^{k-1}+5$ ($k$ odd, $v \equiv 3 \bmod 4$), certified for $k \leq 5$ (every even $s \leq 62$). (iv)* The $s = 2m^2$ mechanism. *$\gamma$ has a square root $g \in \mathrm{SO}(A_s)(\mathbb{Q})$ with $\det g = 1$ iff $\mathrm{tr}\,\gamma + 2 = 2s$ is a perfect square, i.e. iff $s = 2m^2$; then $g = (\gamma + I)/(2m) = \begin{pmatrix} 0 & -m \\ 1/m & 2m \end{pmatrix}$ (in unit language $\varepsilon_s = \eta^2$, $\eta = m + \sqrt{m^2-1}$). Splitting each cone $[r, \gamma r)$ as $[r, gr) \sqcup g[r, gr)$ gives an exact upper bound with gap exactly $m(m-1)$ below the orbit bound for $m = 2, \ldots, 7$. It equals the exact lower bound for $m = 2, \ldots, 6$: $\dim V = 18, 84, 260, 630, 1302$ at $s = 8, 18, 32, 50, 72$. At $s = 98$ the half-cone bound 2408 is not sharp: there are two further relations and $\dim V = 2406$ (Theorem 20.5). (v)* Table *($d_{\mathrm{cusp}} = \dim V - d_{\mathrm{Eis}}$ is the exact dimension of the cuspidal functions in the span):*

| $s$ | $\dim V$ | $d_{\mathrm{Eis}}$ | $d_{\mathrm{cusp}}$ | $s$ | $\dim V$ | $d_{\mathrm{Eis}}$ | $d_{\mathrm{cusp}}$ |
|---|---|---|---|---|---|---|---|
| 5 | 20 | 18 | 2 | 6 | 12 | 12 | 0 |
| 7 | 42 | 36 | 6 | 8 | 18 | 18 | 0 |
| 9 | 72 | 60 | 12 | 10 | 30 | 30 | 0 |
| 11 | 110 | 90 | 20 | 12 | 42 | 36 | 6 |
| 13 | 156 | 126 | 30 | 14 | 56 | 48 | 8 |
| 15 | 210 | 126 | 84 | 16 | 72 | 48 | 24 |
| 17 | 272 | 162 | 110 | 18 | 84 | 60 | 24 |
| 19 | 342 | 270 | 72 | 20 | 110 | 90 | 20 |
| 21 | 420 | 240 | 180 | 22 | 132 | 108 | 24 |
| 23 | 506 | 288 | 218 | 24 | 156 | 120 | 36 |
| 25 | 600 | 468 | 132 | 26 | 182 | 126 | 56 |
| 27 | 702 | 546 | 156 | 28 | 210 | 168 | 42 |
| 29 | 812 | 630 | 182 | 30 | 240 | 144 | 96 |
| 31 | 930 | 720 | 210 | 32 | 260 | 126 | 134 |
| 33 | 1056 | 576 | 480 | 34 | 306 | 198 | 108 |
| 35 | 1190 | 432 | 758 | 36 | 342 | 270 | 72 |
| 37 | 1332 | 684 | 648 | 38 | 380 | 300 | 80 |
| 39 | 1482 | 798 | 684 | 40 | 420 | 270 | 150 |
| 41 | 1640 | 882 | 758 | 42 | 462 | 240 | 222 |
| 43 | 1806 | 1386 | 420 | 44 | 506 | 288 | 218 |
| 45 | 1980 | 990 | 990 | 46 | 552 | 432 | 120 |
| 47 | 2162 | 1080 | 1082 | 48 | 600 | 288 | 312 |
| 49 | 2352 | 1800 | 552 | 50 | 630 | 312 | 318 |
| 51 | 2550 | 1350 | 1200 | 72 | 1302 | 540 | 762 |

*In particular* $d_{\mathrm{cusp}} > 0$ *for every* $s$ *in the table except* $s = 6, 8, 10$*, and at* $s = 11$*:* $\dim V = 110$*,* $d_{\mathrm{Eis}} = 90$*,* $d_{\mathrm{cusp}} = 20$*.*

*Proof notes.* (i) is. (ii): one counts the points of each orbit at each norm in the four cones and takes the exact rank over $\mathbb{Q}$ (FLINT `fmpz_mat`); a rank of evaluations is a lower bound for $\dim V$. (iii), odd $s$: by Theorem 19.5 the constant-term vector (labels with $\alpha \in \mathbb{Z}^2$, i.e. $x = 0$, $\epsilon = 0$) is the pure tensor $\delta_0 \otimes \cdots \otimes \delta_0 \otimes (e_{00} + e_{01})$, with 2-adic part in $A_3$. Since $\rho(\mathrm{SL}_2(\mathbb{Z}))$ is the full product of the local images, the cyclic span of a pure tensor is the tensor product of the local cyclic spans. These are $A_3$ (irreducible, dimension 3) and the cyclic span of $\delta_0$ in $\rho_{q_p}$, of dimension $(p^e + 1)/2$: by hand for $e = 1$, and by exact certification (Krylov rank mod $\ell$ equal to the $O(\Delta)$-orbit upper bound) for the listed $p^e$. Even $s$: the family is literally the Weil representation of $L'/\mathbb{Z}^2$ (an exact identity of the $S, T$ matrices), with local factors certified as stated. (iv): the square root and the half-cone bound are exact; the $s = 8$ relations $N_{O12} = N_{O7} + N_{O9}$ and $N_{O2} = N_{O15} + N_{O19}$ are instances. □

**Status (Category A).** (i); (ii) and the table entries for $\dim V$, exact per $s$; (iii) for the stated $s$, conditional on the cited congruence theorem (odd $s$) and on the exact local certifications; (iv) the square-root criterion and the upper bounds for $m \leq 7$, with equality for $m \leq 6$; $d_{\mathrm{cusp}} = \dim V - d_{\mathrm{Eis}}$ as the dimension of cuspidal functions, given the $\lambda = \frac{1}{4}$ constant-term criterion.

**Category B.** $\dim V = s(s-1)$ for *all* odd $s$ (the lower bound is open); $d_2(k, v)$ for $k \geq 7$ ($k = 6$ is Proposition 20.9). Floating-point singular-value ranks and FFT counts agree with the exact values above, except for visible truncation effects at low $Q$ (e.g. $s = 51$).

**Remark 20.2** (Sector reading of the cusp dimension, odd $s$)**.** By Theorem 19.5(iv), $V \subseteq A_3 \otimes (\text{even} \otimes \text{even}) \oplus B_1 \otimes (\text{odd} \otimes \text{odd})$, where even/odd refer to $Z_{(s)}$ and $Z_{(s-2)}$. The all-even sector $A_3 \otimes \bigotimes_p W_p^+$ has dimension $d_{\text{Eis}}$. Wherever $\dim V = s(s-1)$, the cusp part is $d_{\text{cusp}} = O(s)O(s-2) + 3\big(E(s)E(s-2) - \prod(p^e+1)/2\big) \geq (s-1)(s-3)/4 > 0$, with equality iff $s$ and $s-2$ are both prime powers. This reproduces $2, 6, 12, 20, 30, 84, 110, 72, 180, 218, 132$ at $s = 5, \ldots, 25$ and 758 at $s = 41$. Category A at the certified $s$; for general odd $s$ it is Category B, through the open lower bound. The family cusp part is therefore a local (Weil-parity) phenomenon, not a class-group one.

## 20.2 The case $s = 2m^2$

Let $s = 2m^2$ with $m$ odd and $g = (\gamma + I)/(2m)$, as in Theorem 20.1(iv).

**Proposition 20.3** (Complement trace)**.** *For every odd $m$: $\#\mathrm{Fix}(g) = 2(m-1)$ and $\#\mathrm{Fix}(-g) = 2(m+1)$ on the complement, $\mathrm{tr}(g \mid \mathrm{Fix}(N, \gamma)) = 2m$, and the $g = -1$ eigenspace $X$ of $\mathrm{Fix}(N)$ has dimension $m(m-1)$.*

**Status (Category A).** $A(g \mp 1)^{-1}$ is integral, and $\gamma = 1$ on the complement $C$. Exact at $m = 3, 5, 7$ as well.

**Proposition 20.4** (Half-cone relations)**.** *If $g$ acts on $M_7$-classes (more generally on the classes of $\mathbb{Z}^2 + \mathbb{Z}t$) and $m$ is prime, the span of the half-cone $g$-relations $e_c - e_{gc}$ is exactly $\delta_{\text{iso}} \otimes X$, of dimension $m(m-1)$.*

**Status (Category A).** Proved for prime $m$, given that $g$ is consistent on the classes. That consistency is checked exactly only at $m = 3, 5, 7$. At $m = 3, 5$ the exact half-cone kernel equals $\delta_{\text{iso}} \otimes X$ vector by vector. *Open:* $g$-consistency for all odd $m$, and composite $m$ (the $m$-part is not cyclic). In general the kernel can be larger than the $g$-relations; see Theorem 20.5.

**Theorem 20.5** ($\dim V(98) = 2406$)**.** *At $s = 98$ ($m = 7$), $\dim V = 2406$. The kernel of the lift on the 98-dimensional 7-adic old space has dimension 44. It is spanned by the $42 = m(m-1)$ $g$-relations and two $h$-relations, $R_1$ (at $q = 11/12$) and $R_2$ (at $q = 1/4$). Here $h = (g+I)/4 \in \mathrm{SO}(A)(\mathbb{Q})$, with $h^2 = g$ and $h^{\mathsf{T}}Ah = A$. The half-cone upper bound 2408 is not sharp.*

**Status (Category A).** *Lower bound:* the exact rank is 2406 (flint, $Q_{\max} = 40$; old-piece ranks are stable to $Q_{\max} = 130$). The new pieces are injective, and all 44 missing dimensions lie in the old space. *Upper bound:* take the overlattice $M = \bigcap_{k=0}^{7} h^k M_7$, $M_7 = \mathbb{Z}^2 + \mathbb{Z}t_0$, with basis $(4, 0), (1, 1/7)$, $[M_7 : M] = 4$ and Gram matrix $(2Q)\begin{pmatrix} 32 & 64 \\ 64 & 32 \end{pmatrix}$ (even, $\det -3072$). Exact HNF gives $hM = M$ and $hM^\sharp = M^\sharp$. Moreover $h$ maps the $+$ set of $R_i$ bijectively onto its $-$ set as $M$-cosets ($32 + 32$ for $R_1$, $16 + 16$ for $R_2$). The sets are $\gamma$-invariant ($\gamma = h^4$). Since $h$ has positive eigenvalues $2 \pm \sqrt{3}$, it maps fundamental cones for $\langle\gamma\rangle$ to fundamental cones, so $N_{S^+}(q) = N_{S^-}(q)$ for all $q$, and the theta identities hold. The same argument with $g$ on $M_7$ (eigenvalues $7 \pm \sqrt{48}$) proves the 42 $g$-relations. The 44 relations are independent (exact rank 44), so $\dim V \leq 2450 - 44 = 2406$.

**Category B.** Extra $h$-relations beyond $m(m-1)$ are expected exactly when $2m+2$ is a square, i.e. $m = 2k^2 - 1$ ($m = 7, 17, 31, \ldots$); their number is given by Proposition K1 below. Beyond $m = 7$ the actual lift rank is not computed.

The two $h$-relations at $m = 7$ are the case $k = 2$ of the following count. Fix $k \geq 2$, $m = 2k^2 - 1$, $s = 2m^2$, and let $g = (\gamma + I)/(2m)$ and $h = (g + I)/(2k)$. Let $\mathrm{rank}_g$ and $\mathrm{rank}_h$ be the ranks of the orbit-sum maps $F \to \mathrm{Fun}(\langle X, -1\rangle\text{-orbits on } M^\sharp/M)$, $X \in \{g, h\}$, with $F = \mathrm{Fun}(M_t^\sharp/M_t)^{\langle\gamma, -1\rangle}$, computed by exact orbit sums.

**Proposition 20.6** (K1)**.** $\operatorname{rank}_g = m(m+1)$ *and* $\operatorname{rank}_h = m(m+1) - k(k-1)$*. Hence the number of extra $h$-relations is exactly $k(k-1)$ for every $k \geq 2$. Moreover* $\dim F = 2m^2 = s$*.*

*Proof outline. Order model.* $M_t = \mathbb{Z}e_1 \oplus \mathbb{Z}f$ with Gram matrix $\left(\begin{smallmatrix} 2 & 2m \\ 2m & 2 \end{smallmatrix}\right)$ and $gf = -e_1 + 2mf$, so $(M_t, B, g) \cong (R_m, \operatorname{Tr}(x\bar{y}), \cdot\varepsilon_m)$ with $R_n = \mathbb{Z}[\sqrt{n^2-1}]$, $\varepsilon_n = n + \sqrt{n^2-1}$; hence $C := M_t^\sharp / M_t \cong R_m / 2\sqrt{d_m} R_m$, $d_n = n^2 - 1$. *Pell self-similarity.* With $u = (e_1 + f)/(2k)$ one has $hu = f$, $hf = -u + 2kf$ and $\operatorname{Gram}(u, f) = \left(\begin{smallmatrix} 2 & 2k \\ 2k & 2 \end{smallmatrix}\right)$, so $N_1 = M_t + hM_t = \mathbb{Z}u \oplus \mathbb{Z}f$ is even, $h$-stable, and $(N_1, h) \cong (R_k, \varepsilon_k)$: $h$ is "$g$ at level $k$". $I = N_1/M_t$ is isotropic of order $2k$ and $C_1 = I^\perp / I \cong R_k / 2\sqrt{d_k} R_k$. *Hecke reduction.* On $V_0 = \operatorname{Fun}(C)^{\langle g, -1 \rangle}$ (so $\operatorname{rank}_g = \dim V_0$), $\operatorname{rank}_h$ is the rank of $1 + \tau$ with $\tau = (2k)^{-1} \uparrow \circ \bar{h} \circ \downarrow$, where $\downarrow$ sums over $I$-cosets in $I^\perp$ and $\uparrow$ is its adjoint. *Spectrum.* $\downarrow\uparrow = 2k$, so $\tau^2$ is an orthogonal projection, $\tau^3 = \tau$, and the $(-1)$-eigenspace of $\tau$ has dimension $a - b$, with $a, b$ the numbers of $\langle h^2, -1 \rangle$- and $\langle h, -1 \rangle$-orbits on $C_1$. *Burnside.* On $R_n / 2\sqrt{d} R_n$, $\langle \varepsilon, -1 \rangle$ acts through $\{\pm 1, \pm\varepsilon\}$ with $|\operatorname{Fix}(-1)| = 4$, $|\operatorname{Fix}(\mp\varepsilon)| = 2(n \mp 1)$; so there are $n(n+1)$ orbits of $\langle \varepsilon, -1 \rangle$ and $2n^2$ of $\langle \varepsilon^2, -1 \rangle$. Level $m$ gives $\operatorname{rank}_g = m(m+1)$ and $\dim F = 2m^2$; level $k$ gives $a = 2k^2$, $b = k(k+1)$, so the extra count is $2k^2 - k(k+1) = k(k-1)$. □

**Status (Category A).** Proved for all $k \geq 2$ (the Burnside counts are also brute-forced for $n = 2, \ldots, 40$). *Exact checks of the reformulation:* Exact computation ( over $\mathbb{Q}$, $k = 2, 3, 4$) gives $\operatorname{rank}(I + \tau) = 54, 300, 980$, equal to $\operatorname{rank}_h$, with minimal polynomial $x^3 - x$; the same computation at $k = 5$ ($m = 49$) and $k = 6$ ($m = 71$) gives extra $= 20, 30$ and $\operatorname{rank} \tau = 50, 72$, with ranks computed mod 1000003 and mod 998244353 in agreement. Those mod-$p$ ranks are consistency checks of a proved statement, not independent proofs.

**Corollary 20.7.** *On $V_0$, $\tau$ has eigenvalues $+1$, $-1$, $0$ with multiplicities $k(k+1)$, $k(k-1)$, $m(m+1) - 2k^2$; so $\operatorname{tr} \tau = 2k$ and $\operatorname{rank} \tau = 2k^2$. The extra relations are explicit: $\uparrow$ of the $\bar{h}$-odd $\langle \bar{g}, -1 \rangle$-invariant functions on $C_1 = N_1^\sharp / N_1$. In particular the number of closed orbits is $m(m+1) = \operatorname{rank}_g$.*

**Status (Category A).** Consequence of Proposition K1.

**Proposition 20.8** (Sharpness of Proposition K1 on the old space)**.** *For $k = 2, 3, 4, 5$ ($m = 7, 17, 31, 49$), the theta lift on $F = \operatorname{Fun}(M_t^\sharp / M_t)^{\langle \gamma, -1 \rangle}$ has rank exactly $m(m+1) - k(k-1) =$ 54, 300, 980, 2430. So at these $m$ the old space has no relations beyond the $g$- and $h$-relations.*

**Status (Category A).** The upper bound is Proposition K1. The lower bound is the rank over $\mathbb{F}_{2^{31}-1}$ of the exact (orbit × norm) count matrix of the $x \in M_t^\sharp$ with $0 < |Q(x)| \leq Q_{\max}$, in an exact fundamental domain for $\langle \gamma \rangle$; a rank modulo a prime is a lower bound for the rank over $\mathbb{Q}$. At $k = 2$ this reproduces the old-piece ranks $42 + 12$ of Theorem 20.5. *Not claimed:* $\dim V(578) = 83532$, which would also need injectivity and independence of the new pieces at $s = 578$.

**Status (Category A).** Proposition K1: proved for all $k \geq 2$. The symbolic-in-$k$ ingredients and the Burnside counts (brute-forced for $n = 2, \ldots, 40$) are checked by computer. Consistency checks of the reformulation: exact over $\mathbb{Q}$ for $k = 2, 3, 4$, $\operatorname{rank}(I + \tau) = 54, 300, 980$ with minimal polynomial $x^3 - x$; modulo two primes at $k = 5, 6$ ($m = 49, 71$), extra $= 20, 30$ and $\operatorname{rank} \tau = 50, 72$.

**Dimension consequences (Category A as upper bounds; sharpness open).** Proposition K1 is a statement about the lattice-level rank $\operatorname{rank}_h$, which is a proved upper bound for the old lift rank. Every $h$-relation is a proved theta relation (the argument of Theorem 20.5), so for $s = 2m^2$, $m = 2k^2 - 1$, subtracting the $k(k-1)$ extra relations gives *proved upper bounds*: $\dim V(98) \leq 2406$ (equality, Theorem 20.5) and $\dim V(578) \leq 83532$, and likewise at every $m = 2k^2 - 1$. Whether these bounds are sharp at $m = 17, 31, \ldots$ is *not* proved.

### 20.3 The case $s = 64$

**Proposition 20.9** ($s = 64$)**.** $\dim W_P(64) = 1986$. *The 2-adic cyclic span $d_2$ at $k = 6$ equals 36, so the 2-adic factor of the function span is $36 \cdot 16 = 576$ at $s = 64$.*

**Status (Category A).** $W_P$ is an exact count. For the upper bound, $d_2 \le \#O(\Delta_2)$-orbits $= 42$, minus a 6-dimensional $S, T$-invariant subspace $Y \perp \delta_0$ that is verified exactly in $\mathbb{Z}[\zeta_{64}]$, leaving 36. The lower bound 36 is exact. The value $2^{k-1} + 4$ for general $k$ remains Category B.

### 20.4 Prime-square sectors

For $s = p^2$ the isotropic projector $\Pi$ splits each sector into an *old* piece, the $p$-local invariants, and a *new* piece, their complement; both are subrepresentations.

**Proposition 20.10** (Prime-square sectors: ranks)**.** *Exactly:*

| $s$ | $(+,+)$ *old* | $(+,+)$ *new* | $(-,-)$ *new* | *total* |
|---|---|---|---|---|
| *9* | *12* | *48* | *12* | *72* |
| *11* | *18* | *72* | *20* | *110* |
| *25* | *36* | *432* | *132* | *600* |

*Every piece is injective. The $\nu_0$ coefficient is nonzero on every new piece and zero on every old piece.*

**Status (Category A).** Exact integer ranks. The irreducible decompositions are Proposition 20.12; a uniform proof is open.

Let $s = p^2$ with $p$ an odd prime. Write $W_p^{\pm}$ for the even and odd parts of the Weil representation of $\mathbb{Z}/p^2$ (or of $\mathbb{Z}/q$), and $A_3$ for the irreducible 3-dimensional 2-adic factor.

**Theorem 20.11** (Prime-square structure)**.** *Assume (C1) Theorem 19.5 and (C2) CRT surjectivity, both proved above, and (C3) the cited Nobs–Wolfart / Cliff–McNeilly–Szechtman decomposition $W_p^+ = \mathbf{1} \oplus R$, with $R$ irreducible of dimension $(p^2-1)/2$, and $W_q^{\pm}$ irreducible for primes $q$. Let $\varepsilon = (\varepsilon_q)_{q \mid s-2}$, with $Z_{(s)} = +$. Then:*

*(i) The sector is $\Sigma_\varepsilon = W_p^+ \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$. Its $p$-local invariants are $\mathbf{1} \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$. Summed over the $\varepsilon$ with $Z_{(s-2)} = +$, these have total dimension $3(s-1)/2$.*

*(ii) If $s-2$ is squarefree, then $\Sigma_\varepsilon = (p\text{-local invariants}) \oplus R \otimes (\bigotimes_q W_q^{\varepsilon_q}) \otimes A_3$ is a sum of two non-isomorphic irreducibles.*

**Status (Category A).** Conditional on (C3), which is cited and not re-proved; (C1) and (C2) are proved above; (C3) is Category A for the primes covered by Propositions 20.14 and 20.15. Irreducibility of the complement follows from the outer-tensor argument. *Caveats:* if $s-2$ is composite, the sector splits by prime sign (see $s = 121$ below). If $q^2 \mid s-2$, the complement is reducible.

**Proposition 20.12** (Exact splits)**.** *The "$(+,+)$" sectors split as follows: $s = 9$: $12 + 48$; $s = 11$: $18 + 72$; $s = 25$: $36 + 432$; $s = 49$: $72 + 1728$. In each case the commutant is 2 and the smaller piece is exactly the $p$-local invariant subspace. The $(-,-)$ sectors at $s = 9, 11, 25$ (dimensions $12, 20, 132$) are irreducible, and the mixed sectors vanish.*

**Status (Category A).** The sectors are built with an exact invariance certificate (0 failures). The commutant is computed mod $\ell$ on an exactly verified basis. It is lifted to $\mathbb{C}$ using $\ell \nmid |\mathrm{Mp}_2(\mathbb{Z}/N)|$; this divisibility is machine-checked (at $s = 49$ with the two primes $\ell = 1272914161$ and $\ell = 2036662657$, both giving commutant 2 and pieces of dimensions 72 and 1728). *Standard input, cited:* a Weil representation factors through $\mathrm{Mp}_2(\mathbb{Z}/\mathrm{ord}\, T)$. For $\gamma \in \Gamma(N)$ the cited Borcherds/Scheithauer $\Gamma_0(N)$ formula gives $\rho_D(\gamma) = \chi_D(\gamma)I$. After Proposition 20.14, the splits at $s = 9, 25$ need neither this nor (C3). The $s = 9, 25, 49$ splits agree with Theorem 20.11(ii), where $s - 2 = 7, 23, 47$ is prime.

**Proposition 20.13** (Label-space counts at $s = 121$ and $s = 169$)**.** *Let the $p$-local invariants of a sector be the vectors fixed by $\langle T^a, ST^aS^{-1}\rangle$, where $a \equiv 1 \pmod{p^2}$ and $a \equiv 0 \pmod{n_T/p^2}$; this definition does not use the tensor factorisation.*

*(i)* $s = 121$ *($p = 11$, $s-2 = 7\cdot 17$; signs ordered $Z_7, Z_{11}, Z_{17}$). The nonzero sectors are $(+,+,+)$, $(-,+,-)$, $(+,-,-)$ and $(-,-,+)$, of dimensions* 6588, 4392, 1920 *and* 1620. *They contain at most* 108, 72, 0 *and* 0 *independent $p$-local invariants respectively, for a total of at most $180 = 3(s-1)/2$.*

*(ii)* $s = 169$ *($p = 13$). The $(+,+)$ sector has dimension* 21420 *and contains at most $252 = 3(s-1)/2$ independent $p$-local invariants; the $(-,-)$ sector has dimension* 6972 *and contains none.*

*Given the cited factorisation of a Weil representation through $\mathrm{Mp}_2(\mathbb{Z}/n_T)$, these upper bounds are equalities.*

**Status (Category A).** Sector dimensions: exact orbit counts. Invariant counts: $2k - \mathrm{rank}[X \mid SX]$ computed modulo $\ell$, which bounds the characteristic-0 count from above. At $s = 121$ two primes ($\ell = 19903795297$, $99518976481$) agree exactly; at $s = 169$ one prime was used. Equality holds if $\ell \nmid |H|$, which is guaranteed when $\rho$ factors through $\mathrm{Mp}_2(\mathbb{Z}/n_T)$ (cited), since $\ell$ exceeds every prime dividing $|\mathrm{SL}_2(\mathbb{Z}/n_T)|$. The counts agree with Theorem 20.11(i) and do not use (C1) or (C3). Irreducibility of the complements is not claimed.

**Proposition 20.14** ((C3) for small $p$)**.** *For $p \in \{3, 5, 7, 11, 13, 59\}$, the even Weil representation of $\mathbb{Z}/p^2$ is $\mathbf{1} \oplus R$ with $R$ irreducible, and the odd part is irreducible. Together with (C1), this gives the $(+,+)$ decompositions at $s = 9, 25, 49$ in characteristic 0 directly, without the $\ell \nmid |\mathrm{Mp}_2|$ lift.*

**Status (Category A).** The mod-$\ell$ commutant is 2 (even part) or 1 (odd part), on exactly verified bases, for both square classes. Semicontinuity bounds the characteristic-0 commutant above by the mod-$\ell$ one. The explicit invariant line $\delta_{p\mathbb{Z}/p^2}$ bounds it below by 2. The general-$p$ case remains cited (Nobs–Wolfart / Cliff–McNeilly–Szechtman). The irreducibility of 1728 at $s = 49$ also uses $W_{47}^+$ irreducible (cited; not run).

**Category B.** *Factor dimensions only:* $s = 59^2 = 3481$, $s - 2 = 7^2 \cdot 71$. The prediction is: in $(+,+,+)$, $p$-invariants $108 + 2592$ and complement $187920 + 4510080$ (reducible); in $(+,-,-)$, 2520 and 4384800; total $p$-invariants $5220 = 3(s-1)/2$. No label-space run.

**Proposition 20.15** (Weil representations of $\mathbb{Z}/M$)**.** *The even and odd parts of the Weil representation of $\mathbb{Z}/M$ have the following (dimension, commutant): $M = 47$: $(24,1)$, $(23,1)$; $M = 59$: $(30,1)$, $(29,1)$; $M = 71$: $(36,1)$, $(35,1)$; $M = 17^2$: $(145,2)$, $(144,1)$; $M = 19^2$: $(181,2)$, $(180,1)$; $M = 23^2$: $(265,2)$, $(264,1)$. Hence for $q = 47, 59, 71$ both parts are irreducible, and for $p = 17, 19, 23$ the even part of $\mathbb{Z}/p^2$ is $\mathbf{1} \oplus R$ with $R$ irreducible and the odd part is irreducible.*

**Status (Category A).** Commutants computed mod $\ell \equiv 1 \pmod{M}$ on exactly verified bases, both square classes; the characteristic-0 commutant is bounded above by the mod-$\ell$ one, and for $\mathbb{Z}/p^2$ below by 2 via the invariant line $\delta_{p\mathbb{Z}/p^2}$. With Proposition 20.14, (C3) is Category A for every prime $q \mid s-2$ and every $p^2$ occurring at $s = 9, 25, 49$ (in particular $W_{47}^{\pm}$ irreducible, used for the 1728 at $s = 49$), and for $\mathbb{Z}/17^2, \mathbb{Z}/19^2, \mathbb{Z}/23^2$. The general-$p$ statement remains cited (Scheithauer, IMRN 2009, Prop. 4.5 and Thm. 4.7; Cliff–McNeilly–Szechtman, main theorem; Nobs–Wolfart, theorem number not verified).

**Category B.** At $s = 3481$ the decomposition remains a factor-dimension prediction: the factors are Category A, but the assembled decomposition is conditional on the tensor factorisation and was not computed in label space.

## 21 Cusp content of the vector $u_L$

The family statements of §20 concern the whole span $V$. Whether the single vector $u_L = u_{a,0}$ has cusp content is a different question; the two must be kept apart.

**Proposition 21.1** (Cusp criterion for $u_L$)**.** *Let $G = \mathrm{Cl}^+_{M\infty_1\infty_2}$ with $M = 2dc$, $d = 2(s-2)$, $n_s = c^2 D_0$; let $T \subset G$ be the class set of the coprime part of $u_L$, $S = \mathrm{Stab}(T)$, $j = [-1]$, $\bar{G} = G/(S + \langle j \rangle)$, $H = (1-\sigma)G$ and $\bar{H}$ its image in $\bar{G}$. (i) The coprime part of $u_L$ is Eisenstein (genus-pure) iff $\bar{H} = 1$, i.e. iff $\sigma$ acts trivially on $\bar{G}$; the number of non-genus pure characters is then $|\bar{G}| - |\bar{G}^\sigma|$. (ii) For odd $s$ with $c = 1$ ($n_s$ squarefree), only the coprime part occurs and Theorem 21.3 decides: $u_L$ is Eisenstein iff $h(\mathbb{Q}(\sqrt{n_s})) = 2^{\omega(s)}$. (iii) The verdicts for $5 \le s \le 44$ are those of Table 17. (iv) The rule "$u_L$ has cusp content iff $h^+ \neq$ #genera" is false: $s = 14, 17, 18, 22, 23$ have $h^+ =$ #genera and cusp content (all with $\bar{H}$ of order 2). (v) The two notions differ: $d_{\mathrm{cusp}} > 0$ at $s = 5, 7, 13, 15$, where $u_L$ is Eisenstein. The $u_L$ verdict is class-group data ($\bar{H}$) on top of the local structure.*

**Status (Category A).** (i): reduction (Tate's $\ker \mathrm{N} = C_K^{1-\sigma}$ for cyclic $K/\mathbb{Q}$, plus the dictionary "Maass content of $u_L$ = pure-type character content of its class sets"). (iv): exact finite group computations per $s$ on complete coprime parts (the $s = 18$ coprime part is complete at $B = 150\,000$).

**Category B.** Sufficiency of the modulus $M = 2dc$ for all norms (stable under $M \to 2M$ wherever tested).

Here $\mathbf{W} \subset (\mathcal{O}/M)^\times$ is the unit stabiliser of $\beta_0 + L$ (unrelated to $W_P$), $S_0$ is the coprime-part stabiliser and $\psi(\lambda) = [(\lambda)]$.

**Theorem 21.2** (Coset property)**.** *For every $s$ and every $c$-part $\mathfrak{c}$, $T_\mathfrak{c} + \psi(\mathbf{W}) = T_\mathfrak{c}$, and $S_0 = \psi(\mathbf{W})$. Hence every $T_\mathfrak{c}$ is a union of $S_0$-cosets.*

**Status (Category A).** The lemma $T_\mathfrak{c} + \psi(\mathbf{W}) = T_\mathfrak{c}$ is elementary. $S_0 = \psi(\mathbf{W})$ follows from the local condition $R_p = \mathbf{W}_p$, proved for all $s$ and $p$: it reduces to the fact that ratios of exact-valuation elements of $\beta_0 + L$ lie in the multiplier order of $L$. The cases are $p \nmid f$; odd $p \mid c$ with $p \mid s$ or $p \mid s-2$; and $p = 2$ in each 2-adic case. Also checked exactly: global $S_0 = \psi(\mathbf{W})$ at $s = 5, \ldots, 44$, and $R_p = \mathbf{W}_p$ at $s \le 150$.

**Theorem 21.3** (Squarefree criterion)**.** *Let $s$ be odd with $s(s-2)$ squarefree, and $h = h(\mathbb{Q}(\sqrt{s(s-2)}))$. Then only the coprime $c$-part occurs, $|\bar{G}^\sigma| = 2^{\omega(s)}\varphi(s-2)$, and $u_L$ is Eisenstein $\iff$ $h = 2^{\omega(s)}$. The number of non-genus characters is $\varphi(s-2)(h - 2^{\omega(s)})$.*

**Status (Category A).** The proof uses Chevalley's ambiguous class number formula (ray form, unit-norm index 1) and, independently, Gauss genus theory; these are classical theorems applied to exact data. The exact cross-check covers 47 values $s \le 151$: Eisenstein only at $s = 5, 7, 13, 15$.

| verdict ($5 \le s \le 44$) | $s$ |
|---|---|
| Eisenstein | $5, 6, 7, 8, 9, 10, 12, 13, 15, 16, 20, 24, 28$ |
| cusp content | $11, 14, 17, 18, 19, 21, 22, 23, 25, 26, 27$ and every $s$ with $29 \le s \le 44$ |

Table 17: Verdicts for $u_L$.

**Status (Category A).** Verdicts are per $s$ over all (infinitely many) $c$-parts, using Theorem 21.2 and Proposition 21.4. Coverage at $s = 32, 34, 36, 40$ was completed by enumeration to $B = 200000$. At $s = 24$ there are no mixed-prime parts for any exponents (a local proof), and the parts $(1, 12)$, $(12, 1)$, $(1, 16)$, $(16, 1)$ are genus-pure. Beyond the table, cusp content holds at $s = 53, 55, 57$ (incomplete coverage, matching the prediction of (ii); Category B).

**Proposition 21.4** (Deep bad-prime parts)**.** *At $s = 9$ (the 29 parts $(e_1, e_2)$ with $e_1 \le e_2 \le 7$ at the split prime 3, plus $\mathfrak{c} = 1$), $s = 16$ ($\mathfrak{c} = 1, \mathfrak{p}_2$; and 7) and $s = 24$ (15 parts at 2, up to exponent 8; and 11), every tested $\mathfrak{c}$-part is complete and genus-pure. At $s = 18$ the 3-parts are pure, but the coprime part is not, so $u_L$ has cusp content. If $\bar{H} = 1$ and every $T_{\mathfrak{c}}$ is $S$-invariant, then every $\mathfrak{c}$-part is genus-pure and $u_L$ is Eisenstein; conversely, if $\bar{H} \neq 1$, the coprime part already gives cusp content. By Theorem 21.2 every $T_{\mathfrak{c}}$ is a union of $S$-cosets, hence $S$-invariant, so the coprime criterion decides.*

**Status (Category A).** Per tested part: exact (`matsolvemod` lattice solve, completeness of every class, exact purity test). The reduction lemma, given $S$-invariance: $(1 + j)1_{T_{\mathfrak{c}}}$ is $(S + \langle j \rangle)$-invariant, hence $H$-invariant, and $\sigma H = H$.

### 21.1 Local root numbers at $s = 7$

The 24 cusp characters at $s = 7$ have conductor of norm 140 (ramified at $\mathfrak{p}_2, \mathfrak{p}_5, \mathfrak{p}_7$). The vector-valued components are $\Gamma(140)$-modular and the scalar newforms have level $140^2 = 19600$; the two levels are consistent (Category A).

**Proposition 21.5** (Local root numbers)**.** *For the 24 cusp characters, the local root numbers $\varepsilon_v$ ($\mathfrak{p}_2, \mathfrak{p}_5, \mathfrak{p}_7, \infty$) are exact elements of $\mathbb{Q}(\zeta_{840})$ of modulus 1. $\varepsilon_{\mathfrak{p}_2}$ is a 6th root of unity; $\varepsilon_{\mathfrak{p}_5}$ is never a root of unity. The global root numbers have non-monic minimal polynomials (e.g. $5t^8 + 6t^4 + 5$), so they are not roots of unity. The sector signs are* products *of local root numbers: $Z_p = \psi_{\mathfrak{p}}(-1) = \varepsilon(\psi_{\mathfrak{p}})\varepsilon(\psi_{\mathfrak{p}}^{-1}) = -1$ ($p = 5, 7$), and $S_p^2 = \omega_{\pi_p}(-1)$. They are not individual local root numbers.*

**Status (Category A).** The local factors (Tate's formulas) and the character-side identities are exact.

**Category B.** $\prod_v \varepsilon_v = W$ against PARI's numerical root number ($\le 2.7 \cdot 10^{-63}$). $W_p \leftrightarrow S_p$ holds only up to a fixed unit (the Atkin–Li normalisation, cited and not re-derived).

## 22 $H_s$ in closed form

Throughout, $s \ge 5$ is an integer, $c_1 = (-1, 1)$, $c_2 = (-4, 1)$, $g := \gcd(24, s - 8)$ and $\Delta := (s - 2)(7s - 32)$ (§10.4). All radial expansions rest on Proposition 13.1.

**Theorem 22.1** ($H_s$ in closed form)**.** *(i)* Structure. *$Q(c_1) = -(s - 2)/2$, $Q(c_2) = -(7s - 32)/2$, $B(c_1, c_2) = -4(s - 2)$, so the pair is admissible exactly for $s \ge 5$, and $4Q(c_1)Q(c_2) = \Delta$. $H_s = \frac{1}{2} q^{c_0} \vartheta_{a,0,\mathrm{hol}}^{c_1,c_2} + \Delta_{\mathrm{lines}}(s)$, where $\Delta_{\mathrm{lines}}$ is a finite sum of unary theta series with explicit $s$-dependence,*

*and $H_s$ has an explicit cone double-sum $q$-expansion for every $s \geq 5$; $H_5 \equiv 0$. (ii)* Radial coefficients at $x = 0$. *Along $q = e^{-t}$, $H_s$ has no integer-order terms at all, and for $r \in \{-\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$ its coefficients are $H_r = \sqrt{2\pi s}\,[R_r(s) + S_r(s,g)\sqrt{\Delta}]$ with*

$$
\begin{aligned}
R_{-1/2} &= -\frac{1}{2(s-2)},\\
S_{-1/2} &= \frac{1}{2(s-2)(7s-32)},\\
R_{1/2} &= -\frac{(s-4)(3s-2)}{48(s-2)^2},\\
S_{1/2} &= \frac{s^3-44s^2+192s+2g^2(s-2)}{48(s-2)^2(7s-32)^2},\\
R_{3/2} &= -\frac{15s^4-76s^3+108s^2+96s-128}{3840(s-2)^3},\\
S_{3/2} &= \frac{\Sigma_{3/2}}{3840(s-2)^3(7s-32)^3},\\
R_{5/2} &= -\frac{(s-4)(105s^5-730s^4+2732s^3-5208s^2+5696s-2560)}{645120(s-2)^4},\\
S_{5/2} &= \frac{\Sigma_{5/2}}{645120(s-2)^4(7s-32)^4},
\end{aligned}
$$

*where $\Sigma_{3/2} = 3s^6-152s^5+3872s^4-28160s^3+61440s^2+20g^2s(s^3-18s^2+96s-128)-8g^4(s-2)^2$ and $\Sigma_{5/2} = 15s^9-972s^8+28320s^7-537216s^6+4773888s^5-18923520s^4+27525120s^3+210g^2s^2(s^5-34s^4+448s^3-2816s^2+8192s-8192)-8g^4s(35s^4-504s^3+2940s^2-6832s+5376)+160g^6(s-2)^3$. The denominators are $(s-2)^{r+3/2}(7s-32)^{r+3/2}$, $g$ enters only through $g^{2i}$ with $i \leq r+\frac{1}{2}$, and all four vanish at $s = 5$. (iii) $H_{-1}(x) = 0$ for every rational $x$. (iv) $s = 7$. $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7) = P_{-1/2}(7) - H_{-1/2}(7) = 0.829142617\ldots$ symbolically; these are the radical forms of Table 13. (v) $s = 8$, the $\sqrt{\pi}$-rational scale. $\Delta(8) = 144$, and $H_{-1/2}(8) = -\sqrt{\pi}/6$, $H_{1/2}(8) = 5\sqrt{\pi}/54$, $H_{3/2}(8) = -175\sqrt{\pi}/324$, $H_{5/2}(8) = 995\sqrt{\pi}/972$, $C_{-1/2}(8) = 11\sqrt{\pi}/24$. Among $s < 10^6$, $\Delta$ is a perfect square only at $s = 5, 8, 26, 77, 365, 1178, 5768, 18725, 91877, 298376$. (vi)* Assembly. *The integer orders of $C = P_s^{\mathrm{orth}} - H_s$ at $x = 0$ are those of $(1+E_s)/2$ to all orders, and the half-integer orders are those of $(\theta_3 - \Theta_{\mathrm{hol},s})/2 - H_s$; $C_{-1} = \mathrm{Res}\,/2$, $C_0 = (1+E_0)/2$, $C_1 = E_1/2$.*

**Status (Category A).** (i): the case analysis of the weight and the characteristic correction are exact identities, checked to $q^{400}$ at 27 values of $s$. (ii): the integer orders vanish by the parity argument (the order-$j$ Euler–Maclaurin coefficients of the $\pm a$ cone sums differ by $(-1)^j$; each unary theta contributes only half-integer orders); the closed forms come from symbolic Euler–Maclaurin on the expansion of Proposition 13.1, with exact Bernoulli-sum validation for $s = 5..80, 99, 100, 151, 223, 300$. (iii): cancellation of the residue-class averages of $\mathbf{e}(xQ)$ under $\nu \mapsto -\nu$. (iv), (v): evaluation of the closed forms; the $\Delta$-square list is an exact search. (vi): uses in addition Hecke's continuation theorem.

**Category B.** Supporting numerics: $H_{3/2}$ against small-$t$ fits at 9 values of $s$ (3–4 digits); $H_{5/2}$ only at $s = 6, 8$. The modular completion of $\frac{1}{2}q^{c_0}\vartheta$ is by citation (Zwegers [18], Cor. 2.9).

## 22.1 $H_s$ at $x = \frac{1}{2}$, $x = \frac{1}{3}$ and higher orders at $x = 0$

Let $c_e$ be the coefficient of $t^{e/2}$ in $H_s(\mathbf{e}(x)e^{-t})$.

**Proposition 22.2** (Exact reductions)**.** *For every rational $x$: (i) the corner terms cancel between $Q_+$ and $Q_-$; (ii) every integer order vanishes; (iii) $c_e = \gamma_0 A_e(s) + \sum_r eb_r\, Y_{e,r}(s) + \sum_r ea_r\, X_{e,r}(s,g)$, with universal functions $A_e$, $Y_{e,r}$, $X_{e,r}$ and finite Gauss data $\gamma_0$, $eb_r$, $ea_r$ given by distribution relations.*

**Proposition 22.3** ($x = \frac{1}{2}$)**.** *(i) If $s$ is odd, $\gamma_0 = eb = ea = 0$, so $H_s(-e^{-t}) = O(t^N)$ for every $N$. (ii) If $s$ is even, $\gamma_0 = 0$, $eb_r = 4 \cdot 2^{1-r}(2 - 2^{1-r})B_r$ for even $r \geq 2$ (and 0 otherwise), and $ea_r = [16 \mid s]\, eb_r$. For $16 \nmid s$:*

$$\frac{H_s(-e^{-t})}{\sqrt{2\pi s}} \sim -\tfrac{1}{8}t^{1/2} + \tfrac{s-4}{64}t^{3/2} - \tfrac{31s^2-168s+192}{3072}t^{5/2} + \tfrac{(s-4)(173s^2-632s+544)}{24576}t^{7/2} - \tfrac{25261s^4-245680s^3+845440s^2-1226240s+634880}{3932160}t^{9/2} + \cdots .$$

*For $16 \mid s$ there is an extra $\sqrt{\Delta}$ term at each order; for example $g^2/(8(s-2)(7s-32)^2)$ at $t^{1/2}$*

**Status (Category A).** Exact symbolic computation on the proved expansion (Proposition 13.1). Consistency with an independent exact-term engine to $\leq 7 \cdot 10^{-38}$ through $t^{5/2}$ and to $\leq 5 \cdot 10^{-34}$ through $t^{9/2}$ (these comparisons are support only). Odd $s$: vanishing to all orders means there is no power-law term. The direct sums are not identically zero: outside the window $t \ll 1/(4m)^2$ the terms $e^{-c/(m^2 t)}$ appear (e.g. $H_7(-e^{-0.01}) \approx -7.95$, floating).

**Proposition 22.4** ($x = \frac{1}{3}$)**.** $\gamma_0 = i/\sqrt{3}$, $\mathbf{e}(1/6)/3$, $0$ *for $s \equiv 0, 1, 2 \bmod 3$; $eb_r$ depends only on $s \bmod 3$; $ea_r$ depends only on $(s \bmod 9, \varepsilon)$ with $\varepsilon = (-1)^{v_2(\gcd(s,8))}$ (Theorem 22.5). For $s \equiv 2 \bmod 3$ the $(\gamma_0, eb)$ part of $c_{-1}/\sqrt{2\pi s}$ is $-(1 + i\sqrt{3})/(12s)$, and the higher orders through $t^{5/2}$ are explicit.*

**Status (Category A).** The dependence on $s \bmod 3$ is proved, and that of $ea$ is Theorem 22.5. The closed form is exact computer algebra; consistency with an independent engine to $\leq 2.4 \cdot 10^{-35}$.

**Theorem 22.5** (Period 72 at $x = \frac{1}{3}$)**.** *Let $\varepsilon = (-1)^{v_2(\gcd(s,8))}$. Then $ea$ depends only on $(s \bmod 9, \varepsilon)$, so its period is 72. Explicitly:*

- *for $s \not\equiv 2 \bmod 3$: $ea = eb$ or $ea_r = (-1)^r eb_r$, according to $b \equiv 2$ or $1 \bmod 3$;*
- *for $s \equiv 2 \bmod 9$: $ea = 0$;*
- *for $s \equiv 5 \bmod 9$: $ea = eb$ ($\varepsilon = -1$) or $(-1)^r eb_r$ ($\varepsilon = +1$);*
- *for $s \equiv 8 \bmod 9$: $ea_r = 2^{2-r} i\sqrt{3} \sum_{c<3} \omega^{-\varepsilon c} \tilde{B}_r(c/3)$, which is $V$ for $\varepsilon = -1$ and $(-1)^r V_r$ for $\varepsilon = +1$.*

**Status (Category A).** Elementary proof: reduction mod 3, a reflection lemma, and the mod-3 Gauss sum. Exact check for $s = 5, \ldots, 1000$ and $r \leq 10$. The split is by $\varepsilon$, not by the parity of $s$ (e.g. $s = 28, 52$ have $ea = eb$).

**Proposition 22.6** ($t^{7/2}$, $t^{9/2}$ at $x = 0$)**.** *$H_{9/2} = \sqrt{2\pi s}\,[R_9(s) + S_9(s,g)\sqrt{\Delta}]$, where $R_9$ is an explicit rational function with denominator $5449973760(s-2)^6$ and $S_9$ is of degree 10 in $g$; $H_{9/2}(5) = 0$. The $t^{7/2}$ closed form is likewise explicit.*

**Status (Category A).** Two independent symbolic routes agree to 170 digits at 8 values of $s$; exact computer algebra on the proved expansion.

**Proposition 22.7** ($t^{11/2}$, $t^{13/2}$)**.** *At $x = 0$, $c_e = \sqrt{2\pi s}\,[R_e + S_e\sqrt{\Delta}]$ for $e = 11, 13$, with $R_{11}$ of denominator $5667972710400(s-2)^7$ and $R_{13}$ of denominator $136031345049600(s-2)^8$; both vanish at $s = 5$. At $x = \frac{1}{2}$ (even $s$):*

$$Rh_{11} = \frac{(s-4)(675691s^4 - 5253968s^3 + 14999488s^2 - 18628608s + 8491008)}{94371840},$$

*and $Rh_{13}$ is likewise explicit, with a $[16 \mid s]\sqrt{\Delta}$ part. At $x = \frac{1}{3}$ the coefficients are given by the universal form with the Gauss data of Theorem 22.5. The full expressions are in the data files listed in Appendix C.*

**Status (Category A).** Exact symbolic coefficients at $K = 13$ (identical to the $K = 9$ ones through $t^{9/2}$). The $R + S\sqrt{\Delta}$ split is interpolated and re-verified at 10 held-out values of $s$. *Category B support:* agreement with the term-by-term engine (to $\le 3 \cdot 10^{-41}$ where the quadrature allows), and 60-digit direct sums (relative error $10^{-14}$–$10^{-6}$).

**Proposition 22.8** ($t^{15/2}$ and $t^{17/2}$ at $x = 0$)**.** *For $e = 15, 17$, $c_e = \sqrt{2\pi s}\,[R_e(s) + S_e(s, g)\sqrt{\Delta}]$: the $R + S\sqrt{\Delta}$ split persists. Explicitly*

$$R_{15} = -\frac{109395s^{16} + 101686679952s^{15} - 3187115618544s^{14} + \cdots - 3332235344216064}{148002103413964800\,(s-2)^9},$$

$$R_{17} = -\frac{(s-4)\,(4849845s^{17} - 139775079348190s^{16} + \cdots - 4580182201382993920)}{472422714097375641600\,(s-2)^{10}},$$

*with all coefficients in the data files of Appendix C; $S_e$ is a polynomial in $g^2$ over $(s-2)^{(e+3)/2}(7s-32)^{(e+3)/2}$. Both vanish at $s = 5$. The factor $(s-4)$ occurs in $R_e$ for $e = 9, 13, 17$.*

**Status (Category A).** Exact symbolic runs ($K = 15$, $K = 17$; the $K = 17$ run reuses the exact $e \le 15$ entries). The split is unique at each sample $s$ and both interpolants pass exact checks at 10 held-out $s$. $K = 15$ reproduces every $K = 13$ coefficient exactly; the $e \le 13$ entries of $K = 17$ agree with the independent $K = 13$ run; $c_{15}(5) = c_{17}(5) = 0$.

**Category B.** Independent (floating-point) engine: $c_{15}$ equal at double precision for $s = 6, 7, 9$; $c_{17}$ equal for $s = 6$ and to $\le 9 \cdot 10^{-16}$ for $s = 7, 9$. 60-digit direct sums at $s = 20, 32, 8$: relative errors $2.6 \cdot 10^{-10}$, $4.5 \cdot 10^{-9}$, $8.9 \cdot 10^{-6}$ for $c_{15}$ and $3.5 \cdot 10^{-10}$, $6.1 \cdot 10^{-9}$, $1.2 \cdot 10^{-5}$ for $c_{17}$.

**Structure of $S_e$.** Write $S_e(s-2)^k(7s-32)^k = \sum_j \sigma_{e,2j}(s)g^{2j}$, $k = (e+3)/2$.

**Status (Category A).** Exact factorisations for each of $e = 11, 13, 15, 17$: only even powers $g^0, \ldots, g^{e+1}$ occur; $\sigma_{e,e+1} = \text{const} \cdot (s-2)^{(e+1)/2}$; for $1 \le j \le (e+1)/2$, $\sigma_{e,2j} = \text{const} \cdot s^{(e+1)/2-j}(s-2)^j(s-8)^{\max(0,e+3-4j)}\varphi_{e,j}(s)$; the cofactors pair up ($\varphi_{e,j} = \varphi_{e,j'}$ up to constants when $2j + 2j' = e + 3$); and $\sigma_{e,0} = s^{(e+1)/2}\cdot$(an irreducible polynomial of degree $e+1$).

**Category B.** *Observation, not proved:* that this pattern holds for all odd $e$. Likewise an *observation*: the constant in the top coefficient $\sigma_{e,e+1}$ carries the numerator of the Bernoulli number $B_{e+1}$ (691 for $e = 11$, 3617 for $e = 15$, 43867 for $e = 17$), consistent with the $r = e + 1$ Bernoulli term of the $u_2$-edge.

## 22.2 General $x = a/N$: the CRT Factorization Proposition

**Proposition 22.9** (General $x = a/N$)**.** *(i) (Periodicity) $\gamma_0, G, eb$ depend only on $s \bmod N'$, and $H, ea$ only on $s \bmod 24N'$, where $N' = N$ ($N$ odd) or $2N$ ($N$ even).*

*(ii) (**CRT Factorization Proposition**, CFP) $H_{a/N}(c) = [2 \text{ if } N \text{ odd}]\,T_2 \prod_p T_{p^e}$, with local quadratic sums $T$. The Bernoulli-weighted $ea$ is linear in $H$ but does not factor.*

*(iii)* *(Prime $p \geq 5$, $p \nmid a\, s(s-2)(7s-32)$)* $H(c) = 2\big(\frac{aQ_2(u_2)}{p}\big) g_p\, \mathbf{e}\big(\frac{a}{p}[-g^2(c+\delta)^2/(2(7s-32)) + c_0]\big)$.

*(iv)* *(Degenerate $p \geq 5$)* *If $p \mid s$,* $H(c) = 2p\, \mathbf{e}(an_w^2 c^2/p)$. *If $p \mid s-2$,* $ea = 0$. *If $p \mid 7s-32$, $H$ is a delta at one class $c^*$, and* $ea_r = 2^{2-r} p\, \mathbf{e}(aQ_+(c^* w)/p)\tilde{B}_r(c^*/p)$.

*(v)* *(Even $N$)* *The minimal periods of $s \mapsto H_{a/N}$ are* $16, 16, 144, 96$ *for* $N = 2, 4, 6, 8$. *At* $x = \frac{1}{4}$, $ea = 0$ *unless* $4 \mid s$; $ea = eb = (0, \frac{1}{3} - \frac{i}{6}, 0, -\frac{1}{240} + \frac{7i}{1920}, \dots)$ *for* $s \equiv 0 \bmod 16$; *and* $ea = (0, \frac{1+i}{12}, 0, -\frac{1+i}{3840}, \dots)$ *for* $s \equiv 4, 8, 12 \bmod 16$.

**Lemma 22.10** ($q$-adic reduction)**.** *For any prime $q$, a quadratic sum $\sum_{y \bmod q^n} \mathbf{e}((\alpha y^2 + \beta y + \gamma)/q^n)$ reduces by $q^{v_q(\alpha)}$ to a unit-leading sum. The unit-leading sum is then evaluated by the odd-$q$ Gauss sum, or by the explicit 2-adic formula. This gives a closed form for $H(c)$ for every $N$.*

**Status (Category A).** (i), (ii), (iii), (iv) and Lemma 22.10 have elementary proofs. They were also checked exactly in $\mathbb{Z}[\zeta_L]$ against brute force: about $7 \cdot 10^4$, $6.2 \cdot 10^5$ and $7 \cdot 10^5$ tests, with 0 mismatches. For (v), $144 = \mathrm{lcm}(16, 72)$ is explained by (ii). The periods 16 and 96 are proved conceptually in Propositions 22.12 and 22.13 (and also by exhaustive exact check across the proved bound $48N$). The coefficients at $x = \frac{1}{4}, \frac{1}{6}$ are Proposition 22.14.

**Proposition 22.11** (Checks of the general formula)**.** *The closed form of $H(c)$ given by the CFP together with Lemma 22.10 agrees exactly with brute force for every $N \in \{2, \dots, 13, 15, 16, 21, 25, 27, 32, 35, 45, 49\}$, every $a$ coprime to $N$, every $s < 200$ and every $c$. The runs at $N = 32, 35, 45, 49$ have* 199680, 327600, 421200, 802620 *tests and* 0 *mismatches.*

**Status (Category A).** Exact arithmetic in $\mathbb{Z}[\zeta_L]$, $L = \mathrm{lcm}(8, 2N)$.

**Proposition 22.12** (Even $N$ from the 2-adic reduction)**.** *(i)* $H_{1/2}(c) = 4(-1)^c[16 \mid s]$; *period* 16.

*(ii)* $H_{a/4}(c) = 0$ *if* $4 \nmid s$; $= 4(1 + i^a)$ *if* $s \equiv 4 \bmod 8$ *or* $s \equiv 8 \bmod 16$; $= 8\, \mathbf{e}(ac^2/4)$ *if* $16 \mid s$; *period* 16.

*(iii)* $N = 8$: *there are exactly six classes,* $4 \nmid s$ *(zero),* $s \equiv 0 \bmod 32$, $s \equiv 16 \bmod 32$, $s \equiv 8 \bmod 16$, *and* $s \equiv 4 \bmod 8$ *split by* $b \equiv 3$ *or* $1 \bmod 4$; *the period is* $96 = \mathrm{lcm}(32, 48)$, *the* 48 *coming from* $b \bmod 4$ *through* $g_3^{-1}$.

**Status (Category A).** (i), (ii): conceptual, from the CFP and Lemma 22.10 with the 2-adic valuations of $A2 = s\Delta/g^2$ class by class; (ii) also checked exactly in $\mathbb{Z}[\zeta_8]$ for $s < 2000$ (31920 tests). (iii): the invariants (and the factors 32 and 3) are derived by hand; pairwise distinctness is proved in Proposition 22.13.

**Proposition 22.13** (The six $N = 8$ classes are pairwise distinct)**.** *With* $I_1 = \max_c |H(c)|$, $I_2 = \operatorname{supp} H$, $I_3 =$ *phase of* $H(0)$, $I_4 = H(1)/H(0)$ *(for $a = 1$): class $s \equiv 0 \bmod 32$ has* $(16, \textit{full})$; $s \equiv 16 \bmod 32$ *has* $(16, \textit{odd } c)$; $s \equiv 8 \bmod 16$ *has* $(8, \textit{full})$ *with* $I_3 = \mathbf{e}(1/8)$; *the two $s \equiv 4 \bmod 8$ classes have* $(8, \textit{full})$ *with* $H(0) = +8$ *real, and are separated by* $I_4 = +1$ *($b \equiv 3 \bmod 4$) versus* $I_4 = -1$ *($b \equiv 1 \bmod 4$), for all* $a \in \{1, 3, 5, 7\}$.

**Status (Category A).** Residue reduction ($H_{a/8}$ depends only on $s \bmod 16$ and on $u_2, w \bmod 16$), hand derivation of $I_1$–$I_4$ from Lemma 22.10, and an exact finite computation over all residue tuples satisfying each class's constraints; no scan over $s$.

*ea* **tables at $x = a/8$ (Category A).** For $r = 1, \dots, 4$ and $a = 1$: $s \equiv 0 \bmod 32$: $(0, 1 - \sqrt{2}(1+i)/12, 0, -1/64 + 7\sqrt{2}(1+i)/15360)$; $s \equiv 16 \bmod 32$: $(0, -\sqrt{2}(1+i)/12, 0, 7\sqrt{2}(1+i)/15360)$; $s \equiv 8 \bmod 16$: $(0, \sqrt{2}(1+i)/24, 0, -\sqrt{2}(1+i)/30720)$; $s \equiv 4 \bmod 8$: $(\mp 4, \frac{1}{2}, \pm\frac{3}{64}, -\frac{1}{128})$ for $b \equiv 3, 1 \bmod 4$ (independent of $a$); for $a = 3, 5, 7$ the factor $\sqrt{2}(1+i)$ rotates to $\sqrt{2}(-1+i)$, $\sqrt{2}(-1-i)$, $\sqrt{2}(1-i)$.

**Proposition 22.14** ($t^{11/2}$, $t^{13/2}$ at $x = \frac{1}{4}, \frac{1}{6}$, and vanishing)**.** *For $e = 11, 13$ and $x \in \{\frac{1}{4}, \frac{1}{6}\}$, $c_e(s) = \gamma_0 A_e(s) + \sum_r eb_r Y_{e,r}(s) + \sum_r ea_r X_{e,r}(s, g)$ with the universal functions of the $P = 8$, $P = 12$, $K = 13$ runs and the exact class constants (for $x = \frac{1}{4}$, $4 \mid s$: $\gamma_0 = (1+i)/2$, $eb = (0, \frac{1}{3} - \frac{i}{6}, 0, \dots)$; for $x = \frac{1}{6}$: $\gamma_0 = 0$). Moreover:*

*(i)* *$x = \pm\frac{1}{4}$, $4 \nmid s$: $H_s(\pm i\, e^{-t}) = O(t^M)$ for every $M$;*

*(ii)* *$x = \frac{1}{6}$, $s$ odd: $H_s(\mathbf{e}(1/6)e^{-t}) = O(t^M)$ for every $M$.*

**Status (Category A).** Universal form: exact symbolic runs with the exact class data. (i): a sign-reversing shift kills $G(j)$, so $\gamma_0 = eb = 0$, and $ea = 0$ by Proposition 22.12(ii). (ii): by the CFP (which applies equally to $G$ and $\gamma_0$) the Gauss data contain the $x = \frac{1}{2}$ factor, which vanishes for odd $s$ (Proposition 22.3). Exact zeros at $s = 5, 7, 9, \dots$ confirm both.

**Category B.** The independent (floating-point) engine agrees to double precision at $s = 8, 12, 16$ ($x = \frac{1}{4}$) and $s = 8, 16$ ($x = \frac{1}{6}$). 60-digit direct sums at $s = 8$: relative errors $2.0 \cdot 10^{-13}$, $1.7 \cdot 10^{-9}$ ($x = \frac{1}{4}$, also $\frac{3}{4}$) and $8.0 \cdot 10^{-10}$, $8.3 \cdot 10^{-7}$ ($x = \frac{1}{6}$) for $c_{11}$, $c_{13}$. Larger $s$ were not run (they need $\gtrsim 10^7$ terms).

# 23 The cone: origin of $\Delta$, the choice of walls, and $H_s^{\mathrm{axes}}$

## 23.1 The lattice origin of $\Delta = (s-2)(7s-32)$

Let $A = \left(\begin{smallmatrix} 2 & s \\ s & s \end{smallmatrix}\right)$ be the Gram matrix of $Q_2(n, k) = n^2 + snk + \frac{s}{2}k^2$, so $\det A = -s(s-2)$, and let $c_1 = (-1, 1)$, $c_2 = (-4, 1)$ (the choice) and $e_1 = (1, 0)$.

**Proposition 23.1** (Lattice origin of $\Delta$)**.** *(i) $7s - 32 = -2Q(c_2)$. Indeed $c_2 = c_1 - 3e_1$, $B(c_1, e_1) = s - 2$, $Q(e_1) = 1$, and $-2Q(c_2) = (s-2) + 6(s-2) - 18 = 7s - 32$.*

*(ii) $v := (3s, s-8) = JAc_2$ ($J$ the rotation by $90°$) spans $c_2^{\perp}$, $B(c_2, x) = \det(x, v)$, and $u_2 = v/g$ with $g = \gcd(s-8, 3s) = \gcd(24, s-8)$.*

*(iii) $2Q(v) = \det A \cdot 2Q(c_2) = s(s-2)(7s-32)$, so $Q_2(u_2) = s\Delta/(2g^2)$.*

*(iv) For any $w$ with $\det(w, u_2) = 1$, the residual coefficient in the Gauss phase of Proposition 22.9(iii) is $\det A/(4Q(u_2)) = -g^2/(2(7s-32)) = Q(\pi w)$, where $\pi$ is the $B$-orthogonal projection onto $\mathbb{R}c_2$ along $u_2$.*

*(v) $\Delta = 4Q(c_1)Q(c_2) = B(c_1, c_2)^2 + \det \mathrm{Gram}(c_1, c_2) = 16(s-2)^2 - 9s(s-2)$, with $B(c_1, c_2) = -4(s-2)$ and $\det(c_1, c_2)^2 = 9$.*

**Status (Category A).** All identities are symbolic in $s$. The degenerate prime cases of Proposition 22.9(iv) are read off: $p \mid 7s - 32$ means that $c_2$ is isotropic mod $p$ (the residual sum collapses to a delta), and $p \mid s - 2$ is the $c_1$ analogue. No physical interpretation is offered.

### 23.2 The cone choice: what is intrinsic and what is tied to the fixed wall $c_2$

Write $c_a = (-a, 1)$ and $H_s^{(c_1,c)}$ for the cone series with Zwegers-type weight $w = \frac{1}{2}(\operatorname{sgn} B(c_1, v) - \operatorname{sgn} B(c, v))$, so that $H_s$ is $H_s^{(c_1,c_4)}$. The $c_1$-wall is $n = 0$ (edge $u_1 = (0, 1)$) and the $c_2$-wall is the edge $u_2 \propto (3s, s - 8)$.

**Proposition 23.2** (Edge-class invariance)**.** *$M(n, k) = (-n - sk, k)$ is an integral involution of determinant $-1$ with $Q_+(Mv) = Q_+(v)$ (including the linear term), and $Mc_a = c_{s-a}$. For admissible $a$, $H_s^{(c_1,c_a)} = H_s^{(c_1,c_{s-a})}$. Also $-2Q(c_a) = (2a - 1)s - 2a^2$ is invariant under $a \leftrightarrow s - a$, and $4Q(c_1)Q(c_a) = (s - 2)\big((2a - 1)s - 2a^2\big)$.*

**Status (Category A).** Proof: the difference is $\frac{1}{2}\sum f(v)q^{Q_+(v)}$ with $f = \operatorname{sgn} B(c_{s-a}, \cdot) - \operatorname{sgn} B(c_a, \cdot)$ supported in a sector between two positive lines (finitely many terms per degree); $f \circ M = -f$ and $M$ is a $Q_+$-preserving bijection, so the sum vanishes. Symbolic identities are checked by computer algebra, with an exact $q$-series check at $s = 7, 8, 9, 11, 13$ to $q^{800}$ or $q^{600}$.

**Proposition 23.3** ($c_1$ is forced; the axes cone)**.** *(i) If one wall of the cone is the orthant edge $n = 0$, then $c_1 = \pm(-1, 1)$ (the sign being fixed by orientation), up to its $M$-image $c_{s-1}$, which gives the same series. $Q(c_1) = -(s - 2)/2 < 0$ for all $s > 2$.*

*(ii) (Axes cone.) Let $c_2^{\mathrm{ax}}$ be the primitive vector on the ray of $(-s, 2)$, so that $B(c_2^{\mathrm{ax}}, v) \propto -k$ (the $k = 0$ wall); it is $M$-fixed, with $Q(-s, 2) = -s(s - 2) < 0$. Then $w = \frac{1}{2}(\operatorname{sgn} n + \operatorname{sgn} k)$ and $C_{\mathrm{mixed}} \equiv 0$ identically, i.e. $C = C_{\mathrm{axis}} + C_{\mathrm{third}}$.*

*(iii) For any admissible $c_2$, $C_{\mathrm{mixed}}^{(c_2)} = H_s^{(c_1,c_2^{\mathrm{ax}})} - H_s^{(c_1,c_2)}$, a sector sum with finitely many terms per degree. For $c_2 = (-4, 1)$ at $s = 7$ its first term is the wall point $(21, -1)$, $Q_+ = 294$, weight $-\frac{1}{2}$; this is Theorem 11.1(iv).*

**Status (Category A).** (i): $Ac_1 \propto (1, 0)$. (ii): $P - H = C_{\mathrm{axis}} + C_{\mathrm{third}}$ by direct regrouping (the negative quadrant gives $C_{\mathrm{third}}$ since $Q_+(-n, -k) = Q_-(n, k)$); exact check at $s = 5, \ldots, 13$ to $q^{600}$. (iii): first nonzero degree of $C_{\mathrm{mixed}}$ is $27, 294, 375, 242, 189, 168$ for $s = 6, 7, 10, 11, 12, 13$; $\equiv 0$ at $s = 8$ (where $(-4, 1)$ *is* the axes vector); none through $q^{600}$ at $s = 9$. At $s = 7$, any $c_1 \notin \{(-1, 1), (-6, 1)\}$ tested destroys the $q^0$ normalisation of $C$.

**Proposition 23.4** (Status of $c_2 = (-4, 1)$)**.** *$c_2 = (-4, 1)$ is part of the* definition *of $H_s$ (§10.4): it is fixed at $s = 7$ and kept for all $s$. Within the class $c \in c_1 + \mathbb{Z}e_1$ (second coordinate $b = 1$) at $s = 7$: $Q(c_a) < 0$ iff $a \in \{1, \ldots, 6\}$; the closed orthant lies strictly inside the $w = 1$ region iff $a \geq 4$; and the $C_{\mathrm{mixed}}$ vanishing order is maximal (first nonzero term $q^{294}$) exactly for the $M$-class $\{3, 4\}$. Together these give $a = 4$ uniquely among $b = 1$. Among all primitive $(-a, b)$ with $b \leq 6$ it is* not *unique, even up to $M$: the vanishing order grows as $a/b \to s/2$, and the optimum under the $C_{\mathrm{mixed}}$ criterion is the axes vector $(-7, 2)$. For $s \neq 7$ this choice is not forced (the deepest $b = 1$ class is $a \approx s/2$).*

**Status (Category A).** Exact computations, to $q^{1200}$ at $s = 7$.

**Category B.** Reading the three conditions as the "selection rule" behind the original choice is an interpretation.

**Remark 23.5** (What is intrinsic in $\Delta$)**.** Combining Propositions 23.1–23.4 (all Category A):

- the factor $s-2=-2Q(c_1)$ of $\Delta$ is *intrinsic given* $c_1$, and $c_1$ is itself forced once one wall is the orthant edge $n=0$;
- the factor $7s-32=-2Q(c_2)$, and with it every $\sqrt{\Delta}$ term, every $(7s-32)^r$ denominator and every $g$-dependence in the $H_s$ coefficients of §22, is *tied to the choice* $c_2=(-4,1)$; it is invariant only under $a\leftrightarrow s-a$ (Proposition 23.2), and other admissible $c_2$ give $\Delta(a)=(s-2)((2a-1)s-2a^2)$.

These statements concern $H_s$ as defined; they do not change it.

### 23.3 The parallel object $H_s^{\mathrm{axes}}$: exact closed forms through $t^{13/2}$

**Definition 23.6.** $H_s^{\mathrm{axes}}:=H_s^{(c_1,c_2^{\mathrm{ax}})}$, the cone series with $c_1=(-1,1)$ and $c_2^{\mathrm{ax}}=\mathrm{prim}(-s,2)$ (the $k=0$ wall). It is a separate object from $H_s$; by Proposition 23.3(ii), $H_s^{\mathrm{axes}}=P_s^{\mathrm{orth}}-C_{\mathrm{axis}}-C_{\mathrm{third}}$ exactly.

**Theorem 23.7** (Closed forms for $H_s^{\mathrm{axes}}$)**.** *Let $c_r$ be the coefficient of $t^r$ in $H_s^{\mathrm{axes}}(e^{-t})$ ($x=0$), $s\geq 3$. All integer orders vanish, and for $r\in\{-\frac12,\frac12,\dots,\frac{13}{2}\}$,*

$$c_r=\sqrt{\pi}\left[R_r^{\mathrm{ax}}(s)+S_r^{\mathrm{ax}}(s)\sqrt{2s}\right],\qquad R_r^{\mathrm{ax}},S_r^{\mathrm{ax}}\in\mathbb{Q}(s),$$

*with denominators $(s-2)^{r+3/2}$ times small integers, and no dependence on $g$. The first rows are*

| $r$ | $R_r^{\mathrm{ax}}$ | $S_r^{\mathrm{ax}}$ |
|---|---|---|
| $-\frac12$ | $\dfrac{1}{s-2}$ | $-\dfrac{1}{2(s-2)}$ |
| $\frac12$ | $\dfrac{s^2(s-4)}{24(s-2)^2}$ | $-\dfrac{(s-4)(3s-2)}{48(s-2)^2}$ |
| $\frac32$ | $-\dfrac{s^3(s-4)(s^2-4s+8)}{960(s-2)^3}$ | $-\dfrac{15s^4-76s^3+108s^2+96s-128}{3840(s-2)^3}$ |
| $\frac52$ | $\dfrac{s^4(s-4)(5s^4-40s^3+126s^2-184s+128)}{80640(s-2)^4}$ | $-\dfrac{(s-4)(105s^5-730s^4+2732s^3-5208s^2+5696s-2560)}{645120(s-2)^4}$ |

*and the rows $r=\frac72,\dots,\frac{13}{2}$ are listed in the data files of Appendix C. In particular $c_{-1/2}=\sqrt{\pi}(2-\sqrt{2s})/(2(s-2))=-\sqrt{\pi}/(\sqrt{2s}+2)$, and $R_r^{\mathrm{ax}}$ carries the factors $(s-4)$ and $s^{r+3/2}$ for every $r\geq\frac12$.*

**Status (Category A).** Proof: $H^{\mathrm{axes}}=P-C_{\mathrm{axis}}-C_{\mathrm{third}}$; $P$ and $C_{\mathrm{third}}$ are expanded by the proved two-dimensional quadrant Euler–Maclaurin lemma (Proposition 13.1), $C_{\mathrm{axis}}$ has an exact unary expansion, and the area integrals are evaluated exactly (even moments cancel between $P$ and $C_{\mathrm{third}}$). Everything is symbolic in $\sqrt{s/2}$.

**Category B.** *Numerical support only.* 70-digit engine numerics with PSLQ at $s=7,\dots,13$ recognise the same field $\mathbb{Q}(\sqrt{2s})$ and the leading term; held-out direct summation at $s=20$ converges towards $c_{11/2}+c_{13/2}t$.

### 23.4 The universal piece $R_e$ is wall-independent through $t^{13/2}$

For $H_s$ write, as in Theorem 22.1(ii) and Propositions 22.6, 22.7, $H_r=\sqrt{2\pi s}\,[R_r(s)+S_r(s,g)\sqrt{\Delta}]$.

**Proposition 23.8** (Wall-independence of the universal piece)**.** *For every $r\in\{-\frac12,\frac12,\dots,\frac{13}{2}\}$, $R_r(s)=S_r^{\mathrm{ax}}(s)$ identically in $s$. Equivalently, the $\sqrt{2\pi s}\cdot\mathbb{Q}(s)$ part of the radial expansion is the same for the cone of $H_s$ and for the axes cone through $t^{13/2}$, while $R_r\neq R_r^{\mathrm{ax}}$.*

**Status (Category A).** Exact rational-function identities: $r \le \frac{5}{2}$ by comparison with Theorem 22.1(ii); $r = \frac{7}{2}, \dots, \frac{13}{2}$ by exact comparison (difference 0 at $e = 7, 9, 11, 13$). *Open:* the identity at general order. Comparison at $t^{15/2}$, $t^{17/2}$ needs the axes forms at $K \ge 15$, which were not computed.

**Category B.** *Interpretation:* the $c_1$ wall produces the universal piece and the $c_2$ wall the remainder ($S_r\sqrt{\Delta}$ for the cone of $H_s$, $R_r^{\mathrm{ax}}$ for the axes cone).

# 24 The structure of $C_{\mathrm{third}}$

## 24.1 One coset, one cone weight, one kernel

**Theorem 24.1** ($C_{\mathrm{third}}$: one coset, one cone weight, one kernel, two chains)**.** *Let $F = [e_1, \gamma e_1)$, $Q_1$ the open first quadrant, $F' = F \setminus Q_1$, and $E_\mu^X = \sum_{\nu \in \mu + \mathbb{Z}^2,\, \nu \in X} q^{Q(\nu)}$. (i) For every $s \ge 5$, $2C_{\mathrm{third},s} - 1 + \theta_3 = q^{c_0}\Theta_w(a)$ with $\Theta_w(a) = \sum_{\nu \in a + \mathbb{Z}^2} w(\nu) q^{Q(\nu)}$ and $w = 1_{F'} + 1_{-Q_1}$: one coset, one quadratic form, one non-negative piecewise-constant cone weight. (ii)* Reflection lemma. *With $R_s = \begin{pmatrix} 1 & s \\ 0 & -1 \end{pmatrix}$ ($R_s^{\mathsf{T}} A_s R_s = A_s$, $R_s a = -a$, $R_s F$ again a $\gamma$-fundamental cone): $E_a^F = E_{-a}^F$ for every $s \ge 5$. Hence $\Theta_w(a) = E_L + \vartheta_{\mathrm{hol}}$: the part of $w$ that is even under $\nu \mapsto -\nu$ gives $E_L$, the odd part gives Zwegers' $\vartheta_{\mathrm{hol}}$, and the would-be weight-1 piece on the unit cone is exactly 0. With $1_K = \frac{1}{4}(\rho_{\mathrm{odd}} + \rho_{\mathrm{even}})$, the parity of the cone weight, the radial parity at $x = 0$ and the automorphy type coincide. (iii)* One kernel, two chains. *On the path of lines $c(t)$ in the signature-$(1,1)$ plane let $\Xi_0 = \sqrt{y}\sum_{a+\mathbb{Z}^2} q^{Q(n)} e^{-\pi y B(n,c(t))^2}$ and $\Xi_1 = \sqrt{y}\sum_{a+\mathbb{Z}^2} 2B(n, c'(t)) q^{Q(n)} e^{-\pi y B(n,c(t))^2}$, and $\Xi = \Xi_0 + \Xi_1$. On the chain $\Gamma_{Q_1}$, $\int \Xi_1\, dt$ is exactly Zwegers' 2002 completion, with holomorphic part $\vartheta_{\mathrm{hol}}$. On the unit-cone chain $\Gamma_F = [c_1, \gamma c_1]$, $\int \Xi_0\, dt$ is Zwegers' 2012 mock Maass theta function $\Theta_a$, which here has no shadow and has $E_L = \frac{1}{2}\vartheta_a$; and $\int_{\Gamma_F} \Xi_1\, dt = 0$. (iv) The two outputs come from different operations (holomorphic projection at weight 1; the Lewis–Zagier period correspondence at weight 0) and carry different automorphy factors ($(c\tau + d)$ versus $|c\tau + d|$). So this is a single construction, not a single modular object; Theorems 24.5 and 24.4 still apply.*

**Status (Category A).** (i) proved (elementary, from $2P_s^{\mathrm{orth}} = E_s + 1 + \theta_3 - \Theta_{\mathrm{hol}}$ and $Q_1 \subset F$); checked to $q^{3000}$ at $s = 7$ and to $q^{2000}$ for $s = 5..25, 41$. (ii) proved (symbolic checks of $\gamma a - a$, $R_s a + a$, $R_s^{\mathsf{T}} A_s R_s - A_s$, $R_s \gamma e_1 = \gamma^{-1} e_1$); identically zero to $q^{3000}$ at $s = 7$. The parity/radial dictionary uses Proposition 13.1 and Hecke's theorem. (iii), the weight-1 identity: exact, by the fundamental theorem of calculus with Siegel-majorant domination. (iv): the obstruction of Theorems 24.5 and 24.4.

**Category B.** The weight-0 identification in (iii) is by citation of Zwegers 2012 [19] through the restatement in [2]; numerical agreement $8 \cdot 10^{-16}$ ($s = 7, 9$), $5 \cdot 10^{-15}$ ($s = 13$). Non-$\gamma$ control: the shadow does not cancel ($\approx 0.1$). "No exotic single kernel exists" is open; higher depth is unavailable for the rank-2 lattice.

**Proposition 24.2** (Unit cone)**.** *(i) On the unit cone, both the holomorphic weight-1 $q$-series and its Zwegers completion vanish identically. (ii) $\int_{\Gamma_F} \Xi_0\, dt = \sqrt{y}\sum_{\gamma\text{-orbits } O \subset a+\mathbb{Z}^2} \mathbf{e}(Q(O)x) K_0(2\pi |Q(O)| y)$, which is Zwegers' mock Maass theta $\Theta_a$.*

**Status (Category A).** (i) is proved by the orbit-count symmetry $N(a, +, n) = N(-a, +, n)$, which comes from the automorph $R_s$, together with a telescoping argument for the completion; $\int_{\Gamma_F} \Xi_1 = 0$ follows. (ii) is proved by orbit unfolding. Modularity is by Siegel/Vignéras, which is standard. *Category B:* that no depth-2 rank-3 object carries $C_{\mathrm{third}}$ (a structural argument, not a classification).

**Lemma Prod1.** Finite sums of products of weight-0 holomorphic forms with logarithmic cocycle and holomorphic weight-1 forms have only $t^{-2}, t^{-1}\log t, t^{-1}$ radial terms, so they cannot

carry either sector. (Category A.) **Lemma Prod2.** The same holds for products involving the Zwegers pieces; it is conditional on the same inputs as Theorem 24.5. (Category B.)

## 24.2 Modularity status of $C_{\text{third}}$

The statements of this subsection are proved or exactly checked, conditional where stated; the words "quantum modular form" for the factor $|cx+d|$ are terminology (Category B).
**Identity.** $2C_{\text{third}} = q^8 f_{2,7,7}(-q^{10}, -q^{14}, q) + 1 + E$, checked to $q^{8000}$, and

$$C_{\text{third}} = \tfrac{1}{2}(1-\theta_3) + \tfrac{1}{2}\,\mathbf{e}(7\tau/20)\big[\vartheta_{\text{hol}} + E_L\big].$$

**Lemma 24.3** (Parity splitting)**.** *At every rational $x$, along $q = \mathbf{e}(x)e^{-t}$, $t \to 0^+$: $\mathbf{e}(7\tau/20)\vartheta_{\text{hol}}$, $\theta_3$ and $q^8 f$ contain only half-integer powers of $t$; $E = q^{7/20}E_L$ contains only integer powers $t^{-1}, t^0, t^1, \dots$. Hence the integer-order radial data of $C_{\text{third}}$ are those of $\frac{1}{2}(1+E)$ and the half-integer ones those of $\frac{1}{2}q^8 f$.*

**Status (Category A).** Proved, conditional on two-dimensional Euler–Maclaurin (Proposition 13.1) and Hecke's continuation theorem.

**Theorem 24.4** (Phase obstruction)**.** *Let $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$ have finite index, $\rho$ any finite-dimensional representation, $j(\gamma,\tau) = (c\tau+d)^k$ or $|c\tau+d|^k$, and $V$ holomorphic with $V(\gamma\tau) = j(\gamma,\tau)\rho(\gamma)V(\tau) + o(1/y)$ at every rational $x \neq \gamma^{-1}\infty$. Then $C_{\text{third}} \neq \lambda \cdot V$ for every $\lambda$; likewise for $C$, $P_s^{\text{orth}}$ and $E$.*

**Status (Category A).** Unconditional. The proof uses $G(1/m) = \mathbf{e}(7/(20m))P(m \bmod 140)/m$ for $\gcd(m,70) = 1$ (which also follows from Corollary 14.2(iii)) and a linear-recurrence/essential-singularity argument.

**Theorem 24.5** (Sign obstruction)**.** *If $V$ is holomorphic and finite-dimensional on $\Gamma \ni S, T^N$, $\rho'(T^N)$ is diagonalisable with unimodular eigenvalues, and $V(\gamma\tau) = (c\tau+d)^k\rho'(\gamma)V(\tau) + o(1/y)$, then none of $E_L$, $X = \frac{1}{2}(\vartheta_{\text{hol}} + E_L)$ or the 72-component $E$-vector is $\lambda \cdot V$. Hence $E$ is not the holomorphic part of a weight-1 harmonic Maass form, nor a holomorphic weight-1 form, nor mixed mock with modular factors of weight $< 1$.*

**Status (Category A).** Conditional on the cited analyticity of harmonic-Maass cocycles, which is needed only for the class-membership corollaries; the $|x|$ profile law is Theorem 14.1. Objects whose cocycles are themselves of size $\geq 1/y$ are not excluded (open).

**Theorem 24.6** (Structure of $C_{\text{third}}$)**.** *(i) The decomposition above, with $\vartheta_{\text{hol}}$ a signature-$(1,1)$ Zwegers holomorphic part and $E_L$ the $(a,0)$ component of the 72-dimensional vector. (ii) $E_L$ is the Lewis–Zagier $f$-function of the $\lambda = \frac{1}{4}$ Maass Eisenstein lift; its $|cx+d|$ weight-1 cocycle is real-analytic off $\gamma^{-1}\infty$. (iii) Lemma 24.3. (iv) For $x = h/m$, the renormalised value is $c_C(x) = \frac{1}{2} + \frac{1}{2}\mathbf{e}(7x/20)Z_x(0) - \frac{7}{20}I\,G(x) = \alpha(x) + \beta(x)I$ with $\alpha$ algebraic and $\beta = -\frac{7}{20}G(x)$, which vanishes iff $2 \mid m$ or $5 \mid m$ (Theorem 11.4); the Weil-kernel invariance $G_{a,0}(\gamma x)|cx+1| = G_{a,0}(x)$, $\gamma = \left(\begin{smallmatrix} 1 & 0 \\ 140 & 1 \end{smallmatrix}\right)$, is Corollary 14.2(ii). For example $c_C(0) = \frac{11}{20} - \frac{7}{20}I$, $c_C(1/2) = 0$, $c_C(1/4) = 2 + \frac{3}{2}i$. The cusp-form $L(0)$ values do not enter.*

**Status (Category A).** For (i), (iii) and the formula in (iv); the general genus formula behind $Z_x(0)$ is Category B.

**Verdicts.** $C_{\text{third}}$ is not modular (Theorem 11.5(i) (Baker), Theorem 11.5(ii) and Theorem 24.4); not mock or mixed mock with a holomorphic factor and an $o(1/y)$ cocycle (Theorems 24.5 and 24.4); has no strict quantum-modular values (radial divergence at all 490 rationals with $m \leq 40$; "every rational" is Category B); and is renormalised quantum modular after removing $q^{7/20}$ and the theta pieces.

# 25 Dependence on $s$: the $\zeta$-curve, $g_s$, and the cases $s = 8$, $s = 11$

## 25.1 $\zeta_E(0;s)$ and $\zeta_E(-1;s)$ on one rational curve

**Theorem 25.1** (Rational $\zeta$-curve). *For every integer $s \geq 3$, with $c_0 = s/(4(s-2))$ and $\zeta_E(w;s) = \sum_m e_m(m-c_0)^{-w}$:*

$$\zeta_E(0;s) = \frac{(s-4)(s-5)}{12(s-2)}, \qquad \zeta_E(-1;s) = \frac{2s^5 - 14s^4 - 45s^3 + 425s^2 - 632s - 96}{1440(s-2)^2},$$

*the $t^{-1}$ coefficient of $E_s(e^{-t})e^{c_0 t}$ is $\operatorname{arccosh}(s-1)/\sqrt{s(s-2)}$, and the $t^{-1/2}$ and $t^{1/2}$ coefficients vanish.*

**Status (Category A).** All algebraic steps are symbolic in $s$: in the $\gamma$-adapted basis $Q_2 = S^2 + 2(s-1)ST + T^2$; $E_s$ is the sum over one half-open cone with exactly two lattice classes $(\frac{1}{2}+\delta, 1-\delta)$, $(\delta, \frac{1}{2}-\delta)$, $\delta = 1/(2(s-2))$, via the involution $\varphi(n,k) = (n, -2n-k-1)$ (Remark 25.3(a)). Analytic inputs: Proposition 13.1 and the Mellin dictionary (cf. Zagier, Shintani). Cross-checked against 42 exact cone values ($s = 3, \ldots, 20, 25, 30, 41$); the key steps (shift identity, $\gamma$-invariance, the involution, checked by brute force for $s = 3..30$, the per-class $t^0$ coefficient, both closed forms and the vanishing of the $t^{\pm 1/2}$ sums) were re-derived with independent code.

*Normalisation note.* The ray-class value $\zeta(-1, \mathcal{K}) = 29/60$ and $\zeta_E(-1;7) = 29/1200$ differ only by normalisation: since $\mathrm{N}\mathfrak{a} = 20Q$, $\sum_{\mathfrak{a}\in\mathcal{K}} \mathrm{N}\mathfrak{a}^{-w} = 20^{-w}\zeta_E(w;7)$, and at $w = -1$ this gives $29/60 = 20 \cdot 29/1200$. Likewise $5/12 = 3 \cdot 5/36$ ($\mathrm{N}\mathfrak{a} = 3Q$, $s = 8$) and $1129/36 = 36 \cdot 1129/1296$ ($\mathrm{N}\mathfrak{a} = 36Q$, $s = 11$).

## 25.2 Radial expansion of $g_s$

Let $g_s(q) = \sum_{n\geq 0} q^{n^2}/(-q^s;q^s)_n$.

**Theorem 25.2** (Radial expansion of $g_s$). *For every real $s > 0$, $g_s(e^{-t}) \sim \sum_{k\geq 0} c_k(s)t^k$ as $t \to 0^+$, with $c_k \in \mathbb{Q}[s]$ of exact degree $k$ and no $t^{-1}$ or half-integer terms. In particular $c_0 = 2$, $c_1 = 2(s-3)$, $c_2 = 5s^2 - 44s + 75$, $c_3 = (55s^3 - 930s^2 + 3918s - 4683)/3$, $c_4 = (1073s^4 - 28864s^3 + 214620s^2 - 593128s + 545835)/12$. Consequently $\zeta_g(0) = 1$, $\zeta_g(-1) = 6 - 2s$ and $\zeta_g(-k) = (-1)^k k!\, c_k(s)$.*

**Status (Category A).** Elementary proof: write $1/(-q^s;q^s)_n = 2^{-n}\exp\sum_{j\leq n}\Lambda(sjt)$, expand with Faulhaber polynomials, sum against $2^{-n}$, and bound the remainder by splitting at $N = t^{-1/3}$. Confirmed on held-out $s$; $c_0, \ldots, c_4$ re-derived with independent code (exact match), and $g_s(e^{-t}) - \sum_{k\leq 4} c_k t^k = O(t^5)$ checked numerically at $s = 3, 7.5$. The Dirichlet-series reading of $\zeta_g$ is Category B.

**$P_s^{\mathrm{orth}}$ and $\Theta_{\mathrm{hol}}$ for all $s \geq 3$** (Category A, by Proposition 13.1), with Res $= \operatorname{arccosh}(s-1)/\sqrt{s(s-2)}$: $P_s^{\mathrm{orth}}$ has $t^{-1}$ coefficient Res$/2$ and $t^{-1/2}$ coefficient $\sqrt{\pi}(s^2 - 2\sqrt{2s})/(4s(s-2))$; $\Theta_{\mathrm{hol}}$ has $t^{-1/2}$ coefficient $\sqrt{\pi}(s^2/2 - 2s + \sqrt{2s})/(s(s-2))$ and no integer-order terms through $t^1$. Thus Lemma 24.3 holds at $x = 0$ as an identity in $s$. $g_s$ is not in $\operatorname{span}_{\mathbb{Q}}\{1, \theta_3, R, M_m\}$ (exact rank to $q^{400}$, $s = 3..20, 25, 30, 41$). $H_s$ is Zwegers-admissible only for integer $s \geq 5$. The higher radial coefficients of $H_s$ are given in §22.

## 25.3 The cases $s = 8$ and $s = 11$

- $2P_s^{\mathrm{orth}} = E_s + 1 + \theta_3 - \Theta_{\mathrm{hol},s}$ holds exactly as a $q$-series to $q^{3000}$ for $s = 7, 8, 11$ (Category A).

- Fields and orders: $s = 8$: $\mathbb{Q}(\sqrt{3})$, order $\mathbb{Z}[2\sqrt{3}]$. The maximal order has fundamental unit $2 + \sqrt{3} \notin \mathbb{Z}[2\sqrt{3}]$; the fundamental unit of the order is $7 + 4\sqrt{3} = (2 + \sqrt{3})^2$, which is the $\gamma$-eigenvalue. Multiplicity one (no double counting). $s = 11$: $\mathbb{Q}(\sqrt{11})$, order $\mathbb{Z}[3\sqrt{11}]$, $\varepsilon = 10 + 3\sqrt{11}$. $E_s$ is a single-class ray-class series for each $s$.

- $\zeta_E(0)$, $\zeta_E(-1)$: $1/10, 29/1200$ ($s = 7$); $1/6, 5/36$ ($s = 8$); $7/18, 1129/1296$ ($s = 11$), by Shintani cones (exact) and by PARI Hecke $L$-functions. The $t^{\pm 1/2}$ coefficients of $E_s$ vanish; the Baker obstruction carries over.

- $S/T$ family: $S$ unitary, $S^2 = P$, $(ST)^3 = I$ for all three $s$ (exact). $s = 8$: even lattice, whole family Eisenstein, $\psi = \psi_{\mathrm{Eis}}$, Herglotz form to $3 \cdot 10^{-13}$ (Category B). $s = 11$: $u_L$ has a genuine dihedral cusp component (6 Maass forms with $\lambda = \frac{1}{4}$ from 12 non-genus characters of conductor (3) and (9); exact ray-class test, Proposition 21.1); $\psi = \psi_{\mathrm{Eis}} + \psi_{\mathrm{cusp}}$ with $\psi_{\mathrm{cusp}} \neq 0$ at 8 rationals (floating, Category B). Exactly, $\dim V = 110$, $d_{\mathrm{Eis}} = 90$ and $d_{\mathrm{cusp}} = 20$ (Theorem 20.1). The count "$\geq 90$ dihedral newforms" from conductor norms is not a dimension: as at $s = 7$, the forms occur only in fixed combinations.

- Genus: at $s = 11$ there are 4 genera ($2^{\mu-1}$, $\mu = 3$) and $\mathrm{Cl}^+ \cong \mathbb{Z}/4 \times \mathbb{Z}/2$, so $h^+ = 8 >$ #genera.

**Class numbers versus genera** (Category A for the counts). With $D_s$ the discriminant of the order, $h^+$ equals the number of genera for every $s \in \{3, \ldots, 20, 30\}$ except $s = 11$ (8 vs 4) and $s = 19$ (8 vs 4); also $s = 25$ (12 vs 4) and $s = 41$ (24 vs 8).

**The rule "cusp part $\Leftrightarrow$ $h^+ \neq$ #genera" is false at both levels.** For the family span, $d_{\mathrm{cusp}} > 0$ for every odd $s$ in 5..51, including $s = 5, 7, 13, 15$ where $h^+ =$ #genera (Theorem 20.1). For $u_L$ it fails at $s = 14, 17, 18, 22, 23$ (Proposition 21.1). Both notions are invisible in the radial data at $x = 0$, which are $s$-uniform (§§25.1–25.2).

## 25.4 Structural remarks

**Remark 25.3.**

**(a) One improper automorph (Category A).**
The involution $\varphi(n, k) = (n, -2n - k - 1)$ used in the proof of Theorem 25.1 equals $-\gamma R_s$, and $R_s a = -a$. So $\varphi$, the reflection of Theorem 24.1(ii) and $H_5 \equiv 0$ come from a single symmetry. $R_s c_1 = -c_2$ holds only at $s = 5$, which is why $H_5 \equiv 0$ and $Q(c_1) = Q(c_2)$ there. Adding $R_s$ to $\langle \gamma, -I \rangle$ gives no new label relations for any $s = 5, \ldots, 25$ (exact orbit counts).

**(b) The factor 20 (identity Category A; its meaning Category B).**
The denominator of $c_0$ is $4(s-2)/\gcd(s, 8) = 20, 3, 36$ at $s = 7, 8, 11$, which is the ideal-norm factor $c_s$ in $\mathrm{N}\mathfrak{a} = c_s Q$. For odd $s$ it equals $-B(c_1, c_2) = -8Q(c_1) = 4(s-2)$ of the $H_s$ cone (at $s = 8$ the two differ: 3 versus 24); equivalently $Q_s = (\mathrm{N}\mathfrak{a} + s)/(4(s-2))$ for odd $s$.

**(c) Residue and Pell unit (Category A).**
$\mathrm{Res} = \operatorname{arccosh}(s-1)/\sqrt{n_s} = \log \varepsilon_s / \sqrt{n_s}$ with $\varepsilon_s = (s-1) + \sqrt{n_s}$, the fundamental solution of $x^2 - n_s y^2 = 1$. For even $s$ this is the fundamental unit of the order, not of the maximal order: at $s = 8$, $\varepsilon_8 = 7 + 4\sqrt{3}$.

**(d) $\Delta$ and the $s = 8$ scale (Category A).**
$\Delta = (s-2)(7s-32) = 4Q(c_1)Q(c_2) = 16(s-2)^2 - 9n_s$; $B(c_1, c_2)^2 - 4Q(c_1)Q(c_2) = 9n_s$; $\det(c_1, c_2) = 3$; $\gcd(7s-32, s-8) = g$ and $\gcd(7s-32, s-2) = \gcd(18, s-2)$. $\mathbb{Q}(\sqrt{\Delta}) \neq \mathbb{Q}(\sqrt{n_s})$ for every $s < 10^6$ (exact search), so the $H_s$ half-integer data live in a different quadratic field

from the $E$-residue except where $\Delta$ is a square. At $s = 8$ all the listed half-integer coefficients lie in $\sqrt{\pi}\,\mathbb{Q}$ (Theorem 22.1(v)).

**(e) Three parities are one (algebra Category A; same-$\rho$ identification Category B).** The label parity $P = S_{\mathrm{lab}}^2$, the cone-weight parity $\nu \mapsto -\nu$ and the integer versus half-integer radial orders coincide. A weight-$k$ vector-valued form satisfies $\rho(S)^2 f = (-1)^k f$, so the $E$/Eisenstein sector lies in $W_P$ and the Zwegers/$\vartheta_{\mathrm{hol}}$ sector in $\mathrm{Fix}(-P)$ (dimension $68 = 4(3 \cdot 3 + 2 \cdot 4)$ at $s = 7$). Hence weight-1 vectors cannot realise any part of $V^{\perp} \subset W_P$. Category B: that Zwegers' completed theta transforms under this same 140-dimensional $\rho$ with the $S_{\mathrm{lab}} = \rho(S)^{-1}$ sign.

**(f) The constants $a_H$, $a_C$ are closed forms (Category A).** $a_H = H_{-1/2}(7)$ and $a_C = C_{-1/2}(7)$ symbolically; the $t^{-1/2}$ coefficient of $C_{\mathrm{third}}$ at $x = 0$, $-\sqrt{\pi}(49 - 2\sqrt{14})/140$, is consistent with the all-$s$ formulas of Appendix 25.2.

# 26 K3 lattice realisation of the 140-label representation

This section uses only the label representation of Theorem 11.6 at $s = 7$; nothing in it concerns $H_s$. *Category A (cited)* means proved conditional only on the published theorems named, which we use as stated in the cited sources and do not re-prove.

**Notation.** Throughout, $A = \left(\begin{smallmatrix} 2 & 7 \\ 7 & 7 \end{smallmatrix}\right)$ is the Gram matrix of $Q_7$ (§23.1), $B(u, v) = u^{\mathsf{T}} A v$, and a lattice is written with its *norm* $x \cdot x$ (so $Q_2 = \frac{1}{2} x \cdot x$). We write $N_1$, $N_2$ (italic) for two even lattices.

## 26.1 The even lattice $N_2$ and its discriminant form

The form $(\mathbb{Z}^2, A)$ is odd ($Q_7 \notin \mathbb{Z}$), so it is not itself a K3 lattice. Its even sublattice of index 2 is

$$N_2 := \{x \in \mathbb{Z}^2 : \ x_2 \equiv 0 \bmod 2\}, \qquad \mathrm{Gram}\big((1,0),(0,2)\big) = \begin{pmatrix} 2 & 14 \\ 14 & 28 \end{pmatrix}, \qquad \det = -140.$$

The other natural even version is $N_1 := (\mathbb{Z}^2, 2A)$, with Gram matrix $\left(\begin{smallmatrix} 4 & 14 \\ 14 & 14 \end{smallmatrix}\right)$, also of determinant $-140$.

**Proposition 26.1** (Discriminant form of $N_2$). *(i) In the basis $(1,0), (-7,2)$ of $N_2$ (base change $P = \left(\begin{smallmatrix} 1 & -7 \\ 0 & 1 \end{smallmatrix}\right)$ from $(1,0),(0,2)$), $N_2 \cong \langle 2 \rangle \oplus \langle -70 \rangle$, i.e. $x \cdot x = 2(x_1 + 7y)^2 - 70y^2$. In particular $N_2$ is even, of signature $(1,1)$.*

*(ii) $A_{N_2} = N_2'/N_2 \cong \mathbb{Z}/2 \times \mathbb{Z}/70$, $|A_{N_2}| = 140$, generated by $(\frac{1}{2}, 0)$ and $(0, \frac{1}{70})$ in the $\langle 2 \rangle \oplus \langle -70 \rangle$ basis, with $q_{N_2}(x, y) = \frac{x^2}{4} - \frac{y^2}{140} \bmod 1$.*

*(iii) The local Jordan components and their Gauss-sum signatures are:*

| $p$ | *group* | $q$ *on the $p$-part* | *signature mod* 8 |
|---|---|---|---|
| 2 | $(\mathbb{Z}/2)^2$, *generators* $(\frac{1}{2}, 0), (0, \frac{1}{2})$ | $\langle \frac{1}{4} \rangle \oplus \langle \frac{1}{4} \rangle$, $q = (x^2 + y^2)/4$ | 2 |
| 5 | $\mathbb{Z}/5$ | $\langle \frac{3}{5} \rangle \cong \langle \frac{2}{5} \rangle$ | 4 |
| 7 | $\mathbb{Z}/7$ | $\langle \frac{2}{7} \rangle \cong \langle \frac{1}{7} \rangle$ | 2 |

*The local signatures sum to $2 + 4 + 2 \equiv 0 = \mathrm{sign}(N_2) \bmod 8$ (oddity formula). The 5- and 7-parts are exactly the local forms $q_5 = 2x^2/5$ and $q_7 = x^2/7$ of Theorem 19.3.*

*Proof.* (i) $(1,0)\cdot(1,0)=2$, $(-7,2)\cdot(-7,2)=98-196+28=-70$, $(1,0)\cdot(-7,2)=-14+14=0$, and $\det P=1$. (ii) is read off from (i): $\langle 2\rangle'=\frac{1}{2}\mathbb{Z}$ with $q=\frac{1}{2}\cdot 2\cdot\frac{1}{4}$, $\langle -70\rangle'=\frac{1}{70}\mathbb{Z}$ with $q=-\frac{1}{140}$. (iii) The 2-part is generated by $(\frac{1}{2},0)$ and $(0,\frac{35}{70})$, with $q(0,\frac{1}{2})=-35^2/140\equiv\frac{1}{4}$; the 5-part by $(0,\frac{1}{5})$, $q=-14^2/140\equiv\frac{3}{5}$; the 7-part by $(0,\frac{1}{7})$, $q=-10^2/140\equiv\frac{2}{7}$. The isomorphisms hold because $3\cdot 2^{-1}\equiv 4$ is a square mod 5 and $2\equiv 3^2$ mod 7. The signatures are the standard Gauss-sum values. □

**Status (Category A).** Elementary; exact.

### 26.2 The 140-label representation is the Weil representation of $N_2$

**Conventions.** For an even lattice $L$ we use Borcherds' convention [3]: $\rho_L(T)\mathbf{e}_\gamma=\mathbf{e}(q(\gamma))\mathbf{e}_\gamma$ and $\rho_L(S)\mathbf{e}_\gamma=\mathbf{e}(-\mathrm{sign}(L)/8)\,|A_L|^{-1/2}\sum_\delta\mathbf{e}(-b(\gamma,\delta))\mathbf{e}_\delta$; the dual $\rho_L^*$ is the complex conjugate. Since $\mathrm{sign}(N_2)=0$, $\rho_{N_2}$ is an honest representation of $\mathrm{SL}_2(\mathbb{Z})$ and $\mathbf{e}(\mathrm{sign}/8)=1$. On the label side, the labels are $(\alpha,\beta)$ with $\alpha\in(L'\cup(a+L'))/\mathbb{Z}^2$ (70 classes), $\beta\in\{0,\beta_1\}$, $a=(\frac{7}{10},-\frac{1}{5})$ and $\beta_1$ the stored label of $-a$ mod $\mathbb{Z}^2$; the operators $S_{\mathrm{lab}}$, $T_{\mathrm{lab}}$ are built in the convention of Remark 19.2 (with exact rational phases, all denominators dividing 140). Following Remark 19.1, the label representation in standard normalisation is

$$\rho_{\mathrm{lab}}:\quad S\mapsto S_{\mathrm{lab}}^{-1},\qquad T\mapsto T_{\mathrm{lab}}.$$

**Theorem 26.2** (Weil realisation of the label representation)**.** *(i) $\rho_{\mathrm{lab}}\cong\rho_{N_2}$ as representations of $\mathrm{SL}_2(\mathbb{Z})$. No character twist is needed and no metaplectic sign arises.*

*(ii) An explicit intertwiner is the following. Index the basis of $\rho_{N_2}$ by $z\in N_2'/N_2$, written in $\mathbb{Z}^2$-coordinates, and let $[z]=z$ mod $\mathbb{Z}^2$. Every class $\alpha$ has exactly two lifts $z$, $z+(0,1)$ in $N_2'/N_2$. Put*

$$X\,\mathbf{e}_z\;=\;u_{([z],0)}\;+\;\mathbf{e}\big(B(z,\beta_1)\big)\,u_{([z],\beta_1)}.$$

*Then $X\rho_{N_2}(S)=S_{\mathrm{lab}}^{-1}X$, $X\rho_{N_2}(T)=T_{\mathrm{lab}}X$, and $X$ is invertible.*

*(iii) $\rho_{\mathrm{lab}}\ncong\rho_{N_2}^*$, $\rho_{\mathrm{lab}}\ncong\rho_{N_1}$ and $\rho_{\mathrm{lab}}\ncong\rho_{N_1}^*$.*

*Proof.* (ii) All entries of $\sqrt{140}\,X\rho_{N_2}(S)$, $\sqrt{140}\,S_{\mathrm{lab}}^{-1}X$ (note $\sqrt{140}=2\sqrt{35}$ and $\sqrt{35}\,S_{\mathrm{lab}}^{-1}$ has entries $\mathbf{e}(\cdot)$), $X\rho_{N_2}(T)$ and $T_{\mathrm{lab}}X$ lie in $\mathbb{Z}[\zeta_{280}]$. Each of the $2\cdot 140^2$ entry identities is the vanishing of an explicit integer combination of powers of $\zeta_{280}$; it was checked by exact reduction modulo the cyclotomic polynomial $\Phi_{280}$, with 0 failures for $S$ and 0 for $T$. $X$ is block diagonal with respect to $\alpha$; the block of the two lifts $z,z'=z+(0,1)$ is $\left(\begin{smallmatrix}1 & \\ & \mathbf{e}(B(z,\beta_1))\end{smallmatrix}\ \begin{smallmatrix}1 & \\ & \mathbf{e}(B(z',\beta_1))\end{smallmatrix}\right)$, with determinant $\mathbf{e}(B(z,\beta_1))\big(\mathbf{e}(B((0,1),\beta_1))-1\big)\neq 0$ because $B((0,1),\beta_1)=7(-\frac{7}{10})+7\cdot\frac{1}{5}\equiv\frac{1}{2}$ mod 1. (i) follows from (ii); $X$ intertwines the untwisted $\rho_{N_2}$, and $\mathrm{sign}(N_2)=0$. (iii) $T_{\mathrm{lab}}$ is monomial; its label permutation has 70 fixed points and 35 two-cycles, and its eigenvalue multiset (36 distinct values) equals that of $\rho_{N_2}(T)$ but differs (exactly, as multisets of elements of $\mathbb{Q}/\mathbb{Z}$) from those of $\rho_{N_2}^*(T)$, $\rho_{N_1}(T)$ and $\rho_{N_1}^*(T)$. Conjugate representations have conjugate $T$-spectra. □

**Status (Category A).** Exact computation in $\mathbb{Z}[\zeta_{280}]$ (0 failures among the $2\cdot 140^2$ entry identities; $B(v,\beta_1)\equiv\frac{1}{2}$; $X$ invertible) and exact $T$-spectrum comparison. A second exact solution, with offsets $\{0,\frac{4}{5}\}$, is another element of the Hom space of Proposition 26.3. Further $T$-spectrum tests: $2\langle 5,-7\rangle$, $2\langle -5,7\rangle$, $2\left(\begin{smallmatrix}3 & 1\\ 1 & -12\end{smallmatrix}\right)$ and its negative do not match; $2\langle 1,-35\rangle$ and $2\left(\begin{smallmatrix}1 & 1\\ 1 & -34\end{smallmatrix}\right)$ match only because both are isometric to $N_2$, and their negatives match the dual.

**Proposition 26.3** (Relations and Hom spaces modulo $\ell$)**.** *Let $\ell = 840841 \equiv 1 \bmod 280$. Computing modulo $\ell$ (with a fixed primitive 280-th root of unity in $\mathbb{F}_\ell$):*

*(i)* $(S_{\mathrm{lab}}^{-1}, T_{\mathrm{lab}})$ *satisfies* $S^4 = I$ *and* $(ST)^3 = S^2$*, as* $\rho_{N_2}$ *does;* $(S_{\mathrm{lab}}, T_{\mathrm{lab}})$ *satisfies* $S^4 = I$ *but* fails $(ST)^3 = S^2$*, so it is not a representation of* $\mathrm{SL}_2(\mathbb{Z})$ *in this normalisation (confirming Remark 19.1).*

*(ii)* $\dim_{\mathbb{F}_\ell} \mathrm{Hom}(\rho_{N_2}, (S_{\mathrm{lab}}^{-1}, T_{\mathrm{lab}})) = 8$*, and a random element has rank* 140*.*

*(iii)* $\dim_{\mathbb{F}_\ell} \mathrm{Hom}(\rho_{N_2}, (S_{\mathrm{lab}}, T_{\mathrm{lab}})) = 4$*, and a random element has rank* 72*: the two normalisations agree only on the* 72*-dimensional subspace* $W_P$*.*

**Status (Category A).** Exact computation modulo $\ell$. The isomorphism itself is Theorem 26.2. By semicontinuity the characteristic-0 Hom dimensions are at most 8 and 4. *Category B (reading):* the value 8 is consistent with eight pairwise non-isomorphic irreducible sectors $W_5^{\pm} \otimes W_7^{\pm} \otimes \{A_3, B_1\}$ (§19); that the characteristic-0 dimension equals 8 is not separately proved here.

**Remark 26.4** (Dual, twist, metaplectic sign)**.** $\rho_{\mathrm{lab}} \not\cong \rho_{N_2}^*$ (Theorem 26.2(iii)); no twist by a character of $\mathrm{SL}_2(\mathbb{Z})$ is needed; $\mathrm{sign}(N_2) = 0$, so there is no metaplectic sign. The Weil representation depends only on the discriminant form, so any even lattice with discriminant form $\cong q_{N_2}$ and signature $\equiv 0 \bmod 8$ (e.g. $N_2 \oplus U$, $N_2 \oplus E_8$, and the lattice $T$ of Proposition 26.5, up to dualisation) gives the same representation.

### 26.3 The transcendental lattice

Let $\Lambda_{K3} = U^3 \oplus E_8(-1)^2$ (even unimodular, signature $(3, 19)$), and let $U \oplus U \subset \Lambda_{K3}$ have bases $e_1, f_1, e_2, f_2$ with $e_i \cdot f_i = 1$ and all other products 0.

**Proposition 26.5** (Explicit primitive embedding and $T$)**.** *Put $n_1 = e_1 + f_1$ (norm 2), $n_2 = e_2 - 35 f_2$ (norm $-70$), $t_1 = e_1 - f_1$ (norm $-2$), $t_2 = e_2 + 35 f_2$ (norm 70).*

*(i)* $\mathrm{span}(n_1, n_2) \cong N_2$ *and is primitive in* $U \oplus U$ *(the* $2 \times 2$ *minors of* $(n_1\ n_2)$ *have gcd* 1*).*

*(ii)* $t_1, t_2$ *are orthogonal to* $n_1, n_2$ *and span the full orthogonal complement of* $\mathrm{span}(n_1, n_2)$ *in* $U \oplus U$ *(index* 1*; their Gram determinant is* $-140$*).*

*(iii) Hence the orthogonal complement of* $N_2$ *in* $\Lambda_{K3}$ *is*

$$T := N_2^{\perp} = \langle -2 \rangle \oplus \langle 70 \rangle \oplus U \oplus E_8(-1)^2, \qquad \mathrm{sign}(T) = (2, 18), \qquad A_T \cong \mathbb{Z}/2 \times \mathbb{Z}/70,$$

*with* $q_T(x, y) = -\frac{x^2}{4} + \frac{y^2}{140} = -q_{N_2}(x, y)$ *under the identity identification of generators. The gluing is the one induced by the unimodular overlattice* $U \oplus U$ *(consistent with Nikulin [13], Prop. 1.6.1).*

*Proof.* Direct computation: $n_1 \cdot n_1 = 2$, $n_2 \cdot n_2 = -70$, $n_1 \cdot n_2 = 0$, $t_1 \cdot t_1 = -2$, $t_2 \cdot t_2 = 70$, and $n_i \cdot t_j = 0$. The complement of a rank-2 primitive sublattice of the unimodular $U \oplus U$ has rank 2 and $|\det| = 140$; $\mathrm{span}(t_1, t_2)$ lies in it with $|\det| = 140$, so the index is 1. The other summands $U \oplus E_8(-1)^2$ are orthogonal to $U \oplus U$. The discriminant form of $\langle -2 \rangle \oplus \langle 70 \rangle$ is read off as in Proposition 26.1. □

**Status (Category A).** Elementary; exact. Nikulin's existence theorem is not needed, since the embedding is explicit.

**Corollary 26.6** (The residual law in K3 terms)**.** $\rho_{\mathrm{lab}} \cong \rho_{N_2} \cong \rho_T^* = \overline{\rho_T}$ *(Borcherds' convention; honest* $\mathrm{SL}_2(\mathbb{Z})$*-representations).*

*Proof.* $\mathrm{sign}(T) = -16 \equiv 0 \bmod 8$ and $q_T = -q_{N_2}$ (hence $b_T = -b_{N_2}$) under the identification of Proposition 26.5, so in the formulas for $\rho_T(S)$ and $\rho_T(T)$ every phase is the conjugate of the corresponding phase for $\rho_{N_2}$. Combine with Theorem 26.2. □

**Status (Category A).** Purely lattice-theoretic; no cited geometric theorem is used. In words: the 140-label residual law transforms under the dual Weil representation of a signature-$(2,18)$ lattice, which (Theorem 26.7) is the transcendental lattice of an $N_2$-polarised K3 surface. This is a statement about a shared *representation*; it is not a statement that any residual series is a K3 generating function (see Remark 26.11).

## 26.4 Geometric realisation

**Theorem 26.7** ($N_2$-polarised K3 surfaces)**.** *(i) The primitive embedding* $N_2 \hookrightarrow \Lambda_{K3}$ *is unique up to* $O(\Lambda_{K3})$*; hence* $T$ *is unique up to isometry.*

*(ii) There is a projective K3 surface* $X$ *with* $\mathrm{NS}(X) \cong N_2$ *and* $T_X \cong T$*.*

*(iii)* $N_2$ *represents neither* $-2$ *nor* $0$ *(non-trivially). Hence such an* $X$ *contains no smooth rational curves and admits no elliptic fibration, and its ample cone is the whole component of the positive cone containing an ample class.*

*Proof sketch.* (i) Nikulin [13, Thm. 1.14.4] (p. 126 of the translation): $\Lambda_{K3}$ is even unimodular of signature $(3,19)$, $N_2$ is even of signature $(1,1)$, $3 > 1$, $19 > 1$, and $\mathrm{rk}\,\Lambda_{K3} - \mathrm{rk}\,N_2 = 20 \geq 2 + l(A_{N_2}) = 4$. (ii) Morrison [12, Cor. 2.9(i)] (p. 112; proved there via Cor. 1.9): for $\rho \leq 10$ every even lattice of signature $(1, \rho - 1)$ is the Néron–Severi lattice of some algebraic K3 surface (equivalently, surjectivity of the period map); with (i), $T_X \cong T$. (iii) $x \cdot x = 2\big((x_1 + 7y)^2 - 35y^2\big)$; $-2$ would need $u^2 - 35y^2 = -1$, impossible since $-1$ is not a square mod 7; $0$ would need 35 to be a square. Without $(-2)$-classes the ample cone is a full component of the positive cone, and an elliptic fibration requires a non-zero isotropic class in NS (standard, [9]). □

**Status (Category A). Category A (cited)**: we use Nikulin [13, Thm. 1.14.4] and Morrison [12, Cor. 2.9(i)], and standard facts on ample cones and elliptic fibrations [9]. The arithmetic in (iii) is exact.

## 26.5 Automorphisms

Let $\gamma = \left(\begin{smallmatrix} -1 & -7 \\ 2 & 13 \end{smallmatrix}\right)$, the Pell automorph of Theorem 11.2: $\gamma^{\mathsf{T}} A \gamma = A$, $\det \gamma = 1$, $\mathrm{tr}\,\gamma = 12$, eigenvalues $6 \pm \sqrt{35}$. It preserves $N_2$ (the second coordinate of $\gamma x$ is $2x_1 + 13x_2$). Let $r = \mathrm{diag}(1,-1)$ in the $\langle 2 \rangle \oplus \langle -70 \rangle$ basis; there $\gamma = \left(\begin{smallmatrix} 6 & 35 \\ 1 & 6 \end{smallmatrix}\right)$ (multiplication by $\varepsilon = 6 + \sqrt{35}$).

**Proposition 26.8** (Action on $A_{N_2}$ and $O(N_2)$)**.** *(i) The induced map* $\bar{\gamma}$ *on* $A_{N_2}$ *acts on the local parts as follows: on the 2-part* $(\mathbb{Z}/2)^2$ *as a non-trivial involution (not the identity, which on this part equals* $\pm 1$*); on the 5-part as* $+1$*; on the 7-part as* $-1$*. Hence* $\bar{\gamma} \neq \pm 1$ *and* $\bar{\gamma}^2 = \mathrm{id}$*. This matches the action of* $\gamma$ *in Theorem 19.5(ii) on the 7/5/2 parts.*

*(ii)* $O(N_2) = \{\pm\gamma^k,\ \pm r\gamma^k : k \in \mathbb{Z}\}$*, and* $r\gamma r = \gamma^{-1}$*.*

*(iii) On* $A_{N_2}$*:* $\gamma^{2j}$ *acts as* $+1$ *and* $r\gamma^{2j}$ *as* $-1$*, both preserving the positive cone; odd powers* $\pm\gamma^{2j+1}$*,* $\pm r\gamma^{2j+1}$ *act as neither* $+1$ *nor* $-1$*;* $-\gamma^k$ *and* $-r\gamma^k$ *do not preserve the positive cone.*

*Proof.* (i), (iii): exact computation on the 140 classes. (ii): in the diagonal basis $x\cdot x = 2(u^2-35y^2)$. Proper automorphs of $u^2-35y^2$ correspond to solutions of $t^2-140w^2=4$, i.e. to norm-1 units of $\mathbb{Z}[\sqrt{35}]$, which are $\pm\varepsilon^k$ with $\varepsilon = 6+\sqrt{35}$ fundamental; there is no unit of norm $-1$ ($u^2-35y^2=-1$ is impossible mod 7). Improper automorphs are $r$ times proper ones, and $r\gamma r = \left(\begin{smallmatrix} 6 & -35 \\ -1 & 6 \end{smallmatrix}\right) = \gamma^{-1}$. □

**Status (Category A).** Exact.

**Theorem 26.9** (Automorphisms of $N_2$-polarised K3 surfaces)**.** *Let $X$ be as in Theorem 26.7, with $\mathrm{NS}(X) = N_2$ glued to $T_X = T$ as in Proposition 26.5.*

*(i)* *$\gamma^2\oplus\mathrm{id}_T$ extends to a Hodge isometry of $H^2(X,\mathbb{Z})$ preserving the ample cone, and is induced by a unique $f\in\mathrm{Aut}(X)$. $f$ has infinite order; $f^*$ has spectral radius $(6+\sqrt{35})^2 = 71+12\sqrt{35}$ on $H^2(X,\mathbb{Z})$ (a quadratic unit, not a Salem number of higher degree), so its topological entropy is*

$$h(f) = 2\log(6+\sqrt{35}) = 4\sqrt{35}\,I \approx 4.9558,$$

*where $I = \log(6+\sqrt{35})/(2\sqrt{35})$ is the constant of Theorem 11.3.*

*(ii)* *$r\oplus(-\mathrm{id}_T)$ is induced by a non-symplectic involution $\iota\in\mathrm{Aut}(X)$.*

*(iii)* *If $\mathrm{NS}(X)\cong N_2$ (so that the Hodge isometries of $T_X$ are only $\pm1$, Proposition 26.10(i)), then $\mathrm{Aut}(X) = \langle f,\iota\rangle \cong \mathbb{Z}\rtimes\mathbb{Z}/2$ (the infinite dihedral group $D_\infty$), and $\gamma$ itself does not lift to $\mathrm{Aut}(X)$.*

*Proof sketch.* By Nikulin's gluing criterion [13], Cor. 1.5.2, an isometry $g_{\mathrm{NS}}\oplus g_T$ of $N_2\oplus T$ extends to $H^2(X,\mathbb{Z})$ iff $\bar g_{\mathrm{NS}}$ and $\bar g_T$ agree under the glue isomorphism $A_{N_2}\cong A_T$. By Proposition 26.8, $\bar\gamma^2 = +1$ and $\bar r = -1$, compatible with $\mathrm{id}_T$ and $-\mathrm{id}_T$, which are Hodge isometries. Both preserve the positive cone, hence (Theorem 26.7(iii)) the ample cone. By the strong Torelli theorem (Piatetski-Shapiro–Shafarevich, Burns–Rapoport; e.g. [9], Thm. 7.5.3) they are induced by unique automorphisms $f$, $\iota$; $\iota$ acts by $-1$ on $H^{2,0}$, so it is non-symplectic. The eigenvalues of $f^*$ are $(6\pm\sqrt{35})^2$ on $\mathrm{NS}\otimes\mathbb{R}$ and 1 on $T$, and the entropy is the logarithm of the spectral radius (Gromov–Yomdin [6]). For (iii): every automorphism acts on $T_X$ by $\pm1$, so its restriction to $N_2$ preserves the positive cone and acts on $A_{N_2}$ by $\pm1$; by Proposition 26.8(iii) these restrictions are exactly $\gamma^{2j}$ and $r\gamma^{2j}$, which form $\langle\gamma^2\rangle\rtimes\langle r\rangle\cong D_\infty$; the action on $H^2$ is injective by Torelli. Since $\bar\gamma\neq\pm1$, $\gamma$ does not lift. □

**Status (Category A). Category A (cited)**: we use the strong Torelli theorem, Nikulin's gluing criterion [13, Cor. 1.5.2], the Gromov–Yomdin entropy formula, and Proposition 26.10(i) for (iii). The lattice inputs (Proposition 26.8) are exact. Numerically $2\log(6+\sqrt{35}) = 2\log(11.91608\ldots) = 4.9558\ldots$. The identity $2\log(6+\sqrt{35}) = 4\sqrt{35}\,I$ is definitional; *Category B (reading):* that the regulator driving the Baker obstruction is "the same" as this entropy is an observation about definitions, and no theorem linking $C$ to $X$ is claimed.

**Proposition 26.10** (Lifting $\gamma$)**.** *(i) If $\mathrm{NS}(X)\cong N_2$, then the Hodge isometries of $T_X$ are $\pm1$, and $\gamma$ does not lift to $\mathrm{Aut}(X)$.*

*(ii) If $\gamma$ lifts on some $X$ with $N_2\subset\mathrm{NS}(X)$ primitive and $M = N_2^\perp\subset\mathrm{NS}(X)$, then $M$ carries the 2- and 7-parts of $A_{N_2}$ with $\mathrm{rk}\,M\equiv4\pmod 8$, or the 2- and 5-parts with $\mathrm{rk}\,M\equiv6\pmod 8$. Hence $\rho(X)\geq6$.*

*(iii) Let $M = A_1(-1)^2 \oplus K_7$ with $K_7 = \left(\begin{smallmatrix} -2 & 1 \\ 1 & -4 \end{smallmatrix}\right)$. There is an even rank-6 gluing $N_2 \oplus M \subset L$ and an isometry $g_M$ of $M$ such that $\gamma \oplus g_M \oplus \mathrm{id}_T$ is a Hodge isometry of $H^2(X, \mathbb{Z})$ for every K3 surface $X$ with $\mathrm{NS}(X) \cong L$ (such $X$ exist by [12, Cor. 2.9(i)]). On this rank-6 Noether–Lefschetz locus some $w \circ (\gamma \oplus g_M)$, $w \in W(L)$, is induced by an automorphism, so* Aut *grows there.*

*Proof sketch.* (i) The reduction $\bar{\gamma} \in O(A_{N_2})$ has order 2: it swaps on the 2-part, is $+1$ on the 5-part and $-1$ on the 7-part, so $\bar{\gamma} \neq \pm 1$. The Hodge isometries of $T_X$ form a cyclic group $\mu_m$, with $T_X \otimes \mathbb{Q}$ a $\mathbb{Q}(\zeta_m)$-vector space. If $m > 2$, a prime $p \mid m$ gives, after passing to the kernel of the action on $A_T$, an element of order $p$ without fixed vectors acting trivially on $A_T$. This forces $pA_T = 0$, which is impossible because $A_T \cong \mathbb{Z}/2 \times \mathbb{Z}/70$ is not $p$-elementary. So every automorphism acts on $T_X$ by $\pm 1$, hence on $A_{N_2}$ by $\pm 1$, and $\gamma$ cannot lift. (ii) The glue must carry $\bar{\gamma}$ to an isometry of $A_M$. Milgram's formula, with Gauss-sum signatures $4, 2, 4, 6, 6 \bmod 8$ for the relevant $p$-parts, gives the two congruences. (iii) An exact lattice computation gives the anti-isometric glue and $g_M$ (the swap on $A_1(-1)^2$ and $-1$ on $K_7$) with $\bar{g}_M \circ \phi = \phi \circ \bar{\gamma}$. Because $L$ contains $(-2)$-vectors, $\gamma \oplus g_M \oplus \mathrm{id}$ need not preserve the ample chamber. The strong Torelli theorem [9, Thm. 7.5.3] supplies a Weyl element $w$ for which $w \circ (\gamma \oplus g_M \oplus \mathrm{id})$ is induced by an automorphism. □

**Status (Category A).** (i), (ii): written proofs. (iii): exact lattice computation, together with [12, Cor. 2.9(i)] and the strong Torelli theorem (cited), so it is conditional on these citations. *Open:* whether $w = 1$ can be taken, i.e. whether $\gamma$ itself lifts on this locus.

**Remark 26.11** (Category B and open items for this section)**.** None of the following is a result of this paper.

(a) *Noether–Lefschetz / Heegner series* (Category B). The Noether–Lefschetz/Heegner generating series of the $N_2$-polarised family are vector-valued modular forms of weight $1 + 18/2 = 10$ for $\rho_T$ or $\rho_T^*$ (depending on convention; Borcherds, Kudla–Millson, Maulik–Pandharipande, cited by name only). The residual objects of Theorems 11.6–11.7 are weight-0 Maass-type objects with $\lambda = \frac{1}{4}$. A shared *representation* (Corollary 26.6) is not a shared *form*; no theta-lift or Borcherds-product link has been checked.

(b) *Special members* (open). Whether $\gamma$ itself lifts on the rank-6 locus of Proposition 26.10(iii), or on any $X$ with $\rho(X) \geq 6$.

(c) *Interpretation* (Category B). Any physical or "unification" reading of $X$, $f$ or $\iota$.

(d) *Scope.* Every K3 statement in this paper concerns $N_2$, because the 140-label representation is $\rho_{N_2}$ and not $\rho_{N_1}$ (Theorem 26.2(iii)).

# 27 Uniqueness and normalisation of the vector completion

Throughout this section $s = 7$, $\rho = \rho_{\mathrm{lab}}$ is the 140-label representation ($\rho_{\mathrm{lab}} \cong \rho_{N_2}$, Theorem 26.2), $P = S_{\mathrm{lab}}^2$, and the laws are those of Theorem 11.6: $u(-1/w) = S_{\mathrm{lab}} u(w)$ and $u(w+1) = Tu(w)$. Floating-point checks are Category B and are labelled where they occur.

## 27.1 The adapted basis and the eight sectors

Let $g = (14, 6)/70$ (the generator of $L'/\mathbb{Z}^2$ used in Theorem 19.3), $a_2 = (\frac{1}{2}, 0)$, and put $u'_{x,\varepsilon,\delta} := u_{xg+\varepsilon a_2, -\delta a_2}$ for $x \in \mathbb{Z}/35$ and $\varepsilon, \delta \in \{0, 1\}$, with the phases of the labels in the convention of Remark 19.2.

**Proposition 27.1** (Adapted-basis splitting at $s = 7$)**.** *In the basis $u'_{x,\varepsilon,\delta}$ the* 140 *labels are hit bijectively, and*

$$S' = \Big[\tfrac{1}{\sqrt{35}}\,\mathbf{e}\big(13xy/35\big)\Big]_{x,y} \otimes S_2, \qquad T' = \mathrm{diag}_x\,\mathbf{e}\big(24x^2/35\big) \otimes T_2,$$

*with the same theta-characteristic factors $S_2, T_2$ as in Theorem 19.5. Since $\frac{1}{35} = \frac{3}{5} - \frac{4}{7}$, under $x \leftrightarrow (x \bmod 5, x \bmod 7)$ one has $\mathbf{e}(13xy/35) = \mathbf{e}(4x_5y_5/5)\,\mathbf{e}(4x_7y_7/7)$ and $\mathbf{e}(24x^2/35) = \mathbf{e}(2x_5^2/5)\,\mathbf{e}(2x_7^2/7)$. This is exactly the factorisation $F_5 \otimes F_7$, $T_5 \otimes T_7$ of Theorem 19.3(i).*

**Status (Category A).** Exact check in rational phases: 0 failures over all 4900 $S$-entries and all 140 $T$-entries. The CRT identities are elementary. With the generator $A_7^{-1}e_2$ of Theorem 19.5 one obtains the factors $\mathbf{e}(-2xy/35)$, $\mathbf{e}(-x^2/35)$ instead (Remark 19.6).

Write $W_5^{\pm}$, $W_7^{\pm}$ for the $(\pm1)$-eigenspaces of $x \mapsto -x$ on the $\mathbb{Z}/5$ and $\mathbb{Z}/7$ factors ($\dim W_5^+ = 3$, $\dim W_5^- = 2$, $\dim W_7^+ = 4$, $\dim W_7^- = 3$), and $A_3$, $B_1$ for the 3- and 1-dimensional summands of the 4-dimensional theta-characteristic factor $\rho_2$ (Theorem 19.3(iii)), as in Proposition 19.4. We abbreviate $X^{\pm} = W_5^{\pm} \otimes W_7^{\pm}$; thus $A_3X^+ = A_3 \otimes W_5^+ \otimes W_7^+ = V_{\mathrm{Eis}}$ and $B_1X^- = B_1 \otimes W_5^- \otimes W_7^- = V_{\mathrm{cusp}}$.

**Proposition 27.2** (Sectors, $T$-fixed vectors and the support of $u$)**.** *(i) $\mathbb{C}^{140}$ splits into eight $\rho$-stable sectors $W_5^{\pm} \otimes W_7^{\pm} \otimes \{A_3, B_1\}$:*

| *sector* | *dim* | $P$ | $\dim \mathrm{Fix}(T)$ | *meets $u$* |
|---|---|---|---|---|
| $A_3X^+$ | 36 | $+1$ | 1 | *yes (all* 507 *frequencies)* |
| $B_1X^+$ | 12 | $+1$ | 0 | *no* |
| $A_3X^-$ | 18 | $+1$ | 0 | *no* |
| $B_1X^-$ | 6 | $+1$ | 0 | *yes (*131 *of* 507 *frequencies)* |
| $W_5^+W_7^-A_3$ | 27 | $-1$ | 0 | *no* |
| $W_5^+W_7^-B_1$ | 9 | $-1$ | 0 | *no* |
| $W_5^-W_7^+A_3$ | 24 | $-1$ | 0 | *no* |
| $W_5^-W_7^+B_1$ | 8 | $-1$ | 0 | *no* |

*The dimensions sum to* 140*, and the $P = +1$ sectors give $\dim W_P = 72$.*

*(ii)* $\mathrm{Fix}(T) = \mathbb{C}\,v_0$ *with $v_0 = e_{x=0} \otimes (e_{00} + e_{01}) \in A_3X^+$ (and $Pv_0 = v_0$).*

*(iii) $\rho$ has no nonzero* $\mathrm{SL}_2(\mathbb{Z})$*-invariant vector: $S'v_0 = 35^{-1/2}\sum_y e_y \otimes (e_{00} + e_{10}) \notin \mathbb{C}v_0$.*

*(iv) Projected to the six sectors other than $A_3X^+$ and $B_1X^-$, the positive-frequency coefficients of $u$ vanish at all* 507 *positive frequencies $n \le 40$. The $A_3X^+$ component is nonzero at all* 507*, first at $n = \frac{1}{14}$. The $B_1X^-$ component is nonzero at* 131 *of them, first at $n = \frac{19}{140}$.*

**Status (Category A).** Exact. (i)–(iii): exact linear algebra in the adapted basis. (iv): exact in $\mathbb{Z}[\zeta_{20}]$ (1169 lattice points). That the four $P$-odd sectors carry nothing follows from $Pu = u$. That $u$ lies in $V = A_3X^+ \oplus B_1X^-$ for *all* frequencies is Proposition 19.4; (iv) is an independent exact check up to frequency 40.

## 27.2 Three levels of uniqueness

We separate three levels:

($\kappa$) functions on $\mathbb{Q}$ (radial profiles and constant terms);

($u$) $\lambda = \frac{1}{4}$ Maass vectors and their positive-frequency $q$-series;

($\Phi$) real-analytic weight-1 period functions on $\mathbb{R} \setminus \{0\}$, i.e. the Lewis–Zagier class [10].

In this subsection $\rho$ is the representation in which Corollary 14.2 and Theorem 11.6 are written.

**Proposition 27.3** (The $\kappa$-level gauge). *Let $\mathcal{G}$ be the space of functions $g : \mathbb{Q} \to \mathbb{C}^{140}$ with $g(\gamma x) = |cx + d|\, \rho(\gamma) g(x)$ for all $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and all $x \in \mathbb{Q}$ with $\gamma x \in \mathbb{Q}$. Then $\dim \mathcal{G} = \dim \mathrm{Fix}(T) = 1$. A basis is $g_0(h/m) := m^2\, \mathbf{G}(h/m)$ ($\gcd(h, m) = 1$, $m > 0$), where $\mathbf{G}$ is the profile vector of Corollary 14.2. Moreover $\|g_0(h/m)\| = m\, \|g_0(0)\| = m\sqrt{70}$.*

*Proof. Dimension.* $\mathrm{SL}_2(\mathbb{Z})$ acts transitively on $\mathbb{P}^1(\mathbb{Q})$. So $g$ is determined by $g(0)$, and $g(0)$ must be fixed by the stabiliser of 0, which is $\pm S\langle T\rangle S^{-1}$. On this stabiliser $|c \cdot 0 + d| = 1$. Hence $g(0) \in \rho(S)\big(\mathrm{Fix}(T) \cap \mathrm{Fix}(P)\big) = \rho(S)\mathbb{C}v_0$ (Proposition 27.2(ii)), so $\dim \mathcal{G} \le 1$. Equality holds because $g_0 \neq 0$ lies in $\mathcal{G}$. *Equivariance of $g_0$.* Corollary 14.2(i) gives $\mathbf{G}(\gamma x) = |cx + d|^{-1} \rho(\gamma) \mathbf{G}(x)$. If $x = h/m$ in lowest terms, the denominator of $\gamma x$ is $m' = m|cx + d|$, and $m'^2 = m^2 |cx + d|^2$ turns weight $-1$ into weight $+1$. *Norm.* Choose $\gamma$ with $\gamma(h/m) = 0$; then $|c\, h/m + d| = 1/m$. Corollary 14.2(i) gives $\mathbf{G}(0) = m\, \rho(\gamma) \mathbf{G}(h/m)$, and unitarity gives $\|\mathbf{G}(h/m)\| = \|\mathbf{G}(0)\|/m$. Finally $\mathbf{G}(0) = \sum_\alpha e_{(\alpha,0)}$ has $\|\mathbf{G}(0)\|^2 = 70$. □

**Status (Category A).** Proved, given Theorem 14.1 and Corollary 14.2. Consequence: the only ambiguity at level $\kappa$ is the radial gauge $\mathbb{C}g_0$, which grows like $m$ along $h/m$. Two $\kappa$-level objects with the same law whose difference is $o(m)$ along rationals coincide, and the canonical radial normalisation $q = \mathbf{e}(x)e^{-t}$ removes the gauge. The constant coboundary $\Phi_{v_0}(x) = v_0 - |x| S_{\mathrm{lab}} v_0$ (the analogue of Lewis–Zagier's $1 - z^{-2s}$) solves the homogeneous $\Phi$-level law. It is excluded once the $q^0$ term is prescribed.

**Category B.** The $S$- and $T$-laws for $g_0$ hold to $\le 5 \cdot 10^{-9}$ at 11 rationals; $\|g_0(h/m)\|/(m\|g_0(0)\|) = 1$ to $10^{-11}$; and $ST^{-1}S^{-1} g_0(0) = g_0(0)$ to $4 \cdot 10^{-15}$.

Let $M(\rho)$ be the space of real-analytic $d : \mathbb{H} \to \mathbb{C}^{140}$ of moderate growth at the cusp with $\Delta d = \frac{1}{4} d$, $d(-1/w) = S_{\mathrm{lab}} d(w)$ and $d(w+1) = T d(w)$; such $d$ satisfy $Pd = d$ automatically. Let $S(\rho) \subset M(\rho)$ be the subspace with zero constant term. Write $f_d^+$ for the positive-frequency part of $d$, $a_d$ for the coefficient in its constant term, $u_{\mathrm{lift}} = u$ for the lift of Theorem 11.6, and $c_d(z) = \int_0^{i\infty} [d, R_z^{1/2}]$ with the Green form of Lemma 15.1.

**Theorem 27.4** (Uniqueness of the Maass completion). *(a) $M(\rho) = \mathbb{C}\, u_{\mathrm{lift}} \oplus S(\rho)$.*

*(b) The map $d \mapsto f_d^+$ is injective on $M(\rho)$.*

**Status (Category A).** Proved below. The proof uses Lemma 15.1 (for every real frequency), Lemma 15.2 and the contour argument of §15.4.

**Lemma 27.5** (Estimates for Theorem 27.4(b)). *Let $d \in M(\rho)$. By Theorem 27.4(a) its constant term is $\sqrt{y}\, a_d v_0$. Let $C_d = \sup \|d(w) - \sqrt{\mathrm{Im}\, w}\, a_d v_0\|$ over the standard fundamental domain, which is finite because the nonconstant modes decay. Use the notation of Lemmas 15.1 and 15.2: $R_z(w) = \mathrm{Im}\, w/((w - z)(\bar{w} - z))$ and $\varsigma(z) = \mathrm{sgn}\, \mathrm{Re}\, z$.*

*(i) The Fourier coefficients of every component satisfy* $|c_j(n)| \le K_0(2\pi)^{-1}\big(|a_d|\,|n| + C_d|n|^{1/2}\big)$ *for* $n \neq 0$.

*(ii) Fix* $z \in \mathbb{H}$ *and* $X > 0$, *and let* $Y \ge 2(|z|+X)$. *On the horizontal segment* $\{w = x'+iY : |x' - \operatorname{Re} z| \le X\}$, *the coefficient of* $dw$ *and of* $d\bar w$ *in the form* $[\sqrt{\operatorname{Im} w}, R_z^{1/2}]$ *are bounded in absolute value by* $8/Y$. *So the constant-term contribution of this segment is at most* $16X|a_d|/Y$, *which tends to* $0$ *as* $Y \to \infty$.

*(iii) Near the cusp* $0$, *the pullback by* $w \mapsto -1/w$ *(Lemma 15.2:* $R_{-1/z}(-1/w)^{1/2} = \varsigma(z)\, z\, R_z(w)^{1/2}$*) carries the constant-term contribution to that of (ii), with* $z$ *replaced by* $-1/z$ *and* $v_0$ *by* $S_{\mathrm{lab}}v_0$. *It therefore also tends to* $0$.

*Proof.* (i) We have $c_j(n)\sqrt{y}K_0(2\pi|n|y) = \int_0^1 d_j(x+iy)\mathbf{e}(-nx)\,dx$; take $y = 1/|n| \le 1$. For $\operatorname{Im} w = y \le 1$ the reduced point $w^*$ satisfies $\operatorname{Im} w^* \le 1/y$. Since $\rho$ is unitary, $\|d(w)\| = \|d(w^*)\| \le |a_d|\,y^{-1/2} + C_d$, and dividing by $|n|^{-1/2}K_0(2\pi)$ gives the bound. (ii) On the segment, $Y/2 \le |w - z| \le 3Y/2$, so $|R_z| \le 4/Y$. With $u = \sqrt{\operatorname{Im} w}$ we have $|u_w| = \frac14 Y^{-1/2}$. Also $\partial_{\bar w}R_z = \frac{i}{2}((w-z)(\bar w - z))^{-1} - \operatorname{Im} w\,(w-z)^{-1}(\bar w - z)^{-2}$, so $|\partial_{\bar w}R_z| \le 10/Y^2$, and $|R_z|^{-1/2} \le \frac32 Y^{1/2}$. Hence $|u_w R_z^{1/2}| \le \frac{1}{2Y}$ and $|u\,\partial_{\bar w}R_z^{1/2}| \le \frac12 \cdot \frac32 \cdot 10\,Y^{-1} < 8/Y$. The segment has length $2X$. (iii) We have $-1/w + 1/z = (w-z)/(wz)$ and $\operatorname{Im}(-1/w) = \operatorname{Im} w/|w|^2$. The $S$-law moves the constant term to $\sqrt{\operatorname{Im}(-1/w)}\,a_d S_{\mathrm{lab}}v_0$, so (ii) applies at the point $-1/z$. □

*Proof (outline; the estimates are Lemma 27.5).* (a) Since $\operatorname{Fix}(T) = \mathbb{C}v_0$, the constant term of $d \in M(\rho)$ is $\sqrt{y}\,(a + b\log y)\,v_0$. Green's theorem on the truncated fundamental domain (Maass–Selberg; the Hermitian pairing is legitimate because $\rho$ is unitary) gives $a_1\bar b_2 - b_1\bar a_2 = 0$ for any $d_1, d_2 \in M(\rho)$. The lift has constant term $\sqrt{y}\,(\log\varepsilon)\,v_0$ and no $\log y$ term (§15 and the lift formula). Pairing $d$ with $u_{\mathrm{lift}}$ therefore forces $b_d = 0$. So $d - (a_d/\log\varepsilon)u_{\mathrm{lift}}$ has zero constant term; its nonconstant modes decay exponentially. (b) *Holomorphy.* $c_d(z) = \int_0^{i\infty}[d, R_z^{1/2}]$ is taken along the fixed path $i\mathbb{R}_{>0}$, where $R_z(it) = t/(t^2+z^2)$. It is therefore holomorphic wherever $t^2 + z^2 \neq 0$ for all $t > 0$, i.e. on $\mathbb{C} \setminus i\mathbb{R}$. This needs no contour argument. *Fourier interchange.* By Lemma 27.5(i) the coefficients satisfy $|c(n)| \ll |n|$. Since $\sqrt{y}K_0(2\pi|n|y)$ decays exponentially in $|n|y$, the Fourier series converges absolutely and uniformly on $y \ge y_0 > 0$, so it may be integrated term by term on every segment with $y \ge y_0$. *The $S$-relation.* Lemma 15.1 holds for every real frequency $n \neq 0$ by translation covariance (and gives $\frac12$ for the constant term). Apply it on $\mathbb{H}$ to the modes of $d$, close the contour as in the proof of Theorem 11.7, and pull back by $w \mapsto -1/w$ using $d(-1/w) = S_{\mathrm{lab}}d(w)$ and Lemma 15.2. This gives, on $\mathbb{H} \setminus i\mathbb{R}$,

$$c_d(z) = I_d(z) - \varsigma(z)\,z^{-1}S_{\mathrm{lab}}I_d(-1/z), \qquad I_d = -\tfrac{\pi}{2}f_d^+ + \tfrac12 a_d v_0, \qquad \varsigma(z) = \operatorname{sgn}\operatorname{Re} z.$$

The contour step is not verbatim from Theorem 11.7. There $u_L$ has no constant term and decays at the cusp $0$, but for general $d$ the constant term $\sqrt{y}\,a_d v_0$ is carried to the cusp $0$ by the $S$-law. Its contribution is controlled exactly as at $i\infty$. On the connecting segments the constant-term integrand is $-\frac{i}{2}z\,a\,t^{-2}$ times a fixed vector, i.e. $O(t^{-2})$, so a segment at height $Y$ (or $1/Y$ after the $S$-pullback) contributes $O(1/Y) \to 0$ (Lemma 27.5(ii),(iii)). All other terms decay exponentially. *Reflection.* The reflection $\epsilon(w) = -\bar w$ satisfies $\epsilon^*[u,v] = [\tilde v, \tilde u] = \mathrm{d}(\tilde u\tilde v) - [\tilde u, \tilde v]$ (here d is the exterior derivative, not the Maass vector $d$). Hence $c_d(z) = a_d\big(v_0 - \varsigma(z)z^{-1}S_{\mathrm{lab}}v_0\big) - c_{\tilde d}(-z)$ with $\tilde d(w) = d(-\bar w)$. Here $\tilde d$ obeys the same $S$-law and the $T^{-1}$-twisted $T$-law, and the positive frequencies of $\tilde d$ are the negative frequencies of $d$. *Parity step.* Suppose $f_d^+ = 0$. Then $I_d = \frac12 a_d v_0$ is constant, and the $S$-relation shows that $c_d$ is a rational function on each of the two quadrants of $\mathbb{H} \setminus i\mathbb{R}$. By the

reflection identity, the negative-mode series $F$ of $d$ (the $I$-function of $\tilde{d}$) then satisfies, on the two quadrants,

$$F(w) = \kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) + L_1(w) \qquad \text{(first quadrant)},$$
$$F(w) = -\kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) + L_2(w) \qquad \text{(second quadrant)},$$

for a fixed $\kappa \in \{\pm 1\}$ and rational functions $L_1, L_2$. The relative sign is $\varsigma(z)$. The first-quadrant relation has the holomorphic coefficient $w^{-1}$, so it continues analytically to the whole half-plane on which $F$ is holomorphic. Subtracting the second-quadrant relation there gives $2\kappa\, w^{-1} S_{\mathrm{lab}} F(-1/w) = L_2(w) - L_1(w)$. So $F$ is rational, and since it decays exponentially, $F = 0$. (Using the relations quadrant by quadrant without the continuation only gives $F = PF$, which is vacuous because $Pd = d$.) Then $d = \sqrt{y}(a + b \log y)v_0$, and the $S$-law forces $d = 0$ because $S_{\mathrm{lab}} v_0 \notin \mathbb{C} v_0$ (Proposition 27.2(iii)). □

**Corollary 27.6** (Cocycle and period function determine $u$)**.** *For $d \in M(\rho)$: if $c_d \equiv 0$ on $\mathbb{H}$, then $d = 0$. Moreover, the restriction $\Phi|_{(0,\infty)}$ of the period function of Theorem 11.7 determines $u$, and hence the positive-frequency series $f$.*

**Status (Category A).** Proved below; rests on Theorem 27.4.

*Proof (outline). Key step.* Let $e \in M(\rho)$ and suppose that on the first quadrant $c_e(z) = r(z)$ with $r$ rational. The $S$-relation of Theorem 27.4(b) with $\varsigma(z) = +1$ gives $I_e(z) = z^{-1} S_{\mathrm{lab}} I_e(-1/z) + r(z)$ there. $I_e$ is holomorphic on $\mathbb{H}$, $z \mapsto -1/z$ preserves $\mathbb{H}$, and $z^{-1}$ is holomorphic there, so this relation continues analytically to all of $\mathbb{H}$. Applying it twice, with $I_e(-1/z) = -z\, S_{\mathrm{lab}} I_e(z) + r(-1/z)$, gives $I_e = -PI_e + (\text{rational})$. Since $Pe = e$, $PI_e = I_e$, so $2I_e$ is rational. $f_e^+$ decays exponentially as $\operatorname{Im} z \to \infty$, so $f_e^+ = 0$, and $e = 0$ by Theorem 27.4(b). *First claim.* If $c_d \equiv 0$, apply the key step with $r = 0$. *Second claim.* $\Phi$ (in the normalisation of §15) and $c_u$ differ by an explicit constant-coboundary term, rational in $z$ with coefficient $a_u$, and a scalar. $(0, \infty)$ lies in the right half-plane, where $c_u$ is holomorphic (and $\Omega = \Psi$, Theorem 11.7(i)). Suppose $u, u' \in M(\rho)$ have the same $\Phi$ on $(0, \infty)$. By the identity theorem, $c_{u-u'}$ is rational on the first quadrant, and the key step gives $u = u'$. □

**Theorem 27.7** (Uniqueness and residual ambiguity of the vector completion)**.** *Let $u = u_{\mathrm{lift}}$, with positive-frequency series $f$ and period function $\Phi$ (Theorems 11.6, 11.7).*

*(i)* Uniqueness given strong data. *Among $d \in M(\rho)$, $u$ is the only element with positive-frequency series $f$ and $q^0$-coefficient $a_u$. Moreover, among period functions of elements of $M(\rho)$, $\Phi$ is determined by its restriction to $(0, \infty)$.*

*(ii)* Non-uniqueness under weak conditions. *Consider period functions of elements of $M(\rho)$ with the same $\rho$-law (weight 1, factor $|cx + d|$), the same singular $I/t$ data, and real-analytic on $\mathbb{R} \setminus \{0\}$. They form the affine space $\Phi + \{\Phi_c : c \in S(\rho)\}$, enlarged by $\mathbb{C}\,\Phi_{v_0}$ when the $q^0$ term is not prescribed. The parametrisation $c \mapsto \Phi_c$ is injective.*

*(iii)* The cusp space is nonzero. $\dim S(\rho) \geq 1$. *An explicit witness is $u_{\mathrm{cusp}} := \Pi_{B_1 X^-} u_{\mathrm{lift}}$, the projection to the $\rho$-stable sector $B_1 X^-$. It lies in $M(\rho)$, has zero constant term because $\mathrm{Fix}(T) \cap B_1 X^- = 0$, and is nonzero because its coefficient at $n = \frac{19}{140}$ is nonzero. Hence, by the injectivity in (ii), the class in (ii) contains more than one element.*

*(iv)* Shape of the ambiguity. *The non-cusp part of $M(\rho)$ is exactly one-dimensional and lies in $A_3X^+$, and the $P$-odd sectors contribute nothing. An explicit basis of the residual ambiguity is*

$$\begin{aligned} &\{\Phi_{v_0}\} && (\textit{only if } q^0 \textit{ is free}) \\ \oplus\ &\{\Phi_c :\ c \in \textit{a basis of } S(\rho)\} && (\textit{containing } \Phi_{u_{\mathrm{cusp}}} \neq 0) \\ \oplus\ &\mathbb{C}\, g_0 && (\kappa\textit{-level, Proposition 27.3}). \end{aligned}$$

**Status (Category A).** (i) and the injectivity in (ii): Theorem 27.4 and Corollary 27.6. (iii): $\dim S(\rho) \geq 1$ with witness $u_{\mathrm{cusp}}$ is independent of Theorem 27.4: sector projectors commute with $\rho$, the vanishing of the constant term is Proposition 27.2(i),(ii), and non-vanishing is an exact $\mathbb{Z}[\zeta_{20}]$ value. That the class in (ii) has more than one element, and $\Phi_{u_{\mathrm{cusp}}} \neq 0$ in (iv), use the injectivity (Corollary 27.6). (iv): given Theorem 27.4(a) and Proposition 27.2. *Not claimed:* that every real-analytic $\rho$-cocycle with Lewis–Zagier growth comes from $M(\rho)$ (Remark 27.16(a)); for cusp-type period functions with the growth conditions stated there this follows from Deitmar–Hilgert, but the non-cusp part is not covered.

**Proposition 27.8** (Sector-by-sector uniqueness and the scalars)**.** *$u_{\mathrm{lift}}$ has components only in $\sigma \in \{A_3X^+, B_1X^-\}$. For each such $\sigma$:*

*(i) the $\sigma$-component is the unique element of $\Pi_\sigma M(\rho)$ with positive-frequency series $f_\sigma$;*

*(ii) it is unique up to a scalar given only the line $\mathbb{C}f_\sigma$;*

*(iii) the scalar is fixed by the first nonzero coefficient: $n = \frac{1}{14}$ for $A_3X^+$ and $n = \frac{19}{140}$ for $B_1X^-$, both exact cyclotomic numbers (Appendix C).*

*Only $A_3X^+$ has a non-cusp part, and it is one-dimensional. In $B_1X^-$ the completion is unique given $f$. In the weak class of Theorem 27.7(ii) it is ambiguous by $S(\rho) \cap B_1X^-$, which contains $u_{\mathrm{cusp}}$.*

**Status (Category A).** Support: Propositions 19.4 and 27.2. Uniqueness in (i)–(ii): Theorem 27.4 applied to $\Pi_\sigma d$. (iii): exact values.

**Proposition 27.9** (Sector bounds for $\dim S(\rho)$)**.** *Let $\sigma$ run over the eight $\rho$-stable sectors of Proposition 27.2, and write $M(\rho)_\sigma$, $S(\rho)_\sigma$ for the $\sigma$-components.*

*(i) For the four $P$-odd sectors, $M(\rho)_\sigma = S(\rho)_\sigma = 0$.*

*(ii) $M(\rho)_{A_3X^+} = \mathbb{C}\,\Pi_{A_3X^+}u_{\mathrm{lift}} \oplus S(\rho)_{A_3X^+}$, and $M(\rho)_\sigma = S(\rho)_\sigma$ for $\sigma \in \{B_1X^+, A_3X^-, B_1X^-\}$.*

*(iii) $\dim S(\rho)_{B_1X^-} \geq 1$, with witness $u_{\mathrm{cusp}} = \Pi_{B_1X^-}u_{\mathrm{lift}}$.*

| *sector* | $\dim\sigma$ | $P$ | $\dim S(\rho)_\sigma$ | *status* |
|---|---|---|---|---|
| $A_3X^+$ | 36 | $+1$ | 0 | *A (computer-assisted, Theorem 27.13)* |
| $B_1X^+$ | 12 | $+1$ | 0 | *A (computer-assisted, Theorem 27.13)* |
| $A_3X^-$ | 18 | $+1$ | 0 | *A (computer-assisted, Theorem 27.13)* |
| $B_1X^-$ | 6 | $+1$ | 1 | *A ($\geq 1$: exact witness at $n = \frac{19}{140}$; $\leq 1$: Theorem 27.13)* |
| $W_5^+W_7^-A_3$ | 27 | $-1$ | 0 | *A* |
| $W_5^+W_7^-B_1$ | 9 | $-1$ | 0 | *A* |
| $W_5^-W_7^+A_3$ | 24 | $-1$ | 0 | *A* |
| $W_5^-W_7^+B_1$ | 8 | $-1$ | 0 | *A* |

*The upper bounds in the $P$-even sectors are Theorem 27.13.*

*Proof.* (i) Every $d \in M(\rho)$ satisfies $Pd = d$, because $S_{\mathrm{lab}}^2 = P$ and $d(-1/(-1/w)) = d(w)$. So $d$ has no component in $\mathrm{Fix}(-P)$. (ii) Theorem 27.4(a), together with $\mathrm{Fix}(T) = \mathbb{C}v_0 \subset A_3X^+$, so constant terms vanish outside $A_3X^+$. (iii) Theorem 27.7(iii). The coefficient at $n = \frac{19}{140}$ was also recomputed exactly: 8 lattice points, all phases $\pm 1$, rational projectors, 4 nonzero $B_1X^-$ entries. □

**Status.** (i) and (iii): Category A. (ii): Category A given Theorem 27.4(a). *Irreducibility* of the eight sectors: each has a simple $T$-spectrum with exact exponents in $\frac{1}{140}\mathbb{Z}$ (Category A), and the $S$-graph between $T$-eigenvectors is connected, a zero/nonzero test done in floating point (Category B). *Category B (consistency):* theta lifts of all even binary lattices $ax^2 - by^2$ with discriminant group of order $\le 2000$ give, inside $\rho$, only multiples of the components of $u_{\mathrm{lift}}$, in agreement with Theorem 27.13.

## 27.3 The dimension of $S(\rho)$

Fix a $P$-even sector $\sigma$ of dimension $d$. Its $T$-spectrum is simple, with exponents $r_1, \dots, r_d \in \{0, \dots, 139\}$ (Proposition 27.9). Let $v_1, \dots, v_d$ be the unit $T$-eigenvectors, $Tv_k = \mathbf{e}(r_k/140)v_k$. They are orthonormal, and we write $S_\sigma = (\langle Sv_l, v_k\rangle)_{k,l}$, a unitary matrix with $S_\sigma^2 = I$. For $d \in S(\rho)_\sigma$ put $g_k = \langle d, v_k\rangle$. Then $g(w+1) = \mathrm{diag}(\mathbf{e}(r_k/140))\, g(w)$ and $g(-1/w) = S_\sigma g(w)$. Because the constant term vanishes and $d$ has moderate growth,

$$g_k(x+iy) = \sum_{n \in r_k/140+\mathbb{Z},\ n\neq 0} c_k(n)\sqrt{y}\, K_0(2\pi|n|y)\, \mathbf{e}(nx), \tag{27.1}$$

so $g$ decays exponentially at the cusp and $\|d\|_\infty := \sup_{\mathbb{H}} \|d(w)\|_2 < \infty$. Since $\rho$ is unitary, $\|d(w)\|_2$ is $\Gamma$-invariant, and the supremum is attained at some point $w_0$ of the closed standard fundamental domain. In particular $\Im w_0 \ge \frac{\sqrt{3}}{2}$.

**Lemma 27.10.** *$K_0(x) \le \sqrt{\pi/(2x)}\, e^{-x}$ for $x > 0$.*

*Proof.* $K_0(x) = \int_0^\infty e^{-x\cosh t}\, dt \le \int_0^\infty e^{-x(1+t^2/2)}\, dt = e^{-x}\sqrt{\pi/(2x)}$, since $\cosh t \ge 1 + t^2/2$. □

**Lemma 27.11** (A priori bounds)**.** *Let $d \in S(\rho)_\sigma$ with $\|d\|_\infty = 1$, and let $\mu_0 = \sqrt{1/2}\, K_0(1/2)$. Then:*

*(i)* $|c_k(n)| \le \sqrt{2\pi|n|}/\mu_0$ *for every $k$ and every $n$.*

*(ii) For $M \ge 1$ and $y > 0$, the part of (27.1) with $|n| > M$ is bounded in absolute value by*

$$\tau_M(y) := \frac{\sqrt{\pi/2}}{\mu_0} \cdot \frac{2e^{-2\pi My}}{1 - e^{-2\pi y}}.$$

*(iii) For $y \ge \frac{\sqrt{3}}{2}$,* $\sqrt{y}\, K_0(2\pi|n|y) \le \omega_n := \min\bigl(\sqrt{\pi/2}/\sqrt{2\pi|n|},\ (4|n|)^{-1/2}e^{-\pi\sqrt{3}|n|}\bigr)$.

*Proof.* (i) We have $c_k(n)\sqrt{y}K_0(2\pi|n|y) = \int_0^1 g_k(x+iy)\mathbf{e}(-nx)\, dx$, and $|g_k| \le \|d\|_2 \le 1$. Take $y = 1/(4\pi|n|)$. (ii) By (i) and Lemma 27.10, each term is at most $\sqrt{\pi/2}\,\mu_0^{-1}e^{-2\pi|n|y}$. The frequencies $n > M$ in $r/140 + \mathbb{Z}$ are $> M + j$ for $j = 0, 1, \dots$, and likewise for $n < -M$. (iii) The first bound holds because $\sqrt{t}K_0(t) \le \sqrt{\pi/2}$, by Lemma 27.10. The second follows from Lemma 27.10 and monotonicity in $y$. □

**Proposition 27.12** (Certification criterion). *Fix $M \geq 1$ and let $\mathcal{C} = \{(k,n) : n \in r_k/140+\mathbb{Z},\ n \neq 0,\ n = r_k/140 + j,\ |j| \leq M\}$. Fix sample points $w_m = x_m + iY_m$, $m = 1,\dots,R$, and elements $\gamma_m \in \mathrm{SL}_2(\mathbb{Z})$, and let $w_m^* = \gamma_m^{-1} w_m$. Let $\Gamma_m$ be the matrix with $g(w_m) = \Gamma_m\, g(w_m^*)$; it is a product of $S_\sigma$ and diagonal phase matrices, and it is unitary. Let $G$ be the $(Rd) \times |\mathcal{C}|$ matrix whose block row $m$ is $E_m - \Gamma_m F_m$. Here $E_m$ and $F_m$ evaluate the basis functions $\sqrt{y}K_0(2\pi|n|y)\mathbf{e}(nx)$ of column $(k,n)$, placed in row $k$, at $w_m$ and at $w_m^*$. Let $D = \mathrm{diag}(s_{(k,n)})$ with all $s > 0$, and $H = (GD^{-1})^*(GD^{-1})$. Put*

$$\varrho = \sum_m d\,\big(\tau_M(Y_m) + \tau_M(\Im w_m^*)\big)^2, \quad \Omega = \max_k \sum_{(k,n)\in\mathcal{C}} (\omega_n/s_{(k,n)})^2, \quad t_0 = \sqrt{d}\,\tau_M(\tfrac{\sqrt{3}}{2}).$$

*If $t_0 < 1$ and the $(j+1)$-st smallest eigenvalue of $H$ exceeds $\varrho\,\Omega/(1-t_0)^2$, then $\dim S(\rho)_\sigma \leq j$.*

*Proof.* Let $d \in S(\rho)_\sigma$ and write $c = (c_k(n))_{(k,n)\in\mathcal{C}}$, $c'' = Dc$. The map $d \mapsto c''$ is linear.

*Step 1: the map is injective.* Normalise $\|d\|_\infty = 1$ and evaluate at $w_0$. By Lemma 27.11(ii),(iii) and Cauchy–Schwarz in each component,

$$1 = \|d(w_0)\|_2 \leq \Big(\sum_k \big(\sum_n \omega_n |c_k(n)|\big)^2\Big)^{1/2} + t_0 \leq \Omega^{1/2}\|c''\|_2 + t_0.$$

Hence $\|c''\|_2 \geq (1-t_0)\Omega^{-1/2} > 0$.

*Step 2: the residual is small.* The identity $g(w_m) = \Gamma_m g(w_m^*)$ is exact. Splitting each side of (27.1) into the part in $\mathcal{C}$ and the rest gives $GD^{-1}c'' = \varepsilon$. Here $\varepsilon_m = -(\text{tail of } g(w_m)) + \Gamma_m(\text{tail of } g(w_m^*))$, so $\|\varepsilon_m\|_2 \leq \sqrt{d}(\tau_M(Y_m) + \tau_M(\Im w_m^*))$ by Lemma 27.11(ii) and the unitarity of $\Gamma_m$. Hence $\|GD^{-1}c''\|_2^2 \leq \varrho$.

*Step 3: conclusion.* Suppose $\dim S(\rho)_\sigma \geq j+1$. By Step 1 the image of $d \mapsto c''$ has dimension at least $j+1$. By the Courant–Fischer theorem it contains a vector $c'' \neq 0$ with $\|GD^{-1}c''\|^2 \geq \lambda_{j+1}(H)\|c''\|^2$. Rescale the corresponding $d$ to $\|d\|_\infty = 1$. Steps 1 and 2 then give $\lambda_{j+1}(H)(1-t_0)^2/\Omega \leq \varrho$, which contradicts the hypothesis. □

**Theorem 27.13** (Dimension of the cusp space). $\dim S(\rho)_{A_3X^+} = \dim S(\rho)_{B_1X^+} = \dim S(\rho)_{A_3X^-} = 0$ *and* $\dim S(\rho)_{B_1X^-} = 1$. *Hence* $\dim S(\rho) = 1$ *and* $S(\rho) = \mathbb{C}\,u_{\mathrm{cusp}}$.

*Proof.* We apply Proposition 27.12 with $M = 6$, $Y_m \in \{\frac{4}{5}, \frac{7}{10}, \frac{3}{5}, \frac{1}{2}\}$ and $x_m = \frac{2m+1}{80} - \frac{1}{2}$, $m = 0,\dots,39$. Each $\gamma_m$ is the usual reduction word, chosen in floating point; the identity holds for any word. The law $d(-1/w) = S_{\mathrm{lab}}d(w)$ used here agrees with $\rho(S) = S_{\mathrm{lab}}^{-1}$ (Remark 19.1), since $S_{\mathrm{lab}}^2 = P$ gives $S_{\mathrm{lab}}^{-1} = S_{\mathrm{lab}}$ on every $P$-even sector. All of the following are computed in ball arithmetic (128-bit precision) from the exact rational phase data of $S_{\mathrm{lab}}$, $T_{\mathrm{lab}}$, the adapted basis and the exact sector projectors: $v_k$ (via the exact eigenprojection $\frac{1}{140}\sum_j \mathbf{e}(-r_kj/140)T^j$), $S_\sigma$, $\Gamma_m$, $w_m^*$, the entries of $G$ (with $K_0(x) = \sqrt{\pi}e^{-x}U(\frac{1}{2},1,2x)$), $H$, $\varrho$, $\Omega$ and $t_0$. The ball checks $TV = V\,\mathrm{diag}(\mathbf{e}(r_k/140))$, $V^*V = I$, $SV = VS_\sigma$, $S_\sigma^2 = I$ and $S_\sigma S_\sigma^* = I$ all contain 0. Eigenvalue lower bounds for $H$ come from an approximate unitary $Q$ and two estimates. Weyl's inequality applied to $Q^*HQ - \Lambda$ gives $\lambda_k(Q^*HQ) \geq \Lambda_k - \|Q^*HQ - \Lambda\|_F$. Ostrowski's theorem gives $\lambda_k(H) \geq \lambda_k(Q^*HQ)/(1+\|Q^*Q-I\|_F)$. The resulting certified values are:

| sector | $d$ | $\lvert\mathcal{C}\rvert$ | rows | threshold $\varrho\Omega/(1-t_0)^2 \leq$ | certified lower bound | conclusion |
|---|---|---|---|---|---|---|
| $A_3X^+$ | 36 | 467 | 5760 | $1.04\cdot10^{-14}$ | $\lambda_1 \geq 0.0251$ | $\dim = 0$ |
| $B_1X^+$ | 12 | 156 | 1920 | $3.6\cdot10^{-15}$ | $\lambda_1 \geq 0.0373$ | $\dim = 0$ |
| $A_3X^-$ | 18 | 234 | 2880 | $5.2\cdot10^{-15}$ | $\lambda_1 \geq 0.0154$ | $\dim = 0$ |
| $B_1X^-$ | 6 | 78 | 960 | $4.4\cdot10^{-15}$ | $\lambda_2 \geq 0.1317$ | $\dim \leq 1$ |

A second run with $M = 8$ and $40 \to 48$ points per horocycle gives thresholds $\le 4 \cdot 10^{-20}$ and the same lower bounds to three digits. For $B_1 X^-$, the lower bound $\dim \ge 1$ is Proposition 27.9(iii). The $P$-odd sectors contribute 0 by Proposition 27.9(i). □

**Status: Category A (computer-assisted).** This is a computer-assisted proof in the sense of §10.1. The analytic reduction (Lemmas 27.10, 27.11, Proposition 27.12) is proved in full. It uses no coefficient bound beyond the trivial sup-norm bound and no assumption on $\lambda$ other than $\lambda = \frac{1}{4}$. The finite verification uses ball arithmetic (Arb via python-flint), and every quantity entering the final inequality is a rigorous enclosure. Floating point only chooses the reduction words, column scalings and the approximate eigenbasis $Q$; these choices are not part of the proof. The margin is at least $10^{12}$ in every sector.

**Corollary 27.14.** $\dim M(\rho) = 2$ *and* $M(\rho) = \mathbb{C}\, u_{\mathrm{lift}} \oplus \mathbb{C}\, u_{\mathrm{cusp}}$. *In Theorem 27.7(ii), the ambiguity of period functions under the weak conditions is exactly the line* $\mathbb{C}\, \Phi_{u_{\mathrm{cusp}}}$.

*Proof.* Theorem 27.4(a), which gives $M(\rho) = \mathbb{C} u_{\mathrm{lift}} \oplus S(\rho)$, together with Theorem 27.13. □

**Status.** Category A, given Theorem 27.4(a) and the same conditions as Theorem 27.7(ii).

**Remark 27.15** (Dihedral consistency check)**.** Independently of Theorem 27.13: among the ray class characters $\chi \ne \chi^\sigma$ of $\mathbb{Q}(\sqrt{D})$, $D \in \{2, 5, 7, 10, 14, 35, 70\}$, with infinity type $(+,+)$ or $(-,-)$ and $\mathrm{Ind}\, \chi$ of conductor dividing 19600, there are exactly 120, giving 36 even dihedral representations, and none has projective image field $\mathbb{Q}(\sqrt{7}, \sqrt{10})$. So that field contributes no dihedral $\lambda = \frac{1}{4}$ form at this level, as Theorem 27.13 requires.

**Status (Category A).** Exact enumeration (PARI/GP). Completeness: the sum over primitive characters equals $|\mathrm{Cl}_{\mathfrak{m}}|$ for all seven fields, and split primes $\le 3000$ generate each $\mathrm{Cl}_{\mathfrak{m}}$, so the test $\chi \ne \chi^\sigma$ is exact.

**Remark 27.16** (Further remarks on §27)**.** (a) *Period functions of cusp forms.* We use Deitmar–Hilgert [5, Thm. 3.3 and Thm. 4.4] with $\Gamma = \mathrm{PSL}_2(\mathbb{Z})$, the finite-image representation $\eta = \rho$ and spectral parameter $\nu = 0$ (our $\lambda = \frac{1}{4}$, as $\Delta u = (\frac{1}{4} - \nu^2)u$; the hypotheses $\nu \notin \frac{1}{2} + \mathbb{Z}$ and $\mathrm{Re}\, \nu > -\frac{1}{2}$ hold). The dictionary is $\psi(x) = \rho(S)^{-1}\Phi(x)/x$ on $x > 0$, with $\Phi(x) = \kappa(-1/x) - |x| S_{\mathrm{lab}} \kappa(x)$; $\psi$ satisfies the three-term equation $\eta(T)\psi(z) = \psi(z+1) + (z+1)^{-1}\eta(ST^{-1})\psi(z/(z+1))$, and the growth conditions of [5, Thm. 4.4] read $\Phi(x) = o(1)$ as $x \to 0^+$ and $\Phi(x) = o(x)$ as $x \to \infty$. Under these conditions period functions correspond bijectively to elements of $S(\rho)$, so by Theorem 27.13 they form the line spanned by $\Phi_{u_{\mathrm{cusp}}}$. Lewis–Zagier [10] treats the scalar case for $\mathrm{SL}_2(\mathbb{Z})$ only. The non-cusp form $u_{\mathrm{lift}}$ is not covered by either reference. Status: Category A (cited; hypotheses checked against [5]); the numerical check of the dictionary (residual $7 \cdot 10^{-15}$) is Category B.

(b) *Numerical support for Proposition 27.3*: the $S$- and $T$-laws for $g_0$ hold to $\le 5 \cdot 10^{-9}$ at 11 rationals; $\|g_0(h/m)\|/(m\|g_0(0)\|) = 1$ to $10^{-11}$; and $ST^{-1}S^{-1}g_0(0) = g_0(0)$ to $4 \cdot 10^{-15}$. Floating point: Category B.

## 28 Obstructions to coefficient-emitting maps for $g_s$

One may ask whether the coefficients of $g_s$ (§25.2), or the cone data, can be produced ("emitted") by a natural construction from $E_8$ data or from the integer pattern $\mathrm{PW}(7, 11; 3^5)$: $B' = 7^2 \cdot 11 = 539$, window 18, fibres $68 = 3 \cdot 11 + 5 \cdot 7$ and 67, towers $243 = 3^5$, body 521, auxiliary 8 (the rank of $E_8$). We prove obstructions for several natural classes of such maps (Propositions 28.1–28.4) and

record what is not excluded (Remark 28.5). Data of this kind can select the parameter $s$; once $s$ is fixed, $g_s$ is defined independently of $E_8$ and of PW, so selecting $s$ does not emit coefficients.

**Proposition 28.1** (The $t^1$ obstruction for modular-type sums)**.** *Let $s > 0$ with $s \neq 3$.*

*(i)* $g_s(e^{-t}) = 2 + 2(s-3)\,t + (5s^2 - 44s + 75)\,t^2 + O(t^3)$ *as* $t \to 0^+$ *(Theorem 25.2). For* $s = 1, 7, 8, 11$*:* $(c_1, c_2) = (-4, 36), (8, 12), (10, 43), (16, 196)$.

*(ii) Let $F$ be any finite sum of holomorphic modular or quasimodular forms of nonnegative real weights (any level, any multiplier) that are holomorphic at the cusp 0. Then $F(e^{-t}) = \sum_a C_a t^{-a}$ with all $a \geq 0$, up to an exponentially small error. Hence $g_s$ is not such a sum.*

*(iii) Nor is any single shift $q^r g_s$ ($r \in \mathbb{R}$) such a sum; equivalently, $g_s \neq q^{-r}F$. Cancelling the $t$-term forces $r = s - 3$, and the $t^2$ coefficient of $q^{s-3}g_s$ is then $2(2s^2 - 19s + 33)$, which is nonzero for $s = 7, 8, 11$ (values $-4$, 18, 132) and for $s = 1$ (value 32).*

*The class in (ii) contains $\theta_{E_8} = E_4$, every $\theta_{E_8}(q^m)$, the Eisenstein series built from $\sigma_3$ and their derivatives, and every lattice theta series, of $E_8$ or of any other positive-definite lattice.*

*Proof.* (i) is Theorem 25.2 ($c_0, c_1, c_2$ were also re-derived exactly). (ii) For a form of weight $k$, $F(e^{-t})$ is $(2\pi/t)^k$ times the expansion of $F|_k S$ at $i\infty$, i.e. $C\,t^{-k}$ plus exponentially small terms. A quasimodular form of depth $p \leq k/2$ contributes $t^{-k+j}$ with $0 \leq j \leq p$, so all exponents are $\leq 0$. Finitely many such terms give only $t^{-a}$ with $a \geq 0$, whereas $g_s$ has the coefficient $c_1 = 2(s-3) \neq 0$ of $t^1$. (iii) With $q = e^{-t}$, $q^r g_s = e^{-rt} g_s(e^{-t})$ has $t$-coefficient $2(s-3) - 2r$ and, at $r = s - 3$, $t^2$-coefficient $c_2 - rc_1 + r^2 = 2(2s^2 - 19s + 33)$. □

**Status (Category A).** Proved, given Theorem 25.2. Category B support: $(g_s - c_0 - c_1 t - c_2 t^2)/t^3$ at $t = 10^{-3}, \ldots, 1.25 \cdot 10^{-4}$ is $-1355, -1338, -1329, -1325$ for $s = 7$. This is consistent with the value $c_3(7) = -3962/3 \approx -1320.7$ of Theorem 25.2, and similarly for $s = 1, 8, 11$ ($c_3 = -546.7, -1566.3, -303.3$). *Sanity:* the same test excludes $f_0 = g_1$, consistent with $f_0$ being mock.

**Proposition 28.2** ($E_8$ is invisible to the vector and cone data)**.** *$E_8$ is even unimodular, so $A_{E_8} = 0$ and $\rho_{L \oplus E_8} = \rho_L$ for every even lattice $L$. Consequently the Weil representation of Theorem 26.2, the cone/vector transformation law, the eight sectors of Proposition 27.2, the Hom dimension 8 (Proposition 26.3) and the $q$-exponents modulo 1 of the 140-vector are unchanged by adjoining $E_8$. No construction that factors through discriminant-form data can carry $E_8$ information into them.*

**Status (Category A).** Standard (discriminant forms of orthogonal sums are orthogonal sums). *Arithmetic aside, not a result:* the primes involved are $\{2, 5, 7\}$ at $s = 7$, $\{2, 3\}$ at $s = 8$ and $\{2, 3, 11\}$ at $s = 11$. Only 7 and 11 are also $E_8$ exponents.

**Proposition 28.3** (Exact linear-span test to $q^{1499}$)**.** *Let $\mathcal{F}$ be the following 1036 distinct integer sequences, truncated to $q^0, \ldots, q^{1499}$:*

*F1* $q^r \theta_{E_8}(q^m)$ *for* $1 \leq m \leq 12$ *and* $0 \leq r \leq 20$ *(252 series);*

*F2* $W(E_8)$ *dominant-weight counts per norm, rescaled by* $1 \leq m \leq 6$ *and shifted by* $0 \leq r \leq 20$ *(126 series);*

*F3 the $W(E_8)$ Molien series $\prod_{d \in \{2,8,12,14,18,20,24,30\}} (1 - q^d)^{-1}$, with the same rescalings and shifts (126 series);*

*F4 partition counts into parts from each nonempty subset of* $\{7, 8, 11, 18, 67, 68, 243, 521, 539\}$ *(*511 *series);*

$\delta$ $\delta_{n,r}$ *for* $0 \le r \le 20$ *(*21 *series). No two of the* 1036 *series coincide.*

*None of* $g_7$, $g_8$, $g_{11}$, $g_1$, $g_7 - g_8$, $g_{11} - g_7$ *lies in the* $\mathbb{Q}$*-span of* $\mathcal{F}$*. Moreover* $g_1, g_7, g_8, g_{11}$ *are* $\mathbb{Q}$*-linearly independent.*

*Proof.* For $p = 2^{61} - 1$, the mod-$p$ rank of $\mathcal{F}$ is 880. The exact $\mathbb{Q}$-rank of F4 $\cup\, \delta$ is 376 (532 columns; exact integer rank). F1–F3 have 504 columns, so $\operatorname{rank}_{\mathbb{Q}} \mathcal{F} \le 504 + 376 = 880$. For an integer matrix $\operatorname{rank}_p \le \operatorname{rank}_{\mathbb{Q}}$, so $\operatorname{rank}_{\mathbb{Q}} \mathcal{F} = 880$. Adjoining any one target raises the mod-$p$ rank to 881, so its $\mathbb{Q}$-rank is at least $881 > 880$, and the target is not in the $\mathbb{Q}$-span of the truncated sequences, a fortiori not of the full ones. The independence statement is an exact rank-4 computation. □

**Status (Category A).** Exact. Sanity checks of F2: $d(1..4) = 1, 1, 1, 2$ and $r_{E_8}(4) = 17520 = 17280 + 240$.

**Proposition 28.4** (No low-degree rational generating function)**.** *For* $s \in \{7, 8, 11\}$ *and* $(R, n_0) \in \{(1000, 0), (1000, 999), (600, 1200)\}$*, the* $(R+1) \times (R+1)$ *Hankel matrix* $(a_{n_0+i+j})_{0 \le i,j \le R}$ *of the coefficients of* $g_s$ *is nonsingular modulo* $p = 2^{61} - 1$*, hence over* $\mathbb{Q}$*. Consequently* $g_s \ne P/Q$ *with* $P, Q \in \mathbb{Q}[q]$*,* $\deg Q \le 1000$ *and* $\deg P \le 999$*. The same holds with* $\deg Q \le 600$ *and* $\deg P \le 1200$*.*

*Proof.* Suppose $g_s = P/Q$ with $0 \ne Q = \sum_{k \le D} b_k q^k$. Then $\sum_k b_k a_{n-k} = [q^n]P = 0$ for all $n > \deg P$. Let $R \ge D$ and put $b_k = 0$ for $D < k \le R$ and $c_j = b_{R-j}$ $(0 \le j \le R)$. Row $i$ of the Hankel matrix applied to $c$ gives $\sum_k b_k a_{n_0+i+R-k}$, which vanishes when $n_0 + R > \deg P$; all indices are $\ge n_0 \ge 0$. So the matrix is singular whenever $D \le R$ and $\deg P \le n_0 + R - 1$. Nonsingularity at $(R, n_0) = (1000, 0)$ therefore excludes $\deg Q \le 1000$, $\deg P \le 999$, and at $(600, 1200)$ it excludes $\deg Q \le 600$, $\deg P \le 1799$, which contains the stated range. □

**Status (Category A).** Exact modular arithmetic. This excludes, for example, any finite partition-count or Molien-type map with total part sum $\le 1000$, and any linear combination of such maps with common denominator of degree $\le 1000$. The $W(E_8)$ Molien degree sum is 128.

**Remark 28.5** (Scope: what is *not* excluded)**.** Propositions 28.1–28.4 exclude only the stated classes:

- modular and quasimodular sums of nonnegative weight holomorphic at 0, with at most one $q^r$ shift;
- constructions factoring through discriminant-form data;
- $\mathbb{Q}$-linear combinations of the 1036 tested sequences;
- rational generating functions in the stated degree ranges.

They do **not** exclude:

- nonlinear or non-modular maps;
- mock or false-theta constructions with $E_8$ input;
- sums with several different $q^r$ prefactors;

- weakly holomorphic forms with poles at the cusp 0;
- rational functions of higher degree;
- linear combinations of untested sequences.

Category B (reading): absence of a map in the tested classes is evidence, not proof, that no natural map exists.

# 29 Consequences

## 29.1 No scalar residual law; the law that exists

**Proposition 29.1** (No scalar residual law)**.** *(i) There is no identity $C = F + \mathcal{E}$ with $F$ an algebraic combination of holomorphic modular forms, signature-$(1,1)$ mock (Zwegers) holomorphic parts, unary, partial or false theta series, and $\mathcal{E}(e^{-t}) = o(1/t)$.*

*(ii) There is no quotient law $C \cdot g = F$ with holomorphic modular $g \not\equiv 0$, $F$.*

*(iii) There is no scalar $W_7$ law: for no weight $k$ and no constant $\lambda$ is $C(\tau) - \lambda(7\tau)^{-k} C(-1/(7\tau))$ radially bounded at all rationals.*

**Status (Category A).** This restates Theorem 11.5 and §13.5: (i), (ii) are conditional on Baker's theorem and the $q$-expansion principle, using the leading term $C(e^{-t}) = I/t + a_C t^{-1/2} + \cdots$ and $a_C \neq 0$; (iii) is an exact finite computation: at 178 of the 270 test rationals (denominator $< 30$), including $x = 5/7$, exactly one of $G(x)$, $G(-1/(7x))$ vanishes (Theorem 11.4).

**Proposition 29.2** (The law that exists has weight 1 and is vector-valued)**.** *(i) Where both Gauss means are nonzero, $|G|$ obeys an exact weight-1 relation with $|\lambda| = 7^{-1/2}$ (§13.5); the leading $1/t$ radial profiles obey the $S/T$ law with factor $1/|x|$ (Theorem 14.1); and the vector cocycle has automorphy factor $|cx + d|$ (Theorem 11.7).*

*(ii) The vector law consists of the 140-label Siegel lift to $\lambda = \frac{1}{4}$ Maass forms (Theorem 11.6), the Lewis–Zagier period formula (Theorem 11.7), the 42-dimensional function span $V = V_{\mathrm{Eis}}(36) \oplus V_{\mathrm{cusp}}(6)$ (Theorem 11.8), the Herglotz closed form (Theorems 11.9, 11.10), the identification $\rho_{\mathrm{lab}} \cong \rho_{N_2}$ (Theorem 26.2) and the uniqueness statements of §27. It is a law for the vector of cone sums, not for scalar $C$.*

*(iii) The characteristic $a = A^{-1}(0, \frac{s}{2})^{\mathsf{T}} = \left(\frac{s}{2(s-2)}, -\frac{1}{s-2}\right)$ completes the square: $Q_+(v) = Q_2(v + a) + c_0$ with $c_0 = \frac{s}{4(s-2)}$.*

**Status (Category A).** (i), (ii) with the status of the cited theorems as stated where they appear; Theorem 11.11 is not part of the law and stays Category B. (iii): $A^{-1} = \frac{-1}{s(s-2)} \left(\begin{smallmatrix} s & -s \\ -s & 2 \end{smallmatrix}\right)$ gives $a$, $B(v,a) = \frac{s}{2} k$ and $-Q_2(a) = \frac{s}{4(s-2)}$.

**Remark 29.3** (Weight $\frac{1}{2}$ cannot carry the residue)**.** Any weight-$\frac{1}{2}$ harmonic Maass form has radial growth $O(t^{-1/2})$, so no weight-$\frac{1}{2}$ object can carry the $I/t$ term of $C$ (Category A, conditional on the standard, cited $t^{-1/2}$ radial bound).

**Remark 29.4** (Sample values)**.** Four values 19.80, 1.82, 9.24, 20.90 at $x = 1/7, 3/7, 5/7, 1$ that have been proposed as "residual samples" are not radial values of $C$: $|C(0.995\,\mathbf{e}(x))| \approx 19.5, 18.5, 19.9, 54.0$ (floating point, Category B). The meaningful radial targets are $\lim_{t \to 0} t\, C(\mathbf{e}(x) e^{-t}) = I \cdot G(x)$ at every rational $x$ (Theorem 11.4; Category A).

### 29.2 Uniformity in $s$

**Proposition 29.5** (What depends on $s = 7$)**.** *$7s - 32 = -2Q(c_2)$ for the wall $c_2 = (-4, 1)$ (Proposition 23.1); only the factor $s - 2$ of $\Delta$ is intrinsic given $c_1$ (Remark 23.5). The results are otherwise uniform in $s$: the tensor theorem holds for all odd $s$ (Theorem 19.5), the radial closed forms are rational functions of $s$, and Theorem 25.2 holds for all real $s > 0$. The structurally special values are $s = 5$ (the radial coefficients of §22 vanish; $H_5 \equiv 0$) and $s = 8$ (the cone of $H_s$ is the axes cone, $\Delta = 144 = 12^2$).*

**Status (Category A).** As proved in §§22–23.

### 29.3 The case $s = 1$

$g_1 = f_0$, the fifth-order mock theta function (Hickerson–Mortenson [8]). At $s = 1$ the lattice $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has $\det = -s(s-2) = 1 > 0$ and is positive definite, so the indefinite part of the construction $(H_s, C, \mathbb{Q}(\sqrt{s(s-2)}))$ does not specialise; $H_s$ is Zwegers-admissible only for integer $s \geq 5$. Theorem 25.2 gives a proved radial expansion of $g_s$ uniform in $s$, including $s = 1$. Quantum modularity of $g_s$ is not claimed.

### 29.4 A formal contact with $A_5$

$\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$. At $s = 7$ the label representation has the tensor factor $\rho_{2x^2/5}$ (Theorem 19.3); $V_{\mathrm{Eis}} = A_3 \otimes W_5^+ \otimes W_7^+$ contains $W_5^+$ (dimension 3, factoring projectively through $A_5$) and $V_{\mathrm{cusp}} = B_1 \otimes W_5^- \otimes W_7^-$ contains $W_5^-$ (dimension 2, a faithful representation of $\mathrm{SL}_2(\mathbb{F}_5) = 2 \cdot A_5$). This is standard theory together with Theorem 19.3. Numerically (Category B) the $S, T$ images on $W_5^{\pm}$ generate groups of order 120 with scalar subgroup $\{\pm 1\}$ and quotient of order 60. Any further reading of this contact is Category B.

### 29.5 Status of three questions

For reference we record the status of three questions about $C$: (N1) a map from $E_8$ or PW data that emits the coefficients; (N2) a unique completed period; (N3) a tail. (These labels are unrelated to the lattices $N_1$, $N_2$ of §26.)

| question | status | category |
|---|---|---|
| (N1) emitting map | No coefficient-emitting map in the classes of Propositions 28.1–28.4; such data select $s$ but do not emit coefficients (§28). Not excluded: Remark 28.5. | A for the stated obstructions |
| (N2) completed period | Scalar version excluded (Theorem 11.5); existence as the 140-component vector cocycle (Theorems 11.6, 11.7, 11.9, 11.10). Unique given the positive-frequency $q$-series (with $q^0$) or $\Phi$ on $(0, \infty)$ (Theorem 27.7(i)); not unique under the weak conditions: residual ambiguity $\mathbb{C}\Phi_{v_0} \oplus \{\Phi_c : c \in S(\rho)\}$ with $\dim S(\rho) = 1$ (Theorem 27.13), and the $\kappa$-level gauge $\mathbb{C}g_0$. | A (computer-assisted for $\dim S(\rho) = 1$); a correspondence covering the non-cusp part open |

| question | status | category |
|---|---|---|
| (N3) tail | Exact closed forms at $x = 0$ through $t^{17/2}$, at $x = \frac{1}{2}, \frac{1}{3}$ through $t^{13/2}$, at $x = \frac{1}{4}, \frac{1}{6}$ for $t^{11/2}, t^{13/2}$, the period-72 theorem at $x = \frac{1}{3}$, and the Gauss evaluation and CFP at general $x = a/N$ (§22). Not given in closed form at all rationals; $t^{19/2}$ onward not computed. | A |

## 30 Numerical verification

"Exact" means rational/integer arithmetic. An entry that is exact supports a Category A statement; a floating-point entry is supporting evidence for a proved statement where one exists, and is otherwise Category B.

| check | range / points | result |
|---|---|---|
| Theorem 11.1 identities | through $q^{3000}$ | exact |
| $C_{\text{mixed}} = 0$ | through $q^{293}$ | exact; $-\frac{1}{2}q^{294}$ first term |
| cone sum (12.1), four cones; $E_{-a} = E_a$ | through $q^{3000}$ | exact |
| $H_s$ cone decomposition | through $q^{2000}$ | exact |
| Gauss pattern (Thm. 11.4) | 3374 pairs, $m \le 105$ | 0 mismatches |
| $\zeta_E(0)$, $\zeta_E(-1)$ | $1 + 5$ Shintani cones | exact $\frac{1}{10}$, $\frac{29}{1200}$ |
| Radial fit, $P_s^{\text{orth}}$ | $t \in [0.00125, 0.08]$ | 0.209419 ($I$), 0.52569 ($a_P$) |
| EM remainder (Table 13) | $t \ge 2.5 \cdot 10^{-5}$ | scales as $\sqrt{t}$ ($t$ for $E$) |
| region $\leftrightarrow$ ray class; $u_L$ Eisenstein | $n \le 60\,000$ | exact (max diff 0) |

Table 19: Exact and finite checks.

| check | points | max deviation |
|---|---|---|
| Maass $S$-, $T$-law, $P$-invariance, all 140 rows | 3 points, $y \approx 1$ | $1.2 \cdot 10^{-15}$, $7 \cdot 10^{-16}$, $6 \cdot 10^{-16}$ |
| $S^2 = I$, $(ST)^3 = I$ on 72-dim space | — | $1.3 \cdot 10^{-15}$ |
| Green constant $-\pi/2$ (cross-check) | $n = 1, 2, 3$, two $z$ | 30 digits |
| period formula vs. Bernoulli $\psi$ | 84 rationals | $1.1 \cdot 10^{-13}$ |
| all-row closed form; three-term relation | 7 pts; 4 pts | $2.3 \cdot 10^{-14}$; $7 \cdot 10^{-15}$ |
| component fits ($140 \times 2$) | $n \le 54\,880$ | $4.4 \cdot 10^{-14}$ |
| Thm. 11.9, 34 digits | 44 rationals in $[\frac{1}{8}, 2]$ | $5.7 \cdot 10^{-33}$ |
| Thm. 11.9, random rationals | 40, den. $\le 60$ | $2.5 \cdot 10^{-12}$ |
| Thm. 11.9 on $\mathbb{H}$ (36 digits) | 6 points | $5.5 \cdot 10^{-35}$ |
| Thm. 11.9 off $\mathbb{H}$ (double precision) | $5 + 3$ points | $2.5 \cdot 10^{-16}$; $9.2 \cdot 10^{-16}$ |
| Thm. 11.9 at rationals (36 digits) | 8 rationals | $5.7 \cdot 10^{-35}$ |
| split $G_\chi = 2R_\chi + T_\chi$ | 8 chars, 5 points | $\le 8 \cdot 10^{-34}$ |
| Thm. 11.10 | $N \le 14\,000$ | $1.84 \cdot 10^{-15}$; 627 coeffs equal in $\mathbb{Z}[\phi]$ |
| $\Omega = \Psi$ ($\operatorname{Re} z > 0$), $\Omega = -\Psi^-$ ($\operatorname{Re} z < 0$) | $3 + 3$ points | $\approx 10^{-16}$ |
| cusp rows = $L(0)$ combinations | 24 rows, 7 points | $4 \cdot 10^{-14}$ |

Table 20: Numerical checks. For the rational comparisons at rationals an independent from-scratch period formula at double precision agrees to $5.7 \cdot 10^{-17}$.

| $x$ | $\Phi_c$ dev. (24 rows) | $\Phi_{73}(x)$ |
|---|---|---|
| 1 | $3.8 \cdot 10^{-15}$ | $0.120446470003 + 0.176437422094i$ |
| $-1$ | $4.5 \cdot 10^{-15}$ | $0.006264073515 + 0.213537531676i$ |
| 2 | $9.6 \cdot 10^{-15}$ | $0.206217773055 + 0.263200146342i$ |
| $7/4$ | $3.4 \cdot 10^{-13}$ | $0.188336862451 + 0.245368413466i$ |
| $1/3$ | $6.7 \cdot 10^{-15}$ | $0.038631276713 + 0.077116144808i$ |
| $3/5$ | $4.0 \cdot 10^{-14}$ | $0.073437646967 + 0.123627119436i$ |
| $-2/3$ | $1.2 \cdot 10^{-14}$ | $0.012354928128 + 0.156183157173i$ |

Table 21: Cusp-row cocycle from $L(0)$ values vs. exact Bernoulli sums.

# 31 External dependencies

The following published results are used. Each is cited, not re-proved; "we use [X]" in the text refers to this list.

| result | where used |
|---|---|
| Baker's theorem [1] | Theorem 11.5(i); the obstruction at the cusp pair (§14.1); Proposition 29.1 |
| $q$-expansion principle (Shimura) | Theorem 11.5 |
| Hecke's continuation theorem [7] | Theorem 22.1(vi); Lemma 24.3; §24 |
| Siegel–Klingen [15] | rationality of the partial zeta values in Theorem 11.3 |
| Congruence theorem for Weil representations [16] | Theorems 19.3, 19.5, 20.1(iii); Proposition 19.4(iii) |
| Nobs–Wolfart / Cliff–McNeilly–Szechtman / Scheithauer, decomposition of Weil representations of $\mathbb{Z}/p^2$ | (C3) of Theorem 20.11 for primes not covered by Propositions 20.14, 20.15 |
| Zwegers, thesis Cor. 2.9 [18] | modular completion of $H_s$ (Theorem 22.1) |
| Zwegers 2012 [19] via [2] | identification of $\int \Xi_0$ in Theorem 24.1(iii) (also proved directly, Proposition 24.2) |
| Chevalley's ambiguous class number formula; Gauss genus theory; Tate's $\ker \mathrm{N} = C_K^{1-\sigma}$ | Theorem 21.3; Proposition 21.1(i) |
| Maass's dihedral construction [11] | the identification of $V_{\mathrm{cusp}}$ with dihedral forms (§16.2; Category B parts) |
| Nikulin [13], Thm. 1.14.4, Cor. 1.5.2 | Theorems 26.7, 26.9 |
| Morrison [12], Cor. 2.9(i) | Theorem 26.7; Proposition 26.10(iii) |
| Deitmar–Hilgert [5], Thms. 3.3, 4.4 | Remark 27.16(a) |
| Arb ball arithmetic (python-flint) | Theorem 27.13 (computer-assisted) |
| Strong Torelli theorem [9] | Theorem 26.9 |
| Gromov–Yomdin [6] | entropy in Theorem 26.9 |
| Standard Bessel asymptotics (DLMF 10.40.2) | Lemma 15.1 |

**PARI dependencies.** The following use PARI/GP and are otherwise unverified: the ray-class data at $\mathfrak{p}_2^3\mathfrak{p}_5\mathfrak{p}_7\infty_1\infty_2$ used for the 140-component fits and the Eisenstein/cusp classification of

the components; `lfun` values for $L(\frac{1}{2},\chi)$, $L(1,\chi)$ and root numbers; the numerical side of Theorem 11.11; the dihedral enumeration of Remark 27.15. Not dependent on PARI: $u_L$ Eisenstein, $\mathrm{Cl}(K)$, $|\mathrm{Cl}_{\mathfrak{f}}^{+}| = 16$, the structure of $\mathrm{Cl}^{+}_{(2\sqrt{35})\infty_1\infty_2}$ and the 12 values $L(0,\chi)$ (§18.1).

## 32 Open problems

1. Prove the cusp-row identities of Theorem 11.11 (currently Category B).
2. Prove the symmetry $\psi(-x) = \overline{\psi(x)}$ (observed numerically to $8 \cdot 10^{-14}$; not used).
3. An arithmetic (Gauss-sum or ideal-theoretic) proof of Theorem 11.10.
4. Whether Zwegers' completed theta transforms under the same 140-dimensional $\rho$ (Remark 25.3(e)); whether an exotic single kernel produces both sectors of $C_{\text{third}}$.
5. Objects whose cocycles are of size $\geq 1/y$ are not excluded by Theorems 24.5, 24.4.
6. $\dim V = s(s-1)$ for all odd $s$ (the lower bound); equality $d_{\text{Eis}} = 3\prod(p^e+1)/2$ in general; $d_2(k,v)$ for $k \geq 7$.
7. Sharpness of the Proposition K1 bounds for $m = 2k^2 - 1$ with $k \geq 6$ and on the full space ($\dim V(578) \leq 83532$); composite $m$ and $g$-consistency for all odd $m$ in Proposition 20.4.
8. A uniform proof of the prime-square decompositions; (C3) for general $p$ without citation; irreducibility of the complements at $s = 121$ and $s = 169$; the label-space decomposition at $s = 3481$ (whose $p$-local invariants are out of reach in label space).
9. The ambiguous-class count $|\bar{G}^{\sigma}| = 2^{\omega(s)}\varphi(s-2)$ in general; sufficiency of the modulus $M = 2dc$ for all norms.
10. The pattern of $S_e$ for all odd $e$; the identity $R_r = S_r^{\text{ax}}$ at general order; closed forms at general rationals; the $t^{19/2}$ coefficient and beyond.
11. A period-function correspondence for $\rho$ at $\lambda = \frac{1}{4}$ that also covers the non-cusp element $u_{\text{lift}}$ (Remark 27.16(a)).
12. Whether $\gamma$ itself lifts on the rank-6 Noether–Lefschetz locus of Proposition 26.10(iii) (Remark 26.11(b)); any link between Noether–Lefschetz series and the residual objects beyond the shared representation.
13. Coefficient-emitting maps outside the classes of §28 (Remark 28.5).
14. Quantum modularity of scalar $C$ and of $g_s$ (not claimed).

# Part III
# The 35-component law vector and the normalized Bridge Series

## 33 The 35-component law vector

Notation is that of §2. In Part III, $x$ denotes a rational number $h/m$ with $m \geq 1$, $\gcd(h,m)=1$ (and $m=1$ for $x=0$).

### 33.1 Gauss means

**Definition 33.1.** For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$ the function $\nu \mapsto \mathbf{e}(xQ(\nu)+B(\nu,\beta))$ on $\alpha+\mathbb{Z}^2$ is invariant under $\nu \mapsto \nu+Mu$ ($u \in \mathbb{Z}^2$) for every positive integer $M$ divisible by a suitable $M_0(\alpha,\beta,x)$, because $xQ(\nu+Mu)+B(\nu+Mu,\beta)-xQ(\nu)-B(\nu,\beta) = xMB(\nu,u)+xM^2Q(u)+MB(u,\beta)$ and $B(\nu,u)$, $Q(u)$, $B(u,\beta)$ have bounded denominators. The *Gauss mean* is

$$G_{\alpha,\beta}(x) = M^{-2} \sum_{n \in \mathbb{Z}^2/M\mathbb{Z}^2} \mathbf{e}\big(xQ(\alpha+n)+B(\alpha+n,\beta)\big),$$

which does not depend on the admissible $M$.

Let $h_A = (\tfrac{1}{2},0)$. For $\mu \in D$ we use the representative of $\mu$ in $[0,1)^2$.

**Definition 33.2** (The 35-component law vector)**.** For $\mu \in D$ put

$$K1_\mu = G_{a+\mu,a}, \quad K2_\mu = G_{a+\mu,0}, \quad K3_\mu = G_{\mu,a}, \quad K4_\mu = G_{\mu,0}.$$

The vector $(K2_\mu)_{\mu\in D}$ is the *35-component law vector*; $K3$ and $K4$ are its $S$- and $T$-companions. $\mathcal{V}_{35}$ denotes the complex span of the 105 functions $K2_\mu, K3_\mu, K4_\mu$ on $\mathbb{Q}$.

**Lemma 33.3** (Elementary rules)**.** *For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$:*

*(i)* $G_{\alpha+n,\beta} = G_{\alpha,\beta}$ *for* $n \in \mathbb{Z}^2$*;*

*(ii)* $G_{\alpha,\beta+\lambda} = \mathbf{e}(B(\alpha,\lambda))\, G_{\alpha,\beta}$ *whenever* $A\lambda \in \mathbb{Z}^2$*;*

*(iii)* $G_{g\alpha,g\beta} = G_{\alpha,\beta}$ *for* $g \in \mathrm{GL}_2(\mathbb{Z})$ *with* $g^{\mathsf{T}}Ag = A$.

*Proof.* (i) $\alpha+n+\mathbb{Z}^2 = \alpha+\mathbb{Z}^2$. (ii) For $\nu = \alpha+n$, $B(\nu,\lambda) = B(\alpha,\lambda)+n^{\mathsf{T}}A\lambda \in B(\alpha,\lambda)+\mathbb{Z}$. (iii) $\nu \mapsto g\nu$ maps $\alpha+\mathbb{Z}^2$ bijectively onto $g\alpha+\mathbb{Z}^2$, commutes with translation by $M\mathbb{Z}^2$ up to the bijection $g$ of $\mathbb{Z}^2/M\mathbb{Z}^2$, and $Q(g\nu)=Q(\nu)$, $B(g\nu,g\beta)=B(\nu,\beta)$. □

**Theorem 33.4** ($T$-law and $S$-law)**.** *For $\alpha,\beta \in \mathbb{Q}^2$ and $x \in \mathbb{Q}$,*

$$G_{\alpha,\beta}(x+1) = \mathbf{e}\big(-Q(\alpha)-B(\alpha,h_A)\big)\, G_{\alpha,\beta+\alpha+h_A}(x),$$
$$G_{\alpha,\beta}(-1/x) = \frac{\mathbf{e}(B(\alpha,\beta))}{|x|\sqrt{35}} \sum_{p\in D} G_{\beta+p,-\alpha}(x) \qquad (x \neq 0).$$

*Proof. T-law.* For $n \in \mathbb{Z}^2$, $Q(n) = n_1^2 + 7n_1n_2 + \frac{7}{2}n_2^2 \equiv \frac{7}{2}n_2 \equiv B(n, h_A) \pmod 1$, since $n_2^2 \equiv n_2 \pmod 2$ and $B(n, h_A) = n_1 + \frac{7}{2}n_2$. For $\nu = \alpha + n$, $Q(\nu) = Q(\alpha) + B(n, \alpha) + Q(n) \equiv Q(\alpha) + B(n, \alpha) + B(n, h_A)$, while $B(\nu, \alpha + h_A) = 2Q(\alpha) + B(\alpha, h_A) + B(n, \alpha) + B(n, h_A)$. Hence $Q(\nu) \equiv B(\nu, \alpha + h_A) - Q(\alpha) - B(\alpha, h_A) \pmod 1$, and inserting this into the mean defining $G_{\alpha,\beta}(x + 1)$ gives the law.

*S-law.* Fix $y \in \mathbb{Q} \setminus \{0\}$ and $t > 0$, and let $Z = yA + itI$. Then $Z$ is symmetric with $\operatorname{Im} Z = tI$ positive definite, so $f(v) = \exp(\pi i v^{\mathsf{T}} Z v + 2\pi i v^{\mathsf{T}} A\beta)$ is a Schwartz function on $\mathbb{R}^2$. Its Fourier transform is $\hat{f}(w) = \det(-iZ)^{-1/2} \exp(-\pi i (w - A\beta)^{\mathsf{T}} Z^{-1} (w - A\beta))$, where $\det(-iZ)^{1/2}$ is the branch that is positive for $y = 0$; diagonalising $A$ by an orthogonal matrix, with eigenvalues $\lambda_1 > 0 > \lambda_2$, it equals $\prod_j (t - iy\lambda_j)^{1/2}$ with principal square roots. Poisson summation for $v \mapsto f(\alpha + v)$ gives

$$\sum_{\nu \in \alpha + \mathbb{Z}^2} f(\nu) = \det(-iZ)^{-1/2} \sum_{w \in \mathbb{Z}^2} \mathbf{e}(w^{\mathsf{T}} \alpha) \exp\bigl(-\pi i (w - A\beta)^{\mathsf{T}} Z^{-1} (w - A\beta)\bigr). \tag{33.1}$$

*Left side.* $f(\nu) = \mathbf{e}(yQ(\nu) + B(\nu, \beta)) e^{-\pi t |\nu|^2}$. The phase is $M\mathbb{Z}^2$-periodic on $\alpha + \mathbb{Z}^2$; for each residue class $r + M\mathbb{Z}^2$, Poisson summation gives $\sum_{n \in r + M\mathbb{Z}^2} e^{-\pi t |n|^2} = (M^2 t)^{-1} (1 + O(e^{-c/t}))$ with $c = \pi / M^2$. Hence the left side is $t^{-1} G_{\alpha,\beta}(y) + O(t^{-1} e^{-c/t})$.

*Right side.* Put $w - A\beta = -A\gamma$; as $w$ runs over $\mathbb{Z}^2$, $\gamma = \beta - A^{-1}w$ runs over $\beta + A^{-1}\mathbb{Z}^2$, and $w^{\mathsf{T}}\alpha = (\beta - \gamma)^{\mathsf{T}} A \alpha = B(\alpha, \beta) - B(\gamma, \alpha)$. The exponent is $-\pi i \gamma^{\mathsf{T}} A Z^{-1} A \gamma$. Since $Z^{-1} = (yA)^{-1}(I + it(yA)^{-1})^{-1}$, one has $AZ^{-1}A = y^{-1}A - ity^{-2}I + t^2 R_t$ with $\|R_t\|$ bounded for $t \le t_0$. Hence the summand is

$$\mathbf{e}\bigl(B(\alpha, \beta) - B(\gamma, \alpha) - Q(\gamma)/y\bigr)\, e^{-\pi t |\gamma|^2 / y^2} \exp(-\pi i t^2 \gamma^{\mathsf{T}} R_t \gamma).$$

On the support of the Gaussian the last factor is $1 + O(t^2 |\gamma|^2)$, and $\sum_\gamma t^2 |\gamma|^2 e^{-\pi t |\gamma|^2 / y^2} = O(1)$. The phase $\gamma \mapsto \mathbf{e}(-B(\gamma, \alpha) - Q(\gamma)/y)$ is periodic on $\beta + A^{-1}\mathbb{Z}^2$, a coset of a lattice of covolume $1/35$; its mean over that coset is $\frac{1}{35} \sum_{p \in D} G_{\beta + p, -\alpha}(-1/y)$, because $\beta + A^{-1}\mathbb{Z}^2 = \bigsqcup_{p \in D} (\beta + p + \mathbb{Z}^2)$. The same residue-class argument as on the left gives the right side as $\det(-iZ)^{-1/2}\, \mathbf{e}(B(\alpha, \beta))\, 35 \cdot \frac{y^2}{t} \cdot \frac{1}{35} \sum_p G_{\beta + p, -\alpha}(-1/y) + O(1)$. Finally $\det(-iZ)^{1/2} \to (-iy\lambda_1)^{1/2} (-iy\lambda_2)^{1/2} = |y| \sqrt{|\lambda_1 \lambda_2|} = |y| \sqrt{35}$, since the two arguments are $\mp \pi/2$ and $\pm \pi/2$.

Multiplying (33.1) by $t$ and letting $t \to 0^+$ gives $G_{\alpha,\beta}(y) = \frac{|y|}{\sqrt{35}} \mathbf{e}(B(\alpha, \beta)) \sum_{p \in D} G_{\beta + p, -\alpha}(-1/y)$. Substituting $y = -1/x$ gives the $S$-law. □

**Proposition 33.5** (Closure)**.** *The laws of Theorem 33.4, followed by the reductions of Lemma 33.3(i),(ii), map each of the* 105 *functions $K2_\mu, K3_\mu, K4_\mu$ to a linear combination of them with coefficients in $\mathbb{Q}(\zeta_{9800}, \sqrt{35})$: $T$ maps $K2$ to $K2$ and exchanges $K3$ and $K4$; $S$ maps $K2$ into the span of $K3$, $K3$ into the span of $K2$, and $K4$ into the span of $K4$. Hence $\mathcal{V}_{35}$ is stable under $x \mapsto x + 1$ and under $x \mapsto -1/x$ with the factor $|x|^{-1}$.*

*Proof.* $a - h_A = (\frac{1}{5}, -\frac{1}{5})$ and $A(a - h_A) = (-1, 0)$, so $a \equiv h_A$ modulo $A^{-1}\mathbb{Z}^2$; $A(2a) = (0, 7)$, so $2a \in A^{-1}\mathbb{Z}^2$; and $D \subset A^{-1}\mathbb{Z}^2 / \mathbb{Z}^2$. *$T$.* For $K2_\mu$ the new second argument is $a + \mu + h_A \equiv 2a + \mu \equiv 0$; for $K3_\mu = G_{\mu, a}$ it is $a + \mu + h_A \equiv 0$ (giving $K4_\mu$); for $K4_\mu$ it is $\mu + h_A \equiv a$ (giving $K3_\mu$); all congruences modulo $A^{-1}\mathbb{Z}^2$, where Lemma 33.3(ii) applies. *$S$.* $K2_\mu \mapsto \sum_p G_{p, -a-\mu}$ and $-a - \mu \equiv a$; $K3_\mu \mapsto \sum_p G_{a+p, -\mu}$ and $-\mu \equiv 0$; $K4_\mu \mapsto \sum_p G_{p, -\mu}$ and $-\mu \equiv 0$. The phases introduced are $\mathbf{e}(B(\alpha, \lambda))$ with $\alpha \in \frac{1}{70}\mathbb{Z}^2$ and $\lambda \in A^{-1}\mathbb{Z}^2$, roots of unity of order dividing 9800. □

**Proposition 33.6** (Dimension; computer-assisted)**.** $\dim \mathcal{V}_{35} = 36$.

*Proof. Upper bound.* Lemma 33.3(iii), applied to twelve automorphs $g$ of $A$ (including $\pm I$ and $\gamma = \left(\begin{smallmatrix} -1 & -7 \\ 2 & 13 \end{smallmatrix}\right)$), followed by (i) and (ii), shows that each of the 105 functions equals a root of unity times one of 36 orbit representatives, 12 in each family (exact rational computation, `A35_lawspace_exact/orbits.py`). *Lower bound.* At the 216 rationals $X_{216} = \{h/m : 1 \le m \le 15,\ 1 \le h \le 3m,\ \gcd(h,m) = 1\}$ every value $G_{\alpha,\beta}(h/m)$ is a finite mean over a $4m \times 4m$ box and lies in $R = \mathbb{Z}[\zeta_L][1/(4m)^2]$ with $L = 3\,531\,528\,000$. For a prime $p \equiv 1 \pmod{L}$, sending $\zeta_L$ to a primitive $L$-th root of unity in $\mathbb{F}_p$ defines a ring homomorphism $R \to \mathbb{F}_p$, under which the rank of a matrix cannot increase. The $216 \times 105$ value matrix has rank 36 modulo each of $p = 17\,657\,640\,001$, $21\,189\,168\,001$, $35\,315\,280\,001$, $42\,378\,336\,001$ (`A51_bridgeC_exact/rank216.py`). So the rank in characteristic 0 is at least 36. □

**Proposition 33.7** (Exact $\mathrm{SL}_2$-relations and irreducibility; computer-assisted)**.** *Let $M_S$, $M_T$ be the matrices of $S$ and $T$ on $\mathcal{V}_{35}$ in the basis of orbit representatives.*

*(i)* $M_S^4 = I$ *and* $(M_S M_T)^3 = M_S^2$ *exactly over* $\mathbb{Q}(\zeta_{9800})$*, with* $\sqrt{35} = -i\sum_{k \bmod 35} \left(\frac{k}{35}\right)\zeta_{35}^k$.

*(ii)* $\dim \mathrm{End}_{\mathrm{SL}_2(\mathbb{Z})}(\mathcal{V}_{35}) = 1$*, and $\mathcal{V}_{35}$ is isomorphic to the 36-dimensional summand $\mathcal{E}_{36}$ of the decomposition $\mathcal{E}_{36} \oplus \mathcal{C}_6$ of the 42-dimensional lift representation of Part II at $s = 7$; $\mathcal{E}_{36}$ and $\mathcal{C}_6$ are irreducible.*

*In particular every non-zero vector of $\mathcal{V}_{35}$ generates $\mathcal{V}_{35}$ under $S$ and $T$.*

*Proof.* (i) Exact reduction modulo the cyclotomic polynomial $\Phi_{9800}$ of all 1296 entries (`A51_bridgeC_exact/sl2_exact.py`). (ii) The intertwining equations were solved over $\mathbb{F}_p$ at two primes; a dimension over $\mathbb{F}_p$ bounds the dimension in characteristic 0 from above, and the identity gives $\dim \mathrm{End} \ge 1$. An explicit isomorphism $X : \mathcal{V}_{35} \to \mathcal{E}_{36}$ was verified exactly modulo $\Phi_{58800}$ ($X\,S_V = S_E\,X$ in all 1296 entries, $X$ a product of invertible matrices; `A35_lawspace_exact/exact_intertwiner.py`). The lift representation is unitary, hence semisimple, and its commutant has dimension 2 (exact upper bound, `A32_35_to_rho7/exact_hom.py`); with two non-zero invariant summands $\mathcal{E}_{36}$, $\mathcal{C}_6$ this forces both to be irreducible and non-isomorphic. The final statement holds for any irreducible representation. □

# 34 The Bridge Series and the normalized Bridge Series at leading order

## 34.1 Companion series

**Definition 34.1.** Let $K_+ = \{(n,k) \in \mathbb{R}^2 : n \ge 0,\ n + 21k \ge 0\}$ and $K_- = -K_+$. For $v \in \mathbb{Z}^2$ let $\mathrm{wt}(v) = \mathbf{1}[v_1 \ge 0, v_2 \ge 0] - w(v)$. For each of the 105 pairs $(\alpha,\beta)$ of Definition 33.2 ($\alpha \in \{a+\mu, \mu\}$, $\beta \in \{0, a\}$, excluding the $K1$ pairs) the *companion series* is

$$S_{\alpha,\beta}(\tau) = \sum_{v \in \mathbb{Z}^2} \mathrm{wt}(v)\,\mathbf{e}\big(B(\alpha+v,\beta)\big)\,\mathbf{e}\big(\tau Q(\alpha+v)\big).$$

Since $Q(a+v) = Q_7(v) - \frac{7}{20}$, $S_{a,0} = q^{-7/20}(P_7^{\mathrm{orth}} - H_7) = q^{-7/20}C$: the *normalized Bridge Series*.

**Lemma 34.2** (Weights)**.** $B(c_1,v) = 5v_1$, $B(c_2,v) = -v_1 - 21v_2$, *so* $w(v) = \frac{1}{2}(\mathrm{sgn}\,v_1 + \mathrm{sgn}(v_1 + 21v_2))$. *Hence $w = 1$ on the interior of $K_+$, $w = -1$ on the interior of $K_-$, $w = \pm\frac{1}{2}$ on the boundary rays of $K_\pm$ other than 0, and $w = 0$ elsewhere. The series $S_{\alpha,\beta}$ converges absolutely for $\tau \in \mathbb{H}$.*

*Proof.* The formulas for $B$ are direct. If $w(v) \neq 0$ then $v \in K_+ \cup K_-$, and $Q(\xi u_1 + \eta u_2) = \frac{595}{2}\xi^2 + 140\xi\eta + \frac{7}{2}\eta^2$ with $u_1 = (21, -1)$, $u_2 = (0, 1)$ is positive definite on $\xi, \eta \geq 0$; on the orthant $Q(n, k) \geq n^2 + \frac{7}{2}k^2$. So $\operatorname{Re}(-2\pi i\tau Q) \leq -c|v|^2$ on the support of wt, for some $c > 0$ depending on $\operatorname{Im}\tau$. □

## 34.2 Decomposition into pieces

A *piece* is a triple $(c, U, \sigma)$ with $c \in \mathbb{Z}^2$, $U = (u_1\, u_2) \in M_2(\mathbb{Z})$ and a sign $\sigma$; it stands for the sum over $c + U\mathbb{Z}^2_{\geq 0}$. A *ray* is a pair $(c, u)$ with weight $\sigma$, standing for the sum over $c + u\mathbb{N}$.

**Proposition 34.3** (Exact decomposition)**.** *Let $u_1 = (21, -1)$, $u_2 = (0, 1)$, $e_1 = (1, 0)$. For every function $\Phi$ on $\mathbb{Z}^2$ with $\sum |\operatorname{wt}(v)\Phi(v)| < \infty$ and $\sum_{K_\pm \cap \mathbb{Z}^2} |\Phi| < \infty$,*

$$\sum_v \operatorname{wt}(v)\Phi(v) = \sum_{z \in \mathbb{Z}^2_{\geq 0}} \Phi(z) - \sum_{r=0}^{20}\sum_{z \geq 0} \Phi(re_1 + Uz) + \sum_{r=0}^{20}\sum_{z \geq 0} \Phi(-re_1 - Uz) + \tfrac{1}{2}\sum_{b \geq 0}\big(\Phi(bu_2) + \Phi(bu_1) - \Phi(-bu_2) - \Phi(-bu_1)\big),$$

*with $U = (u_1\, u_2)$.*

*Proof.* For $(n, k) \in \mathbb{Z}^2$ write $n = 21j + r$ with $0 \leq r < 21$. Then $n \geq 0 \iff j \geq 0$, and $n + 21k \geq 0 \iff j + k \geq 0$ because $0 \leq r < 21$. As $(n, k) = re_1 + ju_1 + (j + k)u_2$, the map $(r, j, i) \mapsto re_1 + ju_1 + iu_2$ is a bijection from $\{0, \dots, 20\} \times \mathbb{Z}^2_{\geq 0}$ onto $K_+ \cap \mathbb{Z}^2$. The two boundary rays of $K_+$ are $\mathbb{N}u_2$ ($n = 0$) and $\mathbb{N}u_1$ ($n + 21k = 0$); they meet only at 0. By Lemma 34.2, $\sum_v w(v)\Phi(v) = \big[\sum_{K_+} \Phi - \frac{1}{2}\sum_{\mathbb{N}u_1} \Phi - \frac{1}{2}\sum_{\mathbb{N}u_2} \Phi\big] - \big[\sum_{K_-} \Phi - \frac{1}{2}\sum_{-\mathbb{N}u_1} \Phi - \frac{1}{2}\sum_{-\mathbb{N}u_2} \Phi\big]$: interior points get $\pm 1$, non-zero ray points $\pm\frac{1}{2}$, and 0 gets $(1 - \frac{1}{2} - \frac{1}{2}) - (1 - \frac{1}{2} - \frac{1}{2}) = 0$. Subtracting from the orthant sum gives the identity. □

For a piece $(c, U)$ and a pair $(\alpha, \beta)$ put

$$p = Q(u_1), \quad r = Q(u_2), \quad \mathsf{q} = B(u_1, u_2), \quad \tilde{Q}(z) = Q(Uz) = pz_1^2 + \mathsf{q}z_1z_2 + rz_2^2,$$
$$l_i = B(\alpha + c, u_i), \quad Q_0 = Q(\alpha + c),$$

so that $Q(\alpha + c + Uz) = E(z) := \tilde{Q}(z) + l_1z_1 + l_2z_2 + Q_0$. For the orthant $(p, \mathsf{q}, r) = (1, 7, \frac{7}{2})$; for the pieces of $K_\pm$, $(p, \mathsf{q}, r) = (\frac{595}{2}, 140, \frac{7}{2})$. In both cases $p, r > 0$, $\mathsf{q} \geq 0$ and $\Delta := \mathsf{q}^2 - 4pr > 0$ ($\Delta = 35$ and $\Delta = 15435 = 35 \cdot 441$).

**Lemma 34.4** (Periodicity)**.** *Let $x = h/m$, $N = 2m$, and $\varphi(\nu) = \mathbf{e}(xQ(\nu) + B(\nu, \beta))$ for $\nu \in \alpha + \mathbb{Z}^2$, with $(\alpha, \beta)$ one of the 105 pairs or $(\alpha, \beta) = (a, 0)$. Then $\varphi(\nu + Nu) = \varphi(\nu)$ for all $u \in \mathbb{Z}^2$. Consequently, for every piece, $f(z) := \varphi(\alpha + c + Uz)$ is $N$-periodic in each coordinate, and for every ray $b \mapsto \varphi(\alpha + c + bu)$ is $N$-periodic.*

*Proof.* $\varphi(\nu + Nu)/\varphi(\nu) = \mathbf{e}(xNB(\nu, u) + xN^2Q(u) + NB(u, \beta))$. Here $A\nu \in A\alpha + \mathbb{Z}^2$, and $A\alpha \in \mathbb{Z}^2$ (if $\alpha = \mu$) or $A\alpha \in (0, \frac{7}{2}) + \mathbb{Z}^2$ (if $\alpha = a + \mu$), since $A\mu \in \mathbb{Z}^2$. So $B(\nu, u) = u^{\mathsf{T}}A\nu \in \frac{1}{2}\mathbb{Z}$ and $xNB(\nu, u) \in h\mathbb{Z}$. $Q(u) \in \frac{1}{2}\mathbb{Z}$ gives $xN^2Q(u) \in 2mh\mathbb{Z}$. $B(u, 0) = 0$ and $B(u, a) = \frac{7}{2}u_2$ give $NB(u, \beta) \in m\mathbb{Z}$. □

### 34.3 Leading order

**Theorem 34.5** (Profile of $C$ and of $q^{-7/20}C$)**.** *Let $G(x) = (2m)^{-2}\sum_{v \bmod 2m} \mathbf{e}(xQ_7(v))$. Then $G(x) = \mathbf{e}(7x/20)\,K2_0(x)$ for every rational $x$, and $G \notin \mathcal{V}_{35}$ (computer-assisted).*

*Proof.* $Q_7(v) = Q(v+a) + \frac{7}{20}$ and $v \mapsto v+a$ maps $\mathbb{Z}^2$ onto $a+\mathbb{Z}^2$, so $G(x) = \mathbf{e}(7x/20)G_{a,0}(x) = \mathbf{e}(7x/20)K2_0(x)$. If $G \in \mathcal{V}_{35}$, then on $X_{216}$ the column of values of $G$ lies in the complex span of the 105 columns, and the $216 \times 106$ matrix has rank at most 36 over $\mathbb{Q}(\zeta_L)$, with the notation of the proof of Proposition 33.6 (the values of $G$ also lie in $R$). Reduction modulo $p$ cannot raise this rank. The computed rank modulo each of the four primes of Proposition 33.6 is 37, and each run verifies the identity $G = \mathbf{e}(7x/20)K2_0$ exactly modulo $p$ at all 216 points (`BRIDGE_C_35/scripts/testA.py`). □

**Theorem 34.6** (Radial asymptotics at leading order)**.** *Let $I = \log(6+\sqrt{35})/(2\sqrt{35})$. For every rational $x$, as $t \to 0^+$,*

$$t\,P_7^{\mathrm{orth}}(\mathbf{e}(x)e^{-t}) = I\,G(x) + O(t^{1/2}), \quad t\,H_7(\mathbf{e}(x)e^{-t}) = O(t^{1/2}), \quad t\,C(\mathbf{e}(x)e^{-t}) = I\,G(x) + O(t^{1/2}),$$

*and $t\,S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = I\,G_{\alpha,\beta}(x) + O(t^{1/2})$ for every companion series. In particular the leading profile of $q^{-7/20}C$ is $I\,K2_0 \in \mathcal{V}_{35}$, and that of $C$ is $I\,G \notin \mathcal{V}_{35}$.*

*Proof.* This is the $t^{-1}$ part of Theorem 35.6 below, together with Corollary 35.7 and Theorem 34.5; the statements for $P_7^{\mathrm{orth}}$ and $H_7$ are the orthant term and the $K_\pm$ terms of the same computation. □

**Theorem 34.7** (Vector-valued $S$-law of $C$ at leading order)**.** *Let $\Lambda_C(y) = \lim_{t\to 0^+} t\,C(\mathbf{e}(y)e^{-t})$. For every rational $x > 0$,*

$$\Lambda_C(-1/x) = I\,\mathbf{e}\Big(-\frac{7}{20x}\Big)\frac{1}{x\sqrt{35}}\sum_{p\in D}\mathbf{e}(-B(p,2a))\,K3_p(x).$$

*The leading profile $\Lambda_{a,0} = I\,K2_0$ of $q^{-7/20}C$, together with its images under $x \mapsto x+1$ and under $x \mapsto -1/x$ with the factor $|x|^{-1}$, spans $\mathcal{V}_{35}$; $S$ and $T$ act on it by the matrices of Proposition 33.7.*

*Proof.* By Theorems 34.5 and 34.6, $\Lambda_C(y) = I\mathbf{e}(7y/20)K2_0(y)$. Theorem 33.4 with $(\alpha,\beta) = (a,0)$ gives $K2_0(-1/x) = (x\sqrt{35})^{-1}\sum_p G_{p,-a}(x)$. Since $A(2a) \in \mathbb{Z}^2$, Lemma 33.3(ii) with $\lambda = -2a$ gives $G_{p,-a} = G_{p,a-2a} = \mathbf{e}(-B(p,2a))G_{p,a} = \mathbf{e}(-B(p,2a))K3_p$. Multiply by $I\mathbf{e}(7(-1/x)/20)$. The spanning statement follows from $K2_0 \neq 0$ ($|K2_0(1)| = |G(1)| = 1$ by Theorem 11.4) and Proposition 33.7. □

**Remark 34.8.** By Theorem 11.4, $|\Lambda_C(x)|/I$ equals $\sqrt{7}/7$ at $x = \frac{1}{7}, \frac{3}{7}, \frac{5}{7}$, equals 1 at $x = 1$ and at $x = -1$, equals $\frac{1}{3}$ at $-\frac{1}{3}$, 0 at $-\frac{1}{5}$ and $\sqrt{7}/7$ at $-\frac{1}{7}$. Hence $|\Lambda_C(-1/(7x))|/(|7x|^{-1/2}|\Lambda_C(x)|)$ takes the values $\sqrt{7}, \sqrt{7/3}, 0, 1$ at $x = \frac{1}{7}, \frac{3}{7}, \frac{5}{7}, 1$: no scalar factor $|7x|^{-1/2}$ relates $\Lambda_C$ to its $W_7$-image, in accordance with Theorem 11.5(iii).

# 35 The radial expansion through order $t^0$

## 35.1 One-variable lemmas

**Lemma 35.1** (Euler–Maclaurin with offset)**.** *Let $N \ge 1$, $\rho \in \{0, \dots, N-1\}$, $y_0 = \rho/N$, and let $h \in C^3[0,\infty)$ satisfy $|h^{(i)}(u)| \le c_1 e^{-c_2 u^2}$ for $i \le 3$, with constants $c_1, c_2 > 0$. Then*

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h + R_1, \qquad |R_1| \le \tfrac12 |h(0)| + \tfrac12 \int_0^\infty |h'|,$$

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h - B_1(y_0)h(0) + R_2, \qquad |R_2| \le \tfrac{N}{6}\int_0^\infty |h''|,$$

$$\sum_{j\ge 0} h(\rho + Nj) = N^{-1}\int_0^\infty h - B_1(y_0)h(0) - \tfrac{N}{2}B_2(y_0)h'(0) + R_3, \qquad |R_3| \le \tfrac{N^2}{9}\int_0^\infty |h'''|.$$

*Proof.* Let $\phi(u) = h(Nu)$, so the sum is $\sum_{j\ge0}\phi(y_0+j)$. For $u \ge 0$ the counting function $S(u) = \#\{j \ge 0 : y_0 + j \le u\}$ equals $u + \frac12 - y_0 - \bar B_1(u - y_0) = u - B_1(y_0) - \bar B_1(u-y_0)$, and $S(0^-) = 0$. Stieltjes integration by parts on $[0,\infty)$ gives

$$\sum_{j\ge0}\phi(y_0+j) = -\int_0^\infty \phi'(u)S(u)\,du = \int_0^\infty \phi - B_1(y_0)\phi(0) + \int_0^\infty \phi'(u)\bar B_1(u-y_0)\,du.$$

Since $|\bar B_1| \le \frac12$ and $|B_1(y_0)| \le \frac12$, and $\int\phi = N^{-1}\int h$, $\int|\phi'| = \int|h'|$, the first statement follows. Using $\bar B_1 = \frac12 \bar B_2'$ and continuity of $\bar B_2$, $\int_0^\infty \phi'\bar B_1(u-y_0)du = -\frac12\phi'(0)B_2(y_0) - \frac12\int_0^\infty \phi''\bar B_2(u-y_0)du$ (note $\bar B_2(-y_0) = B_2(y_0)$). Since $|\phi'(0)| \le \int|\phi''|$ and $|\bar B_2| \le \frac16$, the remainder after $-B_1(y_0)\phi(0)$ is at most $\frac16\int|\phi''| = \frac{N}{6}\int|h''|$. One further integration by parts, with $\bar B_2 = \frac13 \bar B_3'$, gives $-\frac12\int\phi''\bar B_2 = \frac16\phi''(0)\bar B_3(-y_0) + \frac16\int\phi'''\bar B_3(u-y_0)du$; with $|\bar B_3| \le \sqrt3/36$ and $|\phi''(0)| \le \int|\phi'''|$ this is at most $\frac{\sqrt3}{108}\int|\phi'''| \le \frac19 N^2\int|h'''|$. Finally $\phi'(0) = Nh'(0)$. □

**Lemma 35.2** (Gaussian bounds)**.** *Let $E(z) = \tilde Q(z) + l_1 z_1 + l_2 z_2 + Q_0$ with $\tilde Q(z) = pz_1^2 + \mathsf{q} z_1 z_2 + r z_2^2$, $p, r > 0$, $\mathsf{q} \ge 0$, and put $\kappa = \min(p, r)$ and $F_t = e^{-tE}$. For every multi-index $a$ with $|a| \le 3$ there is $C_a$ such that, for $0 < t \le 1$ and $z \in \mathbb{R}^2_{\ge0}$,*

$$|\partial^a F_t(z)| \le C_a\, t^{|a|/2}\, e^{-t\kappa|z|^2/4}.$$

*Consequently, uniformly in the remaining variable, $\int_0^\infty |\partial^a F_t|\,dz_i = O(t^{(|a|-1)/2})$, $\iint_{\mathbb{R}^2_{\ge0}}|\partial^a F_t| = O(t^{|a|/2-1})$, and $\sum_{j\ge0} e^{-t\kappa(\rho+Nj)^2/4} = O(t^{-1/2})$.*

*Proof.* On $z \ge 0$, $\tilde Q(z) \ge \kappa|z|^2$, so $tE(z) \ge t\kappa|z|^2 - t|l||z| - t|Q_0| \ge \frac{t\kappa}{2}|z|^2 - c_1$ with $c_1 = |l|^2/(2\kappa) + |Q_0|$. A derivative of order $k$ of $F_t$ is $F_t$ times a sum of products of $k-m$ factors $-t\partial_j E$ (each $\le Ct(1+|z|)$) and $m$ factors $-t\partial_{ij}E$ (each $\le Ct$), with $0 \le m \le k/2$. Such a product is at most $Ct^{k-m}(1+|z|)^{k-2m} = Ct^{k/2}(\sqrt t(1+|z|))^{k-2m}$. Since $\sqrt t(1+|z|) \le 1 + \sqrt t|z|$ and $(1+s)^j e^{-\kappa s^2/4}$ is bounded on $s \ge 0$, the bound follows. The consequences follow from $\int_0^\infty e^{-t\kappa s^2/4}ds = O(t^{-1/2})$ and from the first statement of Lemma 35.1. □

**Lemma 35.3** (Integrals)**.** *With the notation of Lemma 35.2 and $\Delta = \mathsf{q}^2 - 4pr > 0$, as $t \to 0^+$,*

$$\iint_{\mathbb{R}^2_{\ge 0}} F_t = t^{-1}\mathrm{Ar} - t^{-1/2}(l_1 J_a + l_2 J_b) + \big(\tfrac{1}{2}K(l) - Q_0\mathrm{Ar}\big) + O(t^{1/2}),$$

$$\int_0^\infty F_t(s,0)\,ds = \tfrac{1}{2}\sqrt{\pi/(tp)} - \tfrac{l_1}{2p} + O(t^{1/2}), \qquad \int_0^\infty F_t(0,s)\,ds = \tfrac{1}{2}\sqrt{\pi/(tr)} - \tfrac{l_2}{2r} + O(t^{1/2}),$$

$$\int_0^\infty \partial_2 F_t(s,0)\,ds = -\tfrac{\mathsf{q}}{2p} + O(t^{1/2}), \qquad \int_0^\infty \partial_1 F_t(0,s)\,ds = -\tfrac{\mathsf{q}}{2r} + O(t^{1/2}), \qquad F_t(0,0) = 1 + O(t),$$

*where $J_a = \iint u_1 e^{-\tilde{Q}(u)}du$, $J_b = \iint u_2 e^{-\tilde{Q}(u)}du$, $K(l) = \iint (l_1u_1 + l_2u_2)^2 e^{-\tilde{Q}(u)}du$ (over $\mathbb{R}^2_{\ge 0}$), and*

$$\mathrm{Ar} = \iint_{\mathbb{R}^2_{\ge 0}} e^{-\tilde{Q}} = \frac{\log\frac{\mathsf{q}+\sqrt{\Delta}}{\mathsf{q}-\sqrt{\Delta}}}{2\sqrt{\Delta}}, \quad J_a = \frac{\sqrt{\pi}}{4}\cdot\frac{2}{4pr-\mathsf{q}^2}\Big(2\sqrt{r} - \frac{\mathsf{q}}{\sqrt{p}}\Big), \quad J_b = \frac{\sqrt{\pi}}{4}\cdot\frac{2}{4pr-\mathsf{q}^2}\Big(2\sqrt{p} - \frac{\mathsf{q}}{\sqrt{r}}\Big),$$

$$K(l) = \int_0^\infty \frac{(l_1 + l_2\lambda)^2}{2\tilde{Q}(1,\lambda)^2}\,d\lambda.$$

*Proof.* Substituting $z = u/\sqrt{t}$, $\iint F_t = t^{-1}\iint e^{-\tilde{Q}(u)-w(u)}du$ with $w = \sqrt{t}(l_1u_1 + l_2u_2) + tQ_0$. By Taylor's theorem $|e^{-w} - 1 + w - \frac{1}{2}w^2| \le \frac{1}{6}|w|^3 e^{|w|}$, and $e^{-\tilde{Q}(u)+|w(u)|} \le e^{-\kappa|u|^2+|l||u|+|Q_0|}$ is integrable, so the expansion follows with the $t^0$ term $\iint(\frac{1}{2}(l_1u_1 + l_2u_2)^2 - Q_0)e^{-\tilde{Q}}$. The one-variable integrals are treated in the same way with $\int_0^\infty e^{-pu^2}du = \frac{1}{2}\sqrt{\pi/p}$, $\int_0^\infty ue^{-pu^2}du = \frac{1}{2p}$. For the derivative integrals, $\partial_2 F_t(s,0) = -t(\mathsf{q}s + l_2)F_t(s,0)$, and the substitution $s = u/\sqrt{t}$ gives $-t\cdot t^{-1}\mathsf{q}\int ue^{-pu^2}du + O(t^{1/2})$. $F_t(0,0) = e^{-tQ_0}$. For the closed forms write $u = \rho(1,\lambda)$, $du = \rho\,d\rho\,d\lambda$. Then $\mathrm{Ar} = \int_0^\infty d\lambda/(2\tilde{Q}(1,\lambda))$; the roots of $r\lambda^2 + \mathsf{q}\lambda + p$ are $\lambda_\pm = (-\mathsf{q} \pm \sqrt{\Delta})/(2r) < 0$ since $\mathsf{q} > \sqrt{\Delta}$, and partial fractions give $\int_0^\infty d\lambda/(r(\lambda-\lambda_+)(\lambda-\lambda_-)) = \Delta^{-1/2}\log(\lambda_-/\lambda_+)$. Next $J_a = \int_0^\infty\int_0^\infty \rho^2 e^{-\rho^2\tilde{Q}(1,\lambda)}d\rho\,d\lambda = \frac{\sqrt{\pi}}{4}\int_0^\infty \tilde{Q}(1,\lambda)^{-3/2}d\lambda$, and $\frac{d}{d\lambda}\frac{2(2r\lambda+\mathsf{q})}{(4pr-\mathsf{q}^2)\sqrt{r\lambda^2+\mathsf{q}\lambda+p}} = (r\lambda^2+\mathsf{q}\lambda+p)^{-3/2}$; evaluating at 0 and $\infty$ gives $J_a$. $J_b$ follows by the parametrisation $u = \rho(\lambda, 1)$, which exchanges $p$ and $r$. $K(l)$ follows from $\int_0^\infty \rho^3 e^{-\rho^2 c}d\rho = 1/(2c^2)$. □

## 35.2 The expansion

**Theorem 35.4** (Expansion of a piece)**.** *Let $(p,\mathsf{q},r)$, $l_1, l_2, Q_0$ be as above with $p, r > 0$, $\mathsf{q} \ge 0$, $\Delta > 0$, and let $f : \mathbb{Z}^2_{\ge 0} \to \mathbb{C}$ be $N$-periodic in each coordinate with $|f| \le 1$. Put, with $z$ running over $\{0,\dots,N-1\}^2$,*

$$\mathrm{mean}(f) = N^{-2}\sum_z f(z), \qquad W_i = N^{-1}\sum_z f(z)B_1(z_i/N),$$

$$V_i = \sum_z f(z)B_2(z_i/N), \qquad \mathsf{C} = \sum_z f(z)B_1(z_1/N)B_1(z_2/N).$$

*Then, as $t \to 0^+$,*

$$\sum_{z\in\mathbb{Z}^2_{\ge 0}} f(z)e^{-tE(z)} = t^{-1}A_0^P + t^{-1/2}A_1^P + A_2^P + O(t^{1/2}),$$

*with*

$$A_0^P = \mathrm{mean}(f)\,\mathrm{Ar},$$

$$A_1^P = -\,\mathrm{mean}(f)(l_1J_a + l_2J_b) - \tfrac{\sqrt{\pi}}{2}\big(r^{-1/2}W_1 + p^{-1/2}W_2\big),$$

$$A_2^P = \mathrm{mean}(f)\big(\tfrac{1}{2}K(l) - Q_0\mathrm{Ar}\big) + \frac{l_2W_1}{2r} + \frac{l_1W_2}{2p} + \frac{\mathsf{q}}{4}\Big(\frac{V_1}{r} + \frac{V_2}{p}\Big) + \mathsf{C}.$$

*Proof.* Write the sum as $\sum_\rho f(\rho)\Sigma_\rho$ with $\Sigma_\rho = \sum_{j\in\mathbb{Z}^2_{\ge 0}} F_t(\rho+Nj)$, $\rho\in\{0,\dots,N-1\}^2$, and $y_i=\rho_i/N$. Fix $\rho$. For each $z_1=\rho_1+Nj_1$ apply the third statement of Lemma 35.1 in $z_2$:

$$\Sigma_\rho = \sum_{j_1\ge 0}\Big[N^{-1}\int_0^\infty F_t(z_1,s)ds - B_1(y_2)F_t(z_1,0) - \tfrac{N}{2}B_2(y_2)\partial_2F_t(z_1,0) + R_3(z_1)\Big].$$

By Lemma 35.2, $\sum_{j_1}|R_3(z_1)| \le \frac{N^2}{9}\sum_{j_1}\int_0^\infty|\partial_2^3F_t(z_1,s)|ds \le Ct^{3/2}\cdot t^{-1/2}\cdot t^{-1/2} = O(t^{1/2})$. For the first term apply the third statement of Lemma 35.1 in $z_1$ to $g(z_1)=N^{-1}\int_0^\infty F_t(z_1,s)ds$ (differentiation under the integral is justified by Lemma 35.2); its remainder is at most $\frac{N^2}{9}\int|g'''| \le CN\iint|\partial_1^3F_t| = O(t^{1/2})$. This gives $N^{-2}\iint F_t - N^{-1}B_1(y_1)\int_0^\infty F_t(0,s)ds - \frac12 B_2(y_1)\int_0^\infty\partial_1F_t(0,s)ds + O(t^{1/2})$. For the second term apply the second statement of Lemma 35.1 to $F_t(\cdot,0)$; the remainder is at most $\frac{N}{6}\int|\partial_1^2F_t(s,0)|ds = O(t^{1/2})$. This gives $-B_1(y_2)[N^{-1}\int_0^\infty F_t(s,0)ds - B_1(y_1)F_t(0,0)] + O(t^{1/2})$. For the third term apply the first statement of Lemma 35.1 to $\partial_2F_t(\cdot,0)$; the remainder is at most $\frac12|\partial_2F_t(0,0)| + \frac12\int|\partial_1\partial_2F_t(s,0)|ds = O(t^{1/2})$. This gives $-\frac12 B_2(y_2)\int_0^\infty\partial_2F_t(s,0)ds + O(t^{1/2})$. Altogether

$$\begin{aligned}\Sigma_\rho =& N^{-2}\iint F_t - N^{-1}B_1(y_1)\int_0^\infty F_t(0,s)ds - N^{-1}B_1(y_2)\int_0^\infty F_t(s,0)ds + B_1(y_1)B_1(y_2)F_t(0,0)\\ &- \tfrac12 B_2(y_1)\int_0^\infty\partial_1F_t(0,s)ds - \tfrac12 B_2(y_2)\int_0^\infty\partial_2F_t(s,0)ds + O(t^{1/2}).\end{aligned}$$

Insert Lemma 35.3, multiply by $f(\rho)$ and sum over $\rho$. The coefficient of $t^{-1}$ is $N^{-2}\sum f\cdot \mathrm{Ar}$. The coefficient of $t^{-1/2}$ is $-\operatorname{mean}(f)(l_1J_a+l_2J_b) - N^{-1}\sum_\rho f(\rho)B_1(y_1)\frac{\sqrt\pi}{2\sqrt r} - N^{-1}\sum_\rho f(\rho)B_1(y_2)\frac{\sqrt\pi}{2\sqrt p}$. The $t^0$ coefficient is $\operatorname{mean}(f)(\frac12 K - Q_0\mathrm{Ar}) + W_1\frac{l_2}{2r} + W_2\frac{l_1}{2p} + \mathsf{C} + \frac12 V_1\frac{\mathfrak{q}}{2r} + \frac12 V_2\frac{\mathfrak{q}}{2p}$. □

**Lemma 35.5** (Expansion of a ray)**.** *Let $Q(u)>0$, $l_u = B(u,\alpha+c)$, and let $b\mapsto f(b)$ be $N$-periodic with $|f|\le 1$. Then*

$$\sum_{b\ge 0} f(b)e^{-tQ(\alpha+c+bu)} = t^{-1/2}\frac{\sqrt\pi}{2}Q(u)^{-1/2}\operatorname{mean}(f) - \operatorname{mean}(f)\frac{l_u}{2Q(u)} - \sum_{b=0}^{N-1} f(b)B_1(b/N) + O(t^{1/2}),$$

*with* $\operatorname{mean}(f) = N^{-1}\sum_{b<N} f(b)$.

*Proof.* Here $h(b) = e^{-t(Q(u)b^2+l_ub+Q(\alpha+c))}$. Apply the second statement of Lemma 35.1 for each residue $\rho$; the remainder is $O(\int|h''|) = O(t^{1/2})$. Then $\int_0^\infty h = \frac12\sqrt{\pi/(tQ(u))} - \frac{l_u}{2Q(u)} + O(t^{1/2})$ and $h(0) = 1+O(t)$, as in Lemma 35.3. □

**Theorem 35.6** (Radial expansion of the companion series)**.** *For each of the* 105 *companion series and for $q^{-7/20}C = S_{a,0}$, and every rational $x$,*

$$S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = t^{-1}A_0(x) + t^{-1/2}A_1(x) + A_2(x) + O(t^{1/2}) \qquad (t\to 0^+),$$

*where $A_j = \sum_{\text{pieces}}\sigma A_j^P + \sum_{\text{rays}}\sigma A_j^{\text{ray}}$ over the signed pieces and rays of Proposition 34.3, with $f(z) = \mathbf{e}(xQ(\alpha+c+Uz) + B(\alpha+c+Uz,\beta))$ and $N=2m$, $A_j^P$ as in Theorem 35.4, and $A_1^{\text{ray}} = \frac{\sqrt\pi}{2}Q(u)^{-1/2}\operatorname{mean}(f)$, $A_2^{\text{ray}} = -\operatorname{mean}(f)\frac{l_u}{2Q(u)} - \sum_{b<N}f(b)B_1(b/N)$, $A_0^{\text{ray}} = 0$.*

*Proof.* With $\Phi(v) = \mathbf{e}(B(\alpha+v,\beta))\mathbf{e}(xQ(\alpha+v))e^{-tQ(\alpha+v)}$, $S_{\alpha,\beta}(\mathbf{e}(x)e^{-t}) = \sum_v \mathrm{wt}(v)\Phi(v)$, and Proposition 34.3 applies by Lemma 34.2. Each piece is a sum as in Theorem 35.4 with $f$ $N$-periodic by Lemma 34.4, and each ray is as in Lemma 35.5. There are finitely many pieces and rays, so the $O(t^{1/2})$ remainders add up. □

**Corollary 35.7** (Order $t^{-1}$)**.** *$A_0(x) = I\,G_{\alpha,\beta}(x)$ for every companion. In particular $A_0 = I\,K2_0$ for $q^{-7/20}C$, and $t\,H_7(\mathbf{e}(x)e^{-t}) = O(t^{1/2})$.*

*Proof.* The orthant has $U = I$ and $\mathrm{Ar} = \log((7+\sqrt{35})/(7-\sqrt{35}))/(2\sqrt{35}) = I$, because $(7+\sqrt{35})/(7-\sqrt{35}) = 6+\sqrt{35}$; its mean is $G_{\alpha,\beta}(x)$. The 21 pieces of $K_+$ have a common $\mathrm{Ar}_+$. As $(r,j)$ runs over $\{0,\dots,20\}\times\{0,\dots,N-1\}$, $21j+r$ runs over $\{0,\dots,21N-1\}$, which covers each residue modulo $N$ exactly 21 times; for each $(r,j)$, $i-j$ covers each residue once as $i$ does. Hence $\sum_r \mathrm{mean}(f_r) = 21\,G_{\alpha,\beta}(x)$. For $K_-$ the same count applies to $\nu = \alpha - v$, and $\alpha - v$ runs over $\alpha+\mathbb{Z}^2$ modulo $N$ as $v$ does, so $\sum_r \mathrm{mean}(f'_r) = 21\,G_{\alpha,\beta}(x)$. The signs are $-1$ and $+1$, so the cone pieces cancel at order $t^{-1}$. The statement for $H_7$ is the $K_\pm$ part of the same computation for $(\alpha,\beta) = (a,0)$, multiplied by $q^{7/20}$, which changes the expansion by a factor $1+O(t)$ and the phase $\mathbf{e}(7x/20)$. □

**Corollary 35.8** (Order $t^{-1/2}$: algebraicity)**.** *For every companion and every rational $x$, $A_1(x)/\sqrt{\pi}$ is an algebraic number. The number $\log(6+\sqrt{35})$ does not enter $A_1$.*

*Proof.* By Theorems 35.4 and 35.6, $A_1/\sqrt{\pi}$ is a finite sum of products of roots of unity (values of $f$), rational numbers ($B_1$-weights, $l_i$, signs, $\frac{1}{2}$), and the algebraic numbers $J_a/\sqrt{\pi}$, $J_b/\sqrt{\pi}$, $r^{-1/2}$, $p^{-1/2}$, $Q(u)^{-1/2}$ with $p, \mathtt{q}, r, Q(u)$ rational. □

**Remark 35.9.** For the orthant piece alone at $x = 0$ and $(\alpha,\beta) = (a,0)$ the formula gives $0.5256172043\ldots$, the $t^{-1/2}$ coefficient $a_P = \sqrt{\pi}(49-2\sqrt{14})/140$ of $P_7^{\mathrm{orth}}$ in Part II (Table 13).

**Proposition 35.10** (Order $t^{-1/2}$ on the zero set of the leading term; computer-assisted)**.** *Let $Z_0 = \{x \in X_{216} : 2 \mid m \text{ or } 5 \mid m\}$ (105 points). For $q^{-7/20}C$, $A_1(x) \neq 0$ for every $x \in Z_0$ and for $x = -\frac{1}{5}$. Since $A_0(x) = 0$ there (Theorem 11.4), $t\,|q^{-7/20}C(\mathbf{e}(x)e^{-t})| = |A_1(x)|\,t^{1/2} + O(t)$ at these points. The minimum of $|A_1|$ over these points is $0.1229\ldots$, and $|A_1(-\frac{1}{5})| = 0.4285\ldots$.*

*Proof.* The formula of Theorem 35.6 is a finite expression. The phase histograms were computed in exact rational arithmetic, and the trigonometric and algebraic values were enclosed in interval arithmetic at 60 digits (`BRIDGE_C_35/order2/data/map2_interval.json`); the enclosure of $|A_1|^2$ excludes 0 at all 106 points, with maximal width $6\cdot 10^{-59}$. □

## 35.3 Decomposition of $A_1$ by quadratic class

Each term of $A_1$ carries a label: the drift terms $-\mathrm{mean}(f)(l_1J_a + l_2J_b)$ of the pieces; the sawtooth terms $-\frac{\sqrt{\pi}}{2}Q(u_i)^{-1/2}W_i$ of the pieces, labelled by $Q(u_i)$ for the edge direction $u_i$ whose $B_1$-weight they carry (the $W_1$ term carries $r^{-1/2} = Q(u_2)^{-1/2}$, the $W_2$ term $p^{-1/2} = Q(u_1)^{-1/2}$); and the ray terms $\pm\frac{1}{2}\cdot\frac{\sqrt{\pi}}{2}Q(u)^{-1/2}\,\mathrm{mean}(f)$, labelled by $Q(u)$.

**Proposition 35.11** (Exact decomposition of $A_1$)**.** *$Q(e_1) = 1$, $Q(e_2) = Q(0,\pm1) = \frac{7}{2}$ and $Q(\pm(21,-1)) = \frac{595}{2}$. Consequently, for every companion and every rational $x$,*

$$A_1 = D + R + W[1] + W[\tfrac{7}{2}] + W[\tfrac{595}{2}], \qquad R = R[\tfrac{7}{2}] + R[\tfrac{595}{2}],$$

*where $D$ is the sum of the drift terms, $W[\lambda]$ the sum of the sawtooth terms with label $\lambda$, and $R[\lambda]$ the sum of the ray terms with label $\lambda$; $R[\frac{7}{2}]$ collects the rays $\pm\mathbb{N}(0,1)$ and $R[\frac{595}{2}]$ the rays $\pm\mathbb{N}(21,-1)$.*

*Proof.* $Q(21,-1) = \frac{1}{2}(2\cdot 441 - 2\cdot 7\cdot 21 + 7) = \frac{595}{2}$. The edge directions of the pieces are $e_1, e_2$ (orthant) and $\pm u_1, \pm u_2$ (cones), and the rays of Proposition 34.3 are $\pm\mathbb{N}u_1$, $\pm\mathbb{N}u_2$. Every term of $A_1$ in Theorem 35.6 is therefore of exactly one of the listed kinds with one of the listed labels, and the identities are regroupings of a finite sum. □

**Remark 35.12** (Structure of $A_2$)**.** By Theorem 35.6, $A_2$ consists of the moments $\frac{1}{2}K(l) - Q_0\mathrm{Ar}$ of the drift, edge terms $l_i W_j$, terms $\frac{\mathfrak{q}}{4}V_i$ weighted by the cross coefficient, corner terms $\mathsf{C}$ (finite sums of Dedekind-sum type), and the ray constants. Only $K(l)$ and Ar are transcendental integrals.

## 36 Order four: the shadow candidate and the law at order $t^0$

The next two questions are (a) whether $A_1$ becomes $S$-covariant after subtracting an explicit shadow contribution, and (b) whether the $t^0$ coefficient $A_2$ of Theorem 35.6, or $A_2$ minus a completion term, satisfies a weight-0 vector-valued law.

For (a), the candidate tested was $\Xi_\varepsilon = \sum_\lambda \varepsilon_\lambda W[\lambda]$ with $\varepsilon \in \{0,1\}^3$, the sawtooth groups of Proposition 35.11. No Category A statement results; the numerical record (no choice of $\varepsilon$ yields closure) is in Appendix D, item D.8.

Question (b) is not settled in this paper (Problem 2 of §38).

# Part IV
# Foundation

## 37 The foundation attempt: what ties RTA to the framework

This section lists the written RTA objects and, for each, what is proved about its relation to the structures of Parts I–III. References to "RTA" are to the RTA manuscript (Residual Ternary Algebra: Packaging Identities, Residual Series $g_s$, and Bridge Series $C$, draft of 24 August 2026), which has no algebraic axioms: its six numbered rules are process rules.

### 37.1 Proved links

**Proposition 37.1** (Identification of objects)**.** *The following hold.*

*(i) The RTA residual series $g_s$ is the series $g_k$ of Part I with $k = s$.*

*(ii) The RTA double-sum identity for $(-q^s;q^s)_\infty g_s$ holds for every $s \geq 1$ as an identity in $\mathbb{Z}[[q]]$ (Proposition 2.2).*

*(iii) The RTA path quadratic $Q_s$, cones $c_1, c_2$, weight $w$, cone series $H_s$, orthant series and Bridge Series $C = P_7^{\mathrm{orth}} - H_7$ are those of §2 and of Part II; $C_{\mathrm{axis}}$ and $C_{\mathrm{third}}$ are those of Part II, §10; the RTA natural characteristic $\chi_{\mathrm{nat}}(7) = (\frac{7}{10}, -\frac{1}{5})$ is the vector $a$.*

*Proof.* (i) and (iii) are comparisons of definitions: RTA Definition 6.1 and §§9–10 against Definition 2.1, §2 and Part II, §10; $\chi_{\mathrm{nat}}(s) = (s/(2(s-2)), -1/(s-2))$ at $s = 7$ equals $a$. (ii) is Proposition 2.2. □

**Corollary 37.2** (Residue law of $g_7$ and its converse)**.** *$g_7(d) = 0$ whenever $d \equiv 3, 5, 6 \pmod 7$. Conversely, $g_7(d) \neq 0$ for every $d \equiv 0, 1, 2, 4 \pmod 7$ outside the finite set $Z_7$ of 24 elements listed in Appendix B ($\max Z_7 = 247$), and the sign of $g_7(d)$ is $(-1)^{d+1}$ for all such $d \geq D_0(7)$.*

*Proof.* By Proposition 3.2 (Part I), $g_7(d)$ is a sum over $n \geq 0$ with $n^2 \equiv d \pmod 7$; the squares modulo 7 are $0, 1, 2, 4$, so the sum is empty for $d \equiv 3, 5, 6$. The converse is the case $k = 7$ of Theorem 8.1 (Part I, §8.1), parts 1 and 2. □

**Theorem 37.3** (The normalized Bridge Series and the 35-component law vector)**.** *The following hold.*

*(i) The leading radial profile of the RTA Bridge Series $C$ is $I\,\mathbf{e}(7x/20)K2_0(x)$, and it does not lie in $\mathcal{V}_{35}$.*

*(ii) The leading radial profile of $q^{-7/20}C$ is $I\,K2_0$, a non-zero element of $\mathcal{V}_{35}$; together with its $S$- and $T$-images it spans $\mathcal{V}_{35}$, and it obeys the vector-valued $S$-law of Theorem 34.7.*

*(iii) Up to order $t^0$, the radial expansion of $q^{-7/20}C$ at every rational point is given in closed form by Theorem 35.6; at order $t^{-1/2}$ the coefficient divided by $\sqrt{\pi}$ is algebraic, and at every point of $Z_0 \cup \{-\frac{1}{5}\}$ it is non-zero.*

*Proof.* Theorems 34.5, 34.6, 34.7, 35.6, Corollary 35.8 and Proposition 35.10. □

## 37.2 Negative results

**Proposition 37.4** (No scalar residual law)**.** *There is no scalar modular, mock-modular (signature $(1,1)$) or unary/partial/false theta law for $C$ of the kinds listed in Theorem 11.5; in particular for no weight $k$ and constant $\lambda$ is $C(\tau) - \lambda(7\tau)^{-k}C(-1/(7\tau))$ radially bounded at all rationals.*

*Proof.* Theorem 11.5, with the stated dependence on Baker's theorem for part (i). □

**Proposition 37.5** (No vector-valued Fricke law at leading order; computer-assisted)**.** *Let $D_7 f(x) = f(7x)$. Then $\mathcal{V}_{35} \cap D_7\mathcal{V}_{35} = 0$ as spaces of functions on $\mathbb{Q}$. In particular $D_7\mathcal{V}_{35} \not\subset \mathcal{V}_{35}$.*

*Proof.* Both spaces have dimension at most 36 (Proposition 33.6). The $216 \times 210$ matrix of values of the 105 functions and of their dilations by 7 at $X_{216}$ has rank 72 modulo each of the four primes of Proposition 33.6 (`A53_RTA_maps`); by the reduction argument of that proof its rank in characteristic 0 is at least 72. Hence $\dim(\mathcal{V}_{35} + D_7\mathcal{V}_{35}) = 72$ and the sum is direct. □

**Proposition 37.6** (Facts about $T_3$ and 539)**.** *Let $T_3(3k) = k$, $T_3(3k+1) = 4k+2$, $T_3(3k+2) = 2k+1$.*

*(i) If $539 = a_1 + \cdots + a_8$ with positive integers and $\max a_i - \min a_i \le 1$, then $\{a_i\} = \{68, 68, 68, 67, 67, 67, 67, 67\}$.*

*(ii) For $M$ coprime to 6 and $L \ge 1$, as $n$ runs over $\mathbb{Z}/3^L M$, $T_3^L(n) \bmod M$ is well defined and takes each value exactly $3^L$ times.*

*(iii) Extended to $\mathbb{Z}_3$, $T_3$ preserves Haar measure, and the itinerary map $n \mapsto (T_3^j(n) \bmod 3)_{j\ge 0}$ is a homeomorphism $\mathbb{Z}_3 \to \{0,1,2\}^{\mathbb{N}}$ conjugating $T_3$ to the shift.*

*Proof.* (i) Let $q = \min a_i$ and let $b$ be the number of parts equal to $q+1$. If $b = 8$ then $8 \mid 539$, false. So $539 = 8q + b$ with $0 \le b \le 7$, and $q = 67$, $b = 3$. (ii) For $N = 3^i M$ ($i \ge 1$) and $n = 3k + r$, replacing $n$ by $n + N$ changes $T_3(n)$ by $N/3$, $4N/3$ or $2N/3$, all $\equiv 0 \pmod{N/3}$; and for fixed $r$ the maps $k \mapsto k, 4k+2, 2k+1$ are bijections of $\mathbb{Z}/(N/3)$ because $N/3$ is odd. So $n \bmod N \mapsto T_3(n) \bmod N/3$ is well defined and 3-to-1; iterate $L$ times. (iii) Each branch $k \mapsto k, 4k+2, 2k+1$ is an affine bijection of $\mathbb{Z}_3$ with unit linear coefficient, and $k \mapsto 3k + r$ maps $\mathbb{Z}_3$ onto $r + 3\mathbb{Z}_3$ scaling Haar measure by $\frac{1}{3}$; hence $\mu(T_3^{-1}B) = \sum_r \mu(B)/3 = \mu(B)$. By induction on $j$, the points whose itinerary begins with a given word of length $j$ form one class modulo $3^j$; so the itinerary map is a continuous bijection of compact Hausdorff spaces, hence a homeomorphism, and it conjugates $T_3$ to the shift by definition. □

**Remark 37.7** (No written link)**.** Proposition 37.6 is a statement about $T_3$ and the integer 539 alone. No written object (in RTA, in Parts I–III, or in any source used here) defines a map from $T_3$, the carrier $\mathcal{O}_{\mathrm{res}} = \{21 + 9j : 0 \le j \le 538\}$, the packaging identities ($539 = \lfloor 4859/9 \rfloor$, $18 + 521$, $3 \cdot 68 + 5 \cdot 67$) or the linking series $\Lambda_B$ to $g_s$, $C$, $\mathcal{V}_{35}$ or the lift representation. RTA states that its packaging identities "are independent of any dynamical map" and of the modular status of the series. The choice $s \in \{7, 8, 11\}$ is attributed in RTA to the packaging arithmetic, but no derivation of these values from it is written. The statement that the $R = ST$ grading of the cusp space is a foundation of RTA has no proof; the only proved statement of that kind is the grading lemma of Part II's framework (the $\mathbb{Z}/3$ grading of equivariant maps by $R$-degree), which concerns representations and not $T_3$.

### 37.3 Where the shadow $\Xi$ enters

RTA describes a weight-$\frac{3}{2}$ shadow $\Xi = \Xi_{\mathrm{even}} + \Xi_{\mathrm{odd}}$ as "the residual-weighted family of shifted partial thetas of squares whose outer coefficients are the residual disc-35 series", and a weight-$\frac{1}{2}$ period candidate as its Eichler integral. Neither is written as an explicit series with stated coefficients, so no statement about $\Xi$ can be proved from the written definitions. What is proved here bounds the place where $\Xi$ must enter:

(i) at order $t^{-1}$ the normalized series is governed by $\mathcal{V}_{35}$ with no correction term (Theorem 34.7);

(ii) at order $t^{-1/2}$ the coefficient $A_1$ is an explicit finite sum of drift, sawtooth and ray terms (Proposition 35.11) with $A_1/\sqrt{\pi}$ algebraic (Corollary 35.8); any $S$-covariant completion of $A_1$ must therefore come from a term that is not visible at order $t^{-1}$;

(iii) at order $t^0$ the coefficient $A_2$ is explicit (Theorem 35.6).

Whether a completion term of Zwegers type, with coefficients derived from the error-function completion of $H_7$, makes $A_1$ or $A_2$ $S$-covariant is open (Problem 1.). The numerical tests carried out so far (Appendix D) found no closure for any regrouping of the finite sum or for the simplest sawtooth candidate.

## 38 Open problems

1. Derive the completion constants of $H_7$ (erf-completion along the walls $(0, 1)$ and $(21, -1)$), form the completed coefficient $A_1 - \Xi^{\mathrm{compl}}$, and decide whether it satisfies a vector-valued $S$-law of weight $\frac{1}{2}$. This would identify the shadow $\Xi$.

2. Decide whether $A_2$, or $A_2$ minus a completion term, satisfies a weight-0 vector-valued law; this is the period-function level.

3. Prove or disprove $|A_1(\frac{1}{2})| = \sqrt{\pi}/4$ (observed numerically) and find the zero set of $A_1$ on the zero set of $A_0$.

4. Find a written object linking $T_3$, $\mathcal{O}_{\mathrm{res}}$ or 539 to $g_s$, $C$ or $\mathcal{V}_{35}$, or prove that a natural class of such maps does not exist.

5. Extend the parity-dominance theorem of Part I beyond $k = 33$ and to the classes of even $k$ that are not one-sign.

6. The open problems of Part II, §32.

# A Computational reproduction for Part I

All files are in the repository `ArvinHampton/539-Labs-Master` under `reports/s_family_paths/PAPER_gk_v1/` (abbreviated `P/`). Environment: Python 3.13, python-flint 0.9.0, mpmath, gcc, Linux; every command was run with `OMP_NUM_THREADS=1` under `nice`. SHA-256 sums of all files are in `P/SHA256SUMS`.

## A.1 Algebraic checks (Sections 3–7)

`P/code/gklib.py` evaluates $g_k(d)$ exactly (Python integers). `P/code/verify_algebra.py` (log: `P/data/verify_algebra_log.txt`) checks:

- stabilisation, sign rule, $c_m = (-1)^m Q(m)$, the product form (4.1) and the truncation formula (Lemmas 4.1–4.5), for $n \le 80$, $m \le 400$;
- the expansion (3.1) against the direct series ($k \le 20$, $k = 27, 33$, $d \le 1500$);
- Theorem 5.1 for all $k \le 1001$ (608 squarefree $k$: all vanish; 392 non-squarefree $k$: all nonzero);
- formula (5.1) in 5121 cases and the exact failures of Corollary 5.4 for all odd prime powers $k \le 1001$;
- the pair reduction (6.1) for $7 \le k \le 130$ and the values $g_{15}(286) = g_{21}(526) = 0$;
- Theorem 7.7 for even $k \le 130$ (values of $g_{2^a}$);
- the sign law and the zero set of $g_2$ for $d \le 6000$; zero sets for $k = 2, 6, 10, 14$ and the bounds $B_c$.

The checks are consistency tests of the proved statements, not parts of proofs, except the exact term lists quoted in proofs (Corollary 5.4, Sections 6.6, 7.2, 7.3), which are exact integer evaluations. `P/code/even_zero_sets.py` produces Table B.2 (`P/data/even_zero_sets_k2mod4.txt`).

Run: `cd P/code && python3 verify_algebra.py` (about 5 seconds).

## A.2 The composite-zero theorem (Section 6)

Directory `P/code/composite/`. From `P/code/composite/code`:

1. `python3 make_Q.py` builds the exact table $Q(0..35200)$ (`../data/Q_exact_35200.pkl`; sha256 of the decimal list `6a2037fb2c51b525ae73b77b8593bcfaa6ad2868315b7b01be4bb62a24e06a78`).
2. `python3 pairs_certify.py 2500 300` (`../data/pairs_certify.log`): for the 15 616 pairs $3 \le s \le 2500$, $k$ odd, $k > 2s$, $k \mid s^2 - 1$, an integer lower bound $LB \le A_1$ and the exact $A_2$; 15 587 pairs have $LB > A_2$; the remaining 29 have $s \le 32$ and are settled by the exact dynamic programme, which was run for all $s \le 300$ (995 pairs). Only $(4, 15)$ and $(8, 21)$ have $A_1 \le A_2$ (equality).
3. `python3 chern_margins.py` (`../data/chern_margins.log`): interval evaluation (mpmath `iv`, 40 digits) of $C_E$, $K$, $\bar\beta(n)$ at the thresholds quoted in Lemma 6.5, the monotonicity ratios, the comparison $|Q - M| \le \bar B$ on $n \in [300, 2999]$ and every 17th $n \le 35200$ (no violations; this comparison is an extra check, not used in the proof), and the margins of Proposition 6.8 on $2501 \le s \le 60000$ (minimum margins 0.44805 and 0.99364).

## A.3 Parity dominance (Section 8)

Directory `P/code/pd/` holds the programs `arb_compat.py`, `iv56.py` (interval type over Arb), `clt56.py` (Algorithms E and C), `blocks56.py` (Algorithm R), `tails56.py`

(Algorithm T), `params56.py`, `make_cells56.py`, `run_pd56.py`, and the data `params56.json`, `cells56_sharp.json`, `consts56_sharp.json`, `pd56_results.json` (hashes in `P/data/pd/sha256_copied.txt`; the files `*_c15.json` belong to a variant not used in the text).

- Window parameters: `python3 params56.py` (Table 8.3).
- Cells and constants: `python3 make_cells56.py sharp` (Theorem 8.12; Table 8.4).
- Block and tail: `python3 run_pd56.py` evaluates three variants; the text uses `SHARP` (Table 8.2). Output `pd56_results.json`.
- Reproduction: `P/data/pd/regen/regen_all.sh` re-ran all of the above from scratch in a clean directory; `params56.json`, `cells56_sharp.json`, `consts56_sharp.json` and `pd56_results.json` are identical to the stored files except the timing field `secs` of the latter (`regen/compare.log`). `P/code/pd/spot_rerun.py` re-runs cells $0, 100, 200, 311$ and $k = 3, 33$ (`P/data/pd/spot_rerun_log.txt`).
- Tables 8.1–8.4 of the text are produced from the JSON files by `P/code/make_tables.py` (`P/data/tables_pd.md`).

Run time on one core: cells about 20 s, `run_pd56.py` about 75 s for all 16 values of $k$ and three variants.

### A.4 The modular certificate and zero sets

`P/cert/modcert_copy.c` (sha256 `a228eaff1ac42094a28ec7b32ce495e233d3d78007d5ebd33dce4e35482caf08`); compile `gcc -O2 -o modcert modcert_copy.c`; `P/cert/run_cert.sh` runs `modcert k 60000000 p` for all odd $k \le 33$ and $p \in \{2305843009213693951, 4611686018427387847\}$ (outputs `z_k<k>_p<p>.txt`; log `cert_jobs.log`; about 3 min per job, memory below 900 MB). `python3 P/code/zero_sets.py` intersects the two lists, evaluates every candidate exactly, evaluates all allowed $d \le \max C_k$ exactly, and writes `P/data/zero_sets.json` and `zero_sets.md` (Table 8.5, Appendix B.1).

### A.5 Source directories

The programs of A.3 and A.4 are unchanged copies of the following directories of the same repository (`reports/s_family_paths/`): `A56_A49_audit` (block, tail, cell code, modular certificate), `A58_chern_bound_audit` (pair certificate, $Q$ table), `A61_clt_proofs` (written proofs of the lattice-sum, cell, and tail lemmas, `PROOFS.md`), `A59_clt_rederivation` (the independent derivation of Remark 8.9), `A57_gk_31_33` (the cases $k = 31, 33$), `A46_zero_facts_audit`, `A47_composite_zeros`, `A49_gk_runs`, `A50_proofs`, `A54_proofs2`, `A55_small_k_exceptions` (Sections 5–7). The proofs in Part I were written out and checked against these sources.

## B Zero sets of $g_k$

### B.1 Odd $k$, $3 \le k \le 33$

$Z_k = \{d \text{ allowed for } k : \ g_k(d) = 0\}$ (Theorem 8.22). Every element is listed.

$k = 3$ ($\#Z_k = 9$, $\max Z_k = 43$): 3, 4, 6, 7, 15, 28, 31, 40, 43.

$k = 5$ ($\#Z_k = 17$, $\max Z_k = 176$): 5, 9, 10, 15, 16, 19, 20, 21, 24, 31, 35, 41, 46, 71, 86, 161, 176.

$k = 7$ ($\#Z_k = 24$, $\max Z_k = 247$): 2, 7, 14, 16, 18, 21, 28, 32, 35, 36, 42, 43, 44, 57, 60, 63, 78, 79, 85, 100, 107, 135, 191, 247.

$k = 9$ ($\#Z_k = 20$, $\max Z_k = 118$): 7, 22, 25, 27, 31, 36, 49, 52, 63, 64, 67, 70, 73, 76, 85, 88, 91, 97, 112, 118.

$k = 11$ ($\#Z_k = 43$, $\max Z_k = 273$): 3, 5, 11, 14, 22, 26, 31, 33, 36, 37, 38, 44, 49, 55, 60, 64, 66, 69, 70, 71, 75, 77, 80, 82, 88, 92, 93, 99, 100, 102, 104, 110, 111, 126, 133, 135, 136, 143, 147, 148, 166, 196, 273.

$k = 13$ ($\#Z_k = 54$, $\max Z_k = 545$): 3, 10, 12, 13, 23, 26, 30, 35, 39, 42, 43, 49, 51, 52, 64, 65, 77, 78, 82, 88, 91, 94, 95, 101, 104, 114, 116, 117, 120, 121, 130, 143, 144, 146, 147, 156, 157, 159, 160, 173, 179, 183, 195, 205, 212, 222, 224, 256, 269, 324, 334, 480, 519, 545.

$k = 15$ ($\#Z_k = 55$, $\max Z_k = 565$): 6, 10, 15, 16, 21, 30, 31, 34, 39, 45, 55, 60, 61, 66, 75, 76, 81, 90, 96, 100, 105, 106, 109, 120, 126, 135, 150, 154, 165, 180, 184, 186, 195, 199, 201, 210, 211, 216, 231, 241, 246, 255, 259, 276, 286, 291, 324, 369, 376, 384, 414, 436, 460, 520, 565.

$k = 17$ ($\#Z_k = 80$, $\max Z_k = 1137$): 2, 8, 13, 15, 17, 19, 30, 32, 34, 38, 43, 47, 50, 51, 55, 59, 68, 70, 81, 83, 85, 100, 101, 102, 106, 117, 119, 121, 123, 132, 136, 138, 144, 149, 151, 153, 155, 166, 168, 170, 174, 183, 186, 187, 191, 204, 206, 221, 225, 234, 237, 238, 254, 255, 256, 259, 268, 272, 273, 274, 276, 285, 291, 293, 305, 307, 322, 323, 339, 342, 344, 356, 358, 359, 393, 400, 451, 485, 502, 1137.

$k = 19$ ($\#Z_k = 89$, $\max Z_k = 766$): 5, 6, 7, 11, 17, 19, 24, 26, 30, 38, 42, 43, 45, 47, 54, 57, 61, 62, 63, 74, 76, 87, 95, 100, 102, 111, 114, 118, 121, 133, 137, 140, 144, 152, 157, 163, 171, 176, 178, 190, 194, 195, 197, 209, 213, 214, 225, 226, 228, 235, 247, 252, 266, 270, 273, 282, 285, 290, 292, 304, 308, 321, 323, 324, 330, 342, 343, 347, 349, 377, 381, 384, 396, 397, 399, 403, 415, 434, 438, 444, 463, 472, 491, 492, 503, 579, 598, 747, 766.

$k = 21$ ($\#Z_k = 88$, $\max Z_k = 2055$): 7, 15, 18, 21, 25, 28, 39, 42, 51, 58, 60, 63, 64, 67, 78, 84, 85, 88, 91, 105, 109, 121, 123, 126, 127, 142, 144, 147, 148, 165, 168, 172, 183, 189, 196, 207, 210, 211, 228, 231, 232, 246, 252, 273, 274, 289, 294, 295, 310, 312, 315, 316, 336, 337, 357, 373, 378, 394, 399, 420, 436, 442, 457, 463, 483, 498, 499, 520, 526, 561, 582, 603, 610, 624, 639, 645, 660, 681, 702, 723, 862, 1114, 1135, 1141, 1261, 1303, 1534, 2055.

$k = 23$ ($\#Z_k = 105$, $\max Z_k = 837$): 2, 3, 6, 8, 12, 13, 18, 23, 26, 29, 31, 35, 41, 46, 50, 52, 54, 55, 58, 62, 69, 71, 73, 75, 77, 82, 92, 95, 98, 110, 115, 127, 131, 138, 142, 144, 146, 161, 165, 169, 184, 192, 196, 197, 207, 213, 219, 225, 230, 234, 236, 238, 253, 257, 259, 261, 276, 282, 299, 322, 326, 328, 345, 349, 351, 368, 374, 391, 414, 418, 420, 422, 437, 460, 464, 483, 484, 506, 507, 514, 535, 537, 545, 553, 556, 560, 575, 579, 593, 606, 614, 622, 637, 657, 660, 662, 675, 708, 726, 754, 768, 791, 800, 814, 837.

$k = 25$ ($\#Z_k = 76$, $\max Z_k = 936$): 6, 11, 14, 19, 21, 24, 31, 39, 44, 46, 54, 56, 59, 66, 69, 71, 75, 79, 86, 94, 96, 99, 100, 114, 119, 125, 131, 141, 154, 169, 171, 179, 196, 200, 211, 221, 244, 254, 256, 269, 271, 279, 289, 294, 296, 319, 354, 369, 379, 394, 396, 419, 454, 479, 494, 519, 529, 544, 569, 576, 579, 601, 604, 629, 651, 666, 669, 704, 726, 744, 766, 791, 884, 911, 914, 936.

$k = 27$ ($\#Z_k = 68$, $\max Z_k = 1375$): 7, 10, 13, 19, 22, 27, 34, 36, 37, 40, 46, 54, 58, 61, 67, 70, 73, 79, 85, 88, 94, 103, 115, 118, 135, 142, 151, 154, 166, 175, 193, 196, 252, 256, 274, 277, 283, 289, 301, 304, 331, 358, 360, 382, 397, 409, 412, 414, 439, 466, 490, 495, 517, 520, 598, 652, 655, 676, 703, 757, 760, 787, 799, 838, 853, 880, 1143, 1375.

$k = 29$ ($\#Z_k = 121$, $\max Z_k = 1934$): 5, 6, 7, 13, 20, 22, 23, 24, 28, 29, 34, 35, 42, 51, 52, 53, 57, 58, 62, 63, 67, 71, 74, 80, 82, 83, 86, 87, 91, 92, 94, 107, 109, 111, 115, 116, 122, 138, 139, 140, 145, 158, 161, 167, 174, 178, 179, 202, 203, 207, 225, 227, 232, 239, 256, 261, 285, 289, 290, 294, 312, 318, 319, 323, 324, 341, 343, 348, 370, 372, 377, 399, 406, 410, 435, 439, 457, 464, 486, 488, 493, 515, 521, 522, 526, 551, 555, 573, 580, 602, 609, 631, 638, 642, 667, 671, 696, 725, 729, 754, 783, 784, 787, 812, 813, 816, 845, 857, 863, 871, 892, 899, 932, 958, 1002, 1008, 1031, 1060, 1227, 1591, 1934.

$k = 31$ ($\#Z_k = 127$, $\max Z_k = 1712$): 2, 5, 7, 8, 10, 14, 18, 19, 20, 28, 31, 33, 38, 39, 41, 45, 50,

51, 59, 62, 66, 69, 70, 71, 72, 76, 78, 82, 87, 90, 93, 97, 98, 101, 103, 107, 111, 113, 124, 126, 132, 134, 138, 143, 155, 162, 163, 165, 171, 183, 186, 190, 194, 206, 217, 221, 231, 248, 253, 256, 258, 279, 289, 310, 314, 320, 324, 341, 345, 349, 355, 361, 372, 380, 382, 403, 411, 413, 434, 438, 441, 442, 465, 469, 496, 504, 527, 535, 537, 558, 562, 566, 589, 593, 620, 628, 651, 659, 661, 682, 686, 713, 717, 744, 752, 754, 775, 806, 810, 837, 868, 872, 899, 900, 907, 930, 931, 993, 996, 1023, 1027, 1086, 1101, 1163, 1312, 1400, 1712.

$k = 33$ ($\#Z_k = 88$, $\max Z_k = 1587$): 3, 12, 15, 22, 27, 31, 33, 45, 48, 49, 55, 60, 66, 70, 75, 78, 88, 91, 93, 99, 100, 102, 103, 111, 126, 130, 132, 133, 147, 148, 159, 165, 169, 187, 192, 198, 199, 210, 214, 231, 232, 235, 247, 256, 264, 265, 267, 268, 280, 291, 297, 298, 301, 324, 330, 331, 357, 363, 396, 423, 429, 433, 456, 462, 463, 495, 528, 532, 561, 562, 588, 594, 627, 660, 693, 726, 759, 792, 825, 858, 891, 924, 957, 990, 1023, 1056, 1155, 1587.

### B.2 Even $k \equiv 2 \pmod 4$, $2 \le k \le 34$

By Theorem 7.4 and Lemma 7.3 every allowed class is one-sign; $B = \max_c B_c$; the zeros are the allowed $d < B$ with $g_k(d) = 0$ (computed exactly, `code/even_zero_sets.py`).

| $k$ | classes | $B$ | $\#Z_k$ | $\max Z_k$ |
|---|---|---|---|---|
| 2 | 2 | 42 | 4 | 26 |
| 6 | 4 | 186 | 10 | 78 |
| 10 | 6 | 710 | 17 | 120 |
| 14 | 8 | 1778 | 30 | 224 |
| 18 | 8 | 2899 | 19 | 202 |
| 22 | 12 | 6314 | 58 | 528 |
| 26 | 14 | 10166 | 73 | 728 |
| 30 | 12 | 15330 | 59 | 960 |
| 34 | 18 | 21998 | 108 | 1224 |

## C Reproducibility for Part II

All computations were run with Python 3.13.5, python-flint 0.9.0, sympy 1.14.0, mpmath 1.3.0, numpy 2.5.4 and cypari2 2.2.4 (PARI/GP, used only where stated). The scripts, logs and data files are kept in the companion folders below (under `quantum-comp/reports/heavy-daily/`); each folder contains a report and the scripts it describes.

| folder | content |
|---|---|
| `2026-10-17-bridge-c-residual-law`, `2026-10-17-P_s-structure` | Theorems 11.1–11.4; radial fits; Gauss pattern |
| `2026-10-17-psi-cocycle`, `2026-10-17-psi-conditions` | lift, $S/T$ law, period formula, row closed forms |
| `2026-10-17-cusp-part`, `2026-10-17-psiEis-Lvalues`, `2026-10-18-PARI-free` | Eisenstein/cusp split, $L(0)$ identities, PARI-free certification |
| `2026-10-18-ER-cancellation`, `2026-10-18-ER-cancellation-audit` | Herglotz closed form (Theorems 11.9, 11.10) |

| folder | content |
|---|---|
| `2026-10-18-Audit-and-ProfileLaw`, `2026-10-18-S-Eigenvalues`, `2026-10-18-V42-Structure` | profile law; spectra; function span |
| `2026-10-18-Tensor-Pattern-Proof`, `2026-10-18-Tensor-Pattern-Closure`, `2026-10-18-General-s-Tensor`, `2026-10-18-Symmetry-Audit` | tensor theorems, dimensions, label operators |
| `2026-10-18-Cusp-Onset`, `2026-10-18-Cusp-Onset-Closure`, `2026-10-18-uL-Criterion-Closure`, `2026-10-18-s8-s11` | $u_L$ criterion; $s = 8, 11$ |
| `2026-10-18-H_s-Closed-Forms`, `2026-10-18-H_s-Closure`, `2026-10-18-Open-Analytic`, `2026-10-18-Analytic-CatB-Push` | Euler–Maclaurin; $H_s$ closed forms; Gauss data; $t^{11/2}$–$t^{17/2}$; data files of the closed forms |
| `2026-10-18-Algebraic-Round3`, `2026-10-18-Open-Algebraic` | $h$-relations; Proposition K1; Weil inputs; cone and axes cone |
| `2026-10-19-HighRAM-Items` | prime-square splits and commutants (`sec_comm2.py`, transpose convention; $s = 49$ certificate at two primes), $p$-local invariants at $s = 121, 169$ (`famfast.py`, `pinv2.py`, `pinv3.py`) |
| `2026-10-19-Open-Items-Pursuit` | Proposition 20.8 (`item7_k1_sharp.py`); Remark 27.15 (`dihedral_enum.py`, `dihedral_diag.py`, `dihedral_gen.py`); Proposition 26.10(iii) (`item6_candidate.py`); dictionary check in Remark 27.16(a) (`item4_dictionary.py`); Lemma 27.5 checks (`item3_estimates.py`) |
| `2026-10-19-Certified-SRho` | Theorem 27.13 (`certify.py`; outputs `cert_*.json`) |
| `2026-10-18-C_third-Unified`, `2026-10-18-C_third-Closure`, `2026-10-18-C_third-modularity`, `2026-10-18-Cocycle-Closure` | §24; Theorem 15.3 |
| `2026-10-18-Zeta-Rational-Curve-Proof`, `2026-10-18-g_s-Closed-Forms` | Theorems 25.1, 25.2 |
| `2026-10-18-AtkinLehner-Local` | Proposition 21.5 |
| `2026-10-19-Implications-checks` (incl. `k3_140`) | §26; coefficient values of $C$; $A_5$ group closure |
| `2026-10-19-N1-and-Uniqueness` | §27 (sectors, $q$-coefficients to frequency 40, $g_0$); §28 |
| `2026-10-19-Audit-U2-U3-ThmG-SRho` | numerical checks of Theorem 11.7 on both half-planes, Lemma 15.1 quadrature, Theorem 27.4, sector bounds |

To reproduce, run each script with the stated Python environment from its folder (e.g. `python3 n1_tests.py 1500`); scripts write their JSON or log outputs next to themselves. Exact computations are deterministic; floating-point runs agree to the stated tolerances.

# D Category B record for Part III

Everything in this appendix is Category B: floating-point evaluation, least-squares fits, or numerical ranks of real or complex matrices by singular value decomposition with relative threshold $10^{-9}$.

None of it is used in a proof. Data, scripts and logs are in `reports/bridges/BRIDGE_C_35/` and its subfolders `order2`, `order3`, `order3b`, `order3c`, `order4a`; every run was preregistered with a SHA-256 hash recorded before the run.

D.1 *Leading order on the series.* $t\,C(\mathbf{e}(y)e^{-t})$ was fitted at 30 rational points $y$ using up to $3.4\cdot10^6$ lattice points. On a first grid ($t \ge 10^{-3}$ scale) 19 points agree with $I\,G(y)$ to $\le 2\%$ (most $\le 0.1\%$); on $t \in [3.1\cdot10^{-6}, 10^{-4}]$, 10 of the remaining 11 agree to $\le 0.7\%$ and $y = -\frac{5}{12}$ differs by 6.4%. The remaining discrepancies occur at denominators $\ge 9$, where the constant in the $O(t^{1/2})$ term of Theorem 34.6 is large.

D.2 *Order $t^{-1/2}$ on the series.* On grids $t_j = 2^{-j}\cdot 0.5/(297.5\,(2m)^2)$, scaled to the denominator and to the long cone edge $Q(21,-1) = 297.5$, the fitted $t^{-1/2}$ coefficient of $q^{-7/20}C$ agrees with $A_1$ of Theorem 35.6 to relative error $\le 10^{-5}$ at $x = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{5}{6}, \frac{3}{8}, \frac{2}{9}, \frac{3}{10}, \frac{1}{5}, \frac{1}{7}$, and to $\le 7.5\cdot10^{-6}$ at $x = 1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}$ in an independent run. Grids scaled to the denominator only ($t(2m)^2 \le 0.08$, so $t\,Q(u_1)(2m)^2$ up to 24) are outside the regime and do not reproduce $A_1$ at denominators $2, 3, 4, 6, 8, 9$.

D.3 *Order $t^0$ on the series.* On the same cone-scaled grid ($j = 0, \dots, 7$, fit on $1, t^{1/2}, t, t^{3/2}$), the fitted constant term agrees with $A_2$ of Theorem 35.6 at $x \in \{1, \frac{1}{2}, \frac{1}{3}, \frac{2}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}, \frac{3}{7}\}$ with maximal relative error $1.4\cdot10^{-7}$; for example $A_2(1) = -0.3232819 - 0.4449593i$, $A_2(\frac{1}{4}) = 2.4890282 + 0.2339631i$, $A_2(\frac{1}{2}) = 0$ to $10^{-15}$. With the sign of the corner term $\mathsf{C}$ reversed, the formula disagrees by relative errors 0.25 to 2.92 at the seven points where $|\mathsf{C}| > 0.05$.

D.4 *Values.* $|A_1(\frac{1}{2})| = |A_1(\frac{5}{2})| = 0.44311346272638\ldots = \sqrt{\pi}/4$ to 15 digits. The piece decomposition reproduces $A_0 = I\,G_{\alpha,\beta}$ to $1.4\cdot10^{-17}$ in 63 cases.

D.5 *No single-weight law at order $t^{-1/2}$.* Let $\mathbf{A}_1(x)$ be the 105-vector of $A_1$ over the companions, at 138 rational points. With $B_w(x) = |x|^w\mathbf{A}_1(-1/x)$: $\operatorname{rank}\mathbf{A}_1 = 105$ and $\operatorname{rank}[\mathbf{A}_1 \mid B_w] = 138$ for $w = \frac{1}{2}$ and $w = 1$ ($\sigma_{\min}/\sigma_{\max} \approx 0.013$). A law $\mathbf{A}_1(-1/x) = |x|^{-w}M\mathbf{A}_1(x)$ would force rank $\le 105$. The same pipeline applied to $\mathbf{A}_0$ gives rank 36 and joint rank 36 at $w = 1$ (and 72 with the wrong sign of the exponent).

D.6 *Parts of $A_1$.* With the same test on 216 points (ranks; joint ranks at $w = \frac{1}{2}$ / $w = 1$): $D$: 58; 116/98. $R$: 105; 210/210. $W[1]$: 90; 178/156. $W[\frac{7}{2}]$: 60; 116/120. $W[\frac{595}{2}]$: 100; 200/200. $W$: 105; 210/210. None of the parts of Proposition 35.11 closes.

D.7 *Regrouping by quadratic class and by characteristic.* Groups $W[1]$, $W[\frac{7}{2}] + R[\frac{7}{2}]$, $W[\frac{595}{2}] + R[\frac{595}{2}]$, $D$: ranks $90, 60, 100, 58$, joint ranks at $w = \frac{1}{2}$: $178, 116, 200, 116$. Restricted to the characteristic blocks $\Theta_1 = K2 \cup K3$ (70 columns) and $\Theta_2 = K4$ (35 columns), $A_1$ has ranks 70 and 35 with joint ranks 140 and 70. No group closes. $A_0$ restricted to $\Theta_1$ and $\Theta_2$ has ranks 24 and 12, each closed.

D.8 *Sawtooth shadow candidate.* For $A_1 - \sum_\lambda \varepsilon_\lambda W[\lambda]$, $\varepsilon \in \{0,1\}^3$, the ranks are $105, 91, 104, 90, 105, 87, 103, 58$ for $\varepsilon = 000, 001, 010, 011, 100, 101, 110, 111$ (order $W[1], W[\frac{7}{2}], W[\frac{595}{2}]$), with joint ranks at $w = \frac{1}{2}$ equal to twice the rank except 170 for 101, and at $w = 1$ equal to twice the rank except 156 for 101 and 98 for 111. No choice closes.

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*Note on citations.* The cited statements of Nikulin (Thm. 1.14.4) and Morrison (Thm. 2.8, Cor. 2.9(i)) were checked against the original publications; those of Deitmar–Hilgert (Thms. 3.3, 4.4) against the arXiv version, whose numbering is used here.